Schindler 3100/3300/3600/5300/6300
Bionic 5, Rel. 04/05/06/07/09
Bionic 6, Rel. 01

MMS Configuration and Diagnostics
Quick Reference K608208_11
Edition 02-2015

Schindler

Prepared	01.02.15	zurbrure	Lead Office	Classification	Document Owner
Reviewed	02.02.15	nicolaca	EIM	11200	INVENTIO AG CH-6052 Hergiswil
Norms chkd					
Released	07.02.15	staudema			

Modification	01	08	09	10	11
KA No.	107095	107389	107427	107430	252038
KA Date	20.05.05	13.12.09	23.12.10	05.02.11	06.02.15

Contents

1 General Information

1.1 Safety

All persons involved must know and follow all company and local safety regulations. Protective clothing and appropriate safety equipment must be worn.

Safety Equipment

 Safety clothes	 Hardhat or hardcap	 Safety shoes	 Protective gloves
 Safety goggles	 Hearing protection	 Full body safety harness	 Dust mask

DANGER

Hazardous Voltage (during maintenance or diagnostic work)

Contact with live parts will result in electric shock causing serious injury or death.

Take the following precautions during your work:

- Do not touch live parts.
- Switch off the main switch and de-energize the installation completely before removing any protective covers or before starting to work on power components related to voltage > 50 V or heavy currents.
- When the maintenance work is completed, make sure that all the protective covers are back in place.

DANGER

Hazardous voltage

Contact with live parts will result in electric shock

Switch off the main switch and wait at least 10 minutes before starting to work on the installation.

DANGER

Hazardous capacitive energy

Contact with live parts of a power circuit with charged capacitors will result in electric shock.

– Wait at least 10 minutes after switching off the power for a full discharge of the capacitors.
– Keep the capacitive circuit grounded while working on the installation.

WARNING

Unsafe wiring practices

Electrical interconnections of insufficient quality affect safe elevator operation.

- Follow the provided mandatory grounding and shielding instructions.
- Connect only a single wire to each terminal.
- Do not tighten the grounding clamp used to ground the cable shields too much.

NOTICE

Exposure to electrostatic discharge (ESD)

Exposure to ESD destroys ESD-sensitive components.

Strictly follow the ESD-safe procedures when handling ESD-sensitive components.

1.2 About this Quick Reference

The concept of this quick reference is to provide the **service technician (who has attended training before)** with appropriate information to make configuration and diagnostic work easier.

1.2.1 Copyright and Use of this Booklet

The Service Technician is obliged to keep the manual secret and not to disclose it to any third party and to protect it accordingly. The Service Technician is also obliged to return the manual to his line manager whenever requested.

1.2.2 Application Range of this Booklet

This document describes only user interfaces and tools, which are available for the normal service technician. Therefore this document does not describe the use of the service computer (CADI) for example.

1.2.3 Released and available Options

This quick reference guide describes the system as it is delivered to the field.
Some of the available options are not officially released for sales. For the released and available options please refer to the **Product Data Sheets** K 609826, K 609827, K 609828, K 609829 and K 43401267.
For details of new features, enhancements and bug fixes, see J 42106494 Bionic SW release notes and information.
The manual describes the Schindler 3100/3300/3600/5300 and the Schindler 6300 as **delivered in EU** and in those countries which are **supplied by EU**.

1.2.4 Further Information and Support

This quick reference does not claim to include all possibilities. Further information about the Schindler 3100/3300/5300 and 6300:

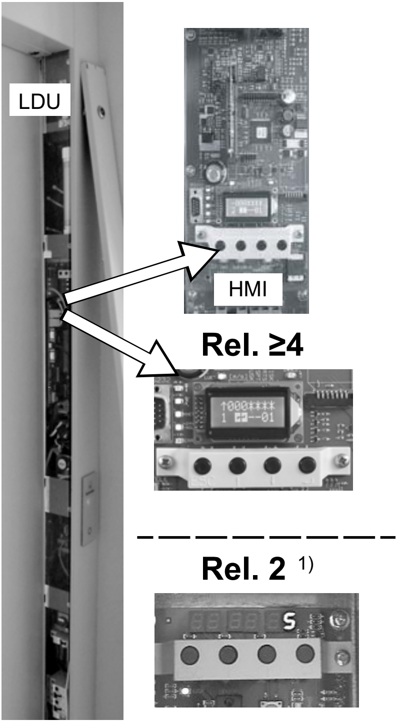
Intranet:

Intranet Product Navigation Center PNC (for technical catalog, specifications etc.: http://pnc.ebi.schindler.com “Elevator Systems” > “Commodity”
Schindler Intranet: http://intranet.eu.schindler.com “Products” > “Elevators Europe” > “Global Commodity Program”

Hotline:

For questions about Control and Electrical Drive: **Hotline Locarno** Schindler Electronics Ltd., Via della pace 22, 6600 Locarno, Switzerland Tel.: +41 91/756 97 85, Fax: +41 91/756 97 54 e-mail: hotline.locarno@ch.schindler.com

1.3 HMI Positions



HMI versions [402002; 02.12.14]

1) Not supported anymore in this document version.

For details about Rel. 2, refer to K 608208 / Version 10.

1.4 OEM Functions (≥ V9.8x)

1.4.1 OEM Levels and Descriptions

OEM software defines three levels of access in the HMI.

Level [0] = Access to adequate maintenance information for third parties and OEM.
Level [1] = Access for third parties and OEM.
Level [2] = Access only for OEM specialist.

OEM Enabling: Enabling means the elevator software starts a countdown of 10K trips before activating the defined levels on the elevator control.

OEM Activation: Activation means the elevator software has introduced the defined levels on the elevator control.

Process: There are two ways to activate the OEM functions on the elevator control:

- **Option 1:** Via a flag set on the SIM card. If this flag is set, the software will count down 10K trips then automatically activates the defined levels on the elevator control.
- **Option 2:** Without setting the flag in the SIM card (no exchange of SIM card), the SPECI tool has a command with two steps to activate the OEM functions:
  - Step 1: enable only → start the 10K trip count, then automatic activation.
  - Step 2: enable AND activate → by-pass the 10K trips and activation is instantaneous.

For option 2, enabling and activating the OEM functions can only be done with the SPECI tool connected. For SPECI on iPhone, refer to section 3.2.
CADI cannot enable or activate OEM functions.

1.4.2 OEM Level Accessibility

For details about the different access OEM levels and their content, refer to the following sections:

- 4.9 Special Modes / Commands
- 5.3.3 Parameter List - Detailed Description

1.5 PCB Overview

PCBs used in Bionic 5 Releases

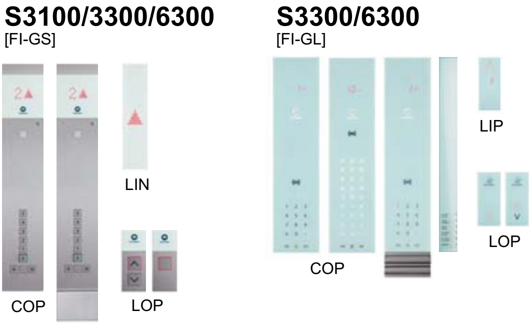
PCB	Rel.					PCB	Rel.				
	4	5	6	7	9		4	5	6	7	9
LDU						**CCU**					
SMIC6x	X					SDIC5x	X	X	X	X	
SMICE6x	1)	X	X	X		SCMI2x					X
SMICE7x					X	SCCI3x					X
SCPU	X	X	X	X	X	SIEU		X	X	X	
SEM2x	X	X	X	X		SUET3	X	X	X	X	X
SEM3x					X	TAM2	X				
CLSD	X					ETMA-CAR		X	X	X	
ETMA-MR		X	X	X	X						

1) possible as spare part

PCBs used in Bionic 6 Rel. 01

PCB	Rel.	PCB	Rel.
	01		01
LDU		**CCU**	
SMICFC	X	SDIC5x	X
SMICHMI	X	SIEU1.Q	X
SCPU	X	SUET3	X
ETMA-MR	X	TAM2	X
		ETMA-CAR	X

1.6 Fixtures



Examples of used Fixtures [402029; 01.12.2014]

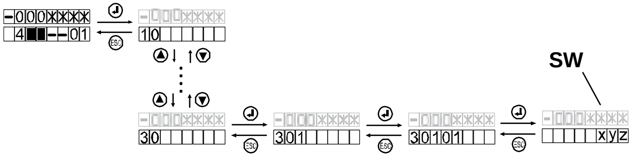
The above showed examples are just a few possible representations.
For detailed information concerning the actual COPs, LOPs, LIPs etc. refer to the TK TICO FI-GS EJ 41320445.

1.7 SW and HW version

HW/SW	Bionic 5 ≥ Rel. 04
Software	– Delivered since January 2008: Software version **9.34** – Delivered since January 2010: Software version **9.53** – Delivered since September 2011: Software version **9.74** – Delivered since December 2011: Software version **9.82** – Delivered since December 2012: Software version **9.85** – Delivered since May 2013: Software version **9.94** – Delivered since December 2013: Software version **10.02 (10.03)** – Delivered since June 2014: Software version **10.06** – Delivered since October 2014: Software version **10.07**
Hardware	Main components: See section 2.2 and 7 (PCBs)

HW/SW	Bionic 6 Rel. 01
Software	Not yet released
Hardware	Main components: See section 2.3 and 7 (PCBs)

How to identify the software version of the system?



[37971; 01.12.2010]

The software version can be read with help of the user interface HMI:

– With help of system info, **Menu 30**, **submenu 301 > 30101** (= SCPU SW version) (see section 9.3).
– Or with the configuration **Menu 40, CF=12, PA=1** (description section 5.3)

The software version of the ≥ Rel.4 system is also displayed on the HMI during every start up.

1.8 Remote Monitoring

A properly connected unit through Telemonitoring (e.g. ETM/ETMA) allows the following benefits:

– To get specific troubleshooting hints via Fieldlink even before the arrival on-site and whenever starting a Job-activity (CBK, MNT, REP).
– To get remote intervention; for example having an elevator expert remotely connected via CADI-GC or programming SIM cards through the KW's services.
– To get remote support because all technical details and the whole elevator's history are available in the tools suite of remote monitoring.

1.9 Documentation and Software

For more details refer to:

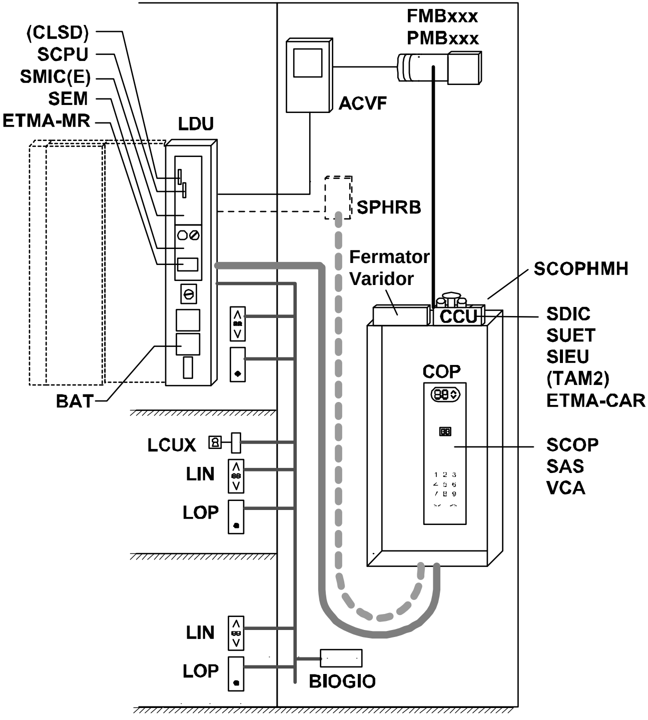
– EJ 604639 - Control Bionic 5 / Bionic 6 Diagnostics
– EJ 604620 - Control Bionic 5 / Bionic 6 Commissioning

2 System Overview

2.1 BIC5 - Rel. 2 Main Components / Bus System

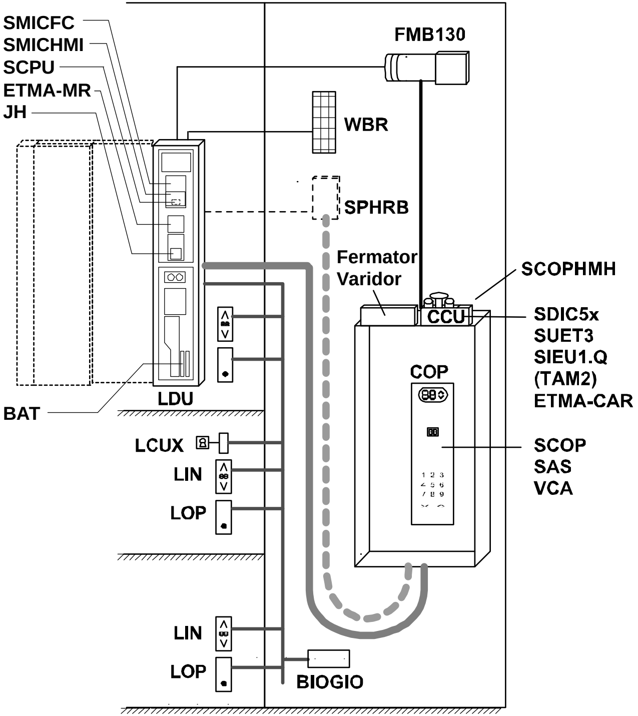
For details about Rel. 2 refer to K 608208 / Version10.

2.2 BIC5 - Rel. ≥ 4 Main Components



Bionic 5 Rel.4/5/6/7/9 [402004; 11.08.2014]

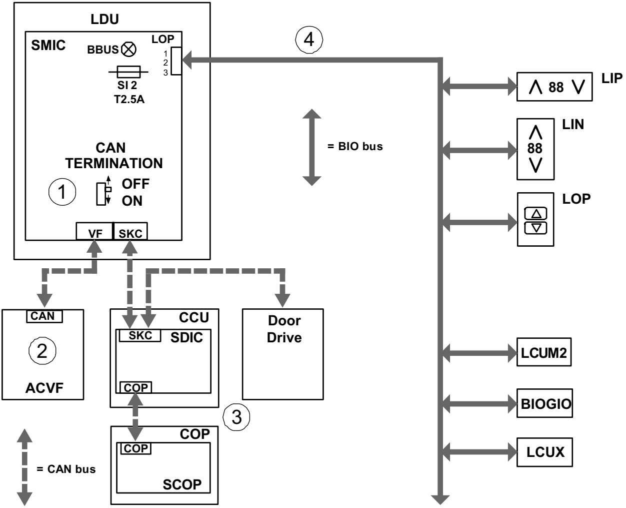
2.3 BIC6 - Rel. 1 Main Components



Bionic 6 Rel.1 [402011; 09.09.2014]

2.4 Bus Systems

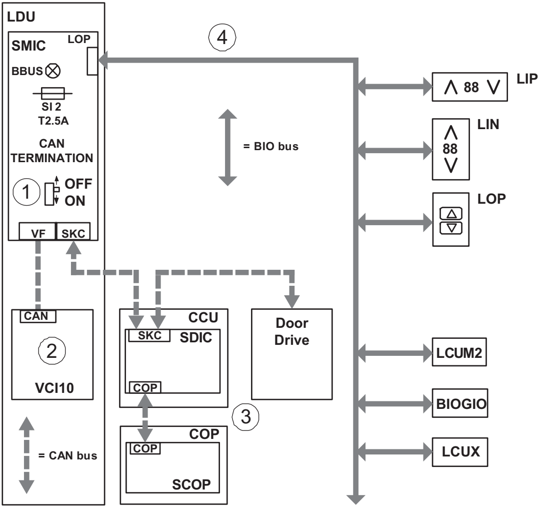
2.4.1 Bionic 5



[402016; 13.10.2014]

1) CAN bus termination switch on SMIC:
   Default position: “OFF”. (ACVF and CCU connected). If CCU is not connected → set switch to “ON” (bus termination on SMIC, for example during installation travel mode.)
2) CAN bus is terminated automatically in ACVF.
3) CAN bus is terminated automatically (either on CCU or on COP)
4) BIO bus 24 V for LOPs/LINs, LIPs and peripherals (LCUX, BIOGIO, etc.)

2.4.2 Bionic 6



[402017; 13.10.2014]

1) CAN bus termination switch on SMICFC:
   Default position: “OFF”. (ACVF and CCU connected). If CCU is not connected → set switch to “ON” (bus termination on SMIC, for example during installation travel mode.)
2) CAN bus is terminated automatically in ACVF.
3) CAN bus is terminated automatically (either on SDIC or) on SCOP
4) BIO bus 24 V for LOPs/LINs, LIPs and peripherals (LCUX, BIOGIO, etc.)

2.5 SIM Card (Chip Card) Options

Remarks:

– To check the options which are available in the system (active on the SIM card): Check the cover page of the schematic wiring diagram.
– The table below shows all options which can be ordered for Schindler 3100/3300/3600/5300 and Schindler 6300.
– A description of most functions (Elevator Systems Standards ESS) can be found on the Intranet, Product Navigation Center.

2

Option	Description	SW
	Basic Functions	
Policy	1 = KA	
	2 = PI	
	3 = KS	
	Fire Service Functions	
BR1	Fire service type 1 Standard	1.0
BR1_LUX	Fire service type 1 Luxemburg (KBFH1,2)	2.1
BR1_NOR	Fire service type 1 Norway (KBF)	5.0
BR1_CH	Fire service type 1 Switzerland (JBF/JBF-A, KBF)	7.1
BR1_MAR	Fire service type 1 Marine (JBF)	9.0
BR1_CN	Fire service type 1 China (JBF)	9.0
BR1_KR	Fire service type 1 Korea (JBF)	9.0
BR1_TW	Fire service type 1 Taiwan (JBF)	9.0
BR1_GB	Fire service type 1 United Kingdom (JBF)	9.0
BR1_HK	Fire service type 1 Hong Kong (JBF)	9.3
BR1_SG	Fire service type 1 Singapore (JBF)	9.3
BR1_MY	Fire service type 1 Malaysia (JBF)	9.3
BR1_AU	Fire service type 1 Australia (JBF)	9.3
BR1_NZ	Fire service type 1 New Zeland (JBF)	9.3

Option	Description	SW
BR1_JP	Fire service type 1 Japan	9.9
BR1_8173A	Fire service type 1 EN8173Type A	9.0
BR1_8173B	Fire service type 1 EN8173Type B	9.3
BR1_8173C	Fire service type 1 EN8173Type C	9.3
BR2	Fire service type 2 Standard (JBF)	9.0
BR2_FR	Fire service type 2 France (JNFF)	1.1
BR2_NL	Fire service type 2 Netherlands (JNFF, KBF)	8.1
BR2_CN	Fire service type 2 China (JBF, JNFF)	9.0
BR2_SG	Fire service type 2 Singapore (JBF, JNFF)	9.3
BR2_HK	Fire service type 2 Hong Kong (JBF, JNFF)	9.3
BR2_8172UK	Fire service type 2 United Kingdom (JBF, JNFF)	9.8
BR3	Fire service type 3 Standard (JBF, JNFF)	8.1
BR3_IN	Fire service type 3 India (JBF)	2.1
BR3_BEL	Fire service type 3 Belgium (JBF, JNFF)	5.0
BR3_KR	Fire service type 3 Korea (JBF, JNFF/ JNFF-S)	9.0
BR3_TW	Fire service type 3 Taiwan (JBF, JNFF)	9.2
BR3_AU	Fire service type 3 Australia (JBF, JNFF/JNFF-S)	9.1
BR3_AU_B	Fire service type 3 Australia Type B (JBF, JNFF/JNFF-S)	9.38
BR3_RUS	Fire service type 3 Russia (JBF, JNFF)	9.7
BR4_HK	Fire service type 4 Hong Kong	9.3
BR4_MY	Fire service type 4 Malaysia	9.3
BR4_NZ	Fire service type 4 New Zealand	9.3
EBR1	Fire on floor (Fire sensor, LCUX required, only in combination with BR2)	6.1

Option	Description	SW
	Signalization	
CPIF (ASE)	Car position indicator on main floor (CF2 PA2)	1.0
CPIAF (ASE)	Car position indicator on all floors	4.2
GA	Car gong type B	4.2
PA1	Alarm by horn	1.0
PA2	Alarm by horn	1.0
PA4	Alarm by horn	9.3
PA5	Alarm by horn	9.3
TDIF (LW, LA)	Travel direction indicator on all floors	1.0
VA	Voice announcement (Requires “Kit voice announcer”)	1.1
VS_D	Voice announcement door	9.3
VS_DIR	Voice announcement direction	9.3
VS_ALARM	Voice announcement alarm	9.3
VS_OL	Voice announcement overload	9.3
VS_RLAB	Voice announcement “out of service”	9.3
VS_BR	Voice announcement fire	9.3
VS_RNO	Voice announcement emergency power	9.3
VS_EQ	Voice announcement earthquake	9.3
DM236	Door opening gong	8.3
	Security	
ZB_LA	Restricted egress	9.3
ZB1	Pin code for restricted access (COP with telephone keypad) (not together with GS on the same floor) (CF10 or CF41 PA1. And CF06 PA3)	1.0
ZB3	Restricted access key JDC (02) (CF05 or CF55. CF83 or CF41 PA2)	8.3

Option	Description	SW
ZBC1	Parallel card reader interface, key	8.3
ZBC2	Parallel card reader interface (Not possible with ZZ2 or ZZ3.)	9.2
ZBCE	Parallel card reader interface, key	9.0
GS	Visitor Control (only simplex, PI, not together with ZB1 on the same floor, needs ZB3 or SAS) (CF17)	8.1
	Capacity	
KL-V	Full load control (only KA or KS)	1.0
RL1	Return to main floor from any floor (CF2 PA3, 4)	1.0
RL2	Return to main floor from floors underneath (CF2 PA3, 5)	1.0
	Comfort	
Duplex	Duplex	
JLC (RLC-A)	Automatic car light (Relay RLC-A) (CF8 PA2)	1.1
BEA	Floor light control (LCUX required) (RFBE: BMK=213)	1.1
VCF	Distribution of free cars (Moscow only. Not possible together with RL1 or RL2)	9.7
VEC	Fan in car type E (DVEC)	9.7
	Special Transport	
BF	Service for Disabled Passengers	9.3
RV1	Independent service without parking (Reservation, JRVC: BMK=59)	1.1
	Emergency	
NF1	Emergency service	9.0
NS21 (type C)	Emergency power operation (1.6 m/s only. Not possible with BIOGIO (GUE/GLT), not possible with TSD systems.)	7.1

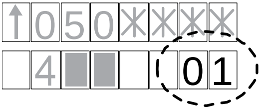
Option	Description	SW
EB	Earthquake standard	9.34
EB (NZ)	Earthquake service New Zealand	9.34
EB (JP)	Earthquake service Japan	9.9
SAFE EXIT	-	9.9
	Door Mode	
DDC	Car call cancellation	9.2
DH	Door hold	2.1
ZZ1	Two car entrances with parallel door operation	9.7
ZZ2	Two car entrances with selective door operation of doors	9.7
ZZ3	Rear openings (interlocked)	9.7
	Door Lock Monitoring	
DLM	Only in RU or UA. According Russian PUBEL code. Not possible with TSD.	9.7
	Misuse	
FT	Final timer, door nudging	9.2
AN1	Car call canceling in empty cabin: ("minimum load")	7.1
AN3	Car call canceling after stop of empty call: ("door protection")	9.0
	Maintenance	
E-RE	Extended inspection service (1.6 m/s only)	9.2
	Miscellaneous	
ASMTL_EMIA	Approach Speed Monitoring at Terminal Landing	9.3
ASMTL_SG	Approach Speed Monitoring at Terminal Landing	9.3
C_xx	xx = Country code	9.3

Option	Description	SW
DRIFTING MITIGATION	See CF=02, PA=19, 20	9.8
Equipment Number	-	9.31
E_RE	Extended inspection to CBD	9.2
GS		8.1
GSC	Car passing a level	9.3
ID	Commissioning number	1.0
ISPT_ Monitoring	Safety chain ISPT (information blocking door) monitoring	9.91
JAB	Out of service (BMK=49)	9.0
KB_KB1	Brake contact monitoring	10.01
LI	Attendant service	9.0
LIFD	Blind floor	9.2
LPC	Low power consumption	9.8
LUB	Maintenance indicator	9.0
LW_A	Next travel direction	9.3
LIFD	Long interfloor distance. Needs blind floor kit. (CF26)	9.2
OEM	Activation of OEM functions	9.8
RetainerPlus	Retainer monitoring	10.01
ROPELC	Rope load compensation	9.3
RVC	Independent service without parking	1.1
RVPC	Independent service with parking	9.0
SAS	Transponder	6.1
SR	Sprinkler recall	9.3
STMM	STM monitoring (including the retainer monitoring)	10.0

Option	Description	SW
SYSTEM TYPE	Traffic control policy	1.0
TT	Car partition door	7.1
	ACVF parameters	
ACVF	System specific parameters (Always)	8.5

2

2.6 Elevator Status (on HMI)



The system status shows the current control status or mode (current service running).

	Description
00	**Out of service operation** The elevator is out of service (SAB or JAB). – Check whether JAB is switched on (LCUX, BIOGIO or LOP). – Check menu 10 > 108 – To reset SAB (remote JAB via Servitel) use menu 10 > 114
01	**Passenger travel operation** The elevator operates in normal mode.
02	**Independent operation** The elevator is in independent service operation (reservation, for example JRVC)
03	**Fire operation** The elevator is in fire recall operation (for example JBF)
04	**Firefighter operation** The elevator is in firefighter operation
05	**Emergency power operation** The elevator operates on emergency power with load measuring disabled.
06	**Earthquake operation** Earthquake service has been activated following the detection of an earthquake condition.
08	**Sprinkler operation** Sprinkler service has been activated following activation of the sprinkler system.

	Description
10	**Attended passenger travel operation** Attendant service has been activated: The elevator operation is controlled by the attendant in the car.
11	**Passenger travel operation without load monitoring** The elevator operates normally (normal operation) but with load measuring disabled.
13	**Power saving mode**
16	**Hospital emergency travel** (SW ≥ V11.0)
19	**Passenger travel full load service** (SW ≥ V11.0)
20	**S in car alarm passenger travel operation** (SW ≥ V11.0)
29	**Move around operation** The elevator performs simulated traffic (move around).
37	**No operation due to stop in car** The elevator is blocked after an emergency stop initiated by passenger action in the car.
39	**No operation due to car overload** The car is blocked after an overload condition lasting for more than 30 seconds.
40	**No operation due to invalid configuration data** The elevator is blocked because the elevator control detected invalid configuration data, for example SIM card missing or defective.
42	**No operation due to invalid load configuration** (SW ≥ V9.83) The elevator is blocked because the configuration of the load measurement is invalid or missing (for example, car load measurement device not calibrated).
44	**Out of service from remote**
45	**Out of service STM monitoring failure** The elevator is blocked for automatic operation due to failure of the STM insufficient residual strength monitoring.

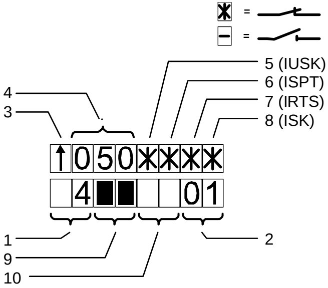
	Description
51	**Installation travel** The elevator operates in Installation Travel mode, also known as Montagefahrt (activation by special command on HMI)
52	**Configuration mode** The elevator operates in Configuration mode (activation by use of the HMI).
53	**Inspection machine room** The elevator operates in Recall Travel mode controlled from a person in the machine room or at the LDU.
54	**Inspection top of car** The elevator operates in Inspection Travel mode controlled from a person on the car roof. (May be indicated during start up of the elevator.)
57	**Test travel** The elevator operates in Test Travel mode, also known as KFM (activation by use of the HMI).
58	**Test mode** The elevator operates in “automatic acceptance test” mode
59	**Learning travel** The elevator operates in Learning Travel mode.
60	**Inspection preparation travel** An automatic car positioning is in progress to provide easy access to the car roof.
61	**Overspeed governor reset travel** The elevator is in a specific overspeed governor reset travel operation which allows manual resetting of the tripped car overspeed governor.
70	**Elevator recovery** An elevator recovery is in progress after a recoverable error. (Or the elevator tries to recover.)
71	**Elevator temperature recovery** An elevator recovery is in progress after an overtemperature condition (hoisting motor, door motor).

	Description
72	**Elevator car position recovery** An elevator recovery is in progress after a car position error (synchronization, ASMTL, etc.) (If the car is not moving, check for the possible error. For example faulty CAN bus connections.)
73	**Elevator door position recovery** An elevator recovery is in progress after a door error or door heart beat error.
75	**No operation due to safety chain open at ISPT** The elevator is blocked because of the interruption of the safety chain at ISPT.
80	**Stop switch** The elevator is blocked after an emergency stop initiated by pressing any stop switch.
97	**Elevator breakdown persistent limited operation** (SW ≥ V10.0)
91	**Elevator startup** (SW ≥ V9.5)
98	**Elevator breakdown** The elevator is blocked following a fatal error. Unblocking after a manual reset or power cycle. A previous evacuation trip was possible.
99	**Elevator breakdown persistent** (SW ≥ V9.72) The elevator is permanently blocked. Unblocking after a special on-site procedure. A previous evacuation trip was possible.

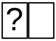
3 User Interfaces

3.1 User Interface HMI

3.1.1 Display during Normal Operation

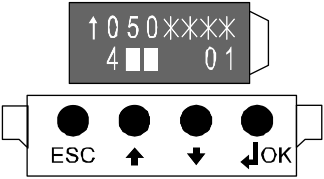


[402005; 12.08.2014]

1	**Number = Actual floor level of the car**			
	SW ≥ V9.7: If the floor position is unknown, the rough position of the car in the hoistway is displayed.			
	Symbol	Car Position	KSE-U	KSE-D
		Invalid	0	0
		Above KSE-U	0	1
		Between KSE-D and KSE-U	1	1
		Below KSE-D	1	0

2	Current control status / mode (Current service running. Information can be found in section 2.6)	
3	Travel direction of the car (up or down), “-” = not defined	
4	Actual car speed [0.01 m/s]	
5	Safety circuit, virtual LED IUSK	[*] = closed [_] = open IUSK blinking = short circuit in safety circuit
6	Safety circuit, virtual LED ISPT	
7	Safety circuit, virtual LED IRTS	
8	Safety circuit, virtual LED ISK	
9	Door status door 1	Detailed description of door status: See section 3.2 (description of SPECI)
10	Door status door 2	

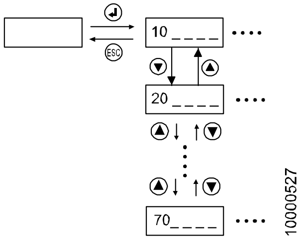
3.1.2 Basic Operation



[402006; 12.08.2014]

Button	Function
ESC	Go one menu level up. Leave menu/item (without saving anything)
UP/DOWN	Move within the menu (change the menu). Change the value
OK (“Enter”)	Go one menu level lower. Confirm the entered value

3.1.3 Main Menu Structure

	To enter the main menus ([10_ _ _ _] ... [70_ _ _ _]): – Press the “OK” button – Choose the menu with the “UP” and “DOWN” buttons

For more information concerning the detailed menu structure and its functions refer to chapter 10 “Main Menu Structure”.

3.2 SPECI on iPhone

SPECI (=Schindler Personal Elevator Communication Interface).

Since SW Version ≥ V9.8x. iSPECI is used with the iPhone.

The app can be downloaded from the Schindler app catalog.

For more details about the management of the app, see the “User Manual” or ask your local SPECI administrator.

> **i** For OEM functions, see section 1.4.

4 Diagnostics and Replacement

4.1 Troubleshooting Procedure

- Do NOT start the diagnostics with resetting the system!
- Do as many as possible checks before a reset to get information about the cause of the problem.

For the diagnostic the below listed sequence is recommend.

1	Before starting
	– Ask the customer about the behavior of the elevator – Check the system by yourself. (If possible: Landing calls, car calls, key switch function, indicators, noise, ...) – If FieldLink and Remote Monitoring is used, check if there are pending “RM Symptoms” (in Fieldlink) linked to the job-activity before arriving on site in order to organize the spare parts in advance.

2	Power Supply and Safety Circuit	Section
	Check the general power supply – Switches (in the lower part of the LDU) – Fuses (SMIC, SEM, Fermator) – LED indication for power supply	4.2 / 4.3
	Safety circuit – HMI indication	4.4

3	Special Mode Active?	Section
	Check with help of the HMI the elevator status or the service running.	2.6
	Check whether there is a special mode activated. (Installation travel, Manual evacuation JEM, Emergency stop button pressed, ...) – Check the push buttons and switches – Special modes are activated with menu 10 – Check the LEDs (Inspection ON? Blinking LEDs?)	 4.9 4.3

4	**LED Indication**	**Section**
	Check all the other LED indications. (KNET=ON?, Inspection? BBUS flickering? WDOG blinking? Emergency power LEDs on SNGL or SEM?	4.3

5	**Error Codes**	**Section**
	– Read the error codes from the control (menu 50). – Read not only the last error but also the prior ones. – Read also time stamps of the errors	4.5 8
	Special error indications	4.6

6	**Fault Detection Tools**	**Section**
	Bionic 5 offers additional menus for diagnostics:	
	Try to give commands with the HMI directly (Car calls, DTO, ...)	9.1
	Check the signals from / to the ACVF (contactors, brake contacts KB/KB1, LUET,...) with help of HMI menu 70 (723,724,725)	4.7
	Service computer (CADI): See document K 608218	
	Using SPECI will provide more details about the errors, causes and actions.	

7	**System Recovery**	**Section**
	Software reset procedures (reset, reset fatal error)	4.8
	Bionic 5 offers special travel modes to recover the system or to move the car with only parts of the electronics. (Open loop travel, travel without car electronics, ...)	4.9
	In certain cases the system can loose some configuration. (For example after a learning travel or after using installation travel mode.) Check the system and repeat the necessary configurations.	5

7	**System Recovery**	**Section**
	Under normal conditions there is no need for an software update. Before doing any software update please contact a specialist or contact the hotline in Locarno.	4.12
	Replacement procedures (PCBs, batteries,...)	4.15

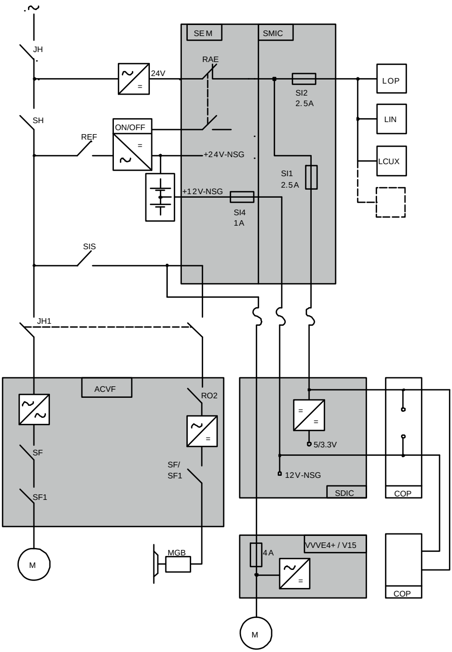
4.2 Power Supply / Fuses

Check:

– Switches (LDU)
– Fuses (SMIC, SNGL, Fermator / VD15)
– LED indication

4.2.1 Power Supply Overview

The following schematic is only an example. Check the schematic on site as it may deviate from this one.



[402008; 12.08.2014]

4.2.2 Fuses

Fuses on SMIC6x, SMIC6Ex, SMICE7x	
SKC (T2.0A)	24 V_{DC} supply to SDIC, car (2.0 AT) (Name in schematic diagram: SI1)
LOP (T2.5A)	24 V_{DC} supply BIO bus, LOPs (2.5 AT) (Name in schematic diagram: SI2)

Fuses on SMICFC	
SKC (T2.0A)	24 V_{DC} supply to SDIC, car (2.0 AT) (Name in schematic diagram: SI1)
LOP (T2.5A)	24 V_{DC} supply BIO bus, LOPs (2.5 AT) (Name in schematic diagram: SI2)

Fuses on SEM2x, SEM3x	
VPUF (T10A)	Internal protection, 24V-NSG (Name in schematic diagram: T10A)
12V-T1 (T1A)	Protection of the 12V-NSG emergency power supply (Name in schematic diagram: SI4)

Fuses on BID	
SIT	Fuse T1.6A_H for the door supply
SIB	Fuse T1.6A_H for the brake supply

Fuses on Fermator Compact VVVF4 Door Drive	
4A 250V	fast, 230VAC power supply input (Name in schematic diagram: FH 4A)

4.3 LED indication

4.3.1 LEDs in LDU (landing door frame)

LEDs on SMIC(E)61/63.Q

	Normal Display	Description
+24V NGL	ON	ON = 24V_{DC} from SEM PCB available
LREC LREC-A	ON/OFF	see extra table “TSD Function Modes” in section 4.3.2
KNET	In TSD systems: ON	OFF = At least one unlocking door contact KNET is activated (TSD option) Without TSD: LED KNET can be ON or OFF permanently
ERR	OFF	ON = Fatal error Blinking = Warning
LUET	ON/OFF	ON = Car position within the door zone
DWNLD	OFF	ON = Software download in progress (MMC) ON = While writing data to the EEPROM. Do not press RESET during this time.
BBUS	Flickering	BIO bus LED – Flickering = Normal operation – ON = Short circuit or reset on BIO bus – OFF = BIO bus communication idle

LEDs on SCPU1.Q

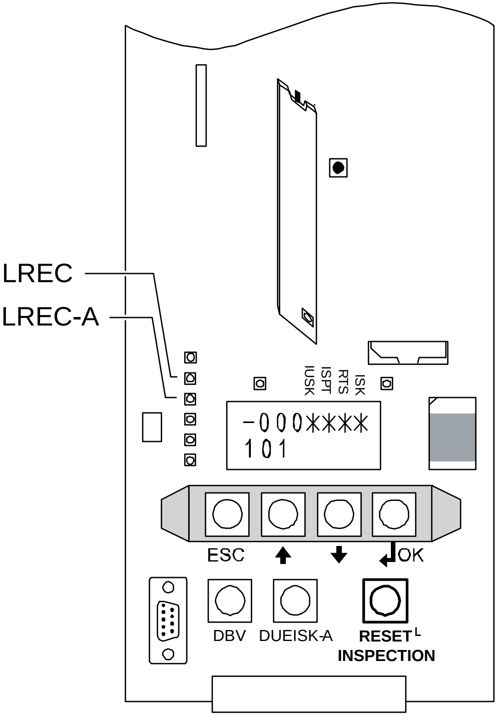
LED	Normal Display	Description
WDG/ DWNLD	Blinking	Blink interval 2 seconds = Microprocessor in normal working mode

LEDs on SEM11/12/21/22.Q

LED	Normal Display	Description
EVAC.ERR	OFF	ON = Internal fault of the SEM1x.Q PCB. Evacuation still possible but danger to damage relay. → Replace SEM PCB as soon as possible.
BATT.WARN **(BATT LOW)**	OFF	ON (while JEM is ON) = Battery capacity is below 10% – Emergency light lasting for up to one hour – Manual and automatic evacuation possible depending on actual available capacity The battery status is updated every ten minutes. If BATT.WARN LED stays ON for more than ten hours (while mains power supply is active): – Battery defective (one or both) – Battery charger defective
VBAT **(12V-NSG)**	ON	ON = Battery voltage available OFF = The batteries are either disconnected or the battery voltage is < 3VDC
BOOST **(BOOSTER)**	OFF/ON	During normal mode: OFF = normal operation During manual evacuation, while pressing DEM: ON = output voltage available
DEM	OFF	LED under yellow button DEM. Blinking LED = JEM is switched on.

4.3.2 LEDs LREC and LREC-A

The table below is valid for TSD system (systems with reduced headroom).



LED Positions [402009; 13.08.2014]

TSD Function Modes

Green LED "Normal Mode" LREC-A	Yellow LED "Inspection" LREC	Car and LDU Buzzer	Functioning Mode
ON	OFF	OFF	**Normal Mode**
OFF	ON	OFF	**Inspection Mode**
Blinking	Blinking	Intermittent fast beeping (only when doors are closed≈)	**STOP Mode Auto Reset** For example: After car maintenance positioning. "Ready to access car roof after ≈ 3 min."
OFF	Blinking	OFF	**Silent STOP Mode** To reset: Press Reset on SMIC. System will return to "STOP Mode Manual Reset after ≈ 3 min."
Blinking	Blinking	Intermittent slow beeping (only when doors are closed)	**STOP Mode Manual Reset** To reset: Press "RESET INSPECTION" on SMIC. System will return to "Normal Mode after ≈ 3 min."
as before the Recall	as before the Recall	OFF	**Recall Mode**
Blinking	OFF	ON	**Pre-Normal Mode**
ON	Blinking	1 beep	**Car maintenance positioning**

4.3.3 LEDs in CCU (car roof)

LEDs on SDICx.Q

LED	Normal Display	Description
24V	ON	ON = 24V_{DC} (P01) supply from the LDU
12V-NSG	ON	ON = 12V_{DC} (VDD) supply from the LDU
3.3V/5V	ON	ON = 3.3V/5V supply (produced on SDIC) for MMC/Internal logic
PHS	ON/OFF	ON = Photocell interrupted (Hoistway information, car in the door zone)
2PHS	ON/OFF	ON = Photocell interrupted (Hoistway information, car in the door zone, 2nd access side
WDOG	Blinking	Blink interval 2 s when software OK
SW DOWN-LOAD	OFF/ Blinking	OFF = Normal display Blinking = During software download
ERROR	OFF	ON = ERROR Blinking = May indicate CAN bus disturbance
LMG	ON	ON = Car load cell frequency available

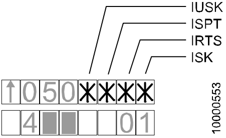
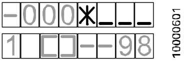
4

4.3.4 LEDs on LCUX, LOP, BIOGIO, COP and VCA

LED description of LCUX, LOP, COP, and VCA can be found in the corresponding section of the PCB.

- LCUX: Section 7.14
- SCOP: Section 7.6 to 7.8
- VCA: Section 7.10
- LOP, BIOGIO, LIN: Section 7.11 to 7.17

4.4 Safety Circuit

The safety circuit signals are indicated with help of the user interface HMI (virtual LEDs). • [X] = closed • [_] = open	
• IUSK blinking = LUEISK (Short circuit in safety circuit)	

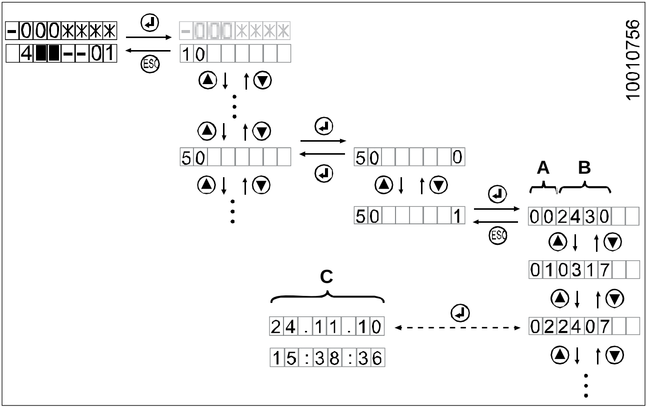
Signal	Normal	Description
LUEISK	OFF	ON = IUSK blinking
		ON = Safety circuit supply is switched off. Possible reason: – Safety circuit current > 800 mA
IUSK	ON	ON= Supply safety circuit 24 .. 55 V_{DC} OK
ISPT	ON	ON = Safety circuit hoistway pit closed
RTS	ON	ON = Safety circuit hoistway pit and landing doors closed
ISK	ON	ON = Safety circuit completely closed

Button	Description
DUEISK-A	Switch safety circuit supply on again (after LUEISK has been activated).

4.5 Event Codes (Menu 50) Error Codes

Error Codes are part of OEM level [0]. See information in section 1.4.

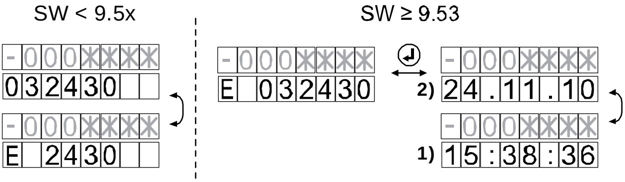
The error codes history can be read with help of the user interface HMI, menu 50:



[32319; 11.11.2009]

- **A** = Error storage
  SW ≥ 9.34: 00 .. 49 = latest error .. oldest error
- **B** = 4 digit error code
- **C** = Date and Time (Only available with SW ≥ 9.53. Can be set manually with CF=04, PA=2, 5. Automatically set if connected to a remote monitoring system).

Additional information



[402013; 09.09.2014]

– Message type:
  E = Error; **F** = Fatal Error; **P** = Persistent Fatal Error; **I** = Information
– Time (1) and Date (2)

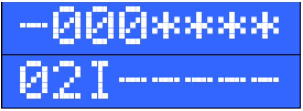
- To have a better overall picture about the error do not only read the latest error. Read always also the older errors.
- To clear the history press the “OK” button until “E-” appears, but it is recommend to leave the errors for troubleshooting reasons.
- Persistent Fatal Errors need a special error recovery procedure (HMI menu 10 > 101, see section 4.8.3)
- The error codes description can be found in **Appendix B, chapter 8** of this document.

4.6 Special Errors

Special Error Indication on the HMI User Interface

Indication	Meaning
SW version (for example "V.9.34.04") **or** **"70" or "72"**	Software version (for example: "V9.34.04") Normal display for a few seconds during start up of the system. If this indication does not disappear after a few seconds, check for the following reasons: – ACVF started up correctly? – No CAN bus connection to ACVF. (Always during start-up. Should disappear after a few seconds). – CAN bus interface on ACVF defective (after wrong connection of encoder / CAN bus) – CAN bus disturbed by defective SDIC or SCOP. To check try to move the car without the car electronics (see section 4.9.9). If car moves, the problem is caused by the car electronics (SDIC or SCOP). – Parameter download SIM card to ACVF failed. Check compatibility ACVF (SW Version) ↔ SIM card (FC parameter file version). For further information please contact the local field specialist.
CF 16	During start-up: Parameters on SIM card differ from parameters stored in ACVF. Possibility 1: After a few seconds system will start up with parameters stored in ACVF and "CF 16" will disappear. (Or press "ESC" to start up immediately.) Possibility 2: Press "OK" on the HMI to see which parameters are different. Confirm or change the parameters.

Indication	Meaning

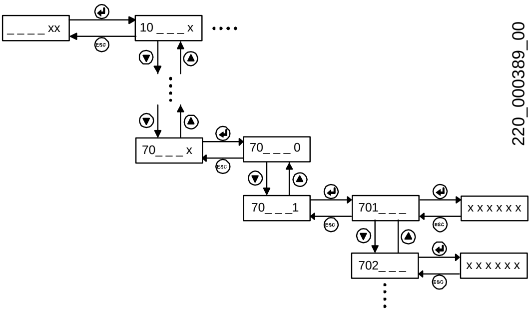
[E _ 000020]	E_ELEVATOR_SAFETY_CHAIN
	The HMI shows in sequence:
	– the date (dd.mm.yy)
	– the time (hh:mm:ss)
	– the safety chain status in the format LLIRPUCS:
	  – LL=level (if 00 the level was unknown)
	  – I=ISK is open
	  – R=RTS is open
	  – P=ISPT is open
	  – U=IUSK is open
	  – C=Safety chain error cause was KTC
	  – S=Safety chain error cause was KTS
	Note: If a dash is shown, the signal or state was zero.
	
	
	For more information go to chapter 8 “Error Codes Descriptions”.

Problems with SIM Cards

If a defective, empty or incorrect SIM card is being used on the SMIC PCB, the elevator will operate for five minutes but only with minimal service, for example Simplex DE, without fireman's control, without parking control etc..
After five minutes the car travels to the bottom floor and remains blocked (DT-O and the photocell remain active).

4.7 ACVF: Monitoring Data, Menu 70 (or 30>306)

Menu 70 is part of the OEM level [0] (see information in section 1.4).



ACVF monitoring, menu 70 [13024; 26.11.2010]

With SPECI the Vacon Monitoring Data is listed in menu 30 > submenu 306.

HMI	ACVF	Meaning	Units
701 **3061**	1.16	Actual elevator speed. Calculated value, based on elevator parameters and encoder input	mm/s
702 **3062**	1.21	Nominal linear speed, calculated	mm/s
703 **3063**	1.17	Encoder speed	rpm
704 **3064**	1.18	Encoder pulses (in SW V8.x: Unit is 0.01 Hz)	mHz
705 **3065**	1.3	Motor speed	rpm
706 **3066**	1.4	Motor current	0.01 A_{rms}
707 **3067**	1.7	Motor voltage	0.1 V
708 **3068**	1.24	Motor Temperature. Measured motor temperature based on KTY84-130 thermal sensor	° C
709 **3069**	1.1	Output frequency of ACVF	mHz
710 **30610**	1.2	Frequency reference FC frequency reference to motor control	mHz
711 **30611**	1.8	DC-link voltage	0.1 V_{DC}
712 **30612**	1.10	Voltage input AI1 AI1 = Thermostat of breaking resistor KTHBR	0.1 V
713 **30613**	1.11	Voltage input AI2 AI2 = Motor Thermistor KTHMH	0.1 V
714 **30614**	1.9	Unit temperature It refers to internal IGBT module temperature measurement	° C

HMI	ACVF	Meaning	Units
715 **30615**	1.23	Test Current Iq Filtered current Iq measured in the middle of the trip in position mode during 16 ms. Iq = output current vector produced by torque.	0.01 A
716 **30616**	1.26	Maximum motor current	mA
717 **30617**	1.28	Position_mm Relative car position from the beginning of the trip. (value calculated after start of trip.)	mm
718 **30618**	1.29	Distance Request Distance requested from control for the next trip. (Value shown after start of trip.)	mm
719 **30619**	1.30	StopDistance_mm It shows the calculated braking distance at each trip	mm
720 **30620**	1.35	FirstFlagCorr ACVF internal position correction when the car leaves the door zone (PHS flag)	mm
721 **30621**	1.36	LastFlagCorr ACVF internal position correction when the car meets the door zone (PHS flag).	mm
722 **30622**	1.31	LastRisingFreq ACVF encoder frequency when the car meets the rising edge of the flag PHS	mHz
723 **30623**	1.12	Digital input DIN1, DIN2, DIN3 states → See extra table below	0...7
724 **30624**	1.13	Digital input DIN4, DIN5, DIN6 states → See extra table below	0...7
725 **30625**	1.14	Digital output DO1, RO1, RO2, RO3 states. At the moment only RO1, RO2 and RO3 are used. → See extra table below	0...15

4

HMI	ACVF	Meaning	Units
726 30626	1.38	CLC Information If parameter “CLC information” is set as default to “0”: Actual CLC information If parameter “CLC information” is set to a value <> “0”: Value of parameter CLC range: -1000 means empty car and +1000 means full load (= parameter “GQN Payload”)	--
727 30627	1.44	Power Mode. 0: Standstill, 1: Motor, 2: Generator	0...2
728 30628	1.45	Shows the actual motor temperature fault reset level. Biodyn xx C BR: ACVF becomes available when the temperature drops below this reset level.	C
729 30629	1.46	Fan speed	%
730 30630	1.47	Brake resistor temperature	C
731 30631	1.49	Actual motor nominal speed during rpm identification run	rpm
732 30632	1.50	U/f curve ID state 0: not used, 1: stand-by, 2: running, 3: OK, 4: not OK	0...4
733 30633	1.51	rpm ID state 0: not used, 1: stand-by, 2: running, 3: OK, 4: not OK	0...4
734 30634	1.25	Motor current mid	0.01A

Explanation 723 / 30623 (Inputs DIN1 ... DIN3)				
Value	**DIN1** **SF** (NC contact) (0 = active)	**DIN2** **SF1** (NC contact) (0 = active)	**DIN3** **Evacuation mode** (from HCU) (1=active)	**Status**
0	0	0	0	Normal trip
1	0	0	1	
2	0	1	0	
3	0	1	1	
4	1	0	0	
5	1	0	1	
6	1	1	0	Stand-by
7	1	1	1	

4

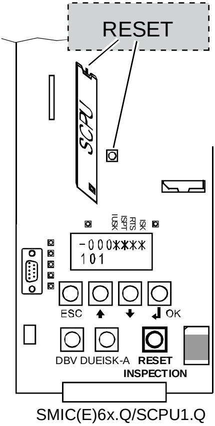
Explanation 724 / 30624 (Inputs DIN4 ... DIN6)				
Value	**DIN4** **"Car on floor"** (1 = active)	**DIN5** **KB** (NC contact) (0 = active)	**DIN6** **KB1** (NO contact) (1=active)	**Status**
0	0	0	0	**KB/KB1 fault**
1	0	0	1	Normal trip car between floors
2	0	1	0	Stand-by, car between floors
3	0	1	1	**KB/KB1 fault**
4	1	0	0	**KB/KB1 fault**
5	1	0	1	Normal trip, car on floor
6	1	1	0	Stand-by, car on floor
7	1	1	1	**KB/KB1 fault**

The status in table 724 is valid for systems with HCU.
For systems without HCU the DIN4 is always 0.

Explanation 725 / 30625 (Outputs R01 ... R03)				
Value	**R01 SF/SF1**	**R02 MGB**	**R03 MVE**	**Status**
0	0	0	0	Stand-by (no MVE)
1	0	0	1	Stand-by (MVE running)
2	0	1	0	
3	0	1	1	
4	1	0	0	Start/End Trip (no MVE)
5	1	0	1	Start/End Trip (MVE running)
6	1	1	0	Normal trip (no MVE)
7	1	1	1	Normal trip (MVE running)

4.8 Resolving Errors

4.8.1 Normal Reset Elevator Control



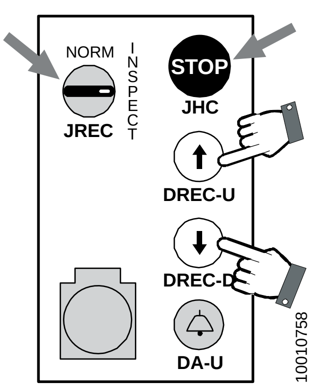
[402014; 10.09.2014]

Press RESET button once on the SCPU (top of SCPU, backside) in the LDU.
(With SMIC**E**6x.Q: Additional reset button on SMICE6x.Q PCB.).

A reset has to be performed after software hang-up or after changing the configuration.

4

4.8.2 Normal Reset Elevator Control from Car Roof

	With **software** ≥ **V9.53** it is possible to reset the control with help of the inspection control on the car roof: – JREC must be on position “INSPECTION” – JHC Stop button must be pressed → Press the UP and DOWN buttons (DREC-U and DREC-D) simultaneously. This will reset the elevator control.

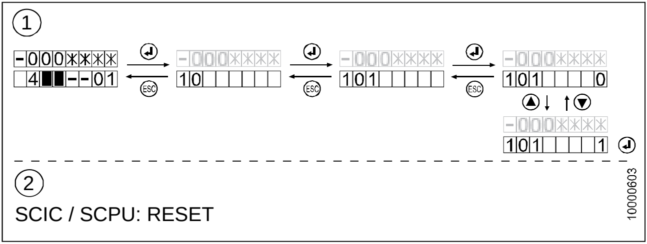
4.8.3 Reset Persistent Fatal Error Control

A persistent fatal error is caused for example by safety circuit problems in pre-opening door zone.

The described procedure will reset the following items:

- Bionic system persistent fatal errors (pre-opening, re-leveling, KSE, KNE, PHSx bridged, safety circuit)
- ACVF (Vacon) fatal errors (for example KB/KB1)
- Clear several states: Fire service state, last fire fighter floor, door lock monitoring condition)

Reset Procedure with User Interface HMI:



[25788; 27.08.2009]

1	On the user interface HMI choose main menu 10 and press OK to confirm → HMI shows [101]
2	Press OK again to confirm submenu 101 → HMI shows [101 0]
3	Change the value from [101 0] to [101 1] and press OK. After a few seconds the HMI will display [101 0] again
4	Leave the menu 10 and press the RESET button on the SMIC or SCPU PCB

4.8.4 Reset Fatal Error Frequency Converter ACVF

Depending on the cause of the error the fatal error status of the ACVF is nowhere indicated.

Reset Procedure with User Interface HMI:

Use the same reset procedure as used to reset a persistent fatal error of the control. See section 4.8.3.

Reset procedure with Vacon user interface panel (optional):

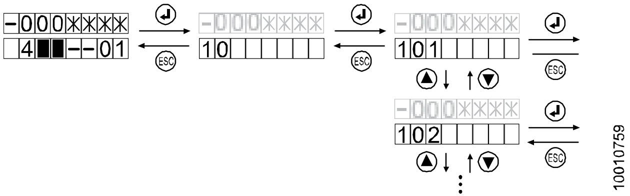
If a Vacon user interface panel (not part of the standard delivery) is available the Vacon ACVF can be reset with help of the Reset button.

4.9 Special Modes, Special Commands (Menu 10)

The HMI menu 10 offers special commands for diagnostics and troubleshooting:

– Execute elevator commands (car calls, door opening, KFM, ...)
– Deactivate special modes (earthquake, SAB, ...)
– Activate special travel modes (ACVF open loop, installation travel, ...)

If OEM is activated, use the "Enable HMI" function in SPECI to enable the HMI.



[32328; 11.11.2009]

Menu	Description	OEM
101	Dispatch a fatal error reset command to ACVF and elevator control. This clears fatal errors related to pre-opening, releveling, KNE, KSE, KB/KB1, safety circuit, PHSx bridging.	2
102	Activate ACVF open loop mode This allows traveling by inspection control or recall control despite the failure of the motor encoder, KB/KB1 brake contact or ACVF/motor temperature sensor.	2
103	Activate GBP reset mode for inspection travel beyond KSE-U in order to reset the overspeed governor contact. Automatically deactivated.	2

Menu	Description	OEM
104	Activate or deactivate KFM mode (also known as maintenance mode, test travel mode)	2
105	Activate or deactivate installation travel mode (this mode is not affected by a reset) Remark: After deactivation a reset is necessary to return to normal mode.	0
107	Activate or deactivate load measuring (CLC) (deactivate 0 → 1)	1
108	Activate or deactivate the key switch function (JAB) (activate 0 → 1, which puts the elevator out of service)	1
109	Activate OEM function. Using this menu it is possible to activate OEM in advance of 10’000 trips.	0
110	Dispatch a car call	1
111	Dispatch a landing call	1
112	Dispatch a door open command (door will not close automatically)	1
113	Dispatch a door close command	1
114	Deactivate SAB The telealarm control center can put the elevator out of service remotely. Command 114 is used to put the elevator into service again.	0
115	Deactivate the Earthquake service (EB) (deactivate 0 → 1)	2
116	Initiate a learning travel	0

Menu	Description	OEM
117	Service Visit ON/OFF (Disable / enable remote monitoring) With this menu it is possible to disable the sending of errors to the RMP for a certain time in case the technician needs to operate on the elevator. – 0 = Errors are not filtered and sent through remote monitoring. Automatic set to '0' after 30 minutes if no detected maintenance activities. – 1 = Errors are not sent through remote monitoring. Automatic set to '1' if maintenance is started (e.g. entering the Inspection mode).	2
123	Activate or deactivate pre-torque calibration (activate 0 → 1) Useful procedure for elevators that are **not** balanced as for example Schindler 3100.	0
124	Deactivate the Sprinkler Recall service (SR) (deactivate 0 → 1)	2
125	Dispatch a remote GBP reset command	2
126	Next Call It forces an immediate TM call, usually helpful to accelerate the auto-configuration. Note: not effective in case of CGW.	0
128	Modem detection To be used after installation, replacement of TM device and in case of not working TM communication (see 5.4.24 ETM Embedded Telemonitoring)".	0
129	COP detection To be used after installation of a new COP.	0
130	LOP detection (like CF=00, LE=00, LOP counting) (Display during LOP counting: [1301])	0

Menu	Description	OEM
133	LMS type confirmation The activation of this menu confirms that the LMS installed is a switch type. This confirmation is needed only if a switch type LMS is installed. – LMS Switch confirmed, showing state 11 – LMS Switch not confirmed, showing state 01	0
134	Temporary disabling of the alarm filter. Used for test alarms. If activated, the alarm is not filtered, even if the car is moving or the doors are open. 134 will be set back to “0” after 30 seconds automatically.	2
135	Stop telealarm activity. Used to reset a pending alarm. (Alarm indication on COP.)	2
136	Overlay detection. (Triplex, Quadruplex) Has to be performed at the of the commissioning of the group. Has to be repeated, if an overlay is removed from the system. Indication [136 _ _ _ 1] = Detection correct Indication [136 _ _ _ 0] = Detection not correct	0
137	Overlay reset. Resets the overlay box. Same function as reset button on the overlay box. Has to be performed after configuration CF=04, PA=01.	0
138	Car light detection/calibration (SW ≥ V11)	0
139	Emergency car light calibration (SW ≥ V11)	0
140	Stationary motor autotuning. (Available with SW ≥ V9.8 and Commodity Mid Rise, Schindler 3600 only.) The HMI signalizes the autotuning phases, at the end shows 0= fail, 1=success.	1

4

Menu	Description	OEM
141	Autotune the encoder offset. (SW ≥ V9.8 and Commodity Mid Rise, Schindler 3600 only.) The HMI signalizes the autotuning phases, at the end shows 0= fail, 1=success.	1
142	Autotune the dynamic motor. (SW ≥ V9.8 and Commodity Mid Rise, Schindler 3600 only.) The HMI signalizes the autotuning phases, at the end shows 0= fail, 1=success.	1
143	High torque trip (SW ≥ V9.9)	1
144	CDD Detection (SW ≥ V10.0)	1
145	ECM Calibration (SW ≥ V10.0) This menu is visible only if menu 190 is triggered	1
146	ECM Clear error (SW ≥ V10.0)	1
147	Door drive detection (SW ≥ V11.0)	-
148	KB/KB1 monitoring bypass (Variodyn)	-
149	Motor overtemp bypass (Variodyn)	-
190	Enable one time STM configuration change (SW ≥ V10.0)	1
199	Call simulator (SW ≥ V10.0) The maintenance person can start a call simulation with a configurable behavior. The maintenance person can enter in this menu the number in hex format to instruct the kind of simulation needed:	2

Menu	Description	OEM
	The parameter uses the XYYZZ format; Parameter X: call pattern: – 0 : Stop simulation (simulation is not running) – 1 : Floor calls (random floor/side) – 2 : Car calls (random floor/side) – 3 : Car and floor calls (random floor/side) – 4 : Stop on each floor and open door operation (ZZ0/1/2: opens all unrestricted doors. Restricted levels are skip. ZZ3 open first side 1 and then side 2) – 5 : Full shaft movement (top->bottom->top->....). Doors shall be kept closed once the car landed. – 6 : Full shaft movement (top->bottom->top->....). Doors shall be opened according the current door policy. Restricted floors shall not be obeyed. Parameter YY: Delay between call (valid only for call pattern 1,2,3) or Delay between the two calls (call pattern 4,5,6): – 0 : Stop the simulation Parameter ZZ: The amount of trip to perform in 100th unit. – 0 : Infinite In case of SW reset, the controller resumes the call simulator until all requested trips are performed. In case of SW reset and the call simulation is with infinite trips, the controller terminate the call simulator.	

Menu	Description	OEM
	Note: A power down of elevator would include errors in the trip count (+/- 100 trips). The behavior of the elevators is according to the configuration of the elevator(s) and according to the current operational mode. Call generated shall not be handicapped and shall not be authenticated (restricted access are not reachable). If case of call simulator running the menu shall show the current snapshot pattern and the ZZ parameter shows the remaining number of trips (in hex format, 100th unit). Example: “2100A” car call simulator running with time between call of 16 seconds and around 1000 trips remaining.	

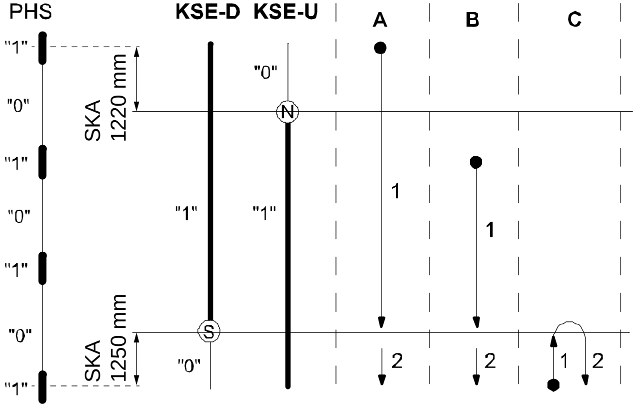
4.9.1 Synchronization Travel

A synchronization travel can not be started manually. After the following situations a synchronization travel will be executed automatically:

- at system power up
- after reset
- after inspection travel
- after installation travel mode
- when a hoistway information recoverable error has occurred
- In TSD/TSD21 systems: KNET supervision activated without opening the landing door (KTS stays closed)

The synchronization travel sequence will differ depending on the starting position of the car. (The start direction depends on the KSE-D and KSE-U status. See drawing below.)

At the end of a synchronization travel the car will remain at the lowest PHS stop.



Sequence of Synchronization Travel, depending on start positions A, B or C
[402037_00; 29.01.2015]

Problems during Synchronization Travel

Possible causes:

- If the door does not close: Check photocell or light curtain.
- Synchronization travel may not start if car load cell is not yet calibrated. → Deactivate the car load cell temporarily.
- Problems with the KSE-D and KSE-U magnets or magnetic switches
- Hoistway information: SKA set incorrectly

4.9.2 Learning Travel

The learning travel is used to:

– Calculate the traction pulley diameter DD and other drive parameters
– Count the number of stops and measure the interfloor distances
– Learn the door sides and door preopening information
– Read HW and SW information (COP, SDIC)

The learning travel is either triggered automatically for example after leaving the installation travel mode or can be forced manually.

Reason why a learning travel should be forced:

- Re-reading of the hoistway information

Learning travel sequence:

- Synchronization travel as described under “Synchronization Travel” above.
- Complete travel to the top stop (calculation of the pulley diameter DD after passing the flags)
- At the top: Pre-torque calculation by the ACVF (opening brake). With SW ≥ V9.71: Unlock / lock the door (to prevent problems with maximum door lock timer in case of long hoistways).
- Complete travel to the bottom stop (reading the number of stops and the complete hoistway information and store it to the EEPROM)
- At the bottom: Pre-torque calculation by the ACVF (opening brake).

After a learning travel the system configuration has to be checked again. Some parameters are changed or reset automatically during a learning travel:

– COP5B-N configuration is reset. CF=15 has to be done again. (Only with SW < V9.5x.)
– Digisens 0 Kg calibration is done at the end of the learning travel. (If the car was not empty this calibration has to be done again with empty car. CF=98) (Or disable the Digisens during the learning travel with HMI menu 10 > 107=1.)
– The ACVF pre-torque calibration is done during the learning travel. (If the car was not empty this calibration

has to be done again with empty car. HMI menu 10 > 123=1. See section 4.9.10.)

– The “blind” floors are activated again. CF=26 has to be done again. (See section 5.4.21.)

SW ≥ V9.34: Forcing a learning travel using the HMI:

1	Activate the HMI menu 10
2	Choose submenu 116
3	Change from [116 0] to [116 1] and press “OK”. → The HMI displays [116 0]. The “0” blinks. → The learning travel starts
4	After the learning travel has finished press ESC several times to leave submenu 116
5	Check the “Note” at the beginning of this section. Tasks after the learning travel.

4

Possible causes for problems during Synchronization Travel and Learning Travel

- If the door does not close: Check photocell or light curtain.
- Synchronization travel may not start if car load cell is not yet calibrated. → Deactivate the car load cell temporarily.
- Problems with the KSE-D and KSE-U magnets or magnetic switches
- Hoistway information: SKA set incorrectly
- ACVF parameters set wrongly. (For example rated speed or leveling speed (to low))

In case of problems with landing accuracy
Some special system configurations can lead to problems with the landing accuracy.

– Systems with big interfloor travel distances:
  In this case the option “blind floors” can be ordered. (Additional PHS flags. For configuration see CF=26)
– Systems with 2 floors only or in case or general problems with the landing accuracy:
  In this case the learning travel can be done with a balanced car (approximately 50% load).
  – Step 1: Learning travel with balanced car
  – Step 2: Manual initiation of the ACVF pre-torque calibration with empty car (0% car load). See section 4.9.10, Menu 123.
  – Step 3: If the Digisens was not disabled during the learning travel, the 0 Kg calibration has to be done again with empty car (0% car load) CF=98, see section 5.4.1.

4.9.3 Open Loop Travel Mode (HMI menu 102)

The open loop travel mode is used to replace defective units in the hoistway head which are used normally to travel in closed travel mode. (Encoder, brake contacts, thermal contact.)

Depending on the reason for traveling in open loop mode there are two different procedures applicable:

– Procedure 1: Open loop travel in case of defective encoder IG or defective brake contacts KB
– Procedure 2: Open loop travel in case of defective thermal supervision THMH.

Open Loop mode is automatically switched OFF as soon as recall Control or Inspection Travel mode is switched OFF!

Do not switch OFF Recall or Inspection while you are travelling in the hoistway outside of a door zone.
You can not switch on Open Loop mode again and you may be blocked on the car roof!

4

Procedure 1: Open loop travel in case of defective encoder IG or defective brake contacts KB:

	Procedure 1
1	Connect the recall control ESE to the SMIC PCB in the LDU.
2	Switch the recall control to “RECALL” travel mode.
3	On the user interface HMI activate the open loop travel mode. – Choose main menu 10 and press OK – Choose submenu menu 102 and press OK – Change from [102 0] to [102 1] and press OK → The HMI shows [102 1] (The “1” blinks)
4	Now the system is in open loop travel mode until recall control is switched off or until the menu 102 is set to “0” again.
5	Use the recall control to move the car to the LDU in order to reach the car roof. Remark: It is advised to place the top of the car short above the LDU floor on KSE level. In this way you can step to the car roof and you can reach the motor and ACVF.

	Procedure 1
	The next steps (6 to 9) apply only if the technician has to travel on the car roof with help inspection control.
6	Open the landing door and activate the inspection control on the car roof. → This may reset the system to closed loop travel mode (HMI indication [102 0]
7	Switch the recall control ESE back to “NORMAL” travel mode.
8	On the user interface HMI activate the open loop travel mode again. (If necessary) – Change [102 0] to [102 1] and press OK. → The HMI shows [102 1] (The “1” blinks)
9	Now you can travel in open loop travel mode to the hoistway head (KSE level) to check the defective part. If necessary use the car blocking device before doing any work on the machine or on the ACVF.

Procedure 2: Open loop travel in case of defective thermal contact THMH:

	Procedure 2 (SW ≥ V9.34)
	In case of problems with the thermal supervision THMH there is time out while no travel is possible.
1	Wait **15 minutes** until the THMH timer has elapsed.
2	After these 15 minutes connect the recall control ESE to the SMIC PCB in the LDU.
3	Switch the recall control to “RECALL” travel mode.
4	On the user interface HMI activate the open loop travel mode. – Choose main menu 10 and press OK – Choose submenu menu 102 and press OK – Change from [102 0] to [102 1] and press OK → The HMI shows [102 1] (The “1” blinks)

	Procedure 2 (SW ≥ V9.34)
5	Press the RESET button on the SCPU and wait until the system has started up again.
6	Continue with step 3 of the procedure 1 above. (Switch on open loop travel mode again [102 1]. Then the recall control ESE can be used.)

4.9.4 GBP Reset Travel Mode (HMI menu 103)

GBP Reset Travel mode is used after the overspeed governor has been released. It allows to travel on the car roof to the top of the hoistway (beyond the KSE-U point) to reset KBV.

1	Unplug SMIC.KBV and plug special jumper plug “GBP Reset” to SMIC.KBV
2	Enter main menu 10 on HMI and choose submenu 103
3	Change value in submenu 103 from “0” to “1” and press OK
4	Switch on inspection control on the car roof. (Recall control has to be switched off).
	The system is now ready to travel in inspection travel mode to the top of the hoistway.

4.9.5 Reset the Safety Gear

1	If the safety gear has been engaged, release the car out of the safety gear with help of the recall control station ESE. – Lower the car by pressing DRH-D for 1 second. – To release the safety gear move the car up by pressing DRH-U – Repeat this procedure 2-4 times
2	If the car does not move after the safety gear acceptance test: – 1st step: carry out a high torque trip (menu 143). – 2nd step: remove part of the test weight from the car and try again.

3	If the car does not move: Try the same procedure (step 1) in open loop travel mode (HMI menu 102. See section 4.9.3)
4	With the recall control ESE move the car to the bottom floor until you can reach from the pit the KF contact below the car. → Reset the KF contact
5	Move the car with help of the recall control ESE to the top of the hoistway. (Car roof just above the LDU level.) In this way you can climb on the car roof and reset the KBV contact. Or use the “GBP Reset Travel mode” (HMI menu 103, see section 4.9.4) to travel with inspection control to the top of the hoistway and to reset the KBV contact.
6	– Verify the car load cell is still working correctly (CF=95). – Check that the load carrying element retainers are positioned securely. – Check that the load carrying elements are positioned correctly on the suspension pulley. – Check the guide rails.

4.9.6 KFM Travel Machine Room Mode (HMI menu 104)

Travel machine room mode can be activated either with HMI menu 104.

Travel machine room mode is used to send the car from the top floor to the lowest floor and back again. (Test trip through the whole hoistway.)

Activation of KFM with the HMI user interface:

1	Enter main menu 10 on HMI
2	Choose submenu 104
3	Change value in submenu 104 from “0” to “1” and press OK
4	Send car down and up: To start the test trip press the “OK” button on the User Interface HMI.

4

4.9.7 Inspection and Recall Travel (ESE)

Inspection Travel

Inspection control is turned on using the JREC switch of the Inspection control station on the roof of the car.

The car can be moved at very low speed. Travel distance will be limited by KSE.

ESE (Recall Control)

The car can be moved at very low speed using the recall control. ESE control is blocked when Inspection travel is turned ON.

Travel distance will **not** be limited by KSE or KNE.
The car can travel right down on to the buffer. (With recall control the following safety contacts are not checked: KF, KF1, KNE, 2KNE, KFG and KBV. Also the hoistway information KSE-U, KSE-D and PHS/PHUET are ignored.)

4.9.8 Car Positioning for Accessing the Car Roof

Schindler 3100/3300/3600/5300/6300 offers the service technician an automatic car positioning to access the car roof. This procedure has to be used before doing any maintenance on the elevator.

	Description
1	In the LDU, on the SMIC, press the “RESET INSPECTION” button for at least 3 seconds. (There is an acknowledge beep.)
2	The car moves to the LDU floor and opens the door. Check that there is no passenger in the car.
3	Press the “RESET INSPECTION” button again.
4	The door closes and the car moves slowly down until the car roof is leveled with the LDU floor. This is indicated by the beeping buzzer and by blinking LEDs LREC and LREC-A.)
5	Open the door with the triangular key, press the STOP button on the car roof and switch on the Inspection. (Either with JREC to “INSPECTION” or in TSD systems with help of the yellow lever.)
	For TSD systems only (systems with reduced headroom), LED behavior is described in section 4.3.2: After finishing the work on the car roof: – Travel with the inspection control to the LDU floor – Leave the car roof and switch back to normal mode with help of the yellow lever – Close the landing door – The buzzer is beeping and LREC and LREC-A are blinking. → Press the RESET INSPECTION button on the SMIC to return to normal mode. (Confirmation that nobody is on the car roof.)

4.9.9 Accessing the Car Roof with Installation Travel

If there is a defect in the electronics on the car roof or on the door drive the car will not move in normal mode anymore.

The following procedures can also be used to verify the ACVF is working.

Procedure 1: Recall control ESE

	Description
1	Connect the recall control ESE in the LDU to the SMIC. Try to move the car in "RECALL" control travel mode. (With recall control the following safety contacts are not checked: KF, KF1, KNE, 2KNE, KFG and KBV. Also the hoistway information KSE-U, KSE-D and PHS/PHUET are ignored.)
2	If the car does not move, continue with procedure 2.

Procedure 2: Recall control ESE with installation travel mode

	Description
1	The recall control ESE is still connected
2	Activate installation travel mode
	– On the HMI choose main menu 10 – Choose submenu 105 – Change [105 0] to [105 1] and press OK
3	Try to move the car with the recall control ESE. (In installation travel mode the door signals are ignored. The safety circuit must be closed.)
4	If the car does not move, continue with procedure 3.

4

Procedure 3: Installation travel mode without car electronics

	Description
1	The recall control ESE is still connected and the system is in installation travel mode
2	Disable the CAN bus communication to the car electronics
	On the top of the SMIC PCB (between the connectors) switch the CAN termination switch to position ON.
3	Try to move the car with the recall control ESE. In this mode the car can be moved in a basic configuration with LDU and ACVF only. If the car can be moved you know there is a problem with the car electronics (SDIC, SUET, SCOP, photocell or door drive)

4.9.10 Manual Pre-torque Calibration (HMI menu 123)

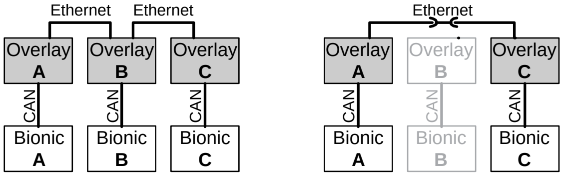
The correct pre-torque calibration is necessary for a high travel quality (jerk at the start of every trip). The pre-torque calibration is automatically performed during the learning travel. If the car was not empty during the learning travel, the pre-torque calibration needs to be done again.

	Description
1	Precondition: The car is fully installed (including all car decoration) and the counterweight is filled according the rules.
2	Make sure the car is empty (0 Kg load).
3	On the user interface HMI activate the manual pre-torque calibration. – Choose main menu 10 and press OK – Choose submenu menu 123 and press OK – Change from [123 0] to [123 1] and press OK
4	→ The car travels to the LDU floor and opens the door. → The HMI displays [123nn 1] (“nn” shows the actual floor, the “1” is blinking.

	Description
5	Verify the car is empty. On the HMI press the “OK” button. → The door closes and the pre-torque calibration starts. – The car travel to the top floor and performs the pre-torque calibration – The car travel to the bottom floor and performs the pre-torque calibration
6	After the pre-torque calibration the HMI displays [123 1] (“1” is blinking)
7	Change from [123 1] to [123 0] and press OK to leave the pre-torque calibration mode.

4.10 Diagnostics for the Overlay

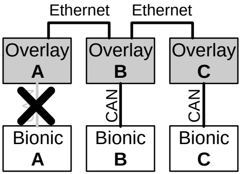
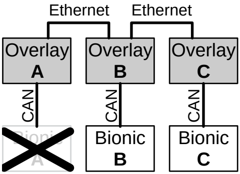
If one elevator has to be taken out of service for maintenance or has to be switched off, it is necessary to connect the remaining elevators of the group with each other.

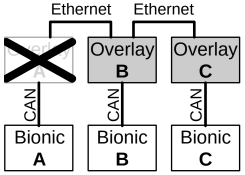


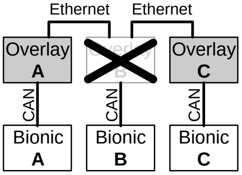
Elevator Group with Overlay, temporary bridge [37579; 08.11.2010]

Typical errors cases and their indication on the HMI:

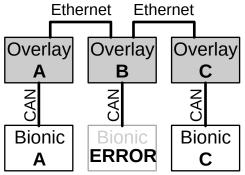
Detailed description of 309-1 to 309-6 (HMI menu 30): Section 9.3

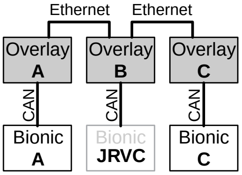
HMI	Case 1			HMI	Case 2		
							
309-1	1	1	1	309-1	-	1	1
309-2	0	1	1	309-2	-	1	1
309-3	x	3	3	309-3	-	3	3
309-4	x	2	2	309-4	-	2	2
309-5	x	2	2	309-5	-	2	2
309-6	3	3	3	309-6	-	3	3
Error	4701	4705	4705	Error	-	4705	4705





HMI	Case 3			HMI	Case 4		
309-1	1	1	1	309-1	1	1	1
309-2	0	1	1	309-2	1	0	1
309-3	x	2	2	309-3	1	x	1
309-4	x	2	2	309-4	1	x	1
309-5	x	2	2	309-5	1	x	1
309-6	3	3	3	309-6	3	3	3
Error	4701	4705	4705	Error	4705	4701	4705





HMI	Case 5			HMI	Case 6		
309-1	1	1	1	309-1	1	1	1
309-2	1	1	1	309-2	1	1	1
309-3	3	3	3	309-3	3	3	3
309-4	3	3	3	309-4	3	3	3
309-5	2	2	2	309-5	2	2	2
309-6	3	3	3	309-6	3	3	3

Case 5 and 6 do not create any overlay errors in the history.

4.11 Communication with Service Computer

4.11.1 Communication with Bionic 5 control

See Quick Reference K 608218 “Diagnostics and Software Update for Experts

4.11.2 Communication with ACVF

See Quick Reference K 608218 “Diagnostics and Software Update for Experts”

4.12 Software Update

4.12.1 Software Update of Control

Necessary General Checks before Executing SW Update:

	Necessary General Checks before Executing SW Update:
1	Verify that the controller hardware is equal or above Bionic 5 Rel. 04 (the display must be the blue LCD). If it is older than BIC 5.4, do not perform the update.
2	Download the current system info to CADI, if applicable.
3	Verify the current installed SW/FW versions of: – SCPU SW (CF12 PA1) – SDIC FW (CF12 PA2) – COP FW (CF12 PA5) – SMIC CPLD (CF12 PA9) and SMICFC BIC 6 – Boot-loader (CF12 PA13) If any of the above mentioned SW/FW is already the required version, that specific SW/FW must not be updated.
4	Verify that the current SCPU SW installed is listed in J 42107990, in case a non-listed SW version, do not perform the update.
5	Read out trip counter (CF11 PA1), CLC parameters (CF96 PA1/2/3) and write down the values in the elevator log book, which shall be restored after the update.

	Necessary General Checks before Executing SW Update:
6	Take the car to where the LDU is located and enter configuration mode (CF 40).
7	Switch OFF the elevator and remove the MMC.

Steps to Update Bionic SW:

The update of Bionic SW includes the following steps:
– Update COP5x
– Update SDIC5x
– Update the boot-loader on SCPU
– Update CPLD and main controller SW on SMIC and SMICFC BIC 6

	Step 1: Update COP5x
1	Access COP and open the COP panel.
2	Insert the MMC card (for example, COP5 V3.4) in the COP and press reset.
3	Wait (a few seconds) until the WDG and DWNLD LEDs are permanently ON.
4	Remove the MMC card and press reset on COP.
5	A test alarm to Call Center is automatically generated.
6	Exit configuration mode (Menu 40).

If alarm horn is present, it will initiate the alarm. In case of disconnecting alarm horn during the update, check to reconnect it after update.

	Step 2: Update SDIC5x (only if FW version ≤ 3.1 upgrading to FW version ≥ 3.2)
1	Position the elevator to easily access the car top (press "reset inspection" button below HMI on SMIC or SMICFC BIC 6).
2	Remove the CCU cover.

	Step 2: Update SDIC5x (only if FW version ≤ 3.1 upgrading to FW version ≥ 3.2)
3	Insert the MMC card for SDIC5x boot-loader (for example, SDIC5x Boot-loader bl08_100) in the SDIC5x and press reset on SDIC5x.
4	Wait and observe the DWNLD LED starts blinking.
5	Wait until the "WDOG" and “ERR” LEDs start to fast blinking.
6	Remove the MMC card.
7	Insert the MMC card for SDIC5x SW (for example, SDIC5 V3.2) in the SDIC5x and press reset on SDIC5x.
8	Wait (± 3 seconds) and press reset again.
9	Wait (± 30 seconds) till DWNLD LED is permanently OFF.
10	Remove the MMC card and press reset on SDIC5x.

In some cases the MMC is not properly detected, in this case just reinsert the MMC several times.

	Step 3: Update the boot-loader on SCPU
1	Insert the MMC card (for example, SCPU boot-loader V2.1) in the SMIC board and press reset on SMIC
2	Wait (± 3 seconds) and press reset again
3	Wait (at least 15 seconds) until the WDG and DWNLD LEDs are permanently OFF
4	Remove the MMC card and press reset
5	Before the boot-loader update starts there is a countdown of 20 seconds, after this countdown do not reset/power off/remove the MMC till the completed download message (“BOOT OK”) appears on the HMI
6	Exit configuration mode (Menu 40).

Once boot-loader V2.1 is installed, downgrading of SW in SCPU is no longer possible.

	Step 4: Main Controller SW on SMIC and/or update the CPLD
1	Insert the MMC card (for example, Main controller SW 10.00 + SMIC CPLD V2.1) in the SMIC board and press reset on SCPU.
2	Before the update starts there is a countdown of 20 seconds, after this countdown do not reset/power off/remove the MMC until the completed download message (Download OK) appears on the display.
3	Remove the MMC card and press reset on SCPU.
4	At start-up, verify that the SW version is updated (for example, V9.85.33)

The backlight of the HMI is on and “MMC detected” => “Downloading” => “Download OK” is shown.

Final Checks:

– Configure LOP counting (one or two access sides).
– Check that required SW/FW versions are installed:
  - SCPU SW (CF12 PA1)
  - SDIC FW (CF12 PA2)
  - COP FW (CF12 PA5)
  - SMIC CPLD (CF12 PA9) and SMICFC BIC 6
  - Boot-loader (CF12 PA13).
– Restore trip counter (CF11 PA1), CLC parameters (CF96 PA1/2/3) values from elevator log book, if necessary.
– Leave all menus and release the elevator for normal operation.
– Verify all the functionalities of the elevator and specific elevator options like GS, Fire Service, ZB3 Keys, Pin Code, Blind Floors etc.
– Remove the stickers “In maintenance” from all floors and COP.

4.12.2 Software Update of ACVF

See Quick Reference K 608218 “Diagnostics and Software Update for Experts

4.12.3 Software Update of Servitel TM4

See Quick Reference K 608218 “Diagnostics and Software Update for Experts

4.13 Sematic C MOD (Schindler 6300)

For the diagnostics of the Sematic C MOD door drive please refer to section 7.23

4.14 Fermator Compact (Schindler 6300)

For the diagnostics of the Fermator Compact door drive please refer to section 7.22

4.15 Replacement Procedures

General rules

- Take ESD Electrostatic Discharge precautions
- Never plug or unplug any connector while the power supply is on
- When you unplug a connector check whether it is labeled correctly (to be plugged afterwards at the correct position again).
- After replacement of any component: Check the system for correct function

4.15.1 Replacement of PCBs

PCB	Tasks
SMIC61	**After replacement**: • Make sure the SIM card is inserted • Check for correct CAN bus termination switch setting • Make sure the SCPU and the CLSD are installed correctly

PCB	Tasks
SCPU	On the SCPU the complete configuration is stored in the EEPROM. Because there is no possibility to do an electronic backup, the whole configuration has to be done after the replacement. Try to read out as much as possible of the existing configuration. **Before replacement:** • Read out the trip counter and travel time (CF=11 or HMI menu 60 > 601, 602) and note the values in the elevator log book. • Read out the car load cell calibration data (see section 5.4.2) – CF=96, PA=1, PA=2 and PA=3 – CF=08, PA=01 and PA=08 • Read the encoder direction and phase sequence – CF=16, PA=14 and 15 • The following parameters need to be read out only if the corresponding items are available in the system – Parking floor, penthouse? → CF=02 – Key switches in car? → CF=05/55, CF=41, CF=17 – PIN code? → CF=41, PA=1 – JDE or JAB on the LOP? → CF=40, L=n, PA=21 – LCUX on floors? → CF=40, L=n
SCPU	**After replacement:** • If necessary chance the encoder direction and phase sequence (CF=16, PA=14 and 15) • Perform a learning travel (see section 4.9.2) • If available enter car load cell data (CF=08, PA=01 and PA=08; CF=97) (see section 5.4.2) • Perform a LOP counting (CF=00, LE=00) (see section 5.4.5) • Enter the parameters which you have read out before • Do the mandatory configurations (see section 5.2) • Set the time and date (CF=04) • Perform the ETM(A) configuration (see section 5.4.24, 5.4.25) • If necessary do the configuration for the other options

<table>
<tr><th>PCB</th><th>Tasks</th></tr>
<tr><td>CLSD</td><td>After replacement:
• Perform the ETM configuration (see section 5.4.24, 5.4.25)</td></tr>
<tr><td>SEM</td><td>After replacement:
• Check the bridge connectors
• Press the DEM button
• Check the manual and automatic evacuation</td></tr>
<tr><td>SDIC</td><td>After replacement:
• Check the bridge connectors
• Check the alarm button (on the car and on the COP)</td></tr>
<tr><td>SUET</td><td>No special actions necessary</td></tr>
<tr><td>COP</td><td>Before replacement:
– To open the COPs please refer to section 7.5
After replacement:
• If available: Make sure the VCA voice announcer and the SAS card reader are installed correctly
• Check for correct function and display
• If necessary repeat the COP configurations. (CF=01, CF=15) (See sections 5.4.3, 5.4.15)
• If available: Check for correct key switch functions
• Check the alarm button
Only in case of problems: It could be that the COP has not been recognized correctly. Perform a COP detection with help of HMI menu 10, submenu 129 ([129 0] to [129 1] and OK), (with SW ≥ V9.34)</td></tr>
</table>

PCB	Tasks
LOPs	**Before replacement**: – To remove the LOP from the door frame: Press the LOP upwards and move it out of the frame. **After replacement**: • If available: Check that LIN, LCUX and key switch contacts are connected again. • Do the LOP configuration on the corresponding floor. (See sections 5.4.4 to 5.4.6) • If a key switch is connected to the LOP check for correct function. If necessary repeat the configuration (see section 5.4.8 to 5.4.10)
LIN	**After replacement**: • Make sure the LIN is connected to the LOP and the BIO bus • Do the LOP configuration on the corresponding floor. (See sections 5.4.4 to 5.4.6)
LCUX	**After replacement**: • Make sure the LCUX is connected to the LOP and the BIO bus • Make sure all inputs and outputs are connected • Do the LOP configuration on the corresponding floor. (See sections 5.4.4 to 5.4.6) • Check the function of all inputs and outputs. If necessary repeat the configurations (see section 5.4.14)

4.15.2 Replacement of other components

Replacement of the ACVF

	Tasks
ACVF	**After replacement**: • If the HMI displays [CF 16] during start up: The ACVF may have been used already in another system. In this case the ACVF parameters on the SIM card must be downloaded to the ACVF manually. Use CF=16, PA=99 for this procedure. • If the elevator stops with an error during the first trip: Check the encoder direction and phase sequence: CF=16, PA=14 and PA=15

Replacement of the Batteries in the LDU

	Sequence
	To replace the batteries the LDU covers must be removed.
1	Unplug the DC-AC connector on the SEM PCB
2	Disconnect the batteries and remove them. (Make sure you remember the correct cable position.)
3	Connect the new batteries to the inverter cables (there may be small sparks on the connectors)
4	After the batteries have been connected again: Plug connector DC-AC to SEM
5	Press the DEM button on the SEM PCB.
6	Check the manual evacuation
7	Check the automatic evacuation

Replacement of the Encoder, KB brake contacts, THMH

To replace the encoder, KB contacts or THMH on the machine the open loop travel has to be used. See section 4.9.3.

5 Commissioning and Configuration

5.1 Commissioning Procedure

A detailed commissioning procedure can be found in the document K 609754 (Quick reference guide “Installation and Commissioning” Schindler 3100/3300/3600/5300/6300).

- Menus and parameters are both on HMI SPECI.
- Not all are visible on HMI depending if OEM is active or not.

	Description and remarks
1	**Installation Travel** Check which have to be performed: Start a trip downwards with help of the recall control. – If the installation travel stops with an error (encoder or shaft speed error) the encoder direction has to be changed with CF=16, PA=14 – If the car travels to the wrong direction the phase sequence has to be changed with CF=16, PA=15
2	**Mechanical installation complete** (Counterweight filled, car decoration completed)
3	**Learning travel** with 0% car load. (Menu 10, Submenu 116=1) (Digisens enabled, 107=0) (see section 4.9.2)
4	**Mandatory Configurations** See section 5.2
5	**Reset** Some changes become active only after a reset. → After you have finished the configuration wait 30s. Then press the reset button (SCPU).

In case of problems with landing accuracy
Some special system configurations can lead to problems with the landing accuracy.

- **Systems with big interfloor travel distances**
  In this case the option “blind floors” can be ordered. (Additional PHS flags. For the configuration see CF=26)
- **Systems with 2 floors only or in case or general problems with the landing accuracy**
  In this case the learning travel can be done with a balanced car (approximately 50% load)
  - Step 1: Learning travel with balanced car
  - Step 2: Manual initiation of the ACVF pre-torque calibration with empty car (0% car load). See section 4.9.10 “Manual Pre-torque Calibration (HMI menu 123)”
  - Step 3: Step 3 applies only if the Digisens was calibrated but was not disabled during the learning travel: Redo the 0 Kg calibration (CF=98) with empty car.)

5.2 Mandatory Configurations and Sequence

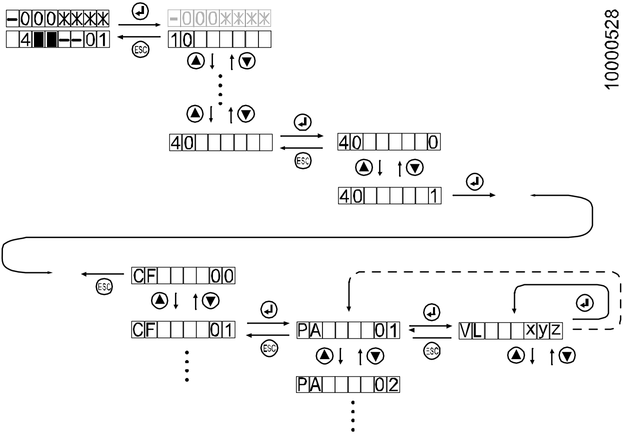
Detailed descriptions can be found in section "**Control Parameters (Menu 40)**" and "**Detailed Configuration Descriptions**".

1	Only in case of Penthouse option (LDU on the second top most floor): Configure the floor where the LDU is installed.	– CF=02, PA=07
2	If necessary (before the learning travel): – Encoder direction – Phase sequence	 – CF=16, PA=14 – CF=16, PA=15
3	Only if blind floor flags are installed: LIFD Long interfloor distance configuration	 – CF = 26
4	Only in case of independent door operation:	– CF = 30
5	Car load cell calibration – GQ definition – Digisens definition – 0 Kg recalibration – Calibration with reference load – Backup of calibration	 – CF=08, PA=01 – CF=08, PA=08 – CF=98 – CF=99 – CF=96
6	COP5B-N, COP4B and FI-GS only: Keypad configuration	 – CF=15
7	Floor designation	– CF=01
8	LOP configuration (If necessary: Manual LOP counting) Remark: In case of duplex systems with only one LOP per floor: refer to section "Duplex, LOP configuration with SBBD board".	– CF=00, LE=-- – (CF=00, LE=00)
9	All the other configurations are depending on the system options. There is no special configuration sequence necessary.	

5.3 Control Parameters (Menu 40)

5.3.1 Basic principle

Elevator Control Configuration with HMI



[24851; 23.11.2007]

Meaning of the abbreviations

CF	Configuration Group (main menu)
C	CF=05 C = COP identification (C1 ... C4; 1st ... 4th)
L	CF=40 and CF=41 only. L = Floor level
S	CF=40 and CF=41 only. S = Door side (1 or 2)
PA	Parameter (submenu) (if PA = 1..n then the VL is defined per floor whereby PA1=Floor1, PA2=Floor2 ..., PAn=highest available floor at learning travel)
VL	Value

To enter the configuration mode

	User Interaction	Display
	HMI shows actual level of elevator status	
1	Press “OK” on the HMI	10 _ _ _ _
2	With the UP/DOWN button change to 40	40 _ _ _ _
3	Press “OK”	40 _ _ _ 0
4	With the UP/DOWN buttons change to “40 1”. (Activation of configuration mode)	40 _ _ _ 1
5	Press “OK” The car travels to the configuration floor and opens the door. System is ready for configuration.	CF _ _ 01

To change a parameter

	User Interaction	Display
	System is ready for configuration.	CF _ _ 01
6	With the UP/DOWN buttons chose the main menu CF you want to configure. Press “OK” → The HMI will show the first submenu PA (Remark, in CF=40 and CF=41 the level L has to be chosen before the submenu PA is shown.)	PA _ _ 01
7	With the UP/DOWN buttons chose the submenu PA you want to configure. Press “OK” → The HMI will show the actual stored value VL	VL _ x y z
8	Press OK to change the first digit of the value → The changeable digit is blinking.	VL _ x y z
9	Press OK to confirm the changed digit and to change the next digit. Repeat step 8 and 9 until the submenu PA is displayed again.	PA _ _ x y

To leave the configuration mode

	User Interaction	Display
10	Press “ESC” until HMI shows 40 _ _ _ 1	40_ _ _ 1
11	With the UP/DOWN buttons change to 40 _ _ _ 0.	40_ _ _ 0
12	Press “OK”. → The top menu level is shown again.	40_ _ _ _
13	Press the ESC button. The HMI shows the actual elevator status again.	
14	After finishing the configuration press RESET on SCPU. (Some of the parameters become active after a reset only.)	

i

Do not change default values until you have to!

After you have finished the configuration wait 30s. Then press the reset button (SCPU). Some changes will be activated only after a reset.

5.3.2 Parameter List - Overview

– This list is valid for software versions V9.34.04, V9.38.08, V9.53, V9.56, V9.73, V9.85 and V10.07 (unless otherwise noticed)
– For most actual parameter check TK Commissioning EJ 604620.

Parameter visibility depends on: Software version, installed hardware and available options on the SIM card.

CF	Details	Remarks
00	LOP configuration (addressing and counting)	-
01	Floor designation (Naming for customer)	-
02	Main floors, parking floors, service floors	-
03	Door timers	-
04	Elevator data (Group (duplex, triplex) configuration, Date and Time)	-
05	COP input key switches	-
06	COP parameters (config - debounce, sensitivity, language, volume)	-
07	LOP/LIN parameters (sensitivity, volume)	-
08	Car parameters (load, light, etc.)	-
09	Alarm filter parameters	-
10	Pin codes for protected car call	→CF41
11	Statistics (Trip counter, hours in service)	-
12	SW versions - read only - (SCIC, SDIC, DRIVE, SEM, 1st ... 4rd COP, ETMA, CPLD, Overlay, Bootloader, Door Drive Side 1/2)	-
13	HW versions - read only - (SCIC, SDIC, DRIVE, SEM, 1st ... 4rd COP, MCCxx, ETMA, SMIC, Overlay, Door Drive Side 1/2)	-
14	NS21 parameters (ZNO, JNO, Door open timers)	-
15	COP configuration	-

CF	Details	Remarks
16	FC parameters	-
17	GS visitor control	1)
18	Car Gong (Type, Volume)	2)
19	ASMTL parameter	3)
20	Hydraulic drive parameters	-
21	Embedded Telemonitoring (and Alarm) ETM(A)	-
25	Mixed doors	-
26	Long interfloor distance LIFD	-
28	STM monitoring	-
30	Independent doors (ZZ2, ZZ3)	3)
31	Custom fire service	2)
32	Power saving	
40	BIO bus nodes configuration (LOP inputs, LIN, LCUX and BIOGIO inputs and outputs)	-
41	Restricted access functions (floor access and car access)	-
49	SIEU inputs and outputs	4)
55	Auxiliary input/output first SCOPH(MH), Dual brand SCOPMX-B	-
56	Auxiliary input/output second SCOPH(MH), Dual brand SCOPMX-B	-
60	Input on Pin 4 on LOP (BIO bus type 1)	→CF40
61..80	SLCUX auxiliary inputs and outputs, LOP input (BIO bus type 2), floor 1 (CF61) .. floor 20 (CF80)	→CF40
83	Level Assignment COP Key 1-4 for ZB3	→CF41
84	Mapping COPH input to CF83 (ZB3)	→CF41
86	Asymmetric group	3)

CF	Details	Remarks
87..89	LCUX in machine room	→CF40
94	BIO bus device address clearing	→CF40
95	Actual cabin load - read only	-
96	CLC calibration values - read only	-
97	CLC configuration	-
98	Zero carload frequency calibration	-
99	Calibration of car load measurement	-

1) not supported with SW ≥ V9.7 (see CF=41, PA=03)
2) not supported yet
3) not used with Schindler 3100/3300/5300 or not used with Schindler 3100/3300/5300 EU version

5

5.3.3 Parameter List - Detailed Description

- For all missing parameters refer to TK Commissioning EJ 604620.
- Parameter visibility depends on: Software version, installed hardware, available options on the SIM card and OEM activation.

Structure of the parameter table

CF	*Name of Parameter Group*		
	PA	*Name* (default = *value*) [**VL**min, **VL**max, step=unit] *Description*	

Abbreviations CF, PA, VL: See section 5.3.1

Parameter table

CF	PA	VL: Values and Description	OEM
00	**LOP configuration and addressing**		
	(See additional explanation in section 5.4 “Detailed configuration descriptions”)		
	- -	[LE - -]: **LOP addressing**. Configure each LOP during 12s countdown by pressing DE-**U**	[2]
	00	[LE 00]: **LOP Counting** (Indicated by “LC” and blinking “- - - -”) Stores addresses of all LOPs in EEPROM.	[2]

CF	PA	VL: Values and Description	OEM
01	**Floor designation (Customer floor naming), if SW < V11** (see additional explanation in section 5.4 “Detailed configuration descriptions”)		
	1 ... n	**Floor Name** (default = floor 1 = 0, 2 = 1 ...), [-9 ... 99, 1] Designation shown on position indicators. Activation of corresponding car call “buttons” on COP5-N. Configuration help: A change of floor level 1 (PA1) will change all floors above automatically. With FIGS100 and COP AP: Also letters (B, G, L, M, P, H, A, R, F) are possible. (Some restrictions may apply.)	[2]
	Floor designation (always), if SW > V11		
	1 ... 35	**Level** (1 ... maximum number of physical levels present, Floor_designation = Floor_level) It is the position of each floor. When the lowest floor is entered, all the others are calculated accordingly (incremental). Only existing floors are displayed (it is also useful to check the number of floors). Possible "Floor Designations" are limited by: displays type on devices (7 segment, 16 segments, matrix, graycode) and to input device (10 keypad or 1 ... n). Not all the alphabet characters can be configured and is limited to B, G, L, M, P, H, A, R, F (some restriction may apply). Examples: "01", "12A","11-F".	[2]

CF	PA	VL: Values and Description	OEM
02	**Main floors, parking floors, service floors**		
	1	**Fire Recall Floor** [1 ... n, 1 = 1] Main floor used for every BR service. The value is assigned automatically during the LOP counting to the first floor found where a JBF/KBF is connected (starting from bottom). In case of Duplex, both elevators must be configured equal. (The elevator with no own riser (that is, SBBD) must be configured manually.) In case of independent doors: Check CF30, PA1.	[0] SIM
	2	**Main Floor Policy (KA, KS)** (default = 1), [1 ... n, 1 = 1] Main floor used for collective control. It contains the same value as the Fire Recall Floor but can be changed. It is the floor used by actuating "star" button on COP. (In systems with CPIF option this parameter has to be set to the floor where the LIN is installed.) In case of selective doors: Set door side with CF30, PA2.	[2] SIM
	3	**Parking Floor** (default = 1), [1 ... n, 1 = 1] Main floor used for both “Return to parking floor” options (CF02, PA4 and PA5). It receives the same value as the Fire Recall Floor, but can be changed. Also used for JAB. In case of independent doors and parking with open doors: Set door side with CF30, PA3.	[2] SIM

CF	PA	VL: Values and Description	OEM
02	**Main floors, parking floors, service floors**		
	4	**Return to Parking Floor Timer** (default = 12 ↔ 120s) [0 ... 90, 1 = 10s] Timer for the option (RL1) “return to parking floor from any floor” (”0” = disable RL1)	[2] SIM
	5	**Return to Parking Floor Timer for Floors Below Parking Floor** (default = 5), [0 ... 90, 1 = 1 s] Time used by the option (RL2) “return to parking floor from any floor below the parking floor” if RL1 and RL2 are enabled: the shorter time is used (”0” = disable RL2).	[2] SIM
	6	**Alternative Fire Recall Floor 2** (default = 1), [1 ... n,1] 2nd main floor, used for every BR-ALT LUX service. The value is assigned automatically during the LOP counting to the second floor found where a JBF/KBF is connected (starting from bottom). If not used, set the same value as in CF02, PA1. In case of duplex, both elevators must be configured identically. (The elevator with no own riser (that is, SBBD) must be configured manually). In case of independent doors: Set door side with CF30, PA6.	[0] SIM
	7	**Configuration Floor** (default = highest floor), [1 ... n,1] Floor level where the car is sent when configuration mode is activated. CF30 PA7: set side for LDU configuration.	[2]

CF	PA	VL: Values and Description	OEM
02	**Main floors, parking floors, service floors**		
	8	**Machine Room Available** (default = 0), [0 ... 1, 1] Distinguishes between MRL and MR elevators. If set to 1 for elevators with MR, KFM does not open the door on the highest floor.	[2]
	9	**Door Hold Open Time After Fire Evacuation** (default = 0) [0 ... 99, 1 = 1s] Door open time after fire evacuation.	[2]
	18	**Call Cancellation Mode** (default = 32), [0 ... 32,1] – 0 = car and floor call cancellation disabled – 11 = car call cancellation, single tap – 12 = car call cancellation, double tap – 21 = floor call cancellation, single tap – 22 = floor call cancellation, double tap – 31 = car and floor call cancellation, single tap – 32 = car and floor call cancellation, double tap	[2] ≥ 9.8
	19	**Drifting to Top Time-out** (default = 5), [0 ... 90, 1 = 1 min] Parks the car away from the topmost floor after the defined time-out. (Sends the car to the parking floor defined by CF02, PA3, or to one floor below the topmost). 0 = Function disabled.	[0] ≥ 9.8 SIM
	20	**Pre-leveling time-out** (default = 2), [0 ... 72, 1 = 10 min] The STM might drift slightly over the traction pulley after a while. When this time is elapsed the elevator does a short correction trip to find the exact floor position again (at 12 cm distance to the PHS-edge). 0 = Function disabled	[1] ≥ 9.8 SIM
	21	Pit car position distance (default = 150), [0 ... 200, 1 = 1 cm] This value defines the distance between first floor and cabin for the pit access.	[2] ≥ 9.9

CF	PA	VL: Values and Description	OEM
02	**Main floors, parking floors, service floors**		
	22	ISPT (default = 3), [2 ... 10, 1] This value defines the maximum number of unexpected ISPT opening before bring to a breakdown.	[2] ≥ 9.9 SIM

CF	PA	VL: Values and Description	OEM
03	**Door timers**		
	1	**Door Hold Open Time for Boarding** (default = 40 ↔ 4s), [10 ... 255,1 = 0.1s] Door open time after a floor call	[0]
	2	**Door Hold Open Time for Exiting** (default = 30 ↔ 3s), [10 ... 255, 1 = 0.1s] Door open time after a car call	[0]
	3	**Minimum Door Open Timer** (default = 20 ↔ 2s), [10 ... 255,1 = 0.1s] Is the minimum door open time used by services (not in normal traffic operation, for example, AAT, Earthquake, NS21, DKFM, Automatic car positioning, Overload….) usually during the conclusion phase. For relay controlled CDD: Default = 30	[2]
	4	**Minimum Door Hold Open Time after DT-O** (default = 10 ↔ 1s), [10 ... 255, 1 = 0.1s] Door hold open time after DT-O For relay controlled CDD: Default = 30	[2]
	5	**Extra Door Hold Open Time** (default = 20 ↔ 2s), [10 ... 255, 1 = 0.1s] Combo Call (Coincidence of car call and floor call) CF3, PA5 is added to CF3, PA1	[2]

CF	PA	VL: Values and Description	OEM
03	**Door timers**		
	6	**Door Pre-opening Delay** (default = 0), [0 ... 99, 1 = 0.1s] Delay between activation of PHUET and start of pre-opening. Preopening problems caused by unaligned PHS-to-PHUET-flags can be compensated by this preopening-delay parameter.	[2]
	7	**Maximum Door Locking Time** (default = 50 ↔ 500s), [6 ... 60, 1 = 10s] Maximum door lock time during an evacuation travel or learning travel. (Does not affect the standard maximum lock time of 3 minutes.)	[2]
	8	**Door Debouncing Time** (default = 3 ↔ 0.3s), [1 ... 30, 1 = 0.1s] Time that the control waits after door is locked (SV active) before it checks the safety circuit (KV). This delay time is used even if no SV/KV is present	[2]
	9	**Door Hold Open Time after Reversing** (default = 20 ↔ 2s), [0 ... 255, 1 = 0.1s] Time of open door after activation of reversing devices (KSKB and RPHT). For relay controlled CDD: Default = 30	[2]
	10	**Door Hold Open Time after KSKB Reversing** (default = 20 ↔ 2s), [0 ... 255, 1 = 0.1s] This parameter is only visible and configurable if CF03, PA9 is set to VL = 0. Time of open door after activation of KSKB. For relay controlled CDD: Default = 30	[2]

CF	PA	VL: Values and Description	OEM
03	**Door timers**		
	11	**Door Hold Open Time after RPHT Reversing** (default = 20 ↔ 2s), [0 ... 255, 1 = 0.1s] This parameter is only visible and configurable if CF03, PA9 is set to VL = 0. Time of open door after interruption of photocell RPHT. For relay controlled CDD: Default = 30	[2]
	14	**DO NOT CHANGE THIS PARAMETER!** (default = 120 ↔ 12s) (Door opening time-out: If the door is not fully opened after this time, the door motor is switched off and an error is logged).	1) [2]
	15	**DO NOT CHANGE THIS PARAMETER!** (default = 120 ↔ 12s) (Door closing time-out: If the door is not fully closed after this time, the door motor is switched off and an error is logged).	1) [2]
	20	**DO NOT CHANGE THIS PARAMETER!** (default = 10 ↔ 1s) Minimum opening time: If the KET-O switch is always active (defective) and the door shall be opened starting from the closed position, the door will move in opening direction for this minimum time.	1) [2]
	21	**Final Timer** (default = 30 ↔ 30s), [0 ... 100, 1 = 1s] **Not available for EU**. (Function needs SIM card option “Final timer”.) If RPHT is interrupted and a call gets pending after the final timer elapses, the elevator ignores the RPHT and tries to close the door with low speed and the elevator serves the call. The Final Timer is exclusive-or to RPHT-Monitoring since SW ≥ V9.5 , activated by CF22, PA2.	1) [2] SIM

5

CF	PA	VL: Values and Description	OEM
03	**Door timers**		
	25	**Parking type 6** (default = 12 ↔ 120s), [1 ... 90, 1 = 10s] In parking type 6 mode, this is the time the door will stay open if the car is parked on the predefined floor. (Used for Moscow only.)	[2] SIM
	26	**Out of Group Time-Out** (default = 3 ↔ 30s), [0 ... 60, 1 = 10s] Defines the time needed for enable out of group function. If value is set to 0 the out of group function is disabled. (To avoid that a door nudging operation takes place while the elevator is working in group but is blocked on a floor by an obstructed photocell, it is advisable to program CF22 PA2 to a value bigger than CF03 PA26 (max 255 s) or to disable the nudging function by setting the parameter CF22 PA2 to VL = 0.)	[2] ≥ 9.7
	27	Door open timer after earthquake evacuation (S00x ≥ V9.9) Default = 15 s [0 ... 60, 1 = 1 sec.] After earthquake, the door shall be opened for a defined time. 0 = infinite time	[0]
	28	Door open timer after automatic evacuation (S00x ≥ V9.9) Default = 15 s [0 ... 60, 1 = 1 sec.] After automatic evacuation, the door shall be opened for a defined time. 0 = infinite time	[0]
	29	Extended Dwell Time for extended door open button (S00x ≥ V9.92, SIM card) Default = 20 s [1 ... 180, 1 = 1 sec.] This parameter is the door open timer applied when dtx-o is pressed. Note: It is recommended to keep this time smaller than final timer CF3PA21.	[2]

CF	PA	VL: Values and Description	OEM
04	**Elevator Data**		
	1	**Elevator ID in a Group** (default = 1), [1 ... 4, 1] 1 = First elevator, ..., 4 = Fourth elevator In case of Duplex systems with **SIM card option "DUPLEX":** The value is set automatically based on commissioning number. (Read only) For SW ≥ V9.7 and in case of **Overlay** (No "Duplex" SIM card option): The value has to be set for each elevator in the group. (See section 5.4 "Detailed configuration description".)	[2] SIM
	2	**Local Time** [00.00.00 ... 23.59.59] Local time in hh.mm.ss. Time stamp. Used for example for the error history.	[0] ≥9.34
	5	**Local Date** [01.01.00 ... 31.12.99] (year 2000 ... 2099) Local date in dd.mm.yy. Date stamp. Used for example for the error history. **Note:** The Remote Monitoring system automatically adjusts the internal clock, but a date too far in the past or future (i.e. years) could be a reason for the not working Remote Monitoring.	[0] ≥9.34

CF	C	PA	VL: Values and Description	OEM
05	**COP5 Input key switches (If SW ≤ V9.8x)**			
	-	**1 ... 4**	**Key1, Key2, Key3, Key4 on COP #1** (default. = 0 ↔ no function), [0 ... 255, 1] Input functions on COP #1, XKey1 ... XKey4: BMK see section 5.3.4 “BMK Function Codes”. (Wiring sequence: SDIC → COP2 → COP1) Remark: Do not use BMKs which are already used in CF05, PA11 ... 14.	[2]
		11 ... 14	**Key1, Key2, Key3, Key4 on COP #2** (default = 0 ↔ no function), [0 ... 255, 1] Input functions on COP #2: see PA1 ... 4 COP# number according to wiring sequence: SDIC → COP2 → COP1) Do not use BMKs which are already used in CF05, PA1 ... 4.	[2] ≥9.34
	If SW ≥ V9.9 and COP detected: COP Input			
	C1	**1 ... 3**	Input key 1 ... 3 (default = 0), [0 ... 999, 1] For the BMK function code, see section 5.3.4 “BMK Function Codes”.	[2] ≥ 9.9
		4	If COP5/COPPI: Input key 4 (default = 0), [0 ... 999, 1] For the BMK function code, see section 5.3.4 “BMK Function Codes”.	[2] ≥ 9.9
		5 ... 6	If COPPI: Input key 5 ... 6 (default = 0), [0 ... 999, 1] For the BMK function code, see section 5.3.4 “BMK Function Codes”.	[2] ≥ 9.9
		51 ... 56	If COPPI: Output 1 ... 6 (default = 0), [0 ... 999, 1] For the BMK function code, see section 5.3.4 “BMK Function Codes”.	[2] ≥ 9.9

CF	C	PA	VL: Values and Description	OEM
	C2	**1 ... 56**	2nd COP for details, see CF05, C1, PA1 ... 56 For the BMK function code, see section 5.3.4 “BMK Function Codes”.	[2] ≥ 9.9

CF	PA	VL: Values and Description	OEM
06	**COP Parameters**		
	1	**Time for 2-digit input** (default = 20 ↔ 2s), [1 ... 50, 1 = 0.1s] Max. time to enter a 2-digit call	[2]
	2	**Car Call Acknowledge Display Time** (default = 5 ↔ 0.5s), [1 ... 50, 1 = 0.1s] Car call acknowledgement display time (for collective system)	[2]
	3	**Time to Enter PIN code** (default = 50 ↔ 5s), [1 ... 99, 1 = 0.1s] Max. time to enter a code-protected call (COP5x with 10-digit keypad). With SAS: Time to enter the call.	[2]
	4	**COP Sensitivity** (default = 5), [0 ... 7, 1] Sensitivity of the COP5x keypad (capacitive) (0 = lowest sensitivity). Best results with VL = 04.	[1]
	6	**COP Gong Volume** (default = 3), [0 ... 10, 1] 0 = minimum ... 10 = maximum The gong is used for DM236 (Italy) only. Changing this parameter has influence since COP software ≥ V3.1 only.	[0] COP ≥ 3.1
	7	**COP Position indicator code** (default = 1), [1 ... 2, 1] For 3rd party position indicator: 1 = Gray code, 2 = Binary code. Not used with Schindler 3100/3300/5300	[2] 1)
	14	**DO NOT CHANGE THIS PARAMETER!** **NF 1, Type 1 or 2** (default = 1), [1 ... 2, 1]	[2] 1)

5

CF	PA	VL: Values and Description	OEM
06	**COP Parameters**		
	15	**DO NOT CHANGE THIS PARAMETER!** **Voice Announcement Door** (default = 3), [0 ... 3, 1] 0 = no announcement, 1 = door closing, 2 = door opening, 3 = door closing and opening.	[2] 1) SIM
	16	**DO NOT CHANGE THIS PARAMETER!** **Timer Voice Announcement Door** (default = 0), [1 ... 255, 1 = 0.1s] Time between the voice announcement and the start of the door movement.	[2] 1) SIM
	20	Volume for Voice Announcement during Specified Time (default = 0), [0 ... 10, 1] 1 = minimum ... 10 = maximum 0 = voice announcement switched off This is the volume applied within time defined by PA21 and PA22. If both PA21 and PA22 are 00:00 the function is disabled. At the moment changing PA20 has no influence. Internally PA20 keeps VL = 0 always.	[2] ≥ 9.7
	21	**Start time for Voice announcement volume change** (default = 00:00), [00:00 ... 23:59, 1] This is the start time when the volume on PA20 is applied. (At the moment: Time when voice announcement is switched off.)	[2] ≥ 9.7
	22	**Stop time for Voice announcement volume change** (default = 00:00), [00:00 ... 23:59, 1] This is the stop time when the volume on PA20 was applied. (At the moment: Time when voice announcement is switched on again.)	[2] ≥ 9.7

CF	PA	VL: Values and Description	OEM
07		**LOP/LIN parameters**	
	1	**LOP Position indicator code** (default = 1), [1 ... 2, 1] For 3rd party position indicator. 1 = Gray code, 2 = Binary code. Not used with Schindler 3100/3300/5300	[2] 1)
	2	**LOP Sensitivity** (default = 4), [1 ... 7, 1] Sensitivity Bionic 5 LOPs (capacitive buttons) (7 = least sensitive)	[0]
	3	**LOP Volume** (default = 3), [0 ... 5, 1] Has only influence on the mechanical LOPs.	[2]
	4	**LIN Volume** (default = 3), [0 ... 5, 1] Adjustment of the volume of the gong, which is connected to the LIN5x.	[2]
	8	**CW preferred elevator** (default = 0), [0 ... 1, 1] Defines which car of the group is used in car preference service (CW). Used for DCW-U and DCW-D landing calls. 0 = Non CW elevator, 1 = CW elevator See additional explanation in section 5.4 “Detailed configuration description, Asymmetric duplex”.	[2]
	10	**LOP configuration countdown time** (default = 12), [6 ... 24, 1s] Countdown timer during LOP configuration	[2] ≥ 9.7

CF	PA	VL: Values and Description	OEM
08		**Car parameters**	
	1	**Rated Load** (default = 0), [20 ... 113, 1 = 10 Kg] (example: 625 Kg = 62) Rated load of the car (GQ)	[2]

5

CF	PA	VL: Values and Description	OEM
08	**Car parameters**		
	2	**Automatic Car Light Timer** – SW <V9.5: (default = 10 min.), [1 ... 40, 1 = 1min] – SW ≥V9.5: (default = 60 min.), [30... 90, 1 = 1min] – SW ≥V9.7: (default = 60 min.), [1 ... 90, 1 = 1min] For the option “automatic car light”. After this time of inactivity, the car lighting is switched OFF.	[0] SIM
	7	**Door Pre-Opening Enable** (default = 1 = enabled), [0 ... 1, 1] Software control of door pre-opening feature with SUET PCB. 1 = enabled, 0 = disabled	[1]
	8	**Car Load Sensor Type** (default = 0), [0 ... 1, 1] – 0 = Red Digisens KL250 – 1 = White Digisens KL66	[1]
	9	**Full Load Threshold** (default = 90 %), [50 ... 90, 1 %] Threshold for full load activation in percent of rated load.	[1]
	10	**Registered Car Call Canceling after Door Reversing** (default = 1), [0 ... 1, 1] In DE and PI policy the car call can be cancelled after door-reversing. 0 = no cancellation, 1 = cancellation after reversing.	[2]
	11	**Action of car light switch JLC** 0 = Type A JLC OFF: turns the car light permanently off; JLC ON: turns the car light on/off according to function ‹RLC› 1 = Type B (default) JLC OFF: turns the car light on/off according to function ‹RLC›; JLC ON: turns the car light permanently on.	[1] 1)

CF	PA	VL: Values and Description	OEM
08	**Car parameters**		
	13	**Car Light and Fan Time-out with Fire Service** (default = 0), [0 ... 300, 1 = 1s] Used in HongKong and Singapore fire service only. (for SW ≥ V9.8: See CF31, PA21)	[0]
	19	**Over Load Threshold** (default = 105), [50 ... 105, 1 = 1%] Threshold for activation of overload in percent of rated load. Limits defined in the EN81-1: If GQ > 750 Kg the maximum = GQ + 5 % IF GQ < 750 Kg the maximum = GQ + 38 Kg (hard-coded)	[1] ≥ 9.8 1
	20	Landing accuracy threshold (default = 1), [0 ... 10, 1 = 1 mm] Minimum step for corrective action.	[1] ≥ 9.9
	21	Landing accuracy correction (default = 6), [0 ... 15, 1 = 1 mm] Number of millimeters that can be added or removed for each corrective action. If this value is set to 0 the landing accuracy improvement function is disabled.	[1] ≥ 9.8 1

CF	PA	VL: Values and Description	OEM
09	**Alarm filter parameters**		
	1	**SW** ≤ **V9.38**: **Alarm Filter** (default = 1), [0 ... 1, 1] Alarm button filtering (while car is moving or door open on floor). 0 = disable, 1 = enable. Systems with SDIC5: Depending on switch JRA-A.	
		SW > V9.38 and < V9.5 (not used with S3300 EU): Alarm Filter (default = 1), [0 ... 3, 1] Alarm button filtering (while car is moving or door open on floor). 0 = disabled, 1 = enable. VL = 2 or 3 must not be used unless instructed by an expert!	1)
		SW ≥ **V9.5**: **Alarm Filter** (default = 1 or 2), [0 ... 2, 1] – VL = 0: Disable – VL = 1: Standard filtering. While the car is moving or the door is open the alarm is filtered. Pushing the alarm button will not release an alarm to the control center. – VL = 2: Advanced filtering. Like VL = 1. But if the elevator is on floor it first tries to open the doors (DTO). If this is unsuccessful an alarm is sent to the control center. In case of COP5 capacitive and COP firmware ≥ V3.1, VL = 2 is default. Alarm filter can be disabled temporarily with menu 10 > 134.	[0] ≥ 9.5

CF	PA	VL: Values and Description	OEM
09	**Alarm filter parameters**		
	3	**DO NOT CHANGE THIS PARAMETER** unless instructed by an expert. **Alarm Button Timer** (default = 30), [0 ... 255, 1=100ms] If the alarm button is pressed for more than this time the alarm request is considered as valid. The alarm monitoring starts to check whether the door opens properly or the car starts a normal trip. Only available with advanced alarm filtering CF09, PA1, VL = 02	[0] ≥ 9.5
	4	**DO NOT CHANGE THIS PARAMETER** unless instructed by an expert. **Alarm Door Open Timer** (default = 20), [0 ... 255, 1 = 1s] After the alarm is considered as valid, this is the maximum time the systems tries to open the door or to start a normal trip. Only available with advanced alarm filtering CF09, PA1, VL = 02	[0] ≥ 9.5
	5	**DO NOT CHANGE THIS PARAMETER** unless instructed by an expert. **Alarm Relay Timer** (default = 10), [0 ... 255, 1 = 1s] Time the alarm relay is switched by the control to generate a real alarm after the filtering time specified in PA4.	[0] ≥ 9.5

5

CF	PA	VL: Values and Description	OEM
11	**Statistics**		
	1	**Trip Counter** [0 ... 999999, 1 = 100] Example: 26 = 2600 trips	[0]

CF	PA	VL: Values and Description	OEM
11	**Statistics**		
	2	**Cumulated Hours in Service** [0 ... 9999, 1 = 1 hour] Total operating hours (car traveling)	[0]

CF	PA	VL: Values and Description	OEM
12	**Software Version (read-only)**		
	1	SW Version **SCPU** (example: 95 ↔ V.9.5)	[2]
	2	SW Version **SDIC** (example: 21 ↔ V.2.1) (Value updated after learning travel)	[2]
	3	SW Version **ACVF** (only closed loop)	[2]
	4	SW Version SEM (SMART MRL SEM)	[2], 2)
	5	SW Version **COP #1**	[2]
	6	SW Version **COP #2** (if available)	[2]
	8	SW Version **CLSD** (example: 1205 ↔ V1.2.05)	[2]
	9	SW Version **SMIC CPLD** (example: 18 ↔ V1.8)	[2] Rel.4
	10	Release Version **Overlay** (xx.xx.xx = “release”.”subrelease”.”revision”)	[2] Rel.6
	11	SW Version **COP #3** (if available)	[2] ≥ 9.7
	12	SW Version **COP #4** (not yet used)	[2]
	13	Bootloader Version	[2]

CF	PA	VL: Values and Description	OEM
13	**Hardware Version (read-only), See table in EJ 604639**		
	1	HW Version **SCPU**	[2]
	2	HW Version **SDIC** (51 ... 58 = SDIC5, 60, 63 = SDIC51, 61, 64 = SDIC52, 62, 65 = SDIC53)	[2]

CF	PA	VL: Values and Description	OEM
13	**Hardware Version (read-only), See table in EJ 604639**		
	3	HW Version **ACVF** (only closed loop)	[2]
	4	HW Version SEM (SMART MRL SEM)	[2], 2)
	5	HW Version **COP #1**(see extra table)	[2]
	6	HW Version **COP #2** (if available) (see extra table)	[2]
	7	HW Version MCCxx	[2], 2)
	8	HW Version **CLSD or ETMA** (65 ... 69 = CLSD11, 49 = ETMA)	[2]
	9	HW Version **SMIC** (5 = SMIC5, 6 = SMIC6)	[2] Rel.4
	10	HW Version **Overlay**	[2] Rel.6
	11	HW Version **COP #3** (if available) (see extra table)	[2] ≥ 9.7
	12	HW Version **COP #4** (not yet used)	[2]

5

0	Unknown HW	**10** **...** **40**	Old COPs, not valid for Schindler 3100/3300/5300
51	COP5_N	**52**	COP5_10
53	COP5B_10 or COP5B_N	**54**	COP4_B (5 floors)
55	COP5 AP (any)	**56**	COP5_N ZLA
57	COP5_10 ZLA	**58**	COP5B_10 ZLA
59	unknown COP5 HW	**75**	SCCI
80	SCOPH3	**81**	SCOPHM3
82	SCOPHMH3	**83**	SCOPMXB3
90	COP5B_10 AU	**93**	COP5 AP with EU fixtures
94	COP4_B_EU_8 (8 floors)	**95**	COP4_B_EU_12 (12 floors)
96	FIGS (any)	**99**	POP1.Q

Values for CF13 PA5, 06, 11, 12, COP HW Version

CF	PA	VL: Values and Description	OEM
14	**NS21 Parameters (SW ≥ V9.38)** **Not possible with BIOGIO (GUE/GLT), not possible with TSD21 systems.**		
	1	**ZNO Timer** (default = 0), [0 ... 99, 1 = 10s] Time between RNO activation and start of evacuation. (To start the evacuation the signal RFEF is necessary)	[1] SIM
	2	**JNO Time-out** (default = 0), [0 ... 99, 10s] Time between the end of the evacuation and the time when the elevator must start monitoring the JNO signal (to release the car for normal service). (0 = no release of this car).	[1] SIM
	3	**Door Open Timer Evacuation** (default = 0), [0 ... 99, 1 = 1] Time to keep the door open at the evacuation floor after evacuation (NS21 Marine)	[1] SIM

CF	PA	VL: Values and Description	OEM
15	**COP5B_N, COP4B, FI-GS Keypad Configuration** (see additional explanation in section 5.4 “Detailed configuration descriptions”)		
	-	**COP Keypad Configuration (SW < V9.5)**, [-3 ... 8, 1] The COP keypad configuration starts with the lowest floor. COP shows FL and n. (“n” stands for the floor which is ready to configure.) COP5B_N: COP with mechanical buttons needs floor assignment before use. To leave CF15 without changing anything: Press DTO for 3 seconds or press <ESC>. SW < V9.5: A learning travel will erase the CF15 configuration.	

CF	PA	VL: Values and Description	OEM
15	**COP5B_N, COP4B, FI-GS Keypad Configuration** (see additional explanation in section 5.4 “Detailed configuration descriptions”)		
	1	**COP#1 Keypad Configuration (SW ≥ V9.5)**, [-3 ... 8, 1] (COP#1 is physical the last COP5 in the chain) The COP keypad configuration starts with the lowest floor. COP shows FL and n. (“n” stands for the floor which is ready to configure.) To leave CF15 without changing anything: Press DTO for 3 seconds or press <ESC>.	[2] ≥9.5x
	2	**COP#2 Keypad Configuration (SW ≥ V9.7)**, [-3 ... 8, 1] (COP#2 is physical the first COP5 in the chain) (see PA1)	[2] ≥9.7
	11	**Clear COP#1 Keypad Configuration (SW ≥ V9.5)**, [0 ... 1, 1] To clear the keypad configuration (= set back to default). VL = 1: Clear command. (With FIGS100: To set back to factory default = Buttons are not available.	[2] ≥9.5x
	12	**Clear COP#2 Keypad Configuration (SW ≥ V9.7)**, [0 ... 1, 1] (see PA11)	[2] ≥9.7

CF	PA	VL: Values and Description	OEM
16	**FC Parameters (Vacon xx C NXP with CAN Bus) (for additional explanation see document J 42101238) Default values are depending on system specification**. **Don’t change any parameter unless you are a system expert. For normal use only PA14 and 15 have to be changed.**		
	01	Leveling speed (default = 20 ↔ 0.20 m/s), [5 ... 30, 1 = 0.01 m/s]	[2]

5

CF	PA	VL: Values and Description	OEM
	02	Inspection speed (default = 25 ↔ 0.25m/s), [10 ... 30, 1 = 0.01 m/s] If Schindler 3600: The maximum inspection speed = 0.6 m/s	[2] SIM
	03	Rated low speed (default = 50 ↔ 0.50m/s), [10 ... 80, 1 = 0.01 m/s]	[2]
	04	Evacuation speed (default = 10 ↔ 0.10m/s), [10 ... 30, 1 = 0.01 m/s]	[2]
	05	Shaft speed limit (default = 30 ↔ 0.30m/s), [0 ... 30, 1 = 0.01 m/s]	[2]
	06	Speed supervision limit (default = 80 ↔ 0.80m/s) [0 ... 80, 1 = 0.01 m/s]	[2]
	07	Acceleration1 (default = 50 ↔ 0.50 m/s2), [10 ... 90, 1 = 0.01 m/s2]	[2]
	08	Deceleration1 (default = 50 ↔ 0.50 m/s2), [10 ... 90, 1 = 0.01 m/s2]	[2]
	09	Stop distance evacuation (default = 100mm), [5 ... 250, 1 mm]	[2]
	10	End distance (default = 120 mm), [5 ... 250, 1 mm]	[2]
	11	PosCorrectLim (default = 30 mm), [0 ... 200, 1 mm]	[2]
	12	Average landing error (default = 0), [-20 ... 20, 1 mm]	[2]
	13	KG Balancing Factor (default = 50%), [0 ... 50, 1%] (depending on SIM card)	[2] SIM
	14	**Encoder direction** (default = 1), [0 ... 1, 1]	[2]
	15	**Phase sequence** (default = 1), [0 ... 1, 1]: 0 = U-V-W, 1 = U-W-V	[2]

CF	PA	VL: Values and Description	OEM
	16	Torque top (default = 0%), [-50 ... 50%] (example: 117 = 11.7%) Do not change this value, it is calculated at the learning travel!	[2]
	17	Torque bottom (default = 0%), [-50 ... 50%] (example: 117 = 11.7%) Do not change this value, it is calculated at the learning travel!	[2]
	18	Shaft Speed Time (default = 40 ↔ 0.40 s), [0 ... 500, 1 = 0.01 s]	[2]
	19	Shaft Service Time (default = 20 ↔ 0.20 s), [0 ... 500, 1 = 0.01 s]	[2]
	20	Shaft Service Speed Limit (default = 8 ↔ 0.08s), [0 ... 30, 1 = 0.01 m/s]	[2]
	21	KB/KB1 monitoring (default = 1 ↔ enable), [0 ... 1, 1]: 1 = enable, 0 = disable. DO NOT DISABLE in a Schindler 3100/3300/5300 system!	[2] 1)
	22	(U/f curve) and stator impedance identification [0 ... 1, 1]	[2]
	23	RPM identification [0 ... 4, 1]	[2]
	24	Brake Closing time-out (default = 125 ↔ 1.25 s), [35 ... 200, 1 = 0.01s]	[2] 1)
	25	Brake Opening time-out (default = 9 ↔ 0.09 s), [5 ... 18, 1 = 0.01s]	[2] 1)
	26	Motor thermistor type (default = 1 ↔ enable), [0 ... 1,1]: 1 = enable, 0 = disable	[2] 1)
	27	Revelling end distance (default = 15 ↔ 15 mm), [0 ... 120, 1 = 1 mm]	[2] 1)
	28	Maximum revelling distance (default = 40 ↔ 40 mm) [0 ... 250, 1 = 1mm]	[2] 1)

5

CF	PA	VL: Values and Description	OEM
	29	Manual pre-torque (default = ...), [-1000 ... 1000, 1]	[2] 1)
	30	Shaft speed limit releveling (default = 0.08m/s), [0 ... 30, 1 = 0.01 m/s]	[2] 1)
	31	Releveling maximum speed (default = 0.02m/s), [1 ... 30, 1 = 0.01 m/s]	[2] 1)
	32	Shaft time releveling (default = 0.02 m/s), [0 ... 100, 1 = 0.01 m/s]	[2] 1)
	33	System inertia (default: CBR:-0.1, PBR-OL = 0, PBR-CL = -0.1) [-1 ... 160, 1 = 0,01 Kgm2]	[2] 1)
	34	Encoder Pulse revolution (default = 4096) (*), [500 ... 9999, 1]	[2] SIM
	35	**Nominal speed** (default = 100 ↔ 1 m/s), [10 ... VKN, 1 = 0.01 m/s]	[2] SIM
	36	Rated motor voltage (default = 340V), [180 ... 420, 1 V]	[2] SIM
	37	Rated motor frequency (default=5000 ↔ 50 Hz) (*) [1000 ... 7000, 1 = 0.01 Hz]	[2] SIM
	38	Rated motor speed (default = 1440 rpm) (*), [100 ... 3000, 1 rpm]	[2] SIM
	39	Rated motor current (default = 12 ↔ 1.2A), [12 ... 330, 1 = 0.1 A] Range depends on Vacon type	[2] SIM
	40	Motor cos phi (default = 85%), [70 ... 90, 1 = 1%]	[2] SIM
	41	Magnetizing current (default = 68 ↔ 6.8 A) (*), [3 ... 300, 1 = 0.1 A] Range depends on PA39	[2] SIM
	42	Rated motor power PMN (default = 75 ↔ 7.5 kW) [0 ... 300, 1 = 0.1 kW]	[2] SIM
	43	Stator impedance (default = 1000 ↔ 1 Ohm) [0 ... 20000, 1 = 0.001 Ohm]	[2] SIM

CF	PA	VL: Values and Description	OEM
	44	IW - Gear ratio (default = 100 ↔ 1) (*), [100 ... 4000, 1 = 0.01] Change or validation of this parameter may modify: PA26, 34, 21 and some speed control parameter.	[2] SIM
	45	KZU - Reeving factor (default = 2 ↔ 2:1) (*), [1 ... 2, 1] 1 = 1:1, 2 = 2:1	[2] SIM
	46	GQN - Rated payload (default = 400 Kg), [0 ... 1300, 1 Kg]	[2] SIM
	47	DD - Traction sheave diameter (default = 870 ↔ 87 mm) [600 ... 30000, 1 = 0.1 mm] (*)	[2] SIM
	48	Motor temperature failure limit (default = 120°C), [5 ... 140, 1°C] Depending on motor type	[2] SIM
	49	ACVF parameter version (Read-only)	[2]
	50	ACVF commissioning number (Read-only)	[2], 1)
	51	Speed Ctrl KP 1 (P gain acceleration), (default: CBR = 20, PBR-OL = 100, PBR-CL = 20), [0 ... 1000, 1 = 0,1x]	[2] V9.8
	52	Speed Ctrl KP 2 (P gain constant speed), (default = 30) [0 ... 1000, 1 = 0,1x]	[2] V9.8
	53	Speed Ctrl KP 3 (P gain deceleration), (default: CBR = 20, PBR-OL = 100, PBR-CL = 20), [0 ... 1000, 1 = 0,1x]	[2] V9.8
	54	Speed Ctrl Ti 1 (Ti integration time acceleration), (default: CBR = 15, PBR-OL = 10, PBR-CL = 15), [0 ... 500, 1 = 0,1x]	[2] V9.8
	55	Speed Ctrl Ti 2 (Ti integration time constant speed), (default: CBR = 45, PBR-OL = 20, PBR-CL = 45), [0 ... 500, 1 = 0,1x]	[2] V9.8
	56	Speed Ctrl Ti 3 (Ti integration time deceleration), (default: CBR = 15, PBR-OL = 10, PBR-CL = 15), [0 ... 500, 1 = 0,1x]	[2] V9.8

CF	PA	VL: Values and Description	OEM
	57	Motor type, 0 = undefined, 1 = asynchronous, 2 = permanent magnet synchronous, (default = 0)	[2] V9.8
	58	Motor d-axis impedance, (default = 0), [0, 65535, 1 = 1 mOhm]	[2] V9.8
	59	Motor q-axis impedance, (default = 0), [0, 65535, 1 = 1 mOhm]	[2] V9.8
	60	Mains supply voltage, (default = 3), [1 ... 6, 1] 1 = 230V, 2 = 380V, 3 = 400V, 4 = 415V, 5 = 460V, 6 = 480V.	[2]
	61	Encoder type, (default = 0), [0 ... 4, 1] 0 = incremental, 1 = sine wave, 2 = no encoder, 3 = sincos ENDAT, 4 = sincos HYPERFACE.	[2]
	62	Load measuring type, (default = 2), [0 ... 130, 1] 0 = no measurement, 1 = contacts, 2 = load sensor, 128 = no measurement and fast speed control, 129 = contacts and fast speed control, 130 = load sensor and fast speed control.	[2]
	63	Gear type, (default = 0), [0 ... 3, 1] 0 = No gear, 1 = worm gear, 2 = planet gear, 3 = semi gearless belt (SGB).	[2]
	95	Restore (EEPROM → SIM card) (Only allowed after a backup with PA96)	[2]
	96	Backup (SIM card → EEPROM)	[2]
	97	Compare ACVF versus SIM card example: [34_ _ _ 0]: PA34 has the same value on ACVF and SIM, [35_ _ _ 1]: PA 35 has different values on SIM and ACVF.	[2]
	98	Upload FC Parameters (ACVF → SIM card)	[2]
	99	Download FC Parameters (SIM card → ACVF)	[2]
	(*) a change of this parameter forces a learning trip		

CF	PA	VL: Values and Description	OEM
17	**Visitor Control GS** **For SW < V9.7 only. (SW ≥ V9.7: See CF=41, PA=03)**		
	1..n	**Acknowledge Time for Guest Calls** (default = 0 = GS disabled) [0..12, 1 = 10s]: Sets the duration of the visitor request signalization and the enable time for the “access granted” button. For GS set to 3 ↔ 30s as standard value. GS is restricted to simplex with PI. See also section 5.4 “Detailed configuration descriptions”	SIM <9.7

CF	PA	VL: Values and Description	OEM
19	**Not used with Schindler 3100/3300/5300 EU** **Approach Speed Monitoring at Terminal Landing (ASMTL)** **For SW ≥ V9.34 only**		
	4	Not used with Schindler 3100/3300/5300 EU KSE distance (default =1250 ↔ 1250mm) [350..3000, 1 = 1mm] KSE distance. This value becomes “read-only” after the learning travel.	L2 1) SIM
	5	Not used with Schindler 3100/3300/5300 EU KSE speed limit (default =985 ↔ 985 mm/s) [350 ... 3000, 1 = 1mm] This value becomes “read-only” after the learning travel.	L2 1) SIM

5

CF	PA	VL: Values and Description	OEM
21	**Embedded Telemonitoring (and Alarm) ETM(A)** **For Rel. 04 only (SW ≥ V9.3x)**		
	1	**Servitel-ID (higher part)** (default = 000000) This parameter is part of the telemonitoring device identification towards the remote monitoring control center.	[1] SIM

CF	PA	VL: Values and Description	OEM
21	**Embedded Telemonitoring (and Alarm) ETM(A)** **For Rel. 04 only (SW ≥ V9.3x)**		
	2	**Servitel-ID (lower part)** (default = 000000) This parameter is part of the telemonitoring device identification towards the remote monitoring control center.	[1] SIM
	3	**Module number** (Default = 10), [1 ... 254, typically 10, 20, 30, ...] It designates a TM module within a group of installations behind the same phone line.	[1]
	4	**Hoistway ID** (default = 1), [1 ... 254, 1], [0 ... 5, 1] It designates an elevator hoistway within a group of installations behind the same phone line.	[1]
	5	**Remote Monitoring phone number (left part)** (default = none), [six digits] Left part of the Remote Monitoring phone number. **For SW ≥ 9.56:** PA5 displays the full phone number. (See additional information at the end of the CF21 table.)	[1]
	6	**Remote Monitoring phone number (middle part)** (default = none), [six digits]	[1] <9.56
	7	**Remote Monitoring phone number (right part)** (default = none), [six digits]	[1] <9.56
	8	**Type of Modem** Modem type (default = 0),[0,1,2,3], Type of modem. (0=No modem detected, 1 = CLSD, 4 = ETMA (Wireless/PSTN), 5 = CGW) It shows the result of the Command "Modem detection" (Menu 10 > 128).	[1]
	33	**Modem country value** Value comes originally from what is stored in the SIM card. If not correct enter the correct one:	[1]

ID	Country			ID	Country	
13	WII	Austria		20	BRU	Belgium
40	SCH	Switzerland (same for ASZ)		53	DEU	Germany
63	ESP	Spain		70	PAR	France
72	UKC	Great Britain		102	MIL	Italy
155	HAG	Netherlands		172	POR	Portugal

For other KG's refer to EJ 604639.
The correct value is strictly required in order to successfully perform the auto-configuration of the ETM(A).

CF	PA	VL: Values and Description	OEM
21	**Embedded Telemonitoring (and Alarm) ETM(A) For Rel. 04 only (SW ≥ V9.3x)**		
	34	**Ring tones volume** (default = 0) [0 ... 3, 1] It set the level volume of the modem (CLSD only) noise. '0' = disabled.	[1]
	35	**External line dialing** (default = _ _ _ ↔ no prefix = no pre-dialing required) This parameter determines the prefix before the phone number to dial an external line if the modem is behind a telephone exchange. • 0_ _ (fetch external phone line with "0", dial immediately) • 0 -_ (fetch external phone line with "0", wait a while before dialing) • 0 - - (fetch external phone line with "0", wait a double while before dialing)	[1]
	50	**TACC phone number #1** (default = none) TACC telephone number for alarms, main number. Used for ETMA. (See additional information at the end of the CF21 table.)	[1] ≥9.56

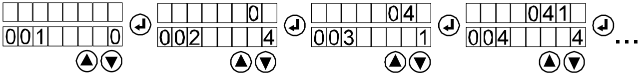
5

CF	PA	VL: Values and Description	OEM
21	**Embedded Telemonitoring (and Alarm) ETM(A)** **For Rel. 04 only (SW ≥ V9.3x)**		
	53	**TACC phone number #2** (default = none) TACC telephone number for alarms, first backup number. Used for ETMA. (See additional information at the end of the CF21 table.)	[1] ≥9.56
	70	**TACC phone number #3** (default = none) TACC telephone number for alarms, second backup number. Used for ETMA. (See additional information at the end of the CF21 table.) – void = no number – 0 ... 9 = DTMF dial tones – , (comma) = wait 2 seconds – W = wait for dial tone	[1] ≥9.56
	73	**TACC phone number #4** (default = none), see PA 70	[1] ≥9.56
	76	**TACC phone number Test Alarm** (default = none) TACC telephone number used for periodic test alarms. (See additional information at the end of the CF21 table.) – void = no number – 0 ... 9 = DTMF dial tones – , (comma) = wait 2 seconds – W = wait for dial tone	[1] ≥9.56

CF	PA	VL: Values and Description	OEM
21	**Embedded Telemonitoring (and Alarm) ETM(A)** **For Rel. 04 only (SW ≥ V9.3x)**		
	79	**TACC phone number Input Monitoring** (default = none) The ETMA unit monitors the alarm button. If the alarm button is broken (alarm button is always active) the ETMA will call this TACC telephone number to transmit the defect. (See additional information at the end of the CF21 table.) – void = no number = no call in case of defect – 0 ... 9 = DTMF dial tones – , (comma) = wait 2 seconds – W = wait for dial tone	[1] ≥9.56
	82	**Time-out Test Line - High part** (default = 0), [0 ... 99999, 1 min.] Time-out (in minutes) for the line test call. VL = void or 0: Re-initiates the time-out with a value of 4320 minutes	[1] ≥9.56
	83	**Time-out Test Line - Low part** (default = 0), [0 ... 99999, 1 min.] Time-out (in minutes) for the line test call. VL = void or 0: Re-initiates the time-out with a value of 4320 minutes	[1] ≥9.56
	84	**Time-out Alarm Debouncing** (default = 3), [0 ... 254, 1s] Debouncing time for the ALARM button inputs	[1] ≥9.56
	84	**Time-out Alarm Debouncing**	[1] ≥9.56
	85	**Car Loudspeaker** (default = 5), [0 ... 8, 1] Loudspeaker volume in car	[1] ≥9.56

CF	PA	VL: Values and Description	OEM
21	**Embedded Telemonitoring (and Alarm) ETM(A)** **For Rel. 04 only (SW ≥ V9.3x)**		
	86	**Car Microphone sensitivity** (default = 5), [0 ... 8, 1] Microphone sensitivity in the car	[1] ≥9.56
	87	**MR Loudspeaker** (default = 5), [0 ... 8, 1] Used for intercom only. Loudspeaker volume in the machine room.	[1] ≥9.56
	88	**MR Microphone sensitivity** (default = 0), [0 ... 8, 1] Used for intercom only. Microphone sensitivity in the machine room.	[1] ≥9.56

Configuration of phone numbers CF=21, PA=5, 50, 53, 70, 73, 76, 79:



Example, configuration phone number 0414..., [38363; 01.02.2011]

CF	PA	VL: Values and Description	OEM
26	**Long Inter-Floor Distance (LIFD)**		
	2 **...** **n-1**	**Blind Floor Assignment** (default = 0), [0 ... 1, 1] 0 = normal floor, 1 = blind floor (no doors) (see additional explanation in section 5.4 “Detailed configuration descriptions”)	[2] SIM

CF	PA	VL: Values and Description	OEM
28	**STM Monitoring** For details, see section: 5.4.29 "STM Configuration" and "STM Re-configuration"		
	1	**Not used**	-

CF	PA	VL: Values and Description	OEM
28	**STM Monitoring** For details, see section: 5.4.29 "STM Configuration" and "STM Re-configuration"		
	2	**STM bend count warning percent (read only)** (default derived from Chipcard), [50 ... 90, 1 %] Percentage of the maximum permitted STM trip count to send the replace STM warning to the service center.	SIM ≥10.0
	3	**STM bend count limit (read only)** (default derived from Chipcard), [0 ... 200000, 1 = 100 trip] Maximum permitted STM trip count. If the STM has reached the service limit, the elevator is out of service.	SIM ≥10.0
	4	**STM manufacturing date** [10100 ... 75399, DWWYY] D = Day of week (1 ... 7) WW = Week of the year (1 ... 53) YY = Year (00 ... 99)	SIM ≥10.0
	5	**Not used**	
	6	**Commissioning number (read only)** [0 ... 99999999, 1] Commissioning number if error 4611. Parameter is editable only after activation of menu 190.	SIM ≥10.0
	7	**STM aging warning level (read only)** (default derived from Chipcard), [0 ... 65000, 1 = 1 month] Number of months before the end of STM life time when the replace STM warning is send to the service center.	SIM ≥10.0

CF	PA	VL: Values and Description	OEM
28	**STM Monitoring** For details, see section: 5.4.29 "STM Configuration" and "STM Re-configuration"		
	8	**STM aging limit (read only)** (default derived from Chipcard), [0 ... 65000, 1 = 1 month] Maximum permitted age of the STM. If the STM has reached the service limit, the elevator is out of service.	SIM ≥10.0
	9	**STM bend counter** [0 ... 4000000, 1 = 1 trip] Parameter is editable only after activation of menu 190.	SIM ≥10.0
	10	**Single bend per trip (read only)** (default derived from Chipcard), [0 ... 10, 1 = 1 bend] Single bend per trip: used to evaluate bend counter.	SIM ≥10.0

CF	PA	VL: Values and Description	OEM
30	**Independent Doors (SW ≥ V9.7x only)**		
	1	**Fire recall floor (main floor), Door side** (default = 3), [0 ... 3, 1] Defines the door side which opens during fire service. VL: 0 = Undefined, 1 = Side 1, 2 = Side 2, 3 = Both sides.	[0] ≥ 9.7 SIM
	2	**Collective policy main floor, Door side** (default = 3), [0 ... 3, 1] Main floor policy side. Used for example for NS21. VL: 0 = Undefined, 1 = Side 1, 2 = Side 2, 3 = Both sides. (With SW ≥ V9.72, the value is set to 3 automatically as soon as CF02, PA2 is changed.)	[2] ≥ 9.7 SIM

CF	PA	VL: Values and Description	OEM
30	**Independent Doors (SW ≥ V9.7x only)**		
	3	**Parking floor, Door side** (default = 3), [0 ... 3, 1] Used for parking with open doors. Used in Moscow only. For parking type 6 (VCF: Distribution of free cars.) 0 = Undefined, 1 = Side 1, 2 = Side 2, 3 = Both sides.	[2] ≥ 9.7 SIM
	6	**Fire recall floor (alternative floor), Door side** (default = 3), [0 ... 3, 1] Defines the door side which opens during fire service. VL: 0 = Undefined, 1 = Side 1, 2 = Side 2, 3 = Both sides.	[0] ≥ 9.7 SIM
	7	**Configuration, Door side** (default = 3), [0 ... 3, 1] Door side which should open during configuration. VL: 0 = Undefined, 1 = Side 1, 2 = Side 2, 3 = Both sides	[2] ≥ 9.7
	21	**Door side served by COP#1** (default = 1), [0 ... 3, 1] Defines to which door side COP#1 belongs. (The numbering of the COP is defined by its physical position: SDIC → COP#2 → **COP#1**) VL: 0 = Undefined, 1 = Side 1, 2 = Side 2, 3 = Both sides.	[2] ≥ 9.7
	22	**Door side served by COP#2** (default = 2), [0 ... 3, 1] Defines to which door side COP#2 belongs. (The numbering of the COP is defined by its physical position: SDIC → **COP#2** → COP#1) VL: 0 = Undefined, 1 = Side 1, 2 = Side 2, 3 = Both sides	[2] ≥ 9.7

CF	L	S	PA	VL: Values and Description	OEM
40	**BIO bus nodes configuration (L = Floor Level)** **For SW ≥ V9.34 only**				
	Availability of the **parameters (PA)** depends on the connected hardware on the respective floor. (For example, PA1 ... 14 are available only, if a LCUX is connected and detected during the LOP configuration. Connected with XCF cable.) The **door side S (1,2)** is available only for SW ≥ V9.7 and in case of independent doors (ZZ2 or ZZ3 on the SIM card). (See additional explanation in section 5.4 “Detailed configuration descriptions”.)				
	0	**L = 0: Affects all nodes on all floors**			
	0	-	**31**	**LIN output at all levels** (default = 0 ↔ no function), [0 ... 999] VL = BMK: See section 5.3.4 “BMK function codes” (VL = 213: Floor light control) (Available with LINV51 or newer)	[2] ≥9.38 3)
	0	-	**99**	**BIO bus node clearing at all levels** (default = 0), [0 ... 1, 1] 1 = clear command (sets address back to pre-defined value)	[2] ≥9.34
	1 ... n	**L = 1 ... n: Affects defined floor level (1 ... n) only**			
	1 ... n	**1,2**	**1**	**LCUX.I/O1 Input 1 function** (default = 0 ↔ no function), [0 ... 999] VL = BMK: See section 5.3.4 “BMK function codes” For group functions in duplex: Both LCUX must be configured equally.	[2] ≥9.34
	1 ... n	**1,2**	**2**	**LCUX.I/O2 Input 2 function** (default = 0 ↔ no function), [0 ... 999]	[2] ≥9.34

CF	L	S	PA	VL: Values and Description	OEM
40	**BIO bus nodes configuration (L = Floor Level)** **For SW ≥ V9.34 only**				
	1 ... n	**1,2**	**3**	**LCUX.I/O3 Input 3 function** (default = 0 ↔ no function), [0 ... 999]	[2] ≥9.34
	1 ... n	**1,2**	**4**	**LCUX.I/O4 Input 4 function** (default = 0 ↔ no function), [0 ... 999]	[2] ≥9.34
	1 ... n	**1,2**	**5 ... 8**	BIOGIO.INPUT1 ... 4, Input 1 ... 4 function (default = 0 ↔ no function), [0 ... 999] Available if BIOGIO was with help of the TEACH-IN button. Not used at the moment. See also PA91 ... 93.	[2] ≥9.34
	1 ... n	**1,2**	**11**	**LCUX.I/O1 Output 1 function** (default = 0 ↔ no function), [0 ... 999] VL = BMK: See section 5.3.4 “BMK function codes” For group functions in duplex: Both LCUX must be configured equally.	[2] ≥9.34
	1 ... n	**1,2**	**12**	**LCUX.I/O2 Output 2 function** (default = 0 ↔ no function), [0 ... 999]	[2] ≥9.34
	1 ... n	**1,2**	**13**	**LCUX.I/O3 Output 3 function** (default = 0 ↔ no function), [0 ... 999]	[2] ≥9.34
	1 ... n	**1,2**	**14**	**LCUX.I/O4 Output 4 function** (default = 0 ↔ no function), [0 ... 999]	[2] ≥9.34
	1 ... n	**1,2**	**15 ... 18**	BIOGIO.OUTPUT1 ... 4, Output function (default = 0 ↔ no function), [0 ... 999] Available if BIOGIO was with help of the TEACH-IN button. Not used at the moment. See also PA91 ... 93.	[2] 1)

CF	L	S	PA	VL: Values and Description	OEM
40	**BIO bus nodes configuration (L = Floor Level) For SW ≥ V9.34 only**				
	1 ... n	**1,2**	**21**	**LOP.XBIO pin 4 input function** (default = 0 ↔ no function), [0 ... 999] BMK: See section 5.3.4 “BMK function codes” (Was with old software: CF60, PA = n or CF61 ... 80, PA17)	[2] ≥9.34
	1 ... n	**1,2**	**31**	**LINV51 output** (default = 0 ↔ no function), [0 ... 999] VL = BMK: See section 5.3.4 “BMK function codes” (VL = 213: Floor light control) (Available with LINV51 or newer)	[2] ≥9.38
	1 ... n	**1,2**	**99**	**BIO bus node clearing at this level** (default = 0), [0 ... 1, 1] 1 = clear command (sets address back to pre-defined value)	[2] ≥9.34
	91 ... 93		**Machine room nodes (LCUX or BIOGIO, floor level independent)**		
	91	-	**1 ... 4**	**LCUX or BIOGIO with predefined address #1, Input function** (CFG1 = ON, CFG2 = ON, section 7.14, 7.15) (default = 0 ↔ no function), [0 ... 999] VL = BMK: See section 5.3.4 “BMK function codes” (PA1: Input I/O1, PA2: Input I/O2,..) **Before doing the configuration:** Set the DIP switches correctly, connect the LCUX or BIOGIO to the BIO bus and initiate a LOP counting manually.	[2] ≥9.34

CF	L	S	PA	VL: Values and Description	OEM
40	**BIO bus nodes configuration (L = Floor Level)** **For SW ≥ V9.34 only**				
	91	-	**5** **...** **8**	**BIOGIO with predefined address #1, Input function** (PA5: Input1, PA6: Input2, etc.) (details see PA1 ... PA4)	[2] ≥9.34
	91	-	**11** **...** **14**	**LCUX or BIOGIO with predefined address #1, Output function** (CFG1 = ON, CFG2 = ON, section 7.14, 7.15) (default = 0 ↔ no function), [0 ... 999] VL = BMK: See section 5.3.4 “BMK function codes” (PA11: Output I/O1, PA12: Output I/O2,..) **Before doing the configuration:** Set the DIP switches correctly, connect the LCUX or BIOGIO to the BIO bus and initiate a LOP counting manually.	[2] ≥9.34
	91	-	**15** **...** **18**	**BIOGIO with predefined address #1, Output function** (details see PA11 ... 14)	[2] ≥9.34
	92	-	**1** **...** **18**	**LCUX or BIOGIO with predefined address #2** (CFG1 = OFF, CFG2 = ON, section 7.14, 7.15) (details see L = 91)	[2] ≥9.34
	93	-	**1** **...** **18**	**LCUX or BIOGIO with predefined address #3** (CFG1 = ON, CFG2 = OFF, section 7.14, 7.15) (details see L = 91)	[2] ≥9.34

CF	L	S	PA	VL: Values and Description	OEM
41	**Restricted Access Functions (L = Floor Level)** **For SW ≥ V9.34 only**				
	The **door side S (1,2)** is available only for SW ≥ V9.7 and in case of independent doors (ZZ2 or ZZ3 on the SIM card). (see additional explanation in section 5.4 "Detailed configuration descriptions")				
	0	**L = 0: Affects all nodes on all floors**			
	0	-	**1**	**CF41, L = 0, PA1: Behavior of Independent Travel, reservation control (JRCV)** (Affecting all levels and all restrictions types of CF41) (default = 0 ↔ no function), [0, 1] – 0 = JRVC can access restricted floors – 1 = JRVC can not access restricted floors	[2] ≥9.34
	1 ... n	**L = 1 ... n: Affects defined floor level (1 ... n) only**			
	1 ... n	**1, 2**	**1**	**ZB1: PIN Code on COP** (default = 0 ↔ no restriction), [-999 ... 9999, 1] A four-digit PIN code for restricted access, disables the normal car call button on this floor. For three-digit code use "-" as first sign. (for example "-123") VL = 0000: No PIN code	[2] ≥9.34 SIM
	1 ... n	**1, 2**	**2**	**ZB3: Key Switch in Car** (default = 0 ↔ no function), [0 ... 4] Example: VL = 1: Key switch connected to XKEY1 in COP (corresponding to CF05 or 55)	[2] ≥9.34 SIM

CF	L	S	PA	VL: Values and Description	OEM
41	**Restricted Access Functions (L = Floor Level)** **For SW ≥ V9.34 only**				
	1 ... n	**1, 2**	**3**	**GS: Visitor Operation** (default = 0 ↔ disabled), [0 ... 12, 1 = 10s] Available with SW ≥ V9.7. (For SW < V9.7 refer to CF17) Sets the duration of the visitor request signalization and the enable time for the “access granted” button. For GS set to 3 ↔ 30s as standard value. GS is restricted to simplex with PI. See also section 5.4 “Detailed configuration descriptions”	[2] ≥ 9.7 SIM
	1 ... n	**1, 2**	**99**	**ZBC2: Used for parallel card reader only.** **Floor restrictions** (default = 0 ↔ no restriction), [0, 1, 2, 3] – 0 = No restriction (free access) – **1 =** Restricted exit on floor – 2 = Restricted exit from floor – 3 = Restricted exit on / from floor (full blocking).	[2] ≥9.38 SIM

5

CF	PA	VL: Values and Description	OEM
49	**SIEU Auxiliary input/output (on SDIC, CCU)**		
	1 ... 8	**SIEU Inputs** (default = 0↔no function), [0 ... 999, 1] BMK see section 5.3.4 “BMK Function Codes”. PA1 = IN1, ... PA8 = IN6 Used for MOS functions only. (Available are inputs 1, 4, 7, 8 only)	[2] ≥ 9.7
	9 ... 10	SIEU Outputs OUT1 (default = 0↔no function), [0 ... 999, 1] BMK see section 5.3.4 “BMK Function Codes”. PA9 = OUT1.1, PA10 = OUT1.2 Not used at the moment	[2] 1)
	11 ... 12	**SIEU Outputs OUT2** (default = 0↔no function), [0 ... 999, 1] BMK see section 5.3.4 “BMK Function Codes”. PA11 = OUT2.1, PA12 = OUT2.2 Used for MOS functions only.	[2] ≥ 9.7

CF	PA	VL: Values and Description	OEM
55	**Auxiliary input/output first SCOPH(MH), Dual brand SCOPMX-B**		
56	**Auxiliary input/output second SCOPH(MH), Dual brand SCOPMX-B**		
	1 ... 6	**COP Inputs** (default = 0↔no function) [0..255, 1] BMK see section 5.3.4 “BMK Function Codes”. PA1 = XIO1, ... PA6 = XIO6 (Pin 2 and 3)	[2]
	7 ... 12	**COP Outputs** (default = 0 = no function) [0..255, 1], BMK see section 5.3.4 “BMK Function Codes”. PA7 = XIO1, ... PA12 = XIO6 (Pin 1 and 2)	[2]

CF	PA	VL: Values and Description	OEM
86	**Asymmetric Group** **(Asymmetric group is supported with SW ≥ V9.7 only.)** **In symmetric duplex systems CF86, PA1 must be VL01.**		
	1	**Starting group level of the elevator** (default = 1), [1 ... n, 1] The lowest floor level of the group is always “1”. In case of asymmetric duplex: On both elevators the starting level (in respect to the group) must be configured. See section 5.4 “Detailed configuration descriptions”	[2] SIM SW <11.0
	Asymmetric Group (1 ... maximum number of physical levels present), if SW ≥ V11		
	1	**Floor Level 1** (default = 1), [1 ... maximum number of levels (≤ 35)] Building level assigned to this floor level 1	[0]
	2	**Floor Level 2** (default = 2), [1 ... maximum number of levels (≤ 35)] Building level assigned to this floor level 2	[0]
	3	**Floor Level 3** (default = 3), [1 ... maximum number of levels (≤ 35)] Building level assigned to this floor level 3	[0]
	...		
	35	**Floor Level 35** (default = 35), [1 ... maximum number of levels (35)] Building level assigned to this floor level 35	[0]

5

CF	PA	VL: Values and Description	OEM
87 ... 89	**LCUX with pre-defined addresses** For SW ≥ V9.3x (with SCPU): Refer to CF41, PA91, 92, 93		
87	**LCUX with pre-defined address #1** (DIP switch setting see section 7.14)		
88	**LCUX with pre-defined address #2** (DIP switch setting see section 7.14)		
89	**LCUX with pre-defined address #3** (DIP switch setting see section 7.14)		
	1 ... 4	**LCUX Input Function** (default = 0 = no function), [0 ... 255, 1] VL = BMK: see separate BMK function code list) PA1: LCUX.I/O1, PA2: LCUX.I/O2, ... **Before doing the configuration: Set the DIP switches correctly, connect the LCUX to the BIO bus and initiate a LOP counting manually.**	-
	9 ... 12	**LCUX Output Function** (default = 0 = no function), [0 ... 255, 1] VL = BMK: see separate BMK function code list) PA9: LCUX.I/O1, PA10: LCUX.I/O2, ... **Before doing the configuration: Set the DIP switches correctly, connect the LCUX to the BIO bus and initiate a LOP counting manually.**	-

CF	PA	VL: Values and Description	OEM
95	**Actual cabin load [Kg] (read-only)**		
	-	**Actual Car Load** (1 = 10 Kg)	[1]

CF	PA	VL: Values and Description	OEM
96	**CLC Values for Backup (read-only)** After the Digisens calibration. Possibility to read our the calibration values of the Digisens. Can be used to reconfigure the Digisens without reference load at a later time. (See additional explanation in section 5.4 “Detailed configuration descriptions”)		
	1	**Zero Carload Frequency** (1 = 10 Hz)	[1]
	2	**Reference Carload Frequency** (1 = 10 Hz)	[1]
	3	**Reference Carload Weight** (1 = 10 Kg)	[1]

CF	PA	VL: Values and Description	OEM
97	**CLC Configuration** To be used to reconfigure the Digisens with the values which have been read out with CF=96 before. (See additional explanation in section 5.4 “Detailed configuration descriptions)		
	1	**Zero Carload Frequency Configuration** – [13400 ... 14800 Hz, 10 Hz]: If KL255 (CF08, PA08, VL=1) – [10000 ... 18000 Hz, 10 Hz]: If KL66 (CF08, PA08, VL=0)	[1]
	2	**Reference Carload Frequency Configuration** – [12100 ... 16100 Hz, 10 Hz]: If KL255 (CF08, PA08, VL=1) – [8000 ... 20000 Hz, 10 Hz]: If KL66 (CF08, PA08, VL=0)	[1]
	3	**Reference Carload Weight Configuration**, [0..113, 10 Kg]	[1]

CF	PA	VL: Values and Description	OEM
98	**Zero carload frequency recalibration**		
	-	Make sure that the car stays empty (0 Kg load) on a floor. Enter CF=98 and press “OK”. (HMI shows actual load.) Press “OK” again to start fully automated recalibration. (A countdown starts and then five measurements are done.) [CF 98] indicates that the recalibration is finished.	[1]

CF	PA	VL: Values and Description	OEM
99	**Calibration of car load measurement** (see additional explanation in section 5.4 “Detailed configuration descriptions”)		
	-	-	[1]

- 1) = not used with Schindler 3100/3300/5300 or not used in EU version
- 2) = not available with current SW
- SIM = SIM card option
- [0] = OEM level 0 according to definition in section 1.4
- [1] = OEM level 1 according to definition in section 1.4
- [2] = OEM level 2 according to definition in section 1.4
- If Schindler 3100/3300/5300: n = 25
- If Schindler 3600: n = 35.

5.3.4 BMK Function Codes

Inputs - Function Codes for CF 05, 40, 55 and 56

VL	BMK	Description	1)	Location
00	-	No function		
02	JDC	Key switch car call [ZB3]	NO	COP
04	DFDC	Push button release push button car call	NO	-
07	DCW-U	Push button car selection up	NO	-
08	DCW-D	Push button car selection down	NO	-
10	JDNF	Key switch / lamp emergency floor call	NO	-
11	KL-V	Contact full load	NC	-
12	KL-X	Contact overload	NC	-
13	KL-M	Contact minimal load	NO	-
17	DE-U, LDE-U	Push button and lamp landing call UP	NO	LOP
18	DE-D, LDE-D	Push button and lamp landing call DOWN	NO	LOP
19	JDE-U	Key switch landing call UP	NO	LOP
20	JDE-D	Key switch landing call DOWN	NO	LOP
22	KTTC	Car partition door	NC	-
23	JBFFG	Fire service firefighter	NO	COP
24	JBFFGH1	Fire service firefighter - recall floor 1	NC	-
25	JBFFGH2	Fire service firefighter - recall floor 2	NC	-
26	JBFH1	Fire service first recall floor	NC	-
27	JBFH2	Fire service second recall floor	NC	-

5

VL	BMK	Description	1)	Location
34	KL-H	Contact half load	NO	-
36	KKE	Contact supervision unlocked	NO	-
39	DDFLI	Bypass landing calls, push button direct travel attendant service [LI]	NO	-
40	DLI	Start travel, attendant service, see CF06 PA12 [LI]	NO	-
41	DLI-U	Change travel direction UP [LI]	NO	-
42	DLI-D	Change travel direction DOWN [LI]	NO	-
48	KTHS	Hoistway thermostat	NO	LCUX
49	JAB / RAB	Switch/relay out of service [JAB]	NO	LOP, LCUX
52	JKLBL	Switch car call locking, parallel card reader [ZBC2]	NO	-
53	JLC	Switch car call locking, parallel card reader [ZBC2]	NO	-
54	JLI	Switch on/off attendant service [LI]	NO	COP
56	JNFF	Switch firefighter service	NC	COP, LOP, LCUX
57	JNFF-S	Switch firefighter service start trip	NC	-
58	JNO	Switch emergency power operation. Release car for travel [NS21]	NO	LCUX
59	JRVC	Switch reservation service [RV1]	NO	COP
61	JRVCP	Switch reservation parking [RV2]	NO	-
66	JVEC	Switch fan on car	NO	-

VL	BMK	Description	1)	Location
69	KGEB	KGEB contact counterweight displaced [EB]	NC	-
77	RFEF	Relay evacuation travel release [NS21]	NO	-
87	JHCL-I	Switch stop car - In car feedback	NC	-
91	DH	Push button stop	NC	COP
97	DEB-A	Push button counterweight earthquake reset	NO	-
99	JBF	Switch fire service (secure input)	NC	LOP, LCUX
100	JBF-A	Switch fire service off (secure input)	NO	LCUX
106	JSDC	Key switch lock car call	NO	-
111	JZH	Switch enforced stop	NO	-
112	KBF	Contact activation fire service fire detector	NC	LCUX
113	KBFH1	Contact activation fire service, first recall floor	NC	LCUX
114	KBFH2	Contact activation fire service, second recall floor	NC	LCUX
116	KEB	Contact earthquake [EB]	NC	-
120	RNO	Relay emergency power service [NS21]	NC	-
121	RSPE / KSPE	Relay blocking floor [EBR1] (secure input)	NC	LCUX
179	JSDC_G	Key switch lock car call group [ZBC1] CF81	NO	LCUX
184	JHCC	Switch stop car - In car	NO	-
185	DVEC	Car fan button	NO	-

VL	BMK	Description	1)	Location
241	DEH-U	Push button hall call handicapped - up	NO	-
242	DEH-D	Push button hall call handicapped - down	NO	-
256	KKE2	Door lock supervision access side 2, used for Haushahn door systems	NO	-
257	JDE-E	Key switch floor lock permission	NO	-
260	KUESG	Contact bridging hoistway pit contacts - first	NC	-
261	KUESG1	Contact bridging hoistway pit contacts - second	NC	-
263	KSG-A	Contact hoistway pit contacts off	NO	-
264	JSPS	Key switch floor locked [ZBCE] CF82	NO	-
265	KTL	Contact mechanical door safety edge	NC	-
266	KSR	Contact sprinkler recall	NC	-
274	JSK-A	Contact smoke recall off	NO	-
276	KOL-M	Contact minimal oil level	NO	-
279	KTEC	Contact door elevator controller box	NC	-
281	KNA-I	Contact emergency exit feedback	NO	-
546	KSKHW	Contact smoke detector hoistway	NC	-
789	KET-O	Contact door open end	NO	-
790	KET-S	Contact door closed end	NO	-
792	KSKB	Contact closing force limiter	NO	-

VL	BMK	Description	1)	Location
848	PHS	PHS	NO	-
871	KEBVL	Contact earthquake very low	NC	-
872	KEBL	Contact earthquake low	NC	-
873	KEBH	Contact earthquake high	NC	-
875	JEB	Key switch earthquake	NC	-
883	JZHPK	Switch force stop panic control	NO	-
886	KLS-D	Contact limit hoistway - down	NC	-
887	KLS-U	Contact limit hoistway - up	NC	-
889	JZHPK-A	Switch force stop panic control - off	NO	-
891	KNET	Contact emergency unlocking landing door	NC	-
892	KSM	Car position relative to middle hoistway	-	-
896	DTX-O	Push button extended door open time	NO	-
900	GPIN1AND	General purpose input 1 with AND logic	NO	-
902	GPIN2OR	General purpose input 2 with OR logic	NO	-
967	DVEC	Toggle switch for car ventilation	NO	-
1091	2PHS	2PHS	NO	-
1095	2KET-O	Contact rear door open end	NO	-
1096	2KET-S	Contact rear door close end	NO	-
1097	2KSKB	Contact rear closing force limiter	NO	-

1) Type of contact: NO = normally open, NC = normally closed

Outputs - Function Codes (VL) for CF 40, 55, 56, 61-80, 87-89

VL	BMK	Description	Location
37	LUB	Lamp maintenance and error, Korea	-
38	LRV	Lamp independent control RESERVATION	-
49	LAB	Lamp out of service (old code, new code is 131)	LCUX
56	LFF	Lamp firefighter	-
131	LAB	Lamp out of service (new code) output lED on LCUX is off when out of service is true	LCUX
134	LAB-E	Lamp out of service active, acknowledgment JAB	LCUX
136	LBFC	Lamp fire service active, on car	-
140	LHC	Lamp car here	LCUX
145	LEF	Lamp evacuation travel [BR4-NZ]	-
146	LGEB	Lamp counterweight displaced [EB]	-
147	LGS	Lamp evacuation travel [BR4-NZ]	-
151	LLI-U	Lamp direction UP, attendant service [LI]	-
152	LLI-D	Lamp direction DOWN, attendant service [LI]	-
154	LNFC-U	Lamp emergency service, on car [NF1]	-
156	LNOC	Lamp emergency power service, on car [NS21]	-
157	LRC-U	Lamp car direction UP	-
158	LRC-D	Lamp car direction DOWN	-
163	LW-U	Lamp further travel up	-

VL	BMK	Description	Location
164	LW-D	Lamp further travel down	-
167	LL-X	Lamp overload	-
170	RBF	Fire service active in machine room [JBF]	-
172	RE-A	Landing calls service suspended (for example reservation)	-
173	REFE1	Relay evacuation travel end	-
174	REFEH1	Relay evacuation travel end recall floor 1	-
175	REFEH2	Relay evacuation travel end recall floor 2	-
177	RIB	Elevator normal mode	-
183	RSM	Elevator fatal or persistent fatal error	-
186	SUMC	Buzzer on car	-
190	GA-k [1)]	Gong arrival (k = floor number)	LCUX
205	LBF	Lamp fire service active on landing, for group [KBF]	-
207	LEB	Lamp earthquake service active, for group [EB]	-
213	RFBE-k [1]	Relay floor lighting (k = floor number)	LCUX
214	SUMP	Buzzer porter	-
225	LL-V	Lamp full load	-
251	GA-D	Gong arrival landing down	-
252	GA-U	Gong arrival landing up	-
258	RUESG	Relay bridging hoistway pit contacts - first	-
259	RUESG1	Relay bridging hoistway pit contacts - second	-

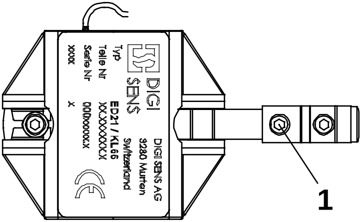
VL	BMK	Description	Location
262	RSG-A	Relay hoistway pit contact off	-
275	LOL-M	Lamp minimal oil level	-
277	RLUO	Relay indicator unauthorized opening of a landing door	-
280	RKTEC	Relay contact door elevator controller cabinet	-
797	LE-D	Lamp acknowledge floor call down	-
798	LE-U	Lamp acknowledge floor call up	-
857	RLC-A	Relay lamp car off	-
859	RPSC-D	Relay standby power mode on car driving signal	-
860	RPSM-D	Relay standby power mode in machine room driving signal	-
868	LAPS	Lamp standby power mode on	-
874	LEBIB	Lamp earthquake evacuation	-
879	LEBEFE	Lamp earthquake evacuation completed	-
880	LEBEFEG	Lamp earthquake evacuation completed group	-
881	LVC	Lamp leave car	-
882	LDTOB	Lamp DTO backlight	-
885	LAEFE	Lamp automatic evacuation travel completed	-
888	LEFFS	Lamp evacuation travel - forced stop each floor	-
890	LNOIB	Lamp emergency power UPS in operation	-
895	LDA	Lamp DA backlight	-

VL	BMK	Description	Location
897	LEH-U	Lamp handicapped acknowledge floor call up	-
898	LEH-D	Lamp handicapped acknowledge floor call down	-
898	LEH-D	Lamp handicapped acknowledge floor call down	-
899	LSI	Lamp safety incident	-
901	GPOUT1	General purpose output 1	-
903	GPOUT2	General purpose output 2	-
906	LDTXOB	Lamp extended DTO backlight	-

1) k is a running number

5.4 Detailed Configuration Descriptions

5.4.1 Calibration of Car Load Cell CLC (CF = 96 ... 99)



CF = 08
PA = 08
VL = 1

White Digisens KL66 [402022: 14.10.2014]

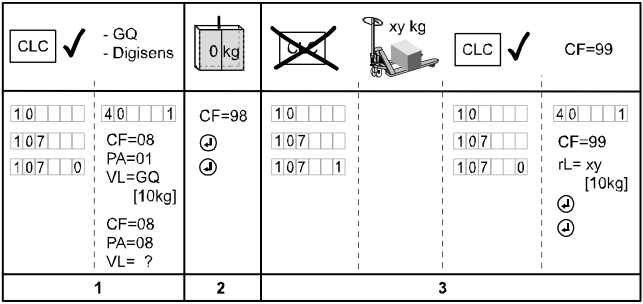
- For the reference load 75% of the rated load (nominal load GQ) is recommended
- Any error displayed during the calibration belongs to error group 11. (Example: er 9 = Error 1109)

Preconditions:

- Counterweight filled completely
- Car installed completely, including car decoration, balustrade, etc.
- Digisens has to be free of pre-stressing (Loose the adjustment screws (1) and tighten them again before starting the calibration.)
- In Schindler 3100: The white Digisens must be mounted with 45° on the pulley (see document K 609754_04)
- Learning travel with activated Digisens has been performed

The calibration of the car load cell is done with 3 steps:

1) Preparation with system relevant values (rated load, ...)
2) 0 Kg calibration (with empty car)
3) Calibration with reference load



Calibration Procedures [402023; 08.10.2014]

Example: 450 Kg nominal load car; reference load rL: 380 Kg.

Calibration with CDD installed (Schindler 3600)

No	Step
1	Enter the car roof.
2	Close the car door using the door operation button.
3	Wait until the CDD is open.
4	Switch off the JH.
5	Unplug the CDD power supply.
6	Close the OKR.
7	Switch on the JH.
8	Proceed with the CLC calibration.

Step 1: Preparation

No	Step
1	Check the preconditions are fulfilled.
2	Make sure the car load cell is enabled. (Menu 10, Submenu 107 = 0)
3	Activate the configuration mode menu 40. Configure the rated (nominal) load of the car (GQ) with help of CF=08, PA=01. (In our example: Rated load 450 Kg: VL=045)
4	Configure the Digisens type with help of CF=08, PA=08. – VL=1: White Digisens KL66

Step 2: Calibration of the 0 Kg point (empty car)

No	Step
5	Make sure the car is empty. (0 Kg load)
6	Choose CF=98 and press “OK”. → HMI shows actual load. [Ld xx] (This value can be wrong because the system is not yet calibrated.)
7	Press “OK” on the HMI to start the 0 Kg calibration. → After a 10 second countdown there are 5 measurements performed. → HMI should show [Ld 0] (= 0 Kg) If an error occurs during this calibration: – Check for correct Digisens type definition (CF=08, PA=08) – Repeat the learning travel (while Digisens is enabled)
8	Choose CF=96, PA=01 and check the 0 Kg frequency: The frequency should be 16500 Hz ± 500 Hz (If the value is out of range, check the mechanical installation of the Digisens. Loose and tight the screws. Check for correct 45° positioning on Schindler 3100 pulley. Redo the 0 Kg calibration.)

Step 3: Calibration with reference load

No	Step	HMI
9	Disable the car load cell. (Menu 10, Submenu 107 = 1). This allows you to travel with the car as long as the car load cell is not yet calibrated.	
10	Load the car with the reference load. (At least 75% of the rated, nominal load GQ. In our example 380 Kg.)	
11	Move the car to the configuration floor.	
12	Enable the car load cell. (Menu 10, Submenu 107 = 0).	
13	Activate the configuration mode menu 40.	CF _ _ 0 0
14	Select CF=99	CF _ _ 9 9
15	Press “OK” → The display shows the default reference load [rL]	r L _ x x x
16	Enter the reference load which is actual in the car. (1 = 10 Kg. In our example 38 = 380 Kg)	r L _ _ 3 8
17	Press “OK” → The display shows the actual measured load of the car. (This value can be wrong or 0, because the car load cell is not yet calibrated.)	Ld _ x x x
18	Press “OK” on the HMI → A 10 second countdown starts.	Cd _ _ 1 0 Cd _ _ _ 0
19	After this 10 seconds there are 5 measurements.	C l _ _ _ 5 C l _ _ _ 1
20	After the 5 measurements the system is calibrated and shows the actual load. (In our example 380 Kg)	Ld _ _ 3 8

No	Step	HMI
21	If the value is correct (corresponds to the weight in the car) confirm calibration by pressing “OK” on the HMI. The display goes back to CF=99 If the displayed value is wrong or a error has occurred, start again with step 9 or redo the complete calibration.	CF _ _ 9 9
22	Read out the calibrated data and write them down. (For example to the cover page of the schematic wiring diagram or with a waterproof pen inside the cover of the LDU.) – Read out CF=96, PA=1 – Read out CF=96, PA=2 – Read out CF=96, PA=3	
23	Leave the configuration mode by pressing “ESC” and changing [40 1] to [40 0].	

The calibrated values have to be read out with the help of CF=96, PA=1,2,3 and written down, for example with a waterproof pen inside the LDU cover or on the schematic wiring diagram (use the delivered sticker).

In case of errors during the 0 Kg calibration (CF=98) try the following procedure:

– Choose CF=97, Pa=01 and enter a typical value.(For the white Digisens 16500.)
– Redo the complete CLC calibration procedure.

5.4.2 Re-Configuration of Car Load Cell (CF = 96 ... 98)

Re-Configuration of CLC without weights

This procedure can be used, if microprocessor PCB (SCPU) has to be exchanged.

Preparation

To be able to re-configure the CLC at a later date the CLC values of the initial calibration have to written down. With the old microprocessor PCB inserted read out the CLC values:

1) Enter configuration mode 96
2) Read out and write down actual data (for example on the cover page of the schematic wiring diagram or inside the LDU cover with a waterproof pen): Example: 1450 = 14500 Hz

CF	PA	VL Meaning (read only)	Actual Value
96	**1**	0 Kg carload frequency	
	2	Reference carload frequency	
	3	Reference carload weight	

5

Re-configuration procedure

With the new SCPU PCB inserted configure the system with the old values:

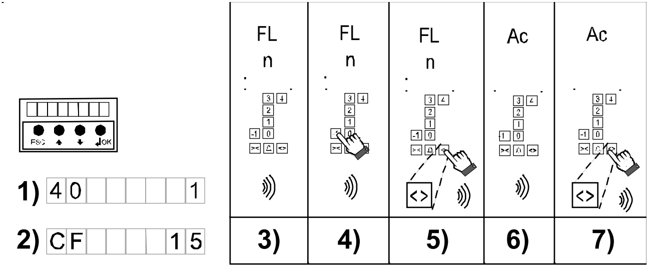
No	Step
	Make sure the Digisens is enabled (107=0)
1	Activate the configuration mode menu 40.
2	Choose **CF=08**, **PA=01** and enter the rated load of the car (GQ)
3	Choose **CF=08**, **PA=08** and enter the Digisens type (VL=1: White Digisens, see drawing in section 5.4.1)
4	Choose CF=97 and enter the values which have been read out with CF=96 before. – **CF=97**, **PA=1**: 0 Kg carload frequency [1=10 Hz] – **CF=97**, **PA=2**: Reference carload frequency [1=10 Hz] – **CF=97**, **PA=3**: Reference carload weight [1 = 10 Kg] (Example: 500 Kg = 50)

No	Step
5	Leave the configuration mode by pressing “ESC” and changing [40 1] to [40 0].

Any error displayed during the calibration belongs to error group 11. For example Er 9 = error 1109.

5.4.3 COP Keypad Configuration (CF=15)

The COP keys have to be assigned to the real floor levels.



COP Configuration [402039_00; 31.1.2015]

The COP keypad configuration starts always with the **lowest floor** level. The configuration sequence is:
Floor level 1→2→3→4→5.

Step	Description
1	Enter the configuration mode menu 40.
2	SW < V9.5x: Choose CF=15
	SW ≥ V9.5x: Choose CF=15, PA=1
3	The HMI displays [CF15_ _ _]. The COP beeps. “FL” and “n” (COP5) is shown. (“n” stands for the floor which is ready to configure). The COP keypad configuration starts always with the **lowest** floor.
4	Press the corresponding push button of the floor which is displayed on the COP or the indicator (e.g. CPI 4 for COP4B).
5	Press “DT-O” to confirm your choice (the COP beeps).
6	The COP or the indicator (e.g. CPI 4 for COP4B) shows “Ac”. The push button is configured.

Step	Description
7	Press “DT-O” or the next higher push button (COP4B) to continue with the next higher floor. **Remark:** On the COP or the indicator (e.g. CPI4) the next higher floor level to be configured is shown.
8	Repeat step 3) to 7) for all the next higher floor levels (1 → 2 → 3 ....).

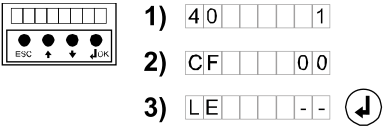
Check whether the button inscription correspond with the floors displayed on the COP or the indicator CPI4 (if COP4B is installed). If they differ the configuration “Floor Designation” CF=01 has to be done.

5.4.4 LOP Configuration (CF=00)

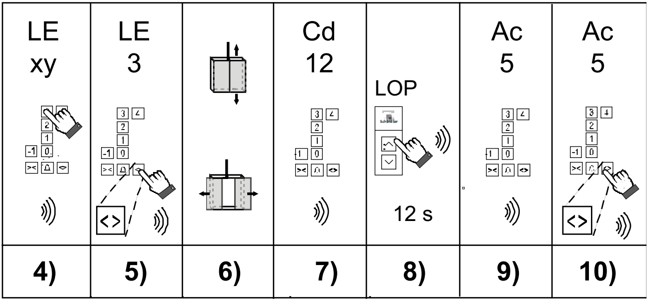
– Each LOP must have an unique address.
– In case of 2 entrance sides the LOP configuration has to be done on the corresponding floor for both sides:
  - SW < V9.7: The same floor level must be entered manually.
  - SW ≥ V9.7: The control offers the same floor level automatically.
– In case of group systems (duplex, triplex and quadruplex) using SBBD PCB, please refer to section 5.4.12
– Description of the error codes during the LOP configuration: Refer to section 5.4.6

The LOP configuration consists of two steps:
– Step 1: LOP addressing
– Step 2: LOP counting

Step 1: LOP addressing [LE - -]



[402042_00; 01.02.2015]

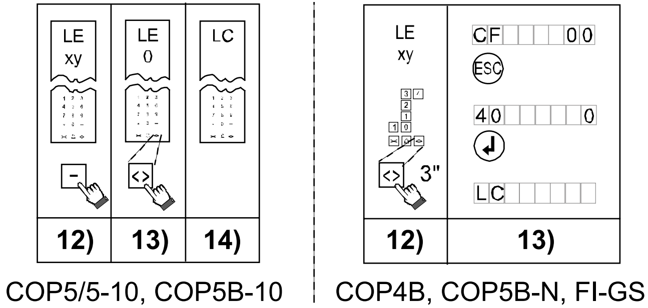


[402041_00; 01.02.2015]

Step	Description
1	Activate the configuration mode menu 40.
2	Choose CF=00 and press “OK”. → HMI displays [LE - -]
3	Press “OK” to enter LOP configuration. → The HMI displays [_ _ _ _ _ _] or [CF00 _ _]
4	The COP displays “LE”. Choose the floor level you want to configure. Press the corresponding button on the COP. (In our example we have chosen floor level 3.)
5	The COP shows the level you want to configure. Press “DT-O” to confirm your choice.
6	The car moves to the chosen floor and opens the door.
7	After the door has opened the COP beeps. The COP or the indicator (e.g. CPI 4 for COP4B) displays “Cd” “12”. This indicates that the 12 seconds countdown has started.
8	During this 12 seconds press the “UP”-button on the LOP and keep it pressed until there is a beep.

Step	Description
9	Wait until the countdown has finished. This is indicated by another beep. The LOP configuration is OK if the COP or the indicator (e.g. CPI4) shows “AC” or “BR” if a JBF has been detected. **Failure indications COP:** – COP indicates by a double beep and displays “Er” or “E0”..”E9”: LOP configuration faulty. Confirm the error by pressing DT-O once and repeat the LOP configuration on the same floor again (step 4). Detailed error description for SW ≥ V9.7: See section 5.4.6 **Failure indications COP4B:** – COP4 with CPI4: A failure is indicated by a double beep. On the CPI4 “Er” is displayed. – COP4 without CPI4: A failure is indicated by a single beep and the acknowledge lamp on the button is off. Confirm the error by pressing DT-O twice and start the countdown again.
10	Press the DT-O to confirm the correct configuration of the just configured LOP. The system will answer with a beep and the COP or the indicator (e.g. CPI4) will show the level “LE” of the next floor.
	In case of 2 access sides: Repeat the configuration for the LOP on the second side (step 4). – SW < V9.7: The same floor level must be entered manually. – SW ≥ V9.7: The control offers the same floor level automatically. Just press DT-O and the second door side will open.
11	Repeat steps 4 ... 10 for all the floors.

Step 2: LOP counting [LC _ _ _ _]



[402043_00; 01.02.2015]

Step	LOP counting on floor level for COP5, 5-10, 5B-10
12	When “LE” is shown on the COP, press “-”.
13	Press “DT-O” to confirm.
14	LOP counting is indicated by “Lc” on COP and HMI. Wait until LOP counting has finished.

Step	LOP counting with HMI
12	When “LE” is shown on the COP or the indicator (e.g. CPI4), press “DT-O” for 3 seconds until there is a beep.
13	– HMI shows CF=00 again – Press ESC and leave the configuration mode. ([40 0] and press “OK”) – LOP counting is indicated by “Lc” on COP and HMI. Wait until LOP counting has finished

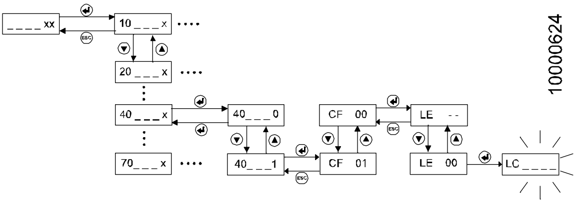
If LOP counting does not start automatically it has to initiated manually. → Manual initiation of LOP counting: See section 5.4.5 “LOP Counting [LE 00]”

5.4.5 LOP Counting [LE 00]

The LOP count polls all biobus devices and store persistently information in EEPROM. This procedure is automatically done after any changes done in CF00 or I/O configuration (CF40 or CF 60-80).

- If the LOP counting does not start automatically, it has to be started manually.
- Manual LOP counting must also be performed after replacing the microprocessor PCB.

The manual initiation of the “LOP counting” has to be started with the User Interface HMI.



[25822; 13.02.2008]

Step	User Interaction	HMI
1	Enter configuration mode menu 40.	
2	Choose CF=00 and confirm with “OK”.	LE - -
3	Change with the UP/DOWN button to [LE 00]	LE 00
4	Press “OK”. → LOP counting is indicated by blinking [LC_ _ _ _]. This can take up to several minutes.	LC _ _ _ _
5	After LOP counting is finished leave configuration mode by pressing “ESC” and by deactivating menu 40. (Change [40_ _ _1] to [40_ _ _0] and press “OK”)	40 1 40 0 40

5.4.6 Errors during LOP Configuration

For SW < V9.7x:
Errors during the LOP configuration are indicated by “Er” on the COP (or CPI).
In case of an error: Confirm the error by pressing DT-O and repeat the configuration of the corresponding floor.

For SW ≥ V9.7x

Errors during the LOP configuration are indicated by “E0” to “E7” on the COP (or CPI).

<table>
<tr><td>E0</td><td>No button has been pressed.</td><td rowspan="3">Confirm the error by pressing DT-O.
Repeat the configuration of the corresponding floor.</td></tr>
<tr><td>E1</td><td>Wrong button has been pressed. (DOWN instead of UP)</td></tr>
<tr><td>E2</td><td>Button has been released too early (before 3 seconds) or pressed twice.</td></tr>
<tr><td>E3</td><td>LOP button state missing</td><td>Confirm the error by pressing DT-O.
Repeat the configuration of the corresponding floor.
If the error happens again, check for loose connectors or replace the LOP.</td></tr>
<tr><td>E4</td><td>Slave (LIN or LCUX) button state missing</td><td rowspan="2">Confirm the error by pressing DT-O.
Repeat the configuration of the corresponding floor.
If the error happens again, check LIN or LCUX wiring for loose connectors or replace LIN or LCUX.</td></tr>
<tr><td>E5</td><td>BIO 1 slave (LIN or LCUX) button state missing</td></tr>
</table>

E6	BIO 2 slave (LIN) button state missing	Confirm the error by pressing DT-O. Repeat the configuration of the corresponding floor. If the error happens again, check LCUX wiring for loose connectors or replace LCUX.
E7	No spare address available (in case of two access sides and separate LOP/LIN teach-in)	Erase the configuration of this floor xy (with CF=40, L=xy, PA=99, VL=1) and start the configuration of the corresponding floor again.

5.4.7 LIN and LCUX configuration

– Also each LIN and each LCUX needs to be configured with an unique address.
– Normally this is done during LOP configuration.
– The configuration of the inputs and outputs of the LCUX has to be done afterwards. It is described in section 5.4.14
– It is possible to use a LCUX also independent from a certain floor. (So called “machine room LCUX”.) In this case the addressing of the LCUX is done with help of the DIP switches on the LCUX. Details see section 7.14

Possible configuration procedures:

Version	LIN5, LIN51 LCUX1.Q	LIN52 LCUX2.Q
SW < V9.7	**A**	**A**
SW ≥ V9.7	**A**	**A** or **B**

Configuration procedure A

1	Make sure the LIN and/or the LCUX is connected to the LOP with help of the synchronization cable (XCF). (On a floor with both LIN and LCUX it may be necessary to do the configuration twice. One time with the LIN connected and a second time with the LCUX connected.)
2	Perform the LOP configuration including the LOP counting for this floor as described in the sections “LOP configuration” (5.4.4. to 5.4.6)

Configuration procedure B:

With the LIN**52** or LCUX**2**.Q installed in a system with software ≥ V9.7 it is possible to configure the LIN and LCUX without the synchronization cable (XCF) to the LOP.

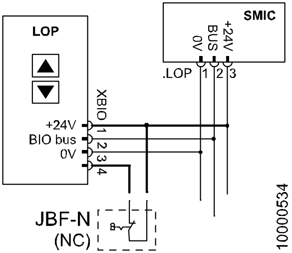
LIN52:

	The LIN52 can be configured with help of a magnet. (For example a hoistway information magnet.)
1	Enter the LOP configuration (CF=00, LE=--) and choose the floor where the LIN is installed.
2	During the 12 second countdown place the magnet in the middle of the LIN on the surface. The correct addressing will be confirmed by a beep of the COP. It is recommended to configure during the same countdown both the LOP (with the button) and the LIN (with the magnet). **Hint**: It is possible to extend the countdown time with help of CF=07, PA=10.
3	After all the LINs are configured the LOP counting has to be performed.

LCUX2.Q:

	The LCUX2.Q can be configured with help of its inputs (IO1 to IO4). **Remark**: The availability of this function is depending on the firmware installed on the LCUX2.Q. Beginning of 2011 this function was not yet released.
1	Enter the LOP configuration (CF=00, LE=--) and choose the floor where the LCUX is installed.
2	During the 12 second countdown close one of the inputs IO1..IO4, pin 3-4. (For example with the (key) switch connected to the LCUX.) The correct addressing will be confirmed by a beep of the COP. It is recommended to configure during the same countdown both the LOP (with the button) and the LCUX (with the input). **Hint**: It is possible to extend the countdown time with help of CF=07, PA=10.
3	After all the LCUXs are configured the LOP counting has to be performed.

5.4.8 Fire Service Switch JBF (BR) on LOP (simplex)



Connection of JBF to LOP in case of simplex systems [24877; 27.11.2007]

Preconditions:

– On the SIM card there must be a fire service BRx defined.
– For JBF a normally closed (NC) key switch has to be used
– JBF has to be connected according schematics to the LOP between XBIO.1 and XBIO.4
– JBF is in normal position (contact is closed).

Configuration:

1	Do the LOP configuration (CF=00, LE=--) on the floor where the JBF is connected. JBF must be in normal (closed) position. (See sections 5.4.4 to 5.4.6)
2	Make sure the LOP counting has been done. During the LOP counting the JBF is detected and configured by the system automatically. There is no additional configuration necessary.
	A correct recognition of the JBF is indicated by “Br” (instead of “Ac”) during the LOP addressing. (In systems without visual indicators (Dual Brand) the correct recognition is indicated by the missing acknowledge beep signal.)
	See extra note for duplex systems.

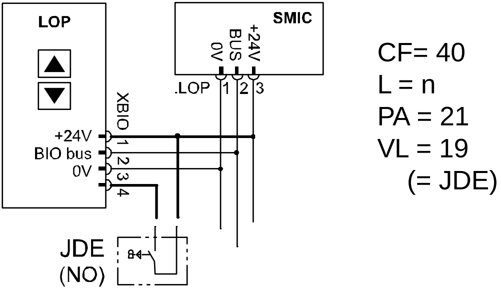
If on **same floor** a **JBF and another switch** (JAB or JDE) are installed then the JBF must be connected to a LCUX (Check the schematics)

In a **Duplex** system with SBBD PCB and only 1 LOP per floor the JBF key switch (with 2 synchronous contacts) must be connected to two LCUX. (Check the schematics.)

In both cases the configuration must be done with CF=40 (with SW ≥ V9.34) manually (see section 5.4.14)

5

5.4.9 Floor Call Key Switch JDE on LOP (CF=40)



JDE configuration [402026; 15.10.2014]

Preconditions:

– No SIM card option necessary
– For JDE a normally open (NO) key switch has to be used.
– JDE has to be connected according schematics to the LOP between XBIO.1 and XBIO.4
– JDE is in normal position (contact is open).

Configuration:

Step	SW ≥ V9.34
1	Enter the configuration mode menu 40

Step	SW ≥ V9.34
2	Do the LOP configuration (CF=00, LE=--) on the floor where JDE is connected. JDE must be in normal (open) position. Make sure the LOP counting has been done.
3	Choose CF=40, L=n (floor level) Choose PA=21 Configure VL: • JDE and JDE-U → VL=019 • JDE-D → VL=020

5.4.10 Out of Service Key Switch JAB on LOP

For the “Out of Service” JAB key switch the same preconditions and the same configuration procedure as for the JDE key switch applies. (See section before.)

The only differences:
– SIM card option “JAB Out of Service”
– Program BMK code VL=049 (= JAB)

JAB parking floor

The floor to which the elevator returns while JAB is active can be configured with CF=02, PA=03. (Independent on the floor where the JAB is mounted.)

5.4.11 Duplex, General information

Direct duplex connection with RS232 cable:

– SIM card option "DUPLEX" must be present on both elevators
– When to connect the elevators:
  - Standard duplex: Connection can be done before or after the complete configuration
  - Asymmetric duplex: Connection must be done **before** the configuration. (See section 5.4.13)
  - Duplex with independent doors (ZZ2 or ZZ3): Connection must be done **after** each elevator has been configured completely independent.
– Duplex with single riser and SBBD: LOP configuration see section 5.4.12

Duplex connection with overlay box:

– SIM card option "DUPLEX" must **not** be present
– Not possible with independent doors (ZZ2 or ZZ3):
– Configuration see section 5.4.12

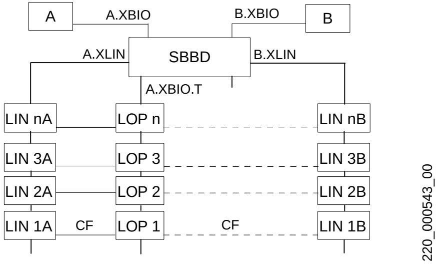
5

5.4.12 Duplex, LOP configuration with SBBD board

For duplex systems with only 1 LOP per floor

If both elevators A and B have power supply, the LOPs are connected to A. If A is without power supply and B has power supply then the LOPs are connected to B automatically.

LINs are always connected to the corresponding elevator.



Duplex with only 1 LOP per floor [17363; 02.02.2011]

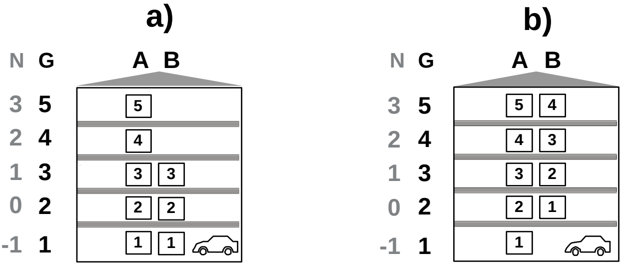
Step	Description
1	Connect all LIN of elevator B to the LOP’s with XCF cable.
2	Power OFF elevator A. (The LOPs are now connected to BIO bus of elevator B)
3	Do the LOP address configuration for elevator B for all the floors. (The LOP counting is not required yet because it has to be done on step 5)
4	Power up elevator A. (The LOPs are now connected to the BIO bus of elevator A.)
5	Do the LOP counting (CF=00, LE=00) for elevator B. Only the devices connected to elevator B are stored in the table of elevator control B.
6	Repeat the LOP address configuration for elevator A for all the floors. (Elevator B could remain with the power on)
7	Repeat the LOP counting (CF=00,LE=00) for elevator A.

On floors with LCUX and LIN it may be necessary to connect the LCUX and LIN to the LOP with a specially made “Y-synchronization XCF cable”. (LCUX and LIN connected parallel to LOP). Otherwise there may be problems to configure the key inputs and outputs.
Or the address configuration on the LIN52 and LCUX2.Q has to be done with the magnet or by using the input during the same countdown the LOP is addressed.

5.4.13 Asymmetric Duplex (CF=86)

Preconditions and restrictions:

– Direct RS232 connection between the two controls must be installed. (At the moment asymmetric duplex is not possible with overlay boxes.)
– KS policy. (To have the possibility to give on the last common floor a call in direction of the extreme floors.)
– For more customer comfort it is advised to install additional landing calls (DCW-U/D key switches or push buttons) to call the CW elevator which serves all floors. Otherwise the customer may have to change the elevator on the last common floor.
– SIM card option “DUPLEX” on both elevators
– Just the following 2 types of asymmetry are allowed, if SW < V11:



Asymmetric layouts [37752; 02.02.2011]

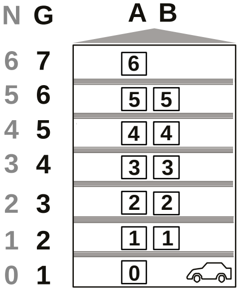
a), b) Possible types of asymmetry
A, B Elevator floor levels of each elevator (used by the control, for example during configuration)
N Customer floor naming (visible on COP)
G Group floor levels (starts always on the lowest position of the whole group)

If SW < V11

Step	Description	
1	Connect the 2 elevators with the RS232 cable.	
2	If available install the additional landing call inputs (DCW-U/D). – These landing calls can be installed on floors which are served by both elevators. (Maybe the customer would like to have the possibility to call the preferred elevator only on some of these floors) – Use an LC UX (or input 4 on the LOP) to connect either an additional push button or a key switch.	
3	Switch on both elevators	
4	Configure **CF=86** on both elevators. CF=86 defines the lowest group floor level for each elevator.	
	Example a) (see picture): • **A**: CF=86, PA=01, **VL=1** • **B**: CF=86, PA=01, **VL=1**	Example b) (see picture): • **A**: CF=86, PA=01, **VL=1** • **B**: CF=86, PA=01, **VL=2**
5	If inputs for DCW-U/D are available, configure the elevator which serves all floors as CW elevator. (**CF=07, PA=08**: preferred elevator).	
	Example a) (see picture): • **A:** CF=07, PA=08, **VL=1** • **B**: CF=07, PA=08, **VL=0**	Example b) (see picture): • **A**: CF=07, PA=08, **VL=1** • **B**: CF=07, PA=08, **VL=0**
6	Perform the LOP configuration for both elevators.	
7	If inputs for DCW-U/D are available, configure these inputs with – **BMK=007** (DCW-U = Preferred car upwards) or – **BMK=008** (DCW-D = Preferred car downwards)	

Since SW V11, following types of asymmetry are also allowed:

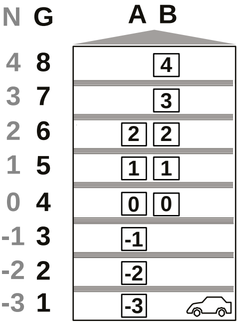
c) Inner asymmetry



[604620_001; 10.09.2014]

- A, B Elevator floor levels of each elevator (used by the control, for example during configuration)
- N Customer floor naming (visible on COP)
- G Group floor levels (starts always on the lowest position of the whole group)

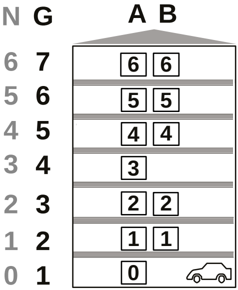
d) Up and down asymmetry (for example, used for both penthouse and garage)



[604620_002; 10.09.2014]

- A, B Elevator floor levels of each elevator (used by the control, for example during configuration)
- N Customer floor naming (visible on COP)
- G Group floor levels (starts always on the lowest position of the whole group)

e) Hole asymmetry (for example, some floors are not available for one elevator)



[604620_003; 10.09.2014]

- A, B Elevator floor levels of each elevator (used by the control, for example during configuration)
- N Customer floor naming (visible on COP)
- G Group floor levels (starts always on the lowest position of the whole group)

Since V11 is possible to set precise building level to cope complex layouts.

Choose the base level via menu CF86.

Configuration example:

Building	Elevator A	Elevator B
15	A3	
14	A2	
13	A1	
12	D7	D7
11	D6	D6
10		5
9		4
8		3
7		2
6		1
5	G	G
4	-1	
3	-2	
2	-3	
1	U	U

– Elevator A has 10 floors, elevator B has 9 floors
– Elevator A and B shares underground (U), lobby (G) and dining room (D)
– Elevator A has exclusive access to attic (A) and garage (-1 ... -3)
– Elevator B has exclusive access to hotel room (1 ... 5)
– The menu parameter CF86 will be configured as:

Elevator A	
CF86 PA	**VL**
1	1
2	2
3	3
4	4
5	5
6	11
7	12
8	13
9	14
10	15

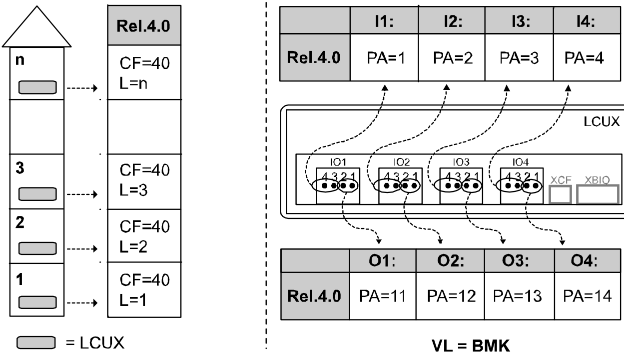
Elevator B	
CF86 PA	**VL**
1	1
2	5
3	6
4	7
5	8
6	9
7	10
8	11
9	12

- If SBBD is available, configure asymmetry before configuring the landing panel
- If blind floor, it is important to set them (CF26) before configuring asymmetry (CF86).

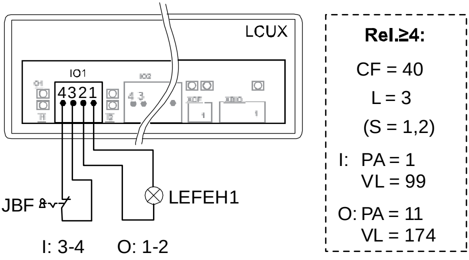
5.4.14 LCUX, Additional Inputs and Outputs

Preconditions:

– For some functions a SIM card option is necessary
– Software must support the configured BMK function code.
– Inputs and outputs connected according schematics. (Additional information: See LCUX, section 7.14.)
– Not possible with LOPB4
– LOP configuration and LOP counting have been performed with LCUX connected according schematics (BIO bus and XCF)



Example: JBF and LEFEH on floor level 3:



Configuration of an Input:

	Description
0	Enter the configuration mode menu 40
1	Definition of floor level where the LCUX is connected
	– CF=40 – L=n (floor level) – (S=1,2 = door side)
2	Definition of input where the contact is connected
	• Input 3: PA=3 • Input 4: PA=4
3	Definition of the input function (BMK)
	– VL=BMK (Code can be found in the section 5.3.4 “BMK function codes”)

Configuration of an Output:

	Description
0	Enter the configuration mode menu 40
1	Definition of floor level where the LCUX is connected
	– CF=40 – L=n (floor level) – (S=1,2 = door side)
2	Definition of output where the signal (24V, 400mA max.) is connected
	– PA=11..14 • Output 1: PA=11 • Output 2: PA=12 • ....
3	Definition of the output function (BMK)
	– VL=BMK (Code can be found in the section 5.3.4 “BMK function codes”)

For group functions in duplex systems: Both LCUX must be configured equally.

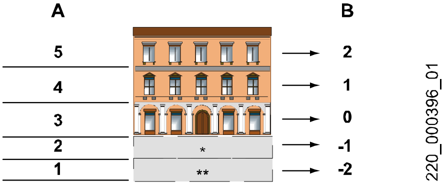
LCUX with pre-defined address which has no XCF connection to a LOP (for example used for hoistway temperature supervision):

– Address definition has to be done with DIP switches on LCUX. See section 7.14.
– Configuration has to be done with CF=40, L=91, L=92, L=93 (SW ≥ 9.34)

5

5.4.15 Floor Designation (CF=01)

Example: The first floor level has to be designated as “-2”. (CF=01, PA=03, VL=-2)



A = Floor level, B = Floor designation [13026; 19.07.2005]

	Description	HMI
1	Enter the configuration mode menu 40.	40_ _ _ _1
2	Choose “Floor designation configuration” CF=01	CF_ _ _01
3	Press “OK”. → The first floor level is displayed.	PA _ _ _ 1
4	Choose the floor level you want to designate. (In our example floor level 1.)	PA _ _ _ 1
5	Press “OK” on the HMI to confirm your choice. → The actual designation of the floor is shown.	VL_ _ _ 1
6	Change the floor designation to the number you prefer. (In our example floor designation “-2”.)	VL_ _ - 2
7	Press “OK” on the HMI to confirm your change. The next floor level is shown.	PA_ _ _ 2
	Repeat step 4) to 7) for all the floor levels. Remark: The floor designations above the lowest floor level are changed ascending automatically.	
	Leave the configuration mode by pressing “ESC”, change [40 1] to [40 0] and press “OK”	40_ _ _ _1 40_ _ _ _0 40

5.4.16 ZB1, PIN Code Protected Car Calls (CF=41, PA=1)

– **CF = 41**
– **L = n** (floor level)
– **(S = door side 1 or 2)**
– **PA = 1** (ZB1)
– **VL = PIN** (PIN code)

Preconditions:
– Only possible with COP5-10 or COP5B-10
– SIM card option “ZB1 Pin Code”
– Not allowed together “GS Visitor Operation” on the same floor

Example: Access to floor level 5 should be protected by Pin code “123”.

	Description	HMI
1	Enter the configuration menu 40	40_ _ _1
2	Choose CF=41 (Restricted access functions)	CF_ _ 41
3	Press “OK”. → The HMI displays the lowest floor level. Choose the floor level you want protect. (In our example L=05)	L_ _ _n
4	Press “OK”. Only in case of independent doors (ZZ2 or ZZ3): Choose the door side you want to restrict the access.	S_ _ _x
5	Press “OK” and choose PA=1 (ZB1 access restriction)	PA_ _ _1
6	Press “OK”. → The HMI displays the actual PIN code for this floor level.	VL 0000
7	Enter the code (in our example [UL-123]). Confirm every digit by pressing “OK” (Afterwards HMI displays PA2)	
8	Leave the configuration mode by pressing “ESC”, change [40 1] to [40 0] and press “OK”	

5

- PIN code can be 3 or 4 digits long
- 3-digit codes must start with a “-” during configuration. In operation, only the three digits will be necessary.
- To erase the code: Enter VL=0000
- It may be necessary to change the time to enter the PIN code. This can be done with CF=06, PA=03.
- In case of independent door operation it is not possible to configure on a certain floor the same PIN code on side S1 as on side S2.

5.4.17 ZB3, Car Call with Key Switch (CF=41, PA=2)

Preconditions:
- SIM card option “ZB3”
- Key switch connected to COP input XKEY (Pin 2-3)

Step	SW ≥ V9.34
1: JDC	– **CF = 05 (=55)** – **PA = Input** (XKEY..) – **VL = 002** (=JDC)
2: ZB3	– **CF = 41** – **L = n** (floor level) – (**S =** door side **1** or **2**) – **PA = 2** (ZB3) – **VL** = Input (XKEY..)
3: JRVC behavior	– **CF = 41** – **L = n** (floor level) – (**S =** door side **1** or **2**) – **PA = 2** (ZB3) – **VL** = Input (XKEY..)

5

Example:
- Floor level 4 should be accessible only with help of a key switch. The key switch is connected to SCOP plug KEY3.

Version COP5, COP5-10, COP5B-N, COP5B-10, COP4B

	Step 1: Key definition as JDC
1	Choose “SCOP5 Key Switch Definition”: CF=05
2	Choose the input (plug) you want to configure: In our example plug “KEY3”: PA=03
3	Assign JDC function to this input: VL=02

	Step 2: Key input assignment to restricted floor level
4	Choose “Access restriction”: CF=41

	Step 2: Key input assignment to restricted floor level
5	Choose the floor level: L = n. In our example L = 4
6	Only in case of independent doors (ZZ2 or ZZ3): Choose the door side S=1 or S=2.
7	Choose ZB3 function: PA = 2
8	Define the same key input (plug) as used in step 1.2. In our example VL=03

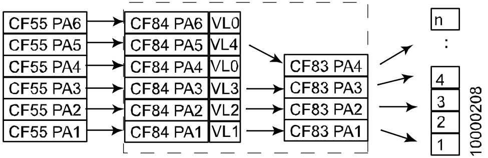
	Step 3: JRVC behavior
	– This step does only apply if a JRVC key switch is in the car – With step 3 it is defined whether JRCV reservation control can access the restricted floors or not.
9	Choose “Access restriction”: CF=41
10	Choose floor level L=0. (JRVC behavior affects all levels)
11	Choose PA=1
12	Assign the JRVC behavior: – VL=0: JRVC reservation control can access the restricted floors – VL=1: JRVC reservation control can **not** access the restricted floors

Handicapped COPH, SCOPH(MH), Dual brand MX-B with Bionic 5

	Step 1: Key definition
1	Choose “SCOPH(MH) Key Switch Definition”: **CF=55**
2	Choose the input (plug) you want to configure: In our example input “IO3”: PA=03
3	Assign JDC function to this input: VL=02

	Step 1b: Input assignment to SCOPH(MH) key input
4	Choose Input-Key Assignment: **CF=84**
5	Define the same input (“IO” plug) as used in step 1.2. In our example again PA=03

	Step 1b: Input assignment to SCOPH(MH) key input
6	Assign this input to a virtual key input of your choice. In our example we choose key input 03: VL=03



[20212; 15.01.2007]

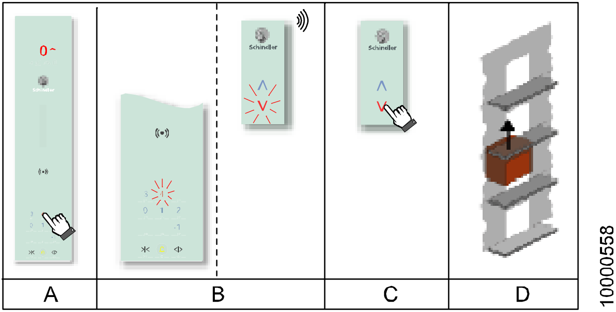
	Step 2: Key assignment to restricted floor level
7	Choose ZB3 key assignment: **CF=83**
8	Define the same (virtual) key input as used in step 1b.6. In our example input 03: PA=03
9	Assign floor level with restricted access to this virtual input. In our example level 4: VL=04

5

5.4.18 GS, Visitor Control (CF=17 or CF=41, PA=03)

	SW < V9.7	SW ≥ V9.73
Step 1: GS	– **CF = 17** – **PA = n** (floor level) – **VL = response time** [1=10s]	– **CF = 41** – **L = n** (floor level) – (**S =** door side **1** or **2**) – **PA = 3** (GS) – **VL = response time** [1=10s]
Step 2: ZB3	see section 5.4.17	see section 5.4.17

Function description:



[25349; 18.12.2007]

Example. Protection of penthouse floor 4:

a) A visitor gives a car call to the protected floor.

b) The car call is acknowledged but the elevator does not yet travel. On the protected floor there is an optical signal (on the LOP) and an acoustical signal (gong connected to LIN)

c) On the protected floor the owner of the apartment has to approve car call in within 30 seconds by pressing the release button.

d) Only after approval the car travels to the protected floor.

Preconditions:

- Only one floor is allowed to have visitor control
- PI control
- Option GS “Visitor Control” on SIM card
- Option ZB3 “Key Restricted Access” on SIM card or a SAS system has to be available
- “Pin Code Restricted Access” ZB1 not allowed on the same floor
- On the protected floor a LIN with a gong must be mounted

Remark: To enhance the security for the customer an intercom connection to the car can installed. (Minimum load function is not available with Schindler 3100/3300/5300.)

(Remark: In the some schematics the floor call button and the acknowledge button may be interchanged.)

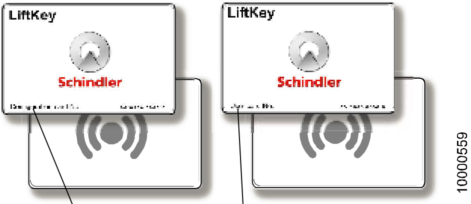
Configuration:

5

	Step 1: Configuration GS (CF=17) SW < 9.7	**Step 1: Configuration GS (CF=17) SW ≥ V9.73**
1	Choose “Visitor Control GS”: **CF=17**	Choose “Access restrictions”: **CF=41**
2	Choose the floor level which has to be protected. In our example floor level 4: PA=04	Choose the floor level which has to be protected. In our example floor level 4: L=04
3	Program the time period while the owner of the apartment can approve the visit. In our example 30 seconds: VL=3	Only in case of independent doors (ZZ2 or ZZ3): Choose the door side S=1 or S=2.
4	-	Choose “Visitor Control GS”: **PA=3**
5	-	Program the time period while the owner of the apartment can approve the visit. In our example 30 seconds: VL=3

	Step 2: **Car call key switch for customer** **SW ≥ 9.34: CF=5 or CF=55, CF=41 PA=2**
5	• To allow the customer to travel to his apartment a key switch for the corresponding apartment floor level has to be programmed in the car (see section 5.4.17). • Instead of a key switch it is also possible to use the Schindler Access System SAS to allow the customer to travel to his apartment (see section 5.4.19).

5.4.19 SAS Schindler Access System (LiftKey)



Configuration card and User card [25395; 20.12.2007]

Preconditions:

- Only possible with COP5, COP5-10 and COP5B-10
- **Not** possible with COP5B-N, COP5-1N 25 EU, COP5B-1N 25 EU or COPB4
- **Not** possible with FI-GS
- Maximum number of floor supported: ZE ≤ 15
- To restrict the access to a certain floor the SIM card option “ZB1 PIN code” must be available
- (SAS card reader itself does not need a SIM card option)
- COP SW ≥ V1.5
- Option SAS (KDCore module and SASA antenna) must be installed inside the COP
- Configuration only possible with designated configuration card
- A set of user cards must be available
- CF=01 has been performed
- Not possible together with independent door operation (ZZ2, ZZ3)

SAS offers the following possibilities

a)	Car call to unprotected floor The customer places the user card to the COP. A car call to the programmed floor is released (without pressing a push button). The access to this floor is also possible by pressing the COP push button. (SAS just offers contactless car calls.)

b)	Car call to floor protected by PIN code The customer places the user card to the COP. A car call to the programmed floor is released (without pressing a push button and without entering the PIN code.) (The access to this floor is also possible without the SAS card by entering the PIN code.)
c)	Access to (many) floors protected by PIN code. The customer places the user card to the COP. The COP shows “FL”. Now the customer can choose the floor he wants to travel by pressing the corresponding COP button. With this option more than one floor can be accessed with the same user card. (The access to the floors is also possible without the SAS card by entering the PIN code.)

Programming of a User Card

	Description
	Before starting the configuration make sure the “Floor designation” CF=01 has been performed. During the SAS configuration the floor **names** (designated with CF=01) are used (and not the floor **levels**).
	For the configuration of a user card the elevator specific configuration card has to be used. (With each SAS system two configuration cards are delivered which work exclusively with this specific COP.) Have both the configuration and the user cards (which should be programmed) ready.
1	Place the **configuration card** close to the receiver sign on the COP. → COP beeps and shows “Cr” (Card received)
2	→ COP beeps again and shows “CF” (Configuration mode) (To exit this menu press DT-O for 3 seconds.)
3	Enter “1” on the COP (1 = User card activation) → COP beeps and shows “FL” (Floor)
4	Enter the floor **name** which has to be configured. (COP beeps.) Push DT-O to confirm the floor level. → COP beeps

	Description
4a	To activate the user card for additional floors, repeat step 4 floor all floors which should be accessible with the user card.
5	To activate the user card for this (these) floors press DT-O for 3 seconds. → COP beeps and shows “CA” (Wait for a card) (To exit this menu press DT-O for 3 seconds.)
6	Place the **user card** close to the receiver sign on the COP. → COP beeps and shows “Cr” (Card received) → COP beeps again and shows “CA” (Wait for a card)
6a	To activate additional user cards with the same rights repeat step 6 for each user card.
7	To exit this menu press DT-O for 3 seconds. → COP beeps and shows “CF” again (Configuration mode) To exit the configuration mode press DT-O for 3 seconds.

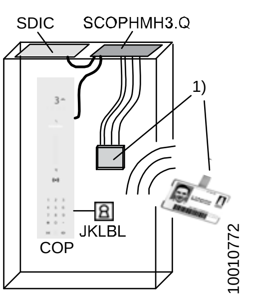
- Before handing over the user cards to the customer write down the used card numbers. (With help of theses numbers the user cards can be deactivated if necessary.)
- For the deactivation of a user card of configuration card please refer to specialist document J 42103073 “SAS General description and user guide”. (English)
- The configuration cards have to be stored on a safe place. Keep in mind that the configuration cards are “married” with COP.

Access restriction

	Description
8	The programmed user card works as a “key switch for car call”. The access to a certain floor is not restricted automatically. → To **restrict** the access to a certain floor a PIN code has to be programmed to this floor. Please refer to section 5.4.16 “**ZB1**, PIN Code Protected Car Calls (CF=10, CF=41)”

5.4.20 Parallel Card Reader CRC, ZBC2

Function description:



1) = 3rd party card reader with 3rd party ID card.

- The 3rd party card reader outputs (contacts) are connected to the SCOPHMH3.Q.
- As soon as the key switch JKLBL is activated (closed), the buttons on the normal COP are deactivated.
- Car calls to restricted floors can only be given with help of a 3rd party card reader. (For example with help of an ID card.)

Preconditions and restrictions:

- SW ≥ 9.38
- Option “ZBC2 Parallel Card Reader Interface” on SIM card
- Not possible together with ZZ2 or ZZ3 (independent doors)
- The SCOPHMH parallel card reader interface installed according to schematics S274199 and S274603. (The power supply for the 3rd party card reader is taken from SDIC.XPOW, 230V. The 3rd party card reader outputs are connected as potential free contacts to SCOPHMH3.Q, connectors XDC1..XDC21, pin 2-3.)
- If the SCOPHMH is installed after the learning travel, it must be assigned to the system manually. Use menu 10, submenu 129 for the COP detection.
- The JKLBL normally open contact is connected either to a COP input or to a LCUX input
- With SW < V9.7 only 1 “regular” COP inside the car is allowed.
- With SW ≥ V9.7 it is possible to have 2 “regular” COPs inside the car (but not with independent doors ZZ2 or ZZ3). (2COPs and ZBC2 may not be released for sales.)
- With SW ≥ V9.6 it is possible to have ZBC2 with multiple car calls. (The customer can choose the floor on the “regular” COP.)

Configuration

	Step 1: Configuration of JKLBL	
	The JKLBL is used to activate the floor access restriction. It can be connected either to the COP or to a LCUX.	
	Variant A: JKLBL at COP	**Variant B: JKLBL at LCUX**
1	– **CF = 05** – **PA = Input** (XKEY.. on COP) – **VL = 052** (=JKLBL)	– **CF = 40** – **L = n** (Floor level, where LCUX is connected) – **PA = Input** (IO.. on LCUX) – **VL = 052** (=JKLBL)

	Step 2: Configuration of the floors where the access should be restricted
2	**CF = 41** (Access restriction)
3	**L = n** (floor level, where the access should be restricted)
4	**PA = 99** (Type of restriction)
5	**VL = 1** (ZBC2. Restricted exit from the car to the floor. Car call disabled.)
	Repeat step 3 to 5 for all floor levels, where the access should be restricted.

5

5.4.21 LIFD, Long Interfloor Distance (CF=26)

The long interfloor distance (LIFD) kit is used to minimize the problems due to slip in systems with big distances between the floors. To reduce this problem an additional PHS flag is installed between the floors. (“Blind” floor.)

During the learning travel the additional flag is recognized as the flags from the normal floors. (Although there is no landing door available.) Therefore this floor has to be defined as “blind” floor after the learning travel.

Preconditions and restrictions:
– SIM card option “LIFD Long Interfloor Distance”
– PHS_B sensor on car and additional flag on “blind” floor installed

– Learning travel executed
– The distance between 2 “real” floors must not exceed 11 meters. (Otherwise an evacuation in case of system failure can not be guaranteed anymore.)

Configuration:

1	Choose “LIFD”: **CF=26**
2	Choose the floor level where no landing door is available. **PA=n**
3	Mark this floor as “blind” floor **VL=1** (= blind floor)

5.4.22 Triplex Configuration with Overlay box

Commissioning: Preconditions and function check:
– There are no cables connected to overlay boxes
– The ethernet cables are routed between the elevators, but not yet connected to the overlay boxes
– SIM card policy must be KS. No special SIM card option for triplex necessary.

1	Do the commissioning of each elevator as if it would be a simplex (or duplex) elevator.
2	Perform the LOP configuration of each elevator. On the elevators where the LOPs and LINs/LCUX are connected via SBBD: Perform the LOP configuration of elevators as described in section 5.4.12.
3	Check the function of each elevator. (Landing calls, car calls, indicators.)
4	If there is a SBBD PCB installed: Switch off the first elevator and check the function of the landing calls and landing indicators. Repeat the check for the second elevator.

Triplex configuration

5	Do the following configurations on the respective elevator: • Elevator **A**: CF=04, PA=01, VL=**1** • Elevator **B**: CF=04, PA=01, VL=**2** • Elevator **C**: CF=04, PA=01, VL=**3**
6	Switch off all elevators.
7	In all elevators connect: • All Y-cables (LOP, VF) • With the Y-cables all overlay boxes (PWR, CAN) • All Ethernet cables (See schematics S277208)
8	Switch on all elevators. (The sequence does not matter.)
9	On each elevator activate menu 10 > submenu 136 (overlay detection)

5.4.23 Independent Doors (ZZ2, ZZ3)

General information, preconditions and restrictions:

- Required software:
  - SCPU software ≥ V10.07 necessary
  - Tested and released with SDIC SW ≥V3.2 and COP SW ≥V3.4
- Necessary SIM card options
  - ZZ2 for selective door function (both doors can be opened independently).
  - ZZ3 for interlock door function (to avoid that both doors are open at the same time).
- Required hardware:
  - 2 COP must be installed in the car
- Restrictions:
  - Not possible with Triplex or Quadruplex
  - In case of Duplex: At the moment only possible with fully symmetric duplex.
  - Dual brand COPs (MX basic) not supported.
  - Schindler Access System SAS (LiftKey) not supported

Configuration

In case of duplex system: Do the following configurations individually for each elevator before connecting the group with the RS232 cable.

	Description
1	Check the assignment of the COPs to the door side: – Give a call on one COP and check, whether the corresponding door side opens. – Repeat the test for the second COP. If the COP assignment is wrong, change it with **CF=30** – **PA=21**: Defines the door side to which **COP#1** belongs. – **PA=22**: Defines the door side to which **COP#2** belongs. VL=0: Undefined, VL=1: Side 1, VL=2: Side 2, VL=3: Both sides The numbering of the COP is defined by its physical position: SDIC → COP#2 → COP#1.

	Description
2	Do the LOP configuration. During the LOP configuration on a floor with two access sides the control will offer automatically first door side 1 for the LOP configuration and directly afterwards door side 2.
3	If necessary configure the door side for special services. Check **CF=30, PA=1, 2, 6, and 7**
4	In case of access restrictions (**CF=41**) check that they are configured for each door side correctly. For that reason in the access restriction menu CF=41 there is a new submenu for the door side selection **S1** (door side 1) and **S2** (door side 2) available.
5	In case of additional inputs and outputs on BIO bus nodes (LOP, LIN, LCUX, BIOGIO) check that they are configured for each door side correctly. For that reason in the access restriction menu **CF=40** there is a new submenu for the door side selection **S1** (door side 1) and **S2** (door side 2) available.

5.4.24 ETM Embedded Telemonitoring

On the SIM card (Chip card) there are all relevant data stored for the automatic ETM configuration. (For example the information about the correct telephone number of the Remote Monitoring Platform RMP.)
Only if the data on the SIM card are wrong or missing, the ETM has to be done manually.

This section describes the following configurations

- **Automatic ETM configuration (default procedure)**
- Manual ETM configuration without PABX
- Manual ETM configuration with PABX

Automatic ETM configuration

Step	Description
	For the correct connection of the CLSD, the TAM2 (GNT alarm device) and the PABX: Refer to the schematic wiring diagrams.
1	Do not yet connect the TAM2 (GNT alarm device). Make sure there is no alarm pending.
2	Do not yet plug the telephone line to the CLSD.
3	On the CLSD PCB set the rotary switch to the correct position: – Position “0”: If the CLSD is connected directly to the telephone line – Position “F”: If the CLSD is connected to the Schindler PABX. (The PABX is connected to the telephone line.) This may be the case in case of duplex systems or in case of multishaft systems.
4	Plug the connector PSNT on the CLSD.
5	Perform the “CLSD detection”. – On the HMI choose menu 10 and press OK – Choose Submenu 128 and press OK – Change from [128 0] to [128 1] and press OK

Step	Description
6	→ Now the CLSD is detected by the system. – HMI displays [1281] during the detection – HMI displays [128 1] when the detection has finished (A correct detection of the CLSD (ETM) can be checked later by CF21, PA8. The value should be different from 0.)
7	→ After the CLSD detection the automatic ETM configuration starts.
8	If the ETM configuration was not successful continue as follows: Perform one of the below listed “Manual configuration”.

The configuration status of the ETM can be checked with help of menu 30, submenu 308. (see section 9.3)

After the ETM configuration connect the TAM2(GNT alarm device). Call the telealarm control center and ask them to configure the TAM2 remotely

5

Manual configuration without PABX

Step	Description
0	This manual configuration has to be done only if the automatic configuration was not successful.
1	On CLSD turn the rotary switch to position “0”

Step	Description
2	In CF21 PA33, enter the appropriate country code, for example VL40 for Switzerland. • Refer to annex “List of Country Codes” in EJ 604639 for the list of country codes. • Setting or modifying PA33 triggers the update of: – CF21 PA15 ... PA19 (country-related modem initialization string) – CF21 PA5 (TACC phone number). • The country parameter is set with the value from the SIM card during learning travel if PA33 has the default setting (“no country defined”) and the SIM card includes the option.
3	Initiate a modem detection on HMI with special command 128. Check CF21 PA8 for the result of the modem type detection.
4	In CF21 PA1 and PA2, enter the Installation ID (Box ID). Example: If the ID is “88999111”, enter: • PA1 → VL000088 • PA2 → VL999111.
5	In CF21 PA3, enter VL1 as module number.
6	In CF21 PA5, PA6 and PA7, enter the appropriate TACC phone number. Example: If the phone number is “0917569785”, enter: • PA5 → VL091756 • PA6 → VL9785 - - • PA7 → VL - - - - - -
7	If applicable, set CF21 PA35 to determine how to dial the trunk (exchange) line.
8	In CF04 PA2 and PA5, set the local time and date. Example: time is 9.38.10 AM, date is 20.06.2007 (20th June, 2007), enter: • PA2 → VL093810 • PA5 → VL200607 (→ will automatically be converted to 20.06.**20**07)

Step	Description
9	Exit configuration mode.
10	Make sure that special command 117, “Service Visit”, is deactivated.

Manual configuration with PABX

Step	Description
0	This manual configuration has to be done only if the automatic configuration was not successful.
1	On CLSD, turn the rotary switch to position “F” (→ with PABX).
2	In CF21 PA33, enter the appropriate country code, for example VL40 for Switzerland. • Refer to annex “List of Country Codes” in EJ 604639 for the country codes • Setting or modifying PA33 triggers the update of: – CF21 PA15 ... PA19 (country-related modem initialization string) – CF21 PA5 (TACC phone number). • The country parameter is set with the value from the SIM card during learning travel if PA33 has the default setting (“no country defined”) and the SIM card includes the option.
3	Initiate a modem detection on HMI with special command 128. Check CF21 PA8 for the result of the modem type detection.
4	In CF21 PA1 and PA2, enter the Installation ID (Box ID). Example: If the ID is “88999111”, enter: • PA1 → VL000088 • PA2 → VL999111

5

Step	Description
5	In CF21 PA3, enter the module number according to the wiring configuration. Example: If the control of: – Elevator A is connected to the TM device on phone line 1 to PABX and – Elevator B is connected to the TM device on phone line 2 to PABX, enter: • On elevator A: PA3 → VL010 • On elevator B: PA3 → VL020.
6	In CF21 PA4, enter the shaft number according to the elevator disposition. Example: If: – Elevator A is the first elevator (that is, located in the first hoistway), and – Elevator B is the second elevator (that is, located in the second hoistway), enter: • On elevator A: PA4 → VL001 • On elevator B: PA4 → VL002.
7	In CF21 PA5, PA6 and PA7, enter the appropriate TACC phone number. Example: If the phone number is “0917569785”, enter: • PA5 → VL091756 • PA6 → VL9785 - - • PA7 → VL - - - - - -
8	If applicable, set CF21 PA35 to determine how to dial the trunk (exchange) line.
9	In CF04 PA2 and PA5, set the local time and date. Example: time is 9.38.10 AM, date is 20.06.2007 (20th June, 2007), enter: • PA2 → VL093810 • PA5 → VL200607 (→ will automatically be converted to 20.06.**20**07)
10	Exit configuration mode.

Step	Description
11	Make sure that special command 117, “Service Visit”, is deactivated.

5

5.4.25 ETMA Embedded Telemonitoring and Alarm

- By following the normal installation procedures, the ETM(A) should configure itself and the elevator can start to communicate through TeleMonitoring.
- In order to verify if the unit is already communicating the "Last successful TM call" date can be checked; through Menu 30 → Command 320-3.
- If the date is within the last 7 days: it means that the unit is already communicating.
- If no date available or older than 7 days: it means that the unit is not able to communicate with the Central and the next sections may help to configure and troubleshooting the ETM(A).

	Prerequisites for the auto-configuration
1	**Unit properly registered in SAP System (YEST):** Usually this is done during the installation phase. In order to verify that and for additional information refer to your Call Center / Backoffice.
2	**Connections done according to wiring schematics:** (in case of ETMA) Check it also with the ETMA LED diagnostic described in section 7.19
3	**Phone line / GSM Coverage available:** Verify it through Menu 30 → Command 308-5 (at least 24 VDC / 14 (in case of ETMA Wireless))
4	**Correct Equipment Number is configured on SIM Card (Chip Card):** Verify it through Menu 30 → Command 320-1 (If the number does not correspond to the expected Equipment number or is 0, a new SIM card that includes the correct Equipment number is required)
5	**Correct Country Code is programmed on SIM Card (Chip Card):** Verify it through Menu 40 → Parameter CF21PA33 (eventually adjust it) (refer to EJ604639 for the complete list of Country Code)

	Prerequisites for the auto-configuration
6	**Correct Clock settings (Date and Time):** Verify it through Menu 40 → Parameter CF4PA5 (eventually adjust it, in iSPECI the parameter is CF4PA98). The Remote Monitoring system automatically adjust the internal clock, but a date too far in the past or future (i.e. years) could be a reason for the not working Remote Monitoring)
7	**TM device properly detected:** Verify it through Menu 40 → Parameter CF21PA8 (0=No modem detect, 1=CLSD, 4=ETMA (Wireless / PSTN), 5=CGW)
8	**Rotary Switch (CLSD only):** Position '0': if CLSD is connected directly to the telephone line Position 'F': if CLSD is connected to the Schindler PABX

	How to re-start the auto-configuration
1	Delete pending Alarmthrough Menu 10 → Command 135-1
2	Reset "Modem type" by setting Menu 40 → Parameter CF21PA8 = 0
3	Perform "Modem detection" through Menu 10 → Command 128-1 (it takes around 60 sec.)
4	Reset the "Servitel ID" by setting Menu 40 → Parameter CF21PA1 and PA2 = 0
5	Perform "Next call"through Menu 10 → Command 126-1 (start auto-configuration)

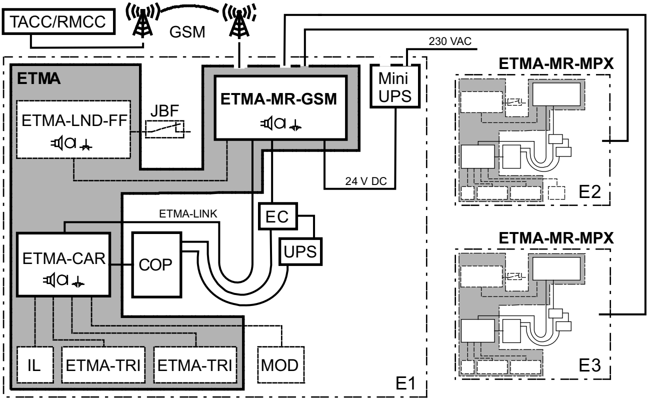
After few minutes the "Last successful TM call" date can be checked again and a TeleAlarm test call have to be performed.

If the system is still not working as expected:

– contact your Call Center / Back office in order to check the situation and eventually try a manual configuration from remote
– check further configuration details and troubleshooting hints by using the following references:

Device type	Document	Document title
All	K 42520120	RM Quick Reference
All	J 42520128	TA/TM registration process in SAP
ETMA PSTN	K 40700049	ETMA PSTN Quick Reference
ETMA PSTN	E J41700380	ETMA PSTN Installation & Commissioning
ETMA PSTN	EJ 41700383	ETMA PSTN Diagnostic
ETMA PSTN	EJ 41700381	ETMA PSTN Technical Information and Configuration
ETMA PSTN	EJ 41700382	ETMA PSTN Maintenance
ETMA Wireless	EJ 41700661	ETMA Wireless Installation
ETMA Wireless	EJ 41700662	ETMA Wireless Commissioning
ETMA Wireless	EJ 41700663	ETMA Wireless Diagnostic
ETMA Wireless	EJ 41700660	ETMA Wireless Technical Information & Configuration
ETMA Wireless	EJ 41700664	ETMA Wireless Maintenance
CGW/ETMA	J 41700485	Installation & Commissioning of ETMA with CGW
CGW	K 40700048	CGW Quick Reference
CGW	EJ 604795	CGW Installation & Commissioning
CGW	EJ 604796	CGW Diagnostic
CGW	EJ 604794	CGW Technical Information & Configuration
CGW	EJ 604797	CGW Maintenance
PABX	S 42102258	Networking with PABX

5.4.26 ETMA Wireless



ETMA Wireless, System Overview Triplex [402035_00; 02.02.2015]

Precondition: ETMA has been installed and pre-wired.

1) Connect antenna
2) Switch on the main power JH

   → ETMA will start. Check that the diagnostics LED of the ANT connector goes to green. If not, find a better position for the antenna using the “Antenna Installation Helpmode”.

   → ETMA configures itself automatically based on the mobile carrier and on the serial number of the device.

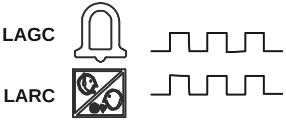
   → The controller will connect to the RMCC with help of the ETMA.
   → The RMCC performs the automatic configuration of the controller.
   → All parameters are stored in EEPROM and a backup has been made in the ETMA.
   → The automatic configuration has been finished successfully when the LAGC (bell) is switched off. (The LARC is still blinking.)
3) Release a test alarm to check the alarm functionality of the ETMA.

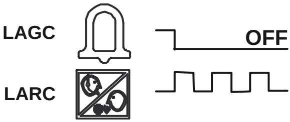
5

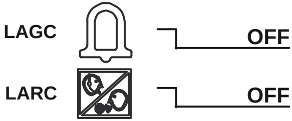
Configuration: Automatic Procedure

Preconditions:

- ETMA-MR and ETMA-CAR are installed and wired correctly
- ETMA-MR is connected to the control
- DAKI is connected (input ETMA-CAR.X1-1-1 is closed)
- Antenna is connected and ANT diagnostic LED is green
- System relevant data like the serial number of ETMA and the equipment number of the elevator are available in SAP OSC transaction YEST of the corresponding installation.

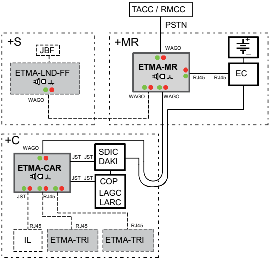
No	Step	Remark
1	Switch on the main switch JH in the controller.	The LARC and LAGC on the COP will blink and indicate that the automatic commissioning has not yet been done. 
2	ETMA-MR-GSM will perform the automatic configuration in background.	It will receive the configuration based on the SIM-card provider, the serial number of the device and the installation data stored in the RMCC database. Note: It is mandatory to have the data in the SAP OSC transaction YEST, otherwise the automatic configuration can not be completed.

No	Step	Remark
3	First step of the automatic commissioning procedure is the “modem detection”: • Either started automatically by the system during the learning travel • Or can be initiated manually with HMI menu 10, submenu 128 (=”modem detection”)	Learning travel: Automatically started if the elevator is switched to normal travel operation the first time. Note: The ETMA must be connected during the learning travel otherwise it is not recognized. The success of the modem detection can be checked with the HMI: CF=21, PA=08. It must have the value VL=4 (=ETMA). ATTENTION: Do not set the value manually to VL=4.
4	After successful modem detection the ETMA will connect to the RMCC as soon as ETMA itself has finished with its own automatic configuration.	Based on the country ID on the chip card, the control will choose the corresponding telephone number which is stored in the EEPROM and will connect to the RMCC with help of the ETMA.
5	The RMCC will automatically • set all the ETMA relevant parameters in the control based on the equipment number stored in the database • the controller will copy the parameters to the ETMA • ETMA will initiate a test call to the TACC (to test the voice channel) This procedure may take several minutes.	The RMCC sets the telephone numbers for the alarm, for the monitoring, for the periodic call, etc. and sets all the other relevant parameters. The successful configuration is indicated on the COP: 

No	Step	Remark
6	Initiate a test alarm to check the function of the ETMA.	Possibilities to disable the alarm filtering: • Either switch off the mains power (JH). • Or temporarily disable alarm filtering: HMI menu 10, submenu 134. • Or (with COP capacitive only) press the alarm button on the COP 5 times within 10 seconds. The systems quits with a double beep. Then press the alarm button again for more than 3 seconds to initiate the test alarm.
7	• ETMA will call the stored alarm phone numbers • Verify the voice quality by talking to the call center operator and verify the equipment number.	
8	The successful test alarm is indicated with LAGC = off and LARC = off.	

Diagnostics: Automatic Procedure

Step by step checklist with HMI for automatic configuration:

No	Checks	Actions in case not OK
1	Check all cable connections on the ETMA-MR and ETMA-CAR. (See also schematic wiring diagrams.)	
2	Check the LEDs on the ETMA-MR and ETMA-CAR (and ETMA-LND-FF).  The system is installed correctly, if • each module has at least one **green** LED lit • no module has any red LED lit	If any red LED is lit or if at any interconnection no LED is lit, refer to the **chapter "LED Diagnosis"** in this document.
3	Verify that the alarm input (**DAKI**) is connected and in normally closed position. (The automatic commissioning will not start when the alarm button is missing.)	Connect the alarm input (DAKI) and make sure it is closed.

No	Checks	Actions in case not OK
4	On the car operating panel **COP** check the **yellow LAGC (bell)** indication. If LAGC is lit, then it is necessary to reset the pending alarm.	Reset the pending alarm with help of the **HMI** user interface. • Choose main menu 10 • Choose submenu 135 • Set **135 = 1**

Check CF21 / PA5 ≠ 0

No	Checks	Actions in case not OK
5	Verify there is the **correct country ID** stored on the chip card. On the HMI check the country ID with menu 40, **CF=21, PA=33**.	
6	With help of the HMI check the equipment number: HMI menu 30, submenu 320, submenu **320-1** Note: If the Submenu 320 is not available, please update the controller software. If the Submenu 320-1 is not SPECI to the controller.	If the equipment number is wrong or missing, a manual configuration of all parameters is required. Please ask the control center responsible person for support.
7	If the control has been a long time without mains power there may be a mismatch in the **date** stored in the EEPROM. Check the current date with help of the HMI: Main menu 40, **CF=04, PA=05** (dd.mm.yy)	If necessary set the correct date with help of the HMI.

<table>
<tr><th>No</th><th>Checks</th><th>Actions in case not OK</th></tr>
<tr><td>8</td><td>Verify that the correct communication protocol is set in the elevator control.
HMI menu 10, submenu 106: Value must be 106 = 0</td><td>Set 106=0.</td></tr>
<tr><td>9</td><td>Verify that the monitoring is not disabled.
HMI menu 10, submenu 117: Value must be 117 = 0</td><td>Set 117=0.</td></tr>
<tr><td>10</td><td>Reset the basic parameters and redo the modem detection:
• SV-ID: Set CF=21, PA=01 and PA=02 to VL=000000
• Prefix: Set CF=21, PA=35 to VL=END (empty)
• Modem type: Set CF=21, PA=08 to VL=0

Restart the modem detection: HMI menu 10, submenu 128, 128=1
→ After the modem detection the modem type (CF=21, PA=08) should be VL=4.
→The automatic commissioning should start again.</td><td></td></tr>
</table>

5

No	Checks	Actions in case not OK
11	Check the data communication between ETMA and the Call Center: • Initiate a data call: HMI menu 10, submenu **126=1** • Verify the connection with HMI menu 30, submenu 308, submenu 308-3: – **308-3=7**: Connection ongoing (PPP negotiation) – **308-3=8**: Online	If the parameter 308-3 remains at value “2”, then the maximum numbers of retries is reached. • Switch off the elevator control • Disconnect the emergency power supply battery • Wait 30 seconds • Reconnect the battery • Switch on the elevator control • Perform again a data call (126=1) If the value goes to "7" but does not reach the "8", then please check the Connectivity Indication LEDs on the ETMA-MR.

5.4.27 STM Configuration

Configuration

Step	Description	
1	Check the controller time.	CF04 PA2
2	Check the controller date.	CF04 PA5
3	Set the manufacturing date of the oldest installed STM. [10100 ... 75399, dwwyy] Example for the back side label on the STM: MEGADYNE STM-PV30-1.73S-PU-42 X**dwwyy**1XXD0 d = day, 1 = Monday ... 7 = Sunday ww = week, 01 ... 52 yy = year, 04 ... 99 If the HMI displays "ER 46xx", the newly set STM manufacturing date is equal to or more recent than the current controller date.	CF28 PA4
4	Push the reset button.	-

Re-Configuration

Step	Description	
1	Check the controller time.	CF04 PA2
2	Check the controller date.	CF04 PA5
3	Activate the menu 190. The CF28 STM configuration menu is activated for 300 seconds.	-

Step	Description	
4	Set the manufacturing date of the oldest installed STM. [10100 ... 75399, dwwyy] Example for the back side label on the STM: MEGADYNE STM-PV30-1.73S-PU-42 X**dwwyy**1XXD0 d = day, 1 = Monday ... 7 = Sunday ww = week, 01 ... 52 yy = year, 04 ... 99 If the HMI displays "ER 46xx", the newly set STM manufacturing date is equal to or more recent than the current controller date.	CF28 PA4
5	Set the STM bending counter.	CF28 PA9
6	Exit the STM configuration menu.	-
7	Push the reset button.	-

5.4.28 Retainer Plus Configuration

Step	Description
1	No configuration required. It is a chip card option.

5.5 Sematic C MOD (Schindler 6300)

For the parameters of the Sematic C MOD door drive please refer to section 7.23.2

5.6 Fermator Compact (Schindler 6300)

For the parameters of the Fermator door drive please refer to section 7.22

5.7 Varidor 15

All the changeable parameters can be found in EJ 41354325.
The parameters can be changed with the aid of the HMI.

6 Preventive Maintenance

6.1 General information

For the preventive maintenance use the instruction K 40700200 “Schindler Expert Plus Maintenance Checklist”

Document K 40700200 [37912; 26.11.2010]

Before doing any maintenance on the elevator make sure the telemonitoring system is deactivated:

– Either use the automatic car positioning (see section 4.9.8) to switch on inspection travel mode
– or disable the telemonitoring with help of the HMI, menu 10, submenu 117=1

6.2 Test alarm

To deactivate temporarily the alarm filtering set HMI menu 10 > 134 to “1” and release a test alarm within 30 seconds.

6

7 Appendix A: PCBs and Components

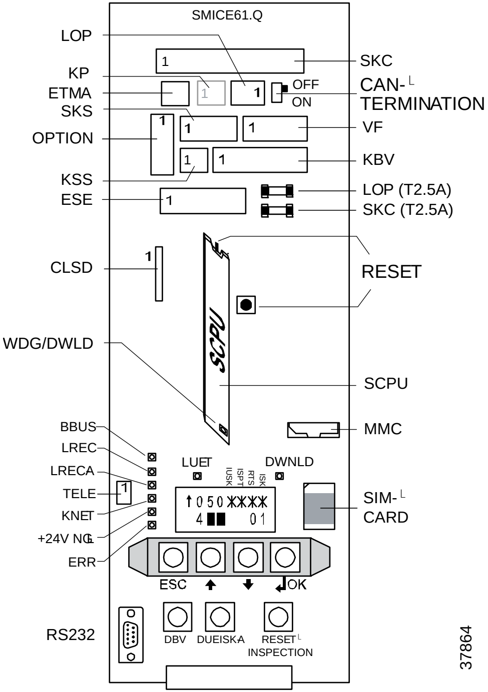
7.1 Relevant PCBs

7.1.1 Main Interface - SMIC(E)61/63.Q, SCPU1.Q

S Main Interface Controller PCB

- SMIC61/63: Rel.4 version with ETM, SMIC63.Q: with KP connector
- SMIC**E**61/63: Rel.≥5 version with ETMA, SMICE63.Q: with KP connector

SCPU microprocessor PCB



[37864; 28.01.2011], SMICE6x.Q PCB

7

LEDs (on SMIC(E)61/63.Q and SCPU1.Q):

LEDs	Description
	LED description see section 4.3

Fuses

Fuse	Description
SKC (SI1)	24 V_{DC} supply to SDIC, car (2.5 AT)
LOP (SI2)	24 V_{DC} supply BIO bus, LOPs (2.5 AT)

Plug Assignments

Plug	Description
SKC	Connection to car (power supply, safety circuit, CAN bus)
LOP	BIO bus, landing fixtures
KP	SMIC(E)**63**.Q only, safety circuit, buffers in the pit
ETMA	SMIC**E**16/63.Q only, connection to ETMA-MR. Data transfer and emergency power supply for alarm.
OPTION	Option connector. Check schematics for I/O function (LAS (DM236), SOA, RNO (NS21))
SKS	Safety circuit landing doors, KNET monitoring
VF	Frequency converter (CAN bus, safety circuit, MVE
KSS	Safety circuit contact slack rope
KBV	Overspeed governor, KBV, MGBV, (KFG)
ESE	Recall control ESE or bridge connector
CLSD	CLSD PCB for ETM (Embedded Telemonitoring system)
RS232	Duplex connection and SPECI or Service PC
TELE	External monitoring device TM4
CHIPCARD	SIM card (control options and ACVF parameters)
MMC	Multi Media Card for main software update (SCPU)

Push buttons (on SMIC(E)61/63.Q)

Button	Description
RESET	SMIC**E**16/63.Q only, Manual reset (see section 4.8)
DBV	Remote trigger over-speed governor GBV during acceptance tests
DUEISK-A	Switch on safety circuit supply (After short circuit in safety circuit (> 1A). Error is indicated by blinking IUSK.)
RESET INSPEC-TION	– Function 1: Automatic positioning of car to access car roof. (See section 4.9.8) – Functions 2: In TSD systems: Reset after TSD activation.

Push buttons (on SCPU1.Q)

Button	Description
RESET	Manual reset (see section 4.8)

Switches

Switch	Description	Remark
CAN TERMIN ATION	Terminating the CAN bus on the SMIC61.Q PCB. – OFF: No termination on SMIC (Normal position) – ON: CAN bus terminated on SMIC	If CAN bus to car (SKC) not connected → Switch to “ON” (For example during installation.) To move the car with disconnected CAN bus: See section 4.9.9

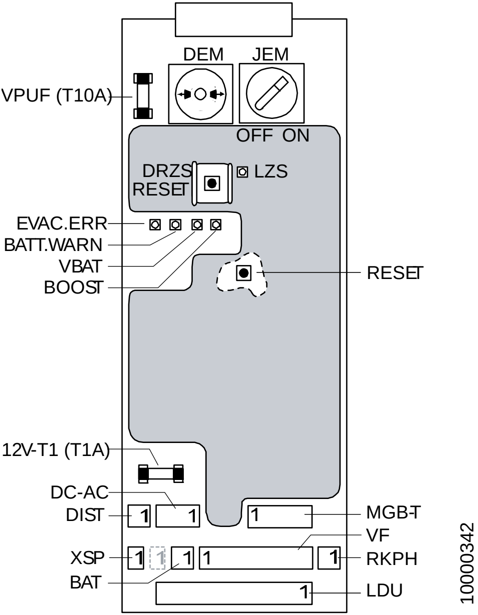
User Interface HMI

The User Interface HMI is explained in section 3.1

7.1.2 Evacuation Module - SEM11/12/21/22.Q

S Evacuation Module PCB

- SEM11/21: Manual and automatic evacuation
- SEM12/22: Manual evacuation only
- 3.3V power supply for Bionic control
- Emergency power supply



[23161; 28.01.2011], SEM11/12/21/22.Q

LEDs:

LEDs	Description
	LED description see section 4.3

Fuses

Fuse	Description
VPUF (T10A)	Internal protection of SEM (24V-NSG)
12V-T1 (T1A)	Protection of the 12V-NSG emergency power supply

Manual Evacuation Interface

	Description	Remark
JEM	Switch manual evacuation.	Counter clockwise position = Manual evacuation ON JEM = ON prevents other types of travel (Safety circuit interrupted)
DEM	Push button manual evacuation	• Opens the brake for a pre-defined time (pulse) • Electronic re-connection of battery. DEM has to be pressed after battery replacement.

Plug Assignments

Plug	Description
DC-AC	SEM11/21.Q only; Battery (12V-NSG, 24V-NSG), Inverter
BAT	SEM12/22.Q only; Battery (12V-NSG)
DIST	Distance sensor. Not used at the moment.
MGB-T	Used for "Half Brake Capability Test" (see J 139452 “Inspector’s Guide”)
XSP	Intercom power supply
VF	ACVF power supply (brake, 24V, Evac. and LUET signal)
RKPH	SEM11.Q only; Optional external line phase detector. Or jumper.
LDU	Power supplies (230VAC, 24VDC)

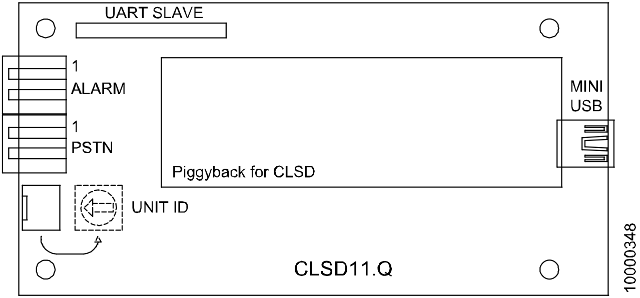
Additional Buttons

	Description
RESET	Only used for diagnostics of battery charger. A reset toggles the battery which will be charged (Without reset toggled every 5 minutes automatically.)

7.1.3 Telemonitoring ETM CLSD11.Q PCB

Communication & Line sharing device, used for ETM in Bionic 5

- Main function: Line manager between external alarm device (TAM2) and embedded telemonitoring
- For correct connection (with or without PABX) refer to schematics



[23169; 09.02.2008]

Plug Assignments

Plug	Description
ALARM	Alarm device (Servitel 10 GNT or GSV)
PSTN	Public telephone line
UART SLAVE	UART interface (Connection of CLSD)
USB	Mini USB connection (for firmware update)

Plug	Description
	All other plugs are for R&D use only.

Switch

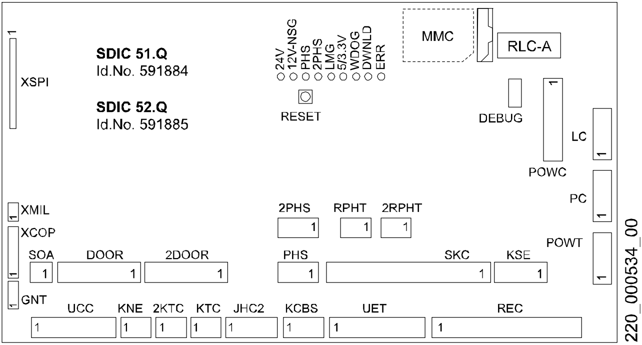
Switch	Description
UNIT ID	Rotary hex switch 0 ... F Unit ID (In case of more than one CLSD connected to the same telephone line.) • If CLSD connected to **direct telephone line**: Set UNIT ID to "0" (default) • If CLSD connected on a **PABX switchboard**: Set UNIT ID to "F"

7

7.2 Car Interface PCB - SDIC 51/52.Q

S Door Interface Controller PCB

- Interface to all car components such as door, hoistway information, car operating panel, safety circuit, alarm,...



SDIC51/52.Q [16521; 09.02.2008]

Type overview

	SDIC51	SDIC52
Main door interface	x	x
Second door interface		x
Door Pre-Opening		x

Remark: In the first systems delivered to the field a SDIC5.Q with a different plug layout has been installed.

LEDs:

LEDs	Description
	LED description see section 4.3

Switches, Push Buttons and Jumpers

Switch	Description
JRA-A	Switch alarm discriminator (SDIC5.Q only. Check together with configuration CF=09

Push Button	Description
RESET	Reset SDIC PCB

Plug Assignments

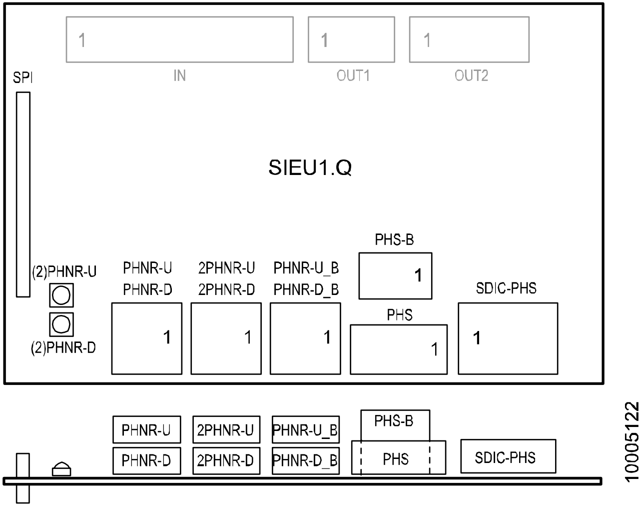
Connector	Description
XSPI	Interface SIEU PCB
XVCA	Not mounted. (Voice Announcer)
MMC	SW download with Multi Media Card
XMIL	Lamp evacuation travel LEFC on COP
XCOP	COP, CAN bus, power supply
GNT	Alarm device, GNT or GSV
SOA	System audible alarm
DOOR	Logic signals door1
2DOOR	Logic signals door 2, SDIC52 only
2PHS	Photo cell floor level (hoistway information), Access side 2, SDIC52 only
PHS	Photo cell floor level (hoistway information), Access side 1
RPHT	Light barrier or light curtain, Door1
2RPHT	Light barrier or light curtain, Door 2, SDIC52 only
SKC	Safety circuit, power supply, alarm, signals, SOA, LAS, TT, CAN bus
KSE	KSE-D and KSE-U
UCC	Car load cell, Alarm below the car DA-D, safety gear contact KF

Connector	Description
KNE	KNE
2KTC	2KTC, Door 2, SDIC52 only
KTC	KTC, Door 1
JHC2	JHC2 (second switch stop car), safety and logic. If car exceeds 1125 Kg. Jumper, if not used, SDIC52 only
KCBS	KCBS. Safety circuit contact car blocking device.
UET	Door over-bridging SUET3.Q. Logic and safety circuit, SDIC52 only
REC	Inspection panel, Logic, safety circuit, DA-U
POWC	230 V_{AC} supply from LDU, door, car light, socket outlet
LC	To the car lighting
PC	Car roof socket outlet, 230 V_{AC}
POWT	230 V_{AC} supply to the door drive(s) VVVF-4

7.3 Re-leveling PCB - SIEU1/11.Q

S Interface Europe

- Mounted on SDIC PCB (Hardware version ≥ “E”, Software ≥ V2.7)
- Re-leveling and blind floors
- SIEU11.Q: Supports special MOS functions (Door lock monitoring)



SIEU1.Q, view from top and view from front [28775; 23.12.2008]

LEDs

LED	Normal Operation	Meaning
(2)PHNR-U	ON/OFF	ON = (2)PHNR-U or PHNR-U_B is active (=24V) = Light beam interrupted
(2)PHNR-D	ON/OFF	ON = (2)PHNR-D or PHNR-D_B is active (=24V) = Light beam interrupted

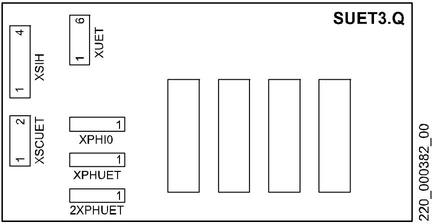
Plug Assignments

Connector	Description
PHNR-U	Sensor re-leveling up (lower sensor)
PHNR-D	Sensor re-leveling down (upper sensor)

Connector	Description
2PHNR-U	Sensor re-leveling up (lower sensor), second door
2PHNR-D	Sensor re-leveling down (upper sensor), second door
PHNR-U_B	Lower sensor blind floor (if blind floor and re-leveling)
PHNR-D_B	Upper sensor blind floor (if blind floor and re-leveling)
PHS-B	Sensor blind floor (if blind floor only; no re-leveling)
PHS	To SDIC.PHS (see safety circuit schematics)
SDIC-PHS	To PHS sensor (see safety circuit schematics)
IN	SIEU11.Q only. Inputs (Used for MOS functions only)
OUT2	SIEU11.Q only. Outputs (Used for MOS functions only.)

7.4 Door Overbridging PCB - SUET3.Q

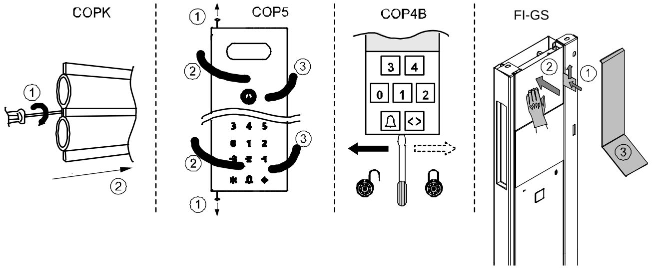
S Ueberbrückung Tür (Door overbridging)



SUET3.Q PCB [13021; 19.05.2005]

Connector	Description
XUET	Control door bridging (SDIC)
XSIH	Not used for Schindler 3100/3300/5300
XSCUET	UET safety circuit (SDIC)
XPHIO	Controller connection (SDIC)
(2)XPHUET	Door zone (optical PHUET photocell)

7.5 COP Opening Procedures



[402036; 28.01.2015]

To open the COPK Key switch panel:

1) Turn the screw on the left side clockwise
2) Slide the COPK to the side and remove it

To open the COP5:

1) Loosen the screws on the top and on the bottom of the COP
2) Open the COP first on the left side
3) Then open it completely from the right side

To open the COP4B:

1) Move the lock in the bottom of the COP4B to the left side
2) Remove COB4B

To open the FI-GS:

1) Insert the plastic card/metal card on right hand side, slide up the card and release the protection hook.
2) Simultaneously push the COP inwards in order to release the protection hook.
3) Insert bended metal sheet or attached special tool on top of COP and open the COP (starting force needed because of the magnets).

7

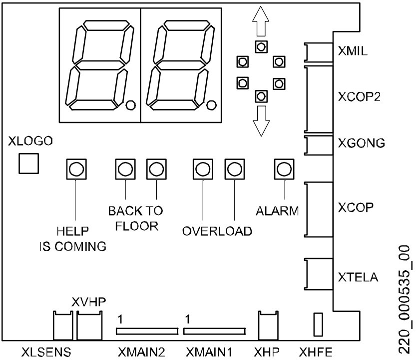
7.6 COP5 with maximum 12 buttons

- COP5-N, COP5-10, COP5-10-PI, COP5B-10, COP5B-N

7.6.1 Indicator PCB - SCOPM 51/53.Q

S Car Operating Panel Main Indicator

- SCOPM51.Q: Used for COP with maximum 12 floors
- SCOPM53.Q: Used for COP with maximum 27 floors.
- SCOPM51.Q and SCOPM53.Q are not interchangeable.



[16523; 15.02.2006]

Plug Assignments

Plug	Description
XMIL	Lamp evacuation travel car
XCOP2	Connection to 2^{nd} COP
XGONG	GONG1.Q PCB (Option, not used with Schindler 3100/3300/5300)
XCOP	Connection SDIC
XTELA	Alarm device GNT (LARC, LAGC)
XHFE	Earth connection (not used)

Plug	Description
XHP	External speaker
XVHP	External speaker (not used)
XMAIN1/2	SCOPD(C)
XLSENS	Photo transistor, Used for emergency light
XLOGO	Logo backlight

LEDs (Illumination of displays)

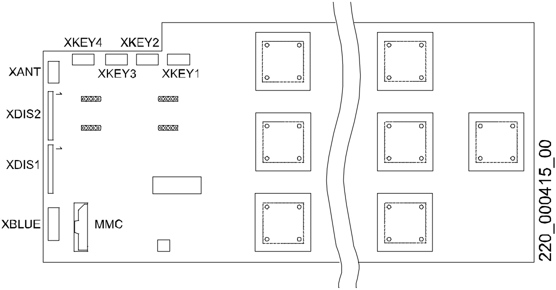
LED	Normal	Description
Arrows	ON/OFF	Travel direction UP/DOWN
“Help is coming”	OFF	ON = LARC
“Back to Floor”	OFF	ON = Evacuation travel
“Overload”	OFF	ON = Overload indication
“Alarm”	OFF	ON = Alarm or LAGC

7.6.2 COP5 PCBs - SCOPC/SCOPK/SCOPBM 5.Q

S Car Operating Panel Configurable/Keys/Button Mechanical

- Main Module of the COP5, used for COP with max.12 buttons
- Microprocessor, SW-Update, Input key switches
- SCOPC: Self configurating capacitive keyboard (blue, red)
- SCOPK: 10 digit capacitive keyboard
- SCOPBM: Mechanical buttons keyboard

7



[13005; 19.05.2005]

LEDs

LED	Normal	Description
WDG	Blinking	Blink interval 2 s when SW OK
SW DOWNLOAD	OFF/ Blinking	OFF = Normal display Blinking = During SW download

Plug Assignments

Connector	Description
XDIS1/2	SCOPD(C)
XANT	Antenna, Schindler Access System SAS
MMCARD	Multi Media Card MMC, SW Update
XKEY1..4	External key inputs (Input: Pin 2 -3)
XBUT1..3	SCOP5B.Q PCB (SCOPBM5.Q only)

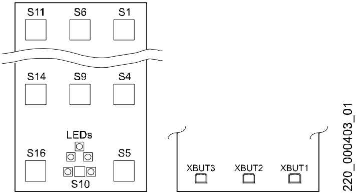
Push Button

Button	Description	Remark
Reset	Reset SCOP PCB	

7.6.3 COP5 PCB - SCOPB 5.Q

S Car Operating Panel Push buttons

- Used together with SCOPBM5.Q



PCB front and rear side [12996; 03.02.2006]

Plug Assignments

Connector	Description
XBUT1..3	SCOPBM 5.Q PCB car operation panel

LEDs

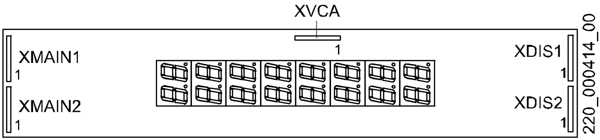
LED	Normal	Description
Five LEDs	OFF	ON = Alarm button (mechanical push button) pressed or emergency light active

7

7.6.4 COP5 PCB - SCOPD 5.Q/SCOPDC 5.Q

S Car Operating Panel Destination (and) Call Indicator

- SCOPD: 8 x 2 digit display (used for KA and KS control)
- SCOPDC: 1 x 2 digit display (used for PI control)



[13004; 03.02.2006]

Plug Assignments

Connector	Description
XMAIN1/2	SCOPM
XVCA	Voice Announcer PCB
XDIS1/2	SCOP C/PK/BM

7.7 COP5 with maximum 27 buttons

- Used in EU for 1.6 m/s systems if necessary
- COP5-1N 25 EU; COP5B-1N 25 EU

7.7.1 Indicator PCB - SCOPM 53.Q

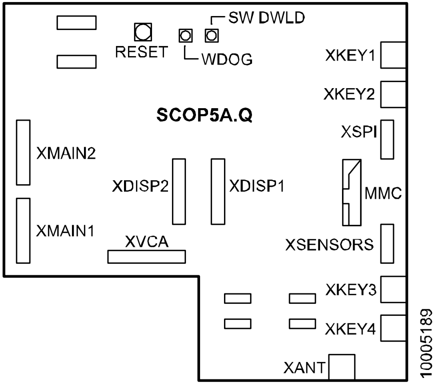
S Car Operating Panel Main Indicator

The description of the SCOPM53.Q can be found in **section 7.7.1**. The SCOPM53.Q must be used with SCOPA5.Q together.

7.7.2 SCOPA5.Q PCB

SCOPA5.Q: S Car Operating Panel Asia

- Main Module of the COP5, used for COP with max.12 buttons
- Microprocessor, SW-Update, Input key switches



[28917; 19.01.2009]

Plug Assignments

Connector	Description
XMAIN1/2	
XVCA	Voice announcer UART interface

7

Connector	Description
XDISP1/2	Display
XKEY1/2/3/4	External key inputs (Input: pin 2-3)
XSENSORS	Interface to buttons
XSPI	Interface to buttons
XANT	SAS Schindler Access System antenna (Lift key)

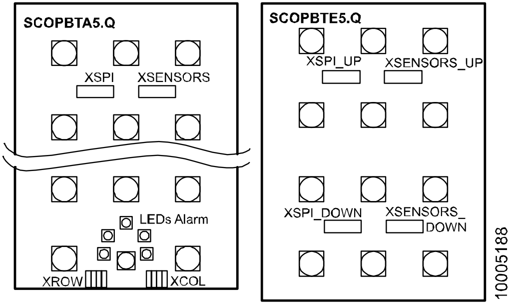
LEDs on SCOPA5.Q.Q PCB

LED	Normal Display	Meaning
WDOG	Blinking	Blinking = Microprocessor on SCOPA5.Q is working
SW DWLD	OFF	ON = Software download in progress (MMC)

Push Buttons

Button	Description
RESET	Reset the microprocessor on SCOPA5.Q

7.7.3 SCOPBTA5.Q and SCOPBTE5.Q PCBs



[28916; 19.01.2009]

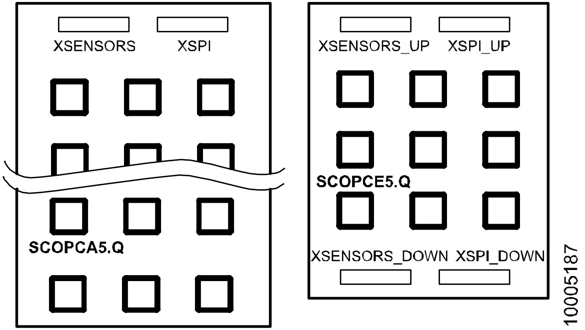
SCOPBTA5.Q: S Car Operating Panel Button Asia

- For COP with mechanical push buttons
- Connected to SCOPA5
- Supports up to 15 push buttons
- Can be extended with SCOPBTE5.Q

SCOPBTE5.Q: S Car Operating Panel Button Extension

- Extension PCB between SCOPA5.Q and SCOPBTA5.Q
- Supports additional 12 push buttons (total 27 buttons)

7.7.4 SCOPCA5.Q and SCOPCE5.Q PCBs



[28913; 19.01.2009]

SCOPCA5.Q: S Car Operating Panel Capacitive Asia

- For COP with self-configurating capacitive buttons
- Connected to SCOPA5
- Supports up to 18 capacitive buttons
- Can be extended with SCOPCE5.Q

SCOPCE5.Q: S Car Operating Panel Capacitive Extension

- Extension PCB between SCOPA5.Q and SCOPCA5.Q
- Supports additional 9 capacitive buttons (total 27 buttons)

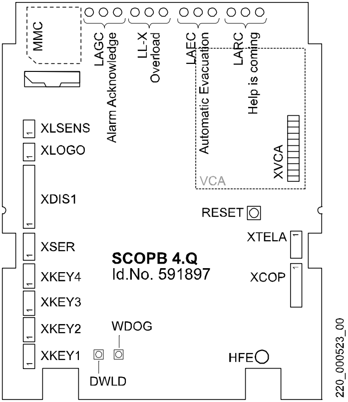
7

7.8 COP4

7.8.1 COP4B PCB, SCOPB4

S Car Operating Panel Button

- Main PCB COP4
- Microprocessor, SW-Update, Power supply, Input key switches



[16410; 09.02.2008]

LEDs (Illumination of displays and Indication)

LED	Normal Display	Description
“Help is coming”	OFF	ON = LARC
“Autom. Evac.”	OFF	ON = Evacuation travel, LAEC
“Overload”	OFF	ON = Overload, LL-X
“Alarm Acknow.”	OFF	ON = LAGC
WDOG	Blinking	Blinks when SW is OK
DWLD	OFF / Blinking	Blinks during SW download

Plug Assignments

Connector	Description
XLSENS	Photo transistor for emergency light (not used)
XLOGO	Logo backlight
XDIS1	External display, SCPI4, option
XSER	Serial connection to external display, option (not used)
XKEY1..4	External key inputs (Input: Pin 2-3)
XTELA	Alarm device GNT (LARC, LAGC)
XCOP	Connection SDIC
XHFE	Earth connection
XVCA	Voice announcer VCA11, option

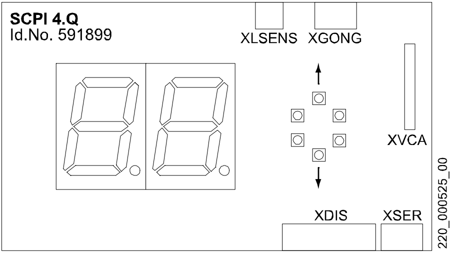
Push Button

Push Button	Description
Reset	Reset SCOP PCB

7.8.2 COP4 PCB - SCPI4

S Car Position Indicator

- Option, Car position indicator with COP4



[16522; 15.02.2006]

LEDs (Illumination of displays)

LED	Normal Display	Description
Up/Down	ON/OFF	(Further) Travel direction indicator

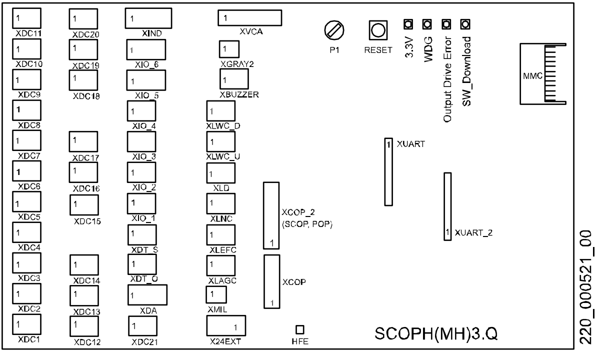
Plug Assignments

Connector	Description	Remark
XLSENS	Photo transistor	Used for emergency light
XGONG	Gong	Option
XDIS1	Connection SCOPB4	
XSER	Connection SCOPB4	
XVCA	Voice announcer	Option

7.9 Handicapped COP and External Card Reader

SCOPH(MH)3.Q: S Car Operating Panel Handicapped

- Used in handicapped COP (SCOPH3.Q)
- Interface for external parallel card reader (CRC) (SCOPHMH3.Q)
- Only one of the above mentioned options is possible (Bionic 5 supports a maximum of 2 COPs only, including the main COP.)



[21780; 24.04.2007]

Plug Assignments

Plug	Description	
XDC 1..21	Call button Push button: Pin 2-3 Acknowledge: Pin 1-2 (max. 28 V, 80 mA)	
XDA	Alarm, Pin 3-5 (normally closed). In case of parallel card reader: Bridge pin 3-5.	
XDT-O	DT-O, Input: Pin 2-3	
XDT-S	DT-S, Input: Pin 2-3	
XCOP	SDIC.COP (Connection to SDIC)	
XCOP2	1st COP (Connection to main COP)	
XIO1..4	Inputs (Pin 2-3), Outputs (Pin 1-2, max. 28V, 80mA)	1)

Plug	Description	
XIO5..6	Inputs (Pin 3-4), Outputs (Pin 1-2, max. 28V, 80mA)	1)
XIND	Gray code indicator, Bit 0..3	1)
XGRAY2	Gray code indicator, Bit 4 (Pin 1 = Gray code bit 4)	1)
X24EXT	24V power supply (from SDIC, used to drive output lamps)	1)
XMIL	SDIC.MIL, Connection to SDIC (Pin 1: input LEFC, Pin 2: input LARC)	1)
XLARC	Output LARC (Pin 1-2, max. 14V, 1.6A)	1)
XLEFC	Output LEFC (Pin 1-2, max. 14V, 1.6A)	1)
XLNC	Emergency lamp LNC (Pin 1-2, Pin 1: 12V, 1.6A)	1)
XLD	Emergency lamp input	1)
XLWC_U/D	Travel direction arrow up/down (max. 28V, 80mA)	1)
XBUZZER	External buzzer output (Pin 1-2, max. 14V, 1.6A, Volume: Trimmer P1)	1)
XVCA	VCA Voice announcer, old version	1)
XUART XUART_2	VCA1.Q, UART interface, Voice announcer, new version	1)

1) = not used with Schindler 3100/3300/5300

LEDs

LED	Normal Operation	Meaning
3V3	ON	ON = 3.3 V internal power supply OK
WDOG	Blinking	Blinking = SW OK
ERROR	OFF	ON = Too high current on Outputs XDC1..21, XIO1..6 or XLWC_U/D
SW_Download	OFF	Blinks during SW update

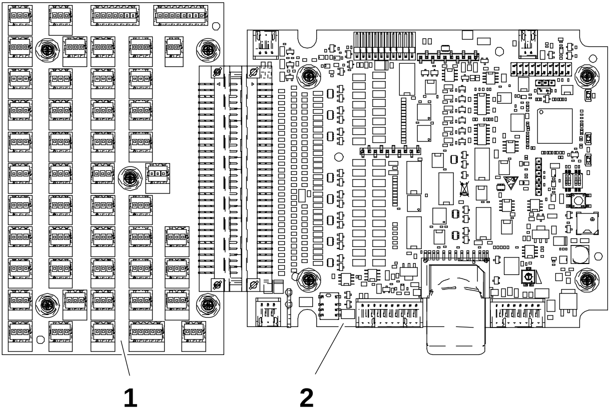
Push buttons and potentiometer

	Function	Remarks
Reset	Reset	
P1	Volume	Adjust volume of buzzer

Extension PCB (PI_P1/PI_P2) and SCOPPI1.Q:

The PI_P1/PI_P2 PCBAs (together with the SCOPPI1 PCBA) replaces the SCOPHM(H)3 PCBA.

The functionality of the board is the adaptation of the connectors required for a parallel COP to the interface connector of the SCOPPI1. Therefore no other components (except connectors) are required on this PCBA.



Extension PCB and SCOPPI1 [402034; 20.01.2015]

1 Extension PCB
2 Parallel PCB SCOPPI1

7

Plug Assignments

Plug	Description	
XDC **1 ... 30**	Call button Push button: Pin 2-3 Acknowledge: Pin 1-2 (max. 28 V, 80 mA)	
XDA	Alarm, Pin 3-5 (normally closed). In case of parallel card reader: Bridge pin 3-5.	
XDT-O	DT-O, Input: Pin 2-3	
XDT-S	DT-S, Input: Pin 2-3	
XIO1..4	Inputs (Pin 2-3), Outputs (Pin 1-2, max. 28V, 80mA)	1)
XIO5..6	Inputs (Pin 3-4), Outputs (Pin 1-2, max. 28V, 80mA)	1)
XIND	Gray code indicator, Bit 0..3	1)
XMIL	SDIC.MIL, Connection to SDIC (Pin 1: input LEFC, Pin 2: input LARC)	1)
XLARC	Output LARC (Pin 1-2, max. 14V, 1.6A)	1)
XLEFC	Output LEFC (Pin 1-2, max. 14V, 1.6A)	1)
XLNC	Emergency lamp LNC (Pin 1-2, Pin 1: 12V, 1.6A)	1)
XLD	Emergency lamp input	1)
XLWC_U/D	Travel direction arrow up/down (max. 28V, 80mA)	1)
XBUZZER	External buzzer output (Pin 1-2, max. 14V, 1.6A, Volume: Trimmer P1)	1)
XVCA2	VCA Voice announcer, old version	1)

1) = not used with Schindler 3100/3300/5300

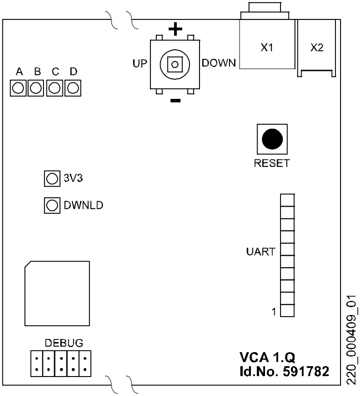
7.10 Voice Announcer PCB - VCA 1/11.Q

Voice Announcement PCB

- Floor name, service- and alarm messages
- Connected on COP

The VCA is delivered pre-configured according to the original order.

For supplementary changes please refer to document J42102314 “VCA commissioning”. (File X42102314)



[12999; 02.02.2006]

Plug Assignments

Connector	Description	Remark
X1	External active speakers with input amplifier	Option, impedance > 4.7 kOhm
X2	Output to the speaker	Speaker 8 Ohm, 1 W
UART	UART interface	Connection SCOP

Connector	Description	Remark
MMCARD	Multi Media Card with mp3 audio files. (Backside of PCB)	MMC must stay inserted (mp3 files can not be downloaded to PCB)

LEDs

LED	Normal Display	Description
DWNLD	OFF	Blinks during SW download (with MMC)
3V3	ON	24V, 5V, 3.3V available
A	OFF	ON = Setting volume (Joystick +/-) Blinks = Main speaker announcement
B	OFF	ON = Setting balance (Joystick +/-) Blinks = Secondary speaker announcement
C	OFF	ON = Setting tremble (Joystick +/-)
D	Blinking	ON = Setting bass (Joystick +/-) Blinks = VCA ready (Watchdog)

Special LED status:

A→B→C→D→A→....	Initializing VCA
AB	Setting main speaker
AC	Setting secondary speaker
AD	Setting general

Joystick and Push Button

Joystick	Description	Remark
Set	Menu activation and “ENTER”	Press down the joystick
UP/DOWN	Change function / menu	Volume, bass, ...
+/-	Increase / decrease value	Set up volume, bass, ...

Push Button	Description	Remark
RESET	Reset VCA PCB	

Setting up the volume

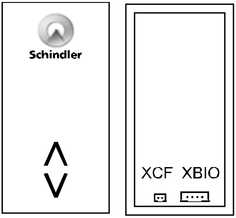
1)	Precondition: LED “3V3” = ON, LED “D” = blinking
2)	Press down the joystick → VCA plays music, LED “A” = ON
3)	Use +/- to change volume
4)	When volume is ok, stop changing value and wait, until music stops automatically. The changed value is stored and the system starts up again. → LEDs blinking, LED “D” = blinking

Language dependent MMC

The order number for the MMC with a certain language can be found in document J 41322160.

7.11 Landing Fixtures LOP5

Landing Operating Panel

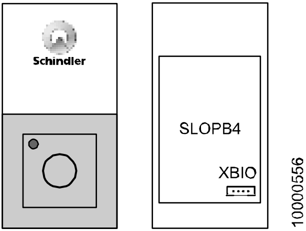


LOP (XBIO)[402038_00; 30.01.2015]

XBIO: 4 pin JST 2.5mm
XCF: 2 pin JST 1.5mm
Delivered since 2006
Remark: Supports BIO bus protocol type 1 or 2, depending on SCPU SW version.

7.12 Landing Fixtures LOPB4

Landing Operating Panel for Schindler 3100



[25140; 08.02.2008]

Plug Assignments

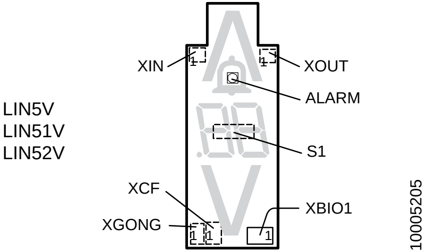
Plug	Description	Remarks
XBIO	BIO bus	BIO bus connection

There is no XCF for the synchronization with LIN/LCUX available. Therefore it is not possible to connect a LIN or a LCUX to a LOPB4.

7

7.13 Landing Indicator PCB - SLINV5/51/52.Q

- Compared to the normal SLINV5.Q the SLINV51/52.Q offer an input and an output.
- SLINV52.Q offers a magnetic reed contact for configuration without a XCF connection to the LOP.



LIN5V, LIN51V, LIN52V, view from backside [28981; 21.10.2010]

Plug	Description
XBIO1	BIO bus connection – LIN5V: WAGO 2.5mm – LIN51V, LIN52V: JST 2.0mm
XCF1	Synchronization with LOP during configuration – LIN5V: JST 2.5mm – LIN51V, LIN52V: JST 1.5mm
XGONG1	Connection of landing gong
XIN	LIN51V, LIN52V only. Input. Not used at the moment
XOUT	LIN51V, LIN52V only. Output. (max. 24V, 100mA) Used for floor light control kit. Configuration: CF=40, L=n, (S=1,2), PA=31, VL=213

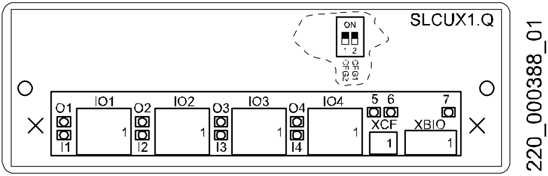
Switch	Description
S1	LIN52V only. Magnetic reed contact. Can be activated by placing a hoistway magnet on the LIN surface. To configure the LIN without the XCF cable to the LOP.

7.14 Landing Input Output PCB - SLCUX1/2.Q

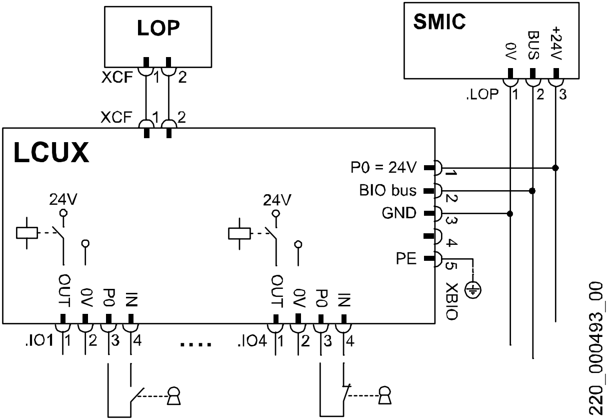
S Landing Call Unit Extension

Main Function:

- Connection additional inputs and outputs in hoistway



[13023; 22.11.2006]



SLCUX1.Q / SLCUX2.Q Connections [13012; 19.05.2005]

7

Plug Assignments

Plug	Function	Remarks
XBIO	BIO bus	
XCF	Synchronization	Connection to LCU(M) or LOP
IO1..IO4	Inputs/outputs	• Input: Pin 3-4 (potential-free NO or NC contact) • Output: Pin1-2 (P0, max 0.4 A)

LEDs on SLCUX1/2.Q PCB

LED	Normal Operation	Description
O1 .. O4	ON/OFF	ON = Output active (depending on BMK configuration)
I1 .. I4	ON/OFF	ON = Input active (contact closed)
5	Blinking	Watchdog
6	OFF	ON = Current overload on output
7	ON	P0, power supply from BIO bus

DIP switch settings

Used for predefined addresses of LCUX if mounted separately (without XCF connection to LOP). **If LCUX is connected to a LOP the DIP switch setting has no influence**.

DIP 1 CFG2	DIP 2 CFG1	Node definition and Configuration menu
ON	**ON**	LCUX has predefined address “node 1”. Configuration with CF=40, L=91 (SW ≥ V9.34)
ON	**OFF**	LCUX has predefined address “node 2”. Configuration with CF=40, L=92 (SW ≥ V9.34)
OFF	**ON**	LCUX has predefined address “node 3”. Configuration with CF=40, L=93 (SW ≥ V9.34)
OFF	**OFF**	LCUX address not pre-defined

Remark 1: To store the predefined address in the EEPROM a LOP counting (CF=00, LE=00) is necessary.

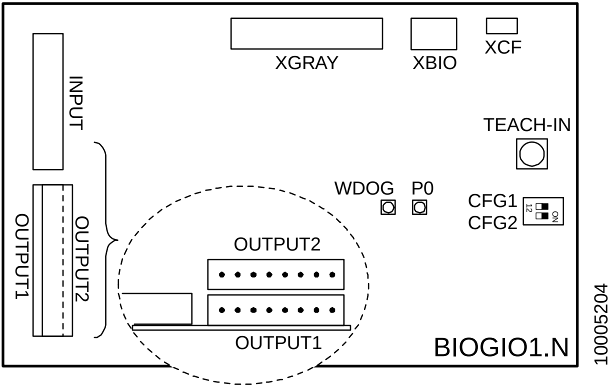
Remark 2: If the LCUX is connected to a LOP on a floor the predefined address will be overwritten by a new address during the LOP configuration. (Independent of the actual DIP switch setting.)

Remark 3: If a LCUX has been addressed during LOP configuration it can be reset to the predefined address with help of CF=94 (Rel.2) or CF=40, L=n, PA=99 (Rel.4)

7.15 Building monitoring GUE/GLT - BIOGIO1.N

BIO bus General Input and Output PCB

- Connected on BIO bus
- 8 free configurable inputs (potential free contacts)
- 8 free configurable outputs (relay contacts, max. 0.2A, 30V)
- Check schematics S274634 for configuration options
- Not possible together with NS21 emergency power supply



[28980; 12.11.2009]

Plug assignment

Plug	Description
INPUT	Pin 1 = 24V, Pin 3..10 = Input IN1 to IN8
OUTPUT1	Pin 1-2 = OUT1, Pin 3-4 = OUT2, .. Pin 7-8 = OUT4; potential free contacts, max. 30V, 200mA
OUTPUT2	Pin 1-2 = OUT5, Pin 3-4 = OUT6, .. Pin 7-8 = OUT8; potential free contacts, max. 30V, 200mA
XGRAY	Gray code and arrow UP/DOWN
XBIO	BIO bus connection

Plug	Description
XCF	Synchronization to LOP (for configuration)

DIP switch settings

DIP 1 CFG1	DIP 2 CFG2	Node definition and Configuration menu
ON	**ON**	BIOGIO has predefined address “node 1”. Configuration with CF=40, L=91
OFF	**ON**	BIOGIO has predefined address “node 2”. Configuration with CF=40, L=92
ON	**OFF**	BIOGIO has predefined address “node 3”. Configuration with CF=40, L=93
OFF	**OFF**	BIOGIO address not pre-defined

Remark 1: To store the predefined address in the EEPROM a LOP counting (CF=00, LE=00) is necessary.

LEDs on BIOGIO1.Q PCB

LED	Normal Operation	Description
WDOG	blinking	Blinking = System in normal operation
P0	ON	ON = 24V (P0) available

Push button

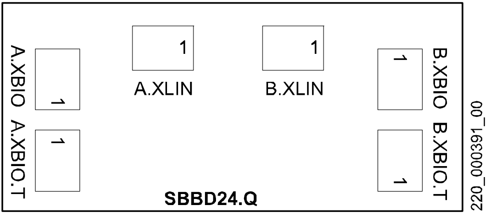
Button	Description
TEACH-IN	Not used with Schindler 3100/3300/5300. (Configures the address of the BIOGIO if the PCB is not connected to a LOP. With Schindler 3100/3300/5300 the address is defined with help of DIP1 and DIP2.)

7.16 Duplex BIO bus PCB - SBBD24.Q

S BIO Bus Duplex 24V

- For Duplex systems with only 1 LOP per floor (one riser)
- For Triplex systems with only 1 or 2 LOPs per floor (one or two risers). (Overlay gateway BBMCM necessary.)
- Allows to switch OFF one elevator without disabling the floor calls. (Switches LOP BIO bus to the elevator which is switched on.)
- Disables the LINs of the elevator which is switched OFF
- Mounted in the hoistway on the LDU floor

– Check schematics for correct cabling and connections (S274125, S274613, S277207, S277208)
– See additional information in section 5.4.12



[13025; 19.05.2005]

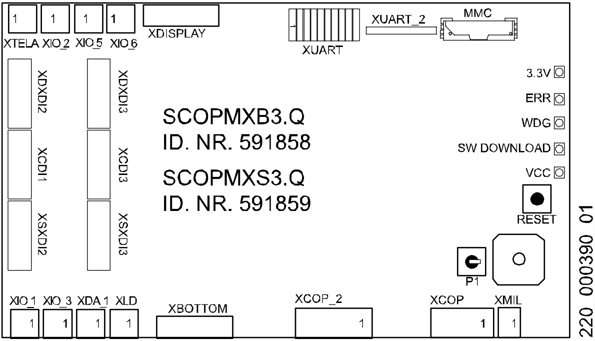
Plug Assignments

Plug	Description	Remarks
A.XBIO	From SMIC.LOP (BIO bus)	Elevator A
B.XBIO	From SMIC.LOP (BIO bus)	Elevator B
A.XLIN	To LINs (BIO bus)	Elevator A
B.XLIN	To LINs (BIO bus)	Elevator B
A.XBIO.T	To LOPs (BIO bus)	single riser LOPs
B.XBIO.T	Normally not used	

7.17 Dual Brand Fixtures

7.17.1 Dual Brand COP PCB - SCOPMXB3.Q

S Car Operating Panel Dual Brand (MX-Basic)



[13980; 13.03.2008]

Plug Assignments

Plug	Description
XCOP	Connection to SDIC (CAN bus, supply)
XCOP_2	2nd COP If no 2nd COP connected: Bridge pin 6 - pin 7!
XMIL	LEFC and LARC indicator (From SDIC)
XDISPLAY	Gray Code, Emergency lamp, LEFC, LAGC, LARC (Emergency light max. 1.2 W)
XDXDI2/3 **XCDI2/3** **XSXDI2/3**	Buttons with acknowledge lamps
XBOTTOM	Signals (DTO, DTS, DA,...)
XDA_1	Alarm

7

Plug	Description
XLD	Emergency power, light detector (Used for Type D Panel)
XTELA	Telealarm LARC, LAGC
XUART	Interface voice announcer (serial) (External VCA box)
XUART_2	Interface voice announcer (serial) (VCA PCB on SCOPMXB PCB)
XIO_1.. **XIO_6**	Additional Inputs (Pin 2-3) and Outputs (Pin 1-2) (24 V, Output max. 350 mA, Input max. 15 mA)
XDBG08	Debug interface (Used for development only)

LEDs on SCOPMXB3.Q PCB

LED	Normal Operation	Meaning
3.3 V	ON	Power supply (3.3 V)
ERR, OVLD	OFF	ON = Too high current on outputs (Out 1..6)
WDG	Blinking	Blinking = SW OK
SW Download	OFF	Blinking during SW update
VCC	ON	Power supply (5 V)

7.17.2 Dual Brand LOP PCBs - SLCU(M)2.Q

S Landing Call Unit (Main)

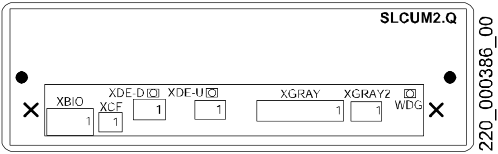
Main Function:

- Parallel Interface BIO bus to Dual brand fixtures

With Bionic 5 systems only LCU(M)2.Q PCBs can be used

(BIO bus data power supply: 24V). Do NOT use SLCUM1.Q PCBs!

SLCU(M)2.Q PCBs can be recognized by the WDOG LED



[13979; 19.12.2006]

LEDs

LED	Meaning
XDE-D	DE-D pressed
XDE-U	DE-U pressed
WDG	Watchdog (only available on SLCU(M)2.Q PCBs)

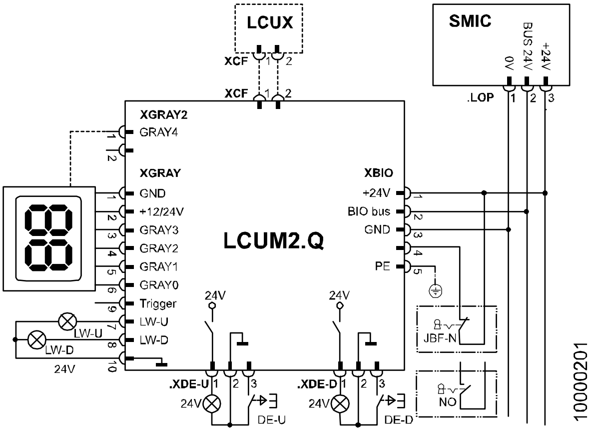
7

Plug Assignments

Plug	Function	Remarks
XBIO	BIO bus	Pin 1: Supply 12 or 24V (red) Pin 2: Data Line (white) Pin 3: GND-0V (black) Pin 4: Input (orange) Pin 5: Earth (Yellow/green)

Plug	Function	Remarks
XDE-U[1)] **XDE-D**	Floor call (UP/DOWN)	Pin 1: Acknowledge (out) Pin 2: GND-0V Pin 3: Floor Call (in)
XCF	Synchronization	connection to LCUX/LIN
XGRAY	Indicator LW-U/D	Gray code/hall lantern (SLCUM2 only)
XGRAY2	Indicator	Gray code bit 4 (SLCUM2 only)

1) 1) during configuration the button connected to XDE-**U** is used!

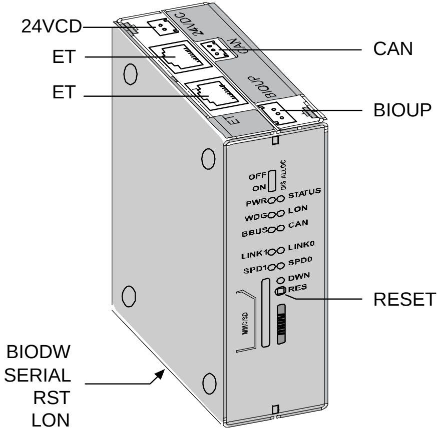


SLCU(M)2.Q connections [20023; 04.01.2007]

Configuration

SLCU(M)2.Q PCBs are recognized as BIO bus type 1 or type 2 fixtures. Configuration of key switch connected to XBIO.4

7.18 Overlay Box, MCM (Triplex, Quadruplex)



Overlay Box, [37488; 04.11.2010]

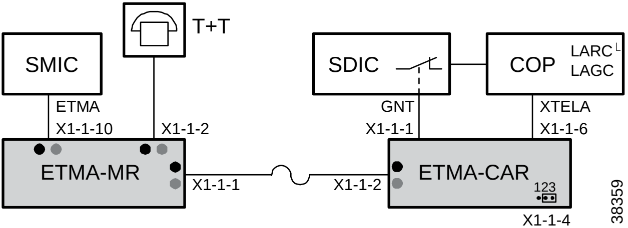
LEDs	Description
PWR, DWN	Only the **PWR** (power) and **DWN** (download) LEDs are operational.

Button	Description
RESET	To reset the overlay box. It has the same function as the HMI special menu 137. The overlay box must be reset if CF=04, PA=01 has been changed.

Plug	Description
24VDC	Power supply (from SMIC.VF and SMIC.LOP)
CAN	CAN bus connection (from SMIC.VF)
ET	Ethernet connection to the overlay box of the other elevators in the group

BIOUP, BIODW, SERIAL, RST, LON: Not used at the moment

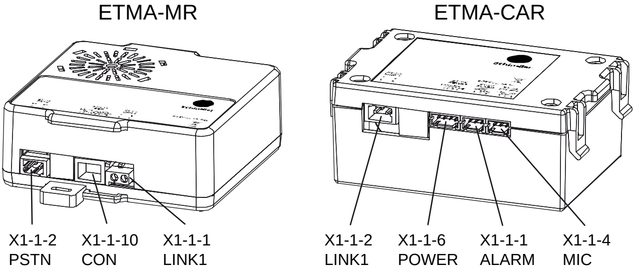
7.19 Embedded Telemonitoring Alarm ETMA



ETMA, overview basic layout [38359; 01.02.2011]

Main documentation for ETMA:

- EJ41700380: ETMA installation and commissioning
- K40700049: ETMA Diagnostics and Maintenance



ETMA-MR, ETMA-CAR, basic modules, [38360; 01.02.2011]

Plug Assignments

Plug	Description
ETMA-MR	
X1-1-2	PSTN: Telephone line

Plug	Description
X1-1-10	CON: Interface to control, SMIC.ETMA. (Includes emergency power supply for alarm.)
X1-1-1	LINK1: Traveling cable ETMA-CAR
ETMA-CAR	
X1-1-2	LINK1: Traveling cable ETMA-MR
X1-1-6	POWER: COP, LARC and LAGC
X1-1-1	ALARM: Alarm contact from SDIC
X1-1-4	MIC: Jumper. Position 2-3 for internal microphone

Diagnostics with help of LEDs

On each main connector there is a green and a red LED which is visible trough the transparent box.

ETMA-MR

LED CON		Description
Green	**Red**	
Off	Off	ETMA CPU not running. → Check power supply, cable connection etc.
Blinks	Off	Data traffic in progress. OK.
On	Off	ETMA OK

LED PSTN		Description
Green	**Red**	
Off	On	PSTN line not connected or no energized.
Blinks	Off	Communication in progress. OK.
On	Off	PSTN line connected and idle. OK.

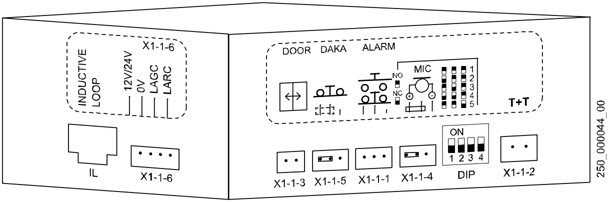
ETMA-MR and ETMA-CAR

ETMA-MR LED LINKx		Description
Green	Red	
Off	Off	ETMA device not powered or defective. → If the power supply is on, the device is defective and must be replaced.
Off	On	Not connected to another ETMA module or not synchronized. → Check the cable and the status of the other ETMA module. If the other ETMA is ok, the device is defective and must be replaced.
Blinks	Off	Communication in progress. OK.
On	Off	Connected to another ETMA module, synchronized and idle. OK.

7.20 Telealarm GNT TAM2 (Servitel 10)

- Telealarm device
- Intercom module optional
- For correct connection see schematics (S274156, S274181)
- Configured remotely from the Telealarm Control Center TACC

For configuration and troubleshooting refer to section 5.4.24



[16428; 07.02.2006]

Plug Assignments

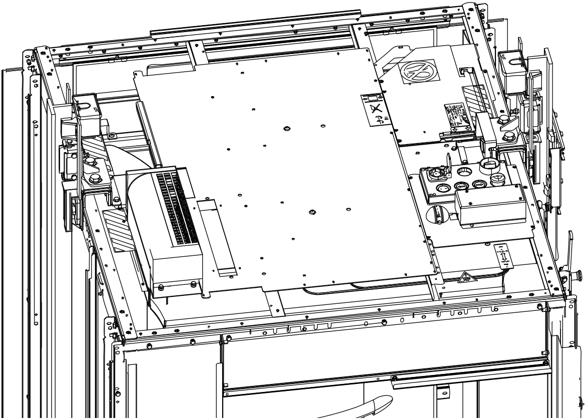
Plug	Description	Remarks
X1-1-1	Alarm button	Connection to SDIC.GNT
X1-1-2	Phone line T+T	Connection to SMIC.SKC
X1-1-3	Alarm misuse discriminator AMD	not used, (integrated in Bionic 5 control)
X1-1-4	External microphone	not used; **Jumper pin 2-3** must be installed
X1-1-5	Alarm button outside car DAKA	not used, (integrated in Bionic 5 control); **Jumper pin 2-3** must be installed
X1-1-6	LARC, LAGC indication	Connection to SCOP.XTELA
IL	Inductive Loop	Option, not yet released; Wireless connection to hearing aid for disabled person
	Triphonie	Option, 6 pin J.S.T. connector inside of TAM2 box

DIP switch settings

DIP1	DIP2	DIP3	DIP4	Description
ON	**ON**	**ON**	**x**	Module ID = 1 (**default for simplex**, always master, Duplex elevator A)
OFF	**ON**	**ON**	**x**	Module ID = 2 (always slave, Duplex elevator B)
ON	**OFF**	**ON**	**x**	Module ID = 3 (always slave)
OFF	**OFF**	**ON**	**x**	Module ID = 4 (always slave)
ON	**ON**	**OFF**	**x**	Module ID = 5 (always slave)
OFF	**OFF**	**OFF**	**x**	Reset all parameters
x	**x**	**x**	**ON**	Alarm button: Normally open
x	**x**	**x**	**OFF**	Alarm button: Normally closed (**default for Schindler 3100/3300/5300**)

7

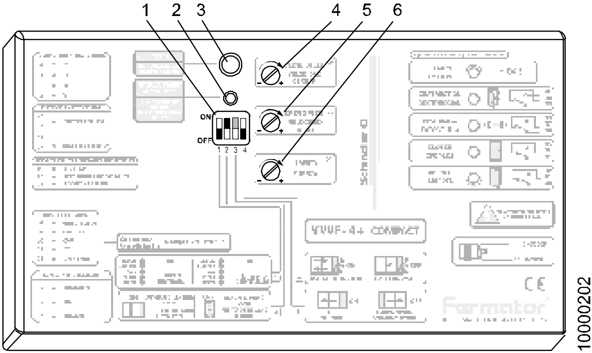
7.21 Short Pit and Headroom System TSD21



For further information concerning the short pit and headroom system TSD21, refer to:

– J 41141066 EMBD TSD21 - Short Headroom - CO BIC 5
– J 41141093 EMBD TSD21 - Short Pit Set - CO BIC 5

7.22 Door Drive Fermator Compact VVVF4+



[20056; 01.02.2007]

DIP switch setting (1)

DIP switch		Normal Position	Description
1	1 & 2 Inputs	**OFF**	• ON = 1 input (open/close) • **OFF = 2 inputs**
2	Type of landing door	**ON**	• **ON = Automatic door** • OFF = Manual door
3	Door opening	**ON or OFF**	• ON = Left opening (TL) • OFF = Right (TR) or Center
4	Master & Slave	**OFF**	• ON= Master • **OFF = Slave**

Remark to door opening:

The setting of DIP switch 3 is different depending on the release of the Fermator VVVF4 control box. (Also some old schematics may be wrong.) → Check the label on the control box for correct setting!

Remark to DIP switches:
Position of DIP switch may be difficult to see. DIP switches of new control device are black. Check position by switching on/off the DIP switches.

Fuse

Fuse	Description
4A 250V	Fast, 230VAC power supply input

Mains Switch

Switch	Description
ON/OFF	Mains switch power supply, side of VVVF4+ box

Commissioning procedure

1	Make sure all plugs are connected correctly. **Car and landing door must be coupled**.
2	Switch ON the door drive (on the side of the Fermator VVVF4 control box)
3	Press the “Auto adjustment” button (2)
4	The test button (3) can be used for test trips
5	Check the function of the force limiter safety. Adjust if necessary with potentiometer (6) to < 150 N
6	If necessary adjust door opening speed (5) and door closing speed (4)

7.23 Door Drive Sematic C MOD

The door drive Sematic C MOD is used for the replacement system Schindler 6300.

7.23.1 Configuration

[25237; 17.12.2007]

Automatic Mode - LED "AUTO"

- After power up or after reset the door stays in the automatic mode
- Key 4 pressed for a while, switches to the Manual Mode

Manual Mode - LED "MAN"

- All signals coming from external (elevator, photocell,...) are ignored
- Door opens with KEY 2 or closes with KEY 3 (permanent pressed)
- Key 1 selects the Self Learning Cycle
- Key 4 pressed for a while, switches to the "AUTO" Mode
- Switches back to "AUTO" mode after 10 min. without key pressed

Programming Mode - LED "PROG"

- Enter "PROG" by pressing KEY 1 and KEY 4 together for a while
- Key 1 acts as "Enter": selects parameter or confirms value
- Key 2 and Key 3: increases or decreases parameter or value
- Key 4: cancel edit mode or switches back to the "AUTO" Mode

Self-learning cycle [SL]

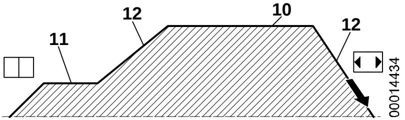
- Must be performed before elevator goes operational
- Important: the door closing with low speed after a power supply interruption (reset) is not a self-learning cycle.
- It is possible in "AUTO" mode also, but we use the "MAN" mode

Self-learning cycle		Indication
1	Press KEY 4 to enter “MAN” mode	LED MAN
2	Press KEY 1 to start the self-learning cycle	“SL”
3	Touch KEY 3: door closes with slow speed or remains closed (If the door opens with KEY 3, change parameter 22)	“CL” flashing
4	Touch KEY 2: door opens with slow speed	“OP” flashing
5	Check carefully that the door panels slide freely and completes its total expected travel	
6	At the end of the opening cycle the learning is complete and automatically finished.	“OP”
7	Change back to “AUTO” mode with KEY 4	LED AUTO

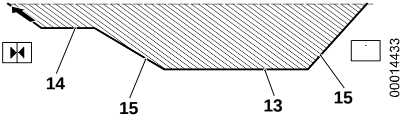
Display indications

- **--** After start-up
- **OP** Door open - if flashing: Door is opening
- **CL** Door closed - if flashing: Door is closing
- **AL** Alarm (flashing with alarm code)
- **SL** Self-Learning
- **FC** Forced closing
- **IM** Reversing system ON

7.23.2 Parameters



Door opening travel curve parameters [18748; 17.10.2006]



Door closing travel curve parameters [18747; 25.11.2009]

Code	Default	Proposed	Range	Parameter
00	0		0/1/2	Reversing system 0: intern 1: external moving, 2: external moving + parking
01	0		0/1/2	Main Lift Controller (MLC) Test 0: moving, 1:moving and parking, 3: OFF
02	0		0/1/2	No MLC signal 0: instant stop, low speed to stop, 2: low speed cycle
03	0		0/1	MLC Input Alarm 0: OFF, 1: ON
04	0		0/1	Limited door reversal effect 0: OFF, 1: ON
05	00	00	0/1	Car door locking device 0: OFF, 1: ON
06	0		0/1	Glass doors 0: OFF, 1: ON
07	0		0/1/2/3	Aux output relay 0: OFF, 1: gong while opening, 2: according % of space, 3: thermic alarm signal
08	00		00..99	Space percentage (for Aux relay) 00 = closing limit

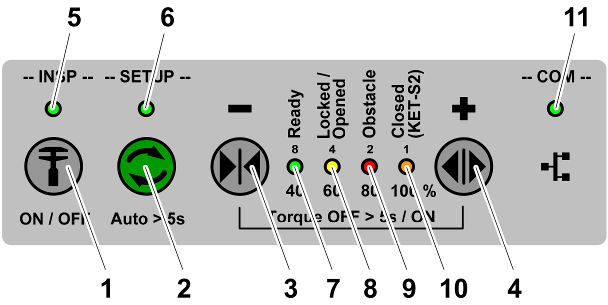
Code	Default	Proposed	Range	Parameter
09	66	66	00..99	**Closing force (reversing trigger)** 10..150 N
10	50	45	01..99	**Opening high speed** (see picture)
11	50	30	01..99	**Opening low speed** (see picture)
12	50	75	01..99	**Opening acc/deceleration** (see picture) (A higher value = lower acc/deceleration)
13	30	50	01..99	**Closing high speed** (see picture)
14	50	20	01..99	**Closing low speed** (see picture)
15	80	80	01..99	**Closing acc/deceleration** (see picture) (A higher value = lower acc/deceleration)
19	1		0/1	Fire fighting mode 0:Reversing OFF, 1:Reversing system reduced sensitivity
20	01		00..05	Opening time on Emergency Opening Device (EOD) contact 0 ... 5 minutes
21	00		0/1	Protective device logic Kn 0:N/O, on obstruction closed, 1: N/C on obstruction open
22	0	00	0/1	**Motor rotation direction** 0: clockwise, 1: anti-clockwise
25	01	00	0/1	**Closed parking mode** 0: closed skate/coupler parking (motor powered), 1: opened skate/coupler parking
26	01	01	0/1	**Skate type** 0:standard alu skate, 1: expansion skate
99	00	00	0/1	**I/O interface** 0: default type, 1: LONIBV

7.23.3 Diagnostics

Code	Alarm Table
1	No Main Lift Controller Signal (only if parameter 01 is not set to OFF and parameter 03 is set to ON)
2	Motor over current protection
3	Reversing system fault. The elevator controller does not send reopening command after the door controller has signaled an obstacle (only if parameter 00 is set to extern).
4	Inverted motor connection or encoder channels
5	Encoder jerk: Interruption of the encoder or the motor cables; Encoder connection inverted
6	Motor over-heating (when internal PTC sensor present)
7	Motor jerk, interruption of the motor cables
8	Over-voltage in the power supply
9	PWM-Trip: Impulse over-current
10	Generic alarm due to an internal malfunction of the door controller
11	Power supply protection (over current due to mechanical strain)

7.24 Door Drive Varidor15

7.24.1 Configuration



HMI Keypad [402032; 15.12.2014]

1 Key INSP
2 Key SETUP
3 Key DRET-S
4 Key DRET-O
5 LED INSP
6 LED SETUP
7 LED Ready
8 LED Locked/Opened
9 LED Obstacle
10 LED Closed (KET-S2)
11 LED COM (CAN communication)

No	Step
1	Make sure JREC is ON (Inspection Top of Car) and switch power ON (JHT).
2	Enable inspection door by pressing the INSP key (1).

No	Step
3	Check the following LED states (see also Appendix A4): – LED INSP **ON** – LED Setup **OFF** – LED Ready **ON** – LED Locked/Opened **Blinking slowly or ON** – LED Obstacle **OFF** – LED Closed (KET-S2) **ON** (if panels closed) – LED COM (communication) **ON** (not flashing)
4	Lock the car door and press the DRET-S key (3) until LED Locked/Opened (8) is ON, if the car door is not locked.
5	While the fitter is still on the car roof, close the landing door(s) manually.
6	Disable inspection door by pressing the INSP key (1) – LED INSP **OFF**
7	On the inspection control station press DRET-U/(D) until the blue LUET LED on the SALSIS / PHS main sensor indicates that the car is in the door zone.
8	Enable inspection door by pressing the INSP key (1) – LED INSP **ON**
9	Run Autosetup Function: – LED 'SETUP' is blinking slowly, indicating that the AutoSetup is in progress. – Door opens for a small distance – Door locks – Door opens, closes 1st time with referencing speed – Door opens, closes 2nd time with red. normal speed – Door opens, closes 3rd time with final normal speed – LED 'SETUP' = OFF, that is, AutoSetup successfully finished If AutoSetup failed due to any reason: – LED '1 … 4' and 'Setup' are blinking fast On the HMI, press Setup key (2) to clear error.

7

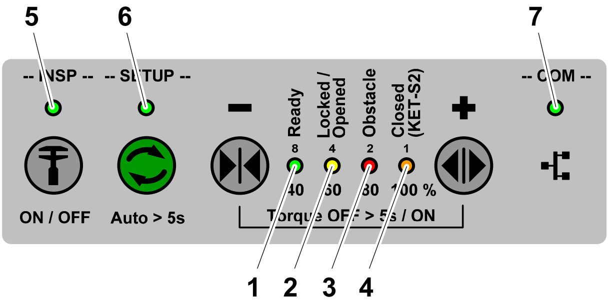
No	Step
10	If AutoSetup failed due to any reason: – LED '1 … 4' and 'Setup' are blinking fast and showing a binary blinking code. On the HMI, press Setup key (2) to clear the error.
11	Check Motion and Performance On the HMI, press DRET-O key (4) / DRET-S key (3) to open, to close or to stop the door. – Check motion while fully opening / locking 2 ... 3 times – Check stopping while opening at full speed (approximately 50 %) – Check stopping while closing at full speed (approximately 50 %).
12	Open the Door to define the opened position – On the HMI, press DRET-O key (4) to fully open the door – On the HMI, press Setup key (2) to setup the opened position – LED 'SETUP' = ON – On the HMI, use key DRET-S key (3) / DRET-O key (4) to place panels flush to frame – Panel moves real time in 1 mm increments / decrements. – On the HMI, press Setup key (2) to store actual opened position. If no action is performed on HMI for 30 seconds, the actual position is not stored. LED 'SETUP' is OFF

No	Step
13	Check the car door to landing door alignment and the clutch motion. – On the HMI, press DRET-S key (3) until LED Locked/Opened is ON – On the HMI, press DRET-O key (4) for < 0.2 s to open the clutch – Check for no motion on landing and car panels – Repeat the locking, unlocking process 2 ... 3 times If the car to landing door alignment is correct, no door motion is visible.
14	Define the door opening and closing speed. – On the HMI, press DRET-S key (3) until LED Locked/Opened is ON – On the HMI, press Setup key (2) to setup speed – On the HMI, use DRET-S (3) / DRET-O (4) to select the opening and closing speed value. – On the HMI, press Setup key (2) to store the defined opening and closing speed. If no action is performed on the HMI for 30 seconds, the actual door position is not stored. – LED 'SETUP' is OFF
15	Disable inspection door by pressing the INSP key (1) – LED INSP **OFF**
16	End of the commissioning procedures.

7.24.2 Parameters

All the changeable parameters can be found in EJ 41354325. The parameters can be changed with the aid of the HMI.

7.24.3 Diagnostics



HMI Keypad [402031; 15.12.2014]

1	LED Ready	5	LED INSP
2	LED Locked/Opened	6	LED SETUP
3	LED Obstacle	7	LED COM (CAN communication)
4	LED Closed (KET-S2)		

1	2	3	4	Description	Fault Rectification
All flashing				Door error	Check error log on HMI.
Blinking slowly				Door is in standby state	-
	Blinking slowly			Referencing not executed	Close door until locked
		Blinking slowly		KSKB / KOKB / RPHT / KTL / KTFP active	Remove obstacle
Flashing				Torque OFF	-

5	6	7	Description	Fault Rectification
Blinking slowly			INSP by EC	-
	Flashing		AutoSetup error	-
		OFF	No CAN communication	Check wiring and EC state
		Blinking slowly	Communication OK. EC not ready	Physical connection is okay. -> Check EC
		Blinking fast	Communication Error during SW Download via CAN interface. SW Download in Preparation (Bootloader Mode)	Check Wiring and restart download
		Single flash	SW Download in Preparation (Bootloader Mode)	-
		Triple flash	SW Download in Progress (Bootloader Mode)	-

8 Appendix B: Error Codes Descriptions

The description how to read the event codes history with help of the user interface HMI can be found in section 4.5.

Legend:

- **C1, C2, C3, ...**: Cause 1, 2, 3, ... of the error
- **A1, A2, A3, ...**: Action 1, 2, 3, ... to solve the error

– Any error displayed during the calibration belongs to error group 11. (Example: er 9 = Error 1109)
– Errors (E0..E9) displayed during the LOP configuration are described in section 5.4.6

Code	General Messages
0001	**E_ELEVATOR_FATAL _ERROR** The elevator is permanently blocked and not operable. Note, this message typically follows another error.
	C1: Different causes **A1:**Check elevator message log for previous reported messages in order to identify the root cause of the problem.
0002	**E_ELEVATOR_SAFETY_CHAIN** The safety circuit has opened unexpected or hasn't closed as expected.
	C1: The safety circuit has opened unexpected (e.g. while the car was moving) **A1:**Check safety circuit for opened contacts. Check 110V fuse on SMIC board.
	C2: Safety chain has not closed when expected. When all doors are closed, the safety circuit is expected to be closed too. **A2:**Check safety circuit contacts at door.Check door parameter 'delay time between door closed and closed safety circuit' (CF03 PA13).

Code	General Messages
	C3: Safety chain is not powered due to UCM (unintended car movement) actuation. **A3:**Check UCM recovery procedures (e.g. remove persistent fatal error).
0003	**E_ELEVATOR_OVERLOAD_MODE** The elevator has detected overload in the car.
	C1: Too high load in the car **A1:**Decrease the load in the car
	C2: The car load measuring signal is faulty **A2:**Check general wiring to the car load measuring device.Redo calibration if signal is present but invalid.
0004	**E_ELEVATOR_NORMAL_MODE** The elevator is in passenger travel mode.
0005	**E_ELEVATOR_POWER_FAIL_MODE** The elevator mains power has failed.
0006	**E_ELEVATOR_INSPECTION_MODE**
0007	**E_ELEVATOR_JBF_MODE**
0008	**E_ELEVATOR_JRVC_MODE**
0009	**E_ELEVATOR_EVACUATION_MODE**
0010	**E_ELEVATOR_RECALL_MODE**
0011	**E_ELEVATOR_JNFF_MODE**
0012	**E_ELEVATOR_NOAUTHORIZATION_MODE** The elevator is blocked because no or an invalid SIM card is inserted at the elevator main control board (PCB).
	C1: No SIM card, no Schindler SIM card, a SIM card of another elevator or a manipulated SIM card is inserted **A1:**Check if SIM card is present or if a invalid SIM card is inserted. Check error log for SIM card errors (#19xx). Get right SIM card.

Code	General Messages
	C2:SIM card with different capacity **A2:**Older version of controller SW does not understand new chipcards with bigger capacity. Upgrade controller SW.
0013	**E_ELEVATOR_FULLOAD_MODE**
0014	**E_ELEVATOR_DKFM_MODE**
0015	**E_ELEVATOR_LEARNING_MODE**
0016	**E_ELEVATOR_SYNCHRONIZATION_MODE**
0017	**E_ELEVATOR_REVISION_NUMBERS_DO_NOT_MATCH** The commissioning number stored on the SIM card doesn't match with the current elevator commissioning number.
	C1: Wrong SIM card inserted (e.g. wrong delivery or SIM card from another elevator). **A1:**Get right SIM card, insert it and perform an elevator main controller reset.
0018	**E_ELEVATOR_CHIP_CARD_DATA_INTEGRITY_ FAILURE** The SIM card could be read but its data are corrupt.
	C1: The SIM card is defective **A1:**Replace defective SIM card
0019	**E_ELEVATOR_WATCHDOG_RESET** Indicates that a watchdog reset (initiated by hardware or software watchdog) has been performed previously. The elevator main control has just started up.
	C1:A main controller software internal problem has occurred **A1:**Verify if a main controller software update is available as fix and update accordingly.
0020	**E_ELEVATOR_S_CHAIN_BRIDGED_PERMANENT** Safety circuit did not open when expected (e.g. while opening door)

Code	General Messages
	C1: The safety circuit is bridged (e.g. at the car or landing door) **A1:**Check safety circuit for bridges (e.g. plugs) and remove them
0021	**E_ELEVATOR_CC_RESET** The car controller was reset (no further action needed).
0022	**E_ELEVATOR_CC_RELEVELING_FAIL** Car controller error during releveling.
0024	**E_ELEVATOR_KNE_U_INTERRUPTED** Safety circuit opened unexpected (e.g. during trip) at tap 'KNE_U'. The information event is sent if rough position is "top" or unknown.
	C1: The car has exceeded the hoistway end limit (e.g. KNE_U) **A1:**Check why the car has exceeded the hoistway end limit. Check log for possible previous reported messages. Release elevator from blocked state (perform reset procedure).
	C2: Safety circuit wiring faulty or safety circuit contact(s) defective or bad adjusted **A2:**Check general wiring safety circuit and safety circuit contacts at corresponding safety circuit section.
0025	**E_ELEVATOR_ISK_PREOPEN_ERROR** Safety circuit opened unexpected at tap 'ISK' during pre-opening of the doors.
	C1:A safety device has tripped **A1:**Check why the corresponding safety device has tripped and resolve problem.

Code	General Messages
	C2: Safety circuit wiring faulty or safety circuit contact(s) defective or bad adjusted **A2:**Check general wiring safety circuit and safety circuit contacts at corresponding safety circuit section.
0026	**E_ELEVATOR_TT_MODE** Car partition door
0027	**E_ELEVATOR_CC_RELEVELING_FAIL_FATAL**
0028	**E_ELEVATOR_NF1_MODE**
0029	**E_ELEVATOR_RV2_MODE**
0030	**E_ELEVATOR_UL3_DISABLED (OEM activated)** Indicates that the user level 3 was disabled
	C1: The user has disabled the user level 3 by a command (e.g. menu 109 or SPECI) **A1:**None
	C2: The elevator performed a auto disabling of the user level 3 for the first time (e.g. if option “OEM” is active on SIM and after 10’000 trips) **A2:**SPECI or CADI is required to access all levels on HMI.
	C3: The elevator performed a auto disabling of the user level 3 for the second and last time (after re-enabling, e.g. after additional 2000 trips) **A3:**none
0031	**E_ELEVATOR_LEARNING_INTERRUPTED** The learning travel has failed.
	C1: Wrong manipulation on elevator. E.g. Recall switched on or any stop switch pressed during learning travel. **A1:**Release corresponding switches. Re-execute the learning travel.
	C2: Safety circuit opened **A2:**Check reason for open safety circuit and get rid of the problem. Re-execute the learning travel.

Code	General Messages
	C3: An error interrupted the learning trip **A3:**Look the errors before in the error history to see which error caused the interruption.
0032	**E_ELEVATOR_EB_MODE**
0033	**E_ELEVATOR_STARTED_UP** Indicates that the main controller has just started up.
	C1: The power got switched off and on again **A1:**None
	C2: The reset button got pressed **A2:**None
	C3:A watchdog reset got performed **A3:**Check first power supply (quality) and possible reset by other staff. If message appears unexpectedly update the main controller with new software release.
0034	**ELEVATOR_SR_MODE** Sprinkler mode
0035	**E_ELEVATOR_SCHOOLHOUSE_MODE**
0036	**E_ELEVATOR_BF_MODE**
0037	**E_ELEVATOR_COP_HEARTBEAT_MISSING** The communication to the cop respectively car user interface node has broken
	C1:COP disconnected **A1:**Connect COP
	C2: Data transmission faulty **A2:**Check data line connection (CAN). Check for correct data line termination (termination switch in ECU or LDU, termination in ACVF)
	C3: Mismatch of car node software and elevator main controller software **A3:**Update the software accordingly

Code	General Messages
0038	**E_ELEVATOR_ECU_HEARTBEAT_MISSING** The car user interface node (COP) has recognized a lost of communication to the elevator main control. Note: This error mainly occurs together with other errors. Check message log first for other reported errors.
	C1:COP disconnected **A1:**Connect COP
	C2: Data transmission faulty **A2:**Check data line connection (CAN). Check for correct data line termination (termination switch in ECU or LDU, termination in ACVF)
	C3: Mismatch of car node software and elevator main controller software **A3:**Update the software accordingly
0062	**E_ELEVATOR_BACKUP_BAT_SUFFICIENT_CHARGE** The charge of the elevator backup battery has resumed to the required minimum level.
0063	**E_ELEVATOR_BACKUP_BAT_INSUFFICIENT_CHARGE** The charge of the elevator backup battery has fallen below the required minimum level or wasn't able to get recharged to the required minimum level after elevator mains power up.
	C1: Battery connection missing or bad **A1:**Check battery's connections
	C2: Battery faulty (e.g. old) **A2:**Replace battery
	C3: Battery charging problem **A3:**Check wiring to charging device. Check function of charging device (charging voltage, fuses).

Code	General Messages
	C4: With battery temperature sensor not on the PCB: external battery temperature sensor faulty or not connected to BC PCB. **A4:**Verify the connection and function of the external battery temperature sensor.
0072	**E_ELEVATOR_BACKUP_BAT_CAR_INSUFFICIENT_ CHARGE** The charge of the car backup battery (e.g. used to power the car backup light) has fallen below the required minimum level or wasn't able to get recharged to the required minimum level after elevator mains power up.
	C1: Battery connection missing or bad **A1:**Check battery's connections
	C2: Battery faulty (e.g. old) **A2:**Replace battery
	C3: Battery charging problem **A3:**Check wiring to charging device. Check function of charging device (charging voltage, fuses).
0073	**E_ELEVATOR_BACKUP_BAT_CAR_SUFFICIENT CHARGE** The charge of the car backup battery (e.g. used to power the car backup light) has resumed to the required minimum level.
0074	**E_ELEVATOR_TRACTION_MEANS_TEMP_EXCEEDED** The temperature of the elevator traction means (e.g. belts) has exceeded the allowable operating temperature
	C1: Too hot ambient air temperature (temperature in hoistway) **A1:**Wait for cool down
	C2: The temperature feedback signal (e.g. KTHS) is faulty **A2:**Check general wiring to temperature sensor

Code	General Messages
	C3: The heat dissipation is not working **A3:**Check operation of heat dissipation device (e.g. fan or forced ventilation) if present
0075	**E_ELEVATOR_TRACTION_MEANS_TEMP_RECOVERY_SUCCESSFUL** The elevator was able to recover from a temperature problem at the elevator traction means.
0076	**E_ELEVATOR_CAR_LIGHT_BROKEN** The car light has broken. Recognized by the alarm button backlight which is switched on continuously.
	C1: The car light is broken. **A1:**Replace car light and verify that the COP alarm button backlight, when available, is properly switched off
	C2: The car light sensor on COP is defective **A2:**Check car light sensor on COP for proper working. Illuminating the sensor should switch the alarm button backlight off, covering the sensor should switch the alarm button backlight on.
	C3:A light absorbing or dark car interior cladding has been installed after commissioning and the available light is not enough to activate the sensor. **A3:**If the car light condition is OK perform again a learning trip to check and store the new working condition of the sensor. (if light is not enough the error 0077 might be reported)
0077	**E_ELEVATOR_NO_CAR_LIGHT_SENSOR** Car light sensor not detected

Code	General Messages
	C1: During learning trip the COP car light sensor is reporting erratic values or is reporting light switched off. **A1:**Check proper working of car light. Check that the backlight of the COP alarm button lights up when light sensor covered and switches off when light sensor illuminated. If light is considered to be working correctly ignore the error (the car light monitoring will remain disabled-no error 0076 will ever be generated)
0078	**E_ELEVATOR_CAR_LIGHT_OK** Whenever the controller notices that the car light is broken, it sends an E_ELEVATOR_CAR_LIGHT_BROKEN event. Once the light is working again, it signals it with an ELEVATOR_CAR_LIGHT_OK event.
0079	**E_ELEVATOR_CAR_BLOCKED** The car was stuck in down direction. After more than 3 times in on hour the error is classified from the controller as Fatal.
	C1: The elevator was blocked in down-direction. **A1:**Inspect the elevator and traction media to find what have stuck the car.
	C2: The elevator was blocked in down-direction. **A2:**Elevator recovers automatically from this error by user (or recall) traveling in up-direction.
	C3: The elevator was blocked in down-direction too many times. **A3:**The threshold depends on the carload calibration. Increase the thresholds set by CF08PA15/16/17.
0080	**E_ELEVATOR_CAR_EMERGENCY_EXIT_MISUSE_ DETECTED** The controller detects a misuse of the emergency exit trap.
0081	**E_ELEVATOR_OEM_POLICY_DISABLED** The OEM policy status is set to "disabled", the service interface uses the standard menu tree.

Code	General Messages
0082	**E_ELEVATOR_OEM_POLICY_STATUS_CHANGE**
	The OEM policy status changed. The message is sent to RMP but is not saved to the local control log.
0084	**E_ELEVATOR_BACKUP_BAT_INFO_NOT_SUPPORTED_BY_HW**
	The battery cannot be monitored due to missing HW
0091	**E_ELEVATOR_KNE_D_INTERRUPTED**
	Safety circuit opened unexpected (e.g. during trip) at tap 'KNE' with rough shaft position "bottom"
0099	**E_ELEVATOR_DIAGNOSTIC_CALL**
	The elevator generates this message when a diagnostic specific message has been logged. This msg is necessary for old TMx device to trigger a call to the central.

Code	Car Door Messages (see also section 7.22, 7.23 and 7.24)
0301	**E_DOOR_CLOSING**
	The door wasn't able to close successfully within a certain time limit. As a typical response a door recovery gets started (repetitive opening and closing of the door).
	C1: The door could not be closed fully in spite of repeated attempts. The KET-S signal was not set. In the Fermator door drive, the KET-S is implemented electronically, the door control measures the door position and sets the KET-S signal if the preset value is reached.
	A1: Mechanical blocking
	Measure: - Check the door threshold and clean if necessary - Visual check for mechanical damage Tips & tricks: It is essential to move to all floors and to check the door.

Code	Car Door Messages (see also section 7.22, 7.23 and 7.24)
	C2: Door closed position feedback signal is faulty. **A2:**Check door closed position contact (e.g. KET-S, if present and not emulated, depending on door type) for dirt or bad mechanical adjustment. Check general wiring door closed position contact.
	C3: Door doesn't move or moves to slow **A3:**Check for obstacle or dirt at door shoes and tracks. Check door mechanism (broken belt, mechanical coupling, dirt). Check door motor power supply (fuses).
	C4: The door could not be closed fully in spite of repeated attempts. The KET-S signal was not set. In the Fermator door drive, the KET-S is implemented electronically, the door control measures the door position and sets the KET-S signal if the preset value is reached. **A4:**Check KET-S signal Measure: --Check signal "...OK!" » should flash, self-test OK » Lights without flashing » Control unit defective, go to cause 3 --Use the test button to operate the door and observe whether the door closed diode light up --Perform the door teach-in travel --Use the test button to operate the door --Closed diode lights up again, successful measure Tips & tricks: It is essential to move to all floors and to check the door Closed functions normally but the fault is still not yet rectified: Notify field technician and put a cross in "other" as measure.
	C5: Door close parameter set wrong **A5:**Check door close parameter for correct setting

8

Code	Car Door Messages (see also section 7.22, 7.23 and 7.24)
	C6: The door could not be closed fully in spite of repeated attempts. The KET-S signal was not set. In the Fermator door drive, the KET-S is implemented electronically, the door control measures the door position and sets the KET-S signal if the preset value is reached. **A6:**The door could not be closed fully in spite of repeated attempts. The KET-S signal was not set. In the Fermator door drive, the KET-S is implemented electronically, the door control measures the door position and sets the KET-S signal if the preset value is reached.
0302	**E_DOOR_OPENING** The door wasn't able to open successfully within a certain time limit. As a typical response a door recovery gets started (repetitive closing and opening of the door).
	C1: The door could not be closed fully in spite of repeated attempts. The KET-O signal was not set. In the Fermator door drive, the KET-O is implemented electronically, the door control measures the door position and sets the KET-O signal if the preset value is reached. **A1:**Mechanical blocking Measure: - Check the door threshold and clean if necessary - Visual check for mechanical damage Tips & tricks: It is essential to move to all floors and to check the door opening.
	C2: Door opened position feedback signal is faulty. **A2:**Check door opened position contact (e.g. KET-O, if present and not emulated, depending on door type) for dirt or bad mechanical adjustment. Check general wiring door opened position contact.

Code	Car Door Messages (see also section 7.22, 7.23 and 7.24)
	C3: Door doesn't move or moves to slow **A3:**Check for obstacle or dirt at door shoes and tracks. Check door mechanism (broken belt, mechanical coupling, dirt). Check door motor power supply (fuses).
	C4: The door could not be opened fully in spite of repeated attempts. The KET-O signal was not set. In the Fermator door drive, the KET-O is implemented electronically, the door control measures the door position and sets the KET-O signal if the preset value is reached. **A4:**Measure: --Check signal "...OK!" » should flash, self-test OK » Lights without flashing » Control unit defective, go to step 3 --Use the test button to operate the door and observe whether the door opened diode lights up --Perform the door teach-in travel --Use the test button to operate the door --Opened diode lights up again, successful measure Tips & tricks: It is essential to move to all floors and to check the door opening. Opened functions normally but the fault is still not yet rectified: Notify field technician and put a cross in "other" as measure.

Code	Car Door Messages (see also section 7.22, 7.23 and 7.24)
	C5: The door could not be opened fully in spite of repeated attempts. The KET-O signal was not set. In the Fermator door drive, the KET-O is implemented electronically, the door control measures the door position and sets the KET-O signal if the preset value is reached. **A5:**Control unit defective Measure: --If the OK lamp does not light up or does not flash: » Check the fuse on the Fermator control unit » Check the power supply on the control unit --KET-O diode still does not light up » Renew device Tips & tricks: Obtain the right device (left or right opening).
	C6: Door open parameter set wrong **A6:**Check door open parameter for correct setting
0303	**E_DOOR_MAX_LOCK_TIME** A door was for a too long period of time in locked position. Note, monitoring the time the door is in locked position prevents the motor from burning. Note, this error might occur while the car is moving slowly (e.g. during learning travel or emergency power recall travel)
	C1: Parameter “max door lock time” is set wrong **A1:**Check corresponding parameter (CF03 PA07)
0304	**E_DOOR KSKB** The door wasn't able to close due to an mechanical blockade. This error occurs if the number of door re-openings due to an activated door closing force limiter (KSKB) exceeds the predefined maximum (e.g. 20).
	C1: Obstructions/barriers in the door zone or in the slit/gap **A1:**Remove obstacle

Code	Car Door Messages (see also section 7.22, 7.23 and 7.24)
	C2: Door closing force limiter feedback signal KSKB is faulty **A2:**Check contact KSKB for dirt or bad mechanical adjustment. Check general wiring KSKB.
0305	**E_DOOR_WRONG_EVENT**
	C1: The controller misses a door lock state **A1:**None
0306	**E_DOOR_WRONG_EVENT_2**
	C1: Same as event 0305 but while a service is running **A1:**None
0307	**E_DOOR_WRONG_EVENT_3**
0308	**E_DOOR_WRONG_EVENT_4**
0309	**E_DOOR_WRONG_CMD_VALUE**
0310	**E_DOOR_WRONG_MOTION_VALUE**
0311	**E_DOOR_WRONG_DRIVER_EVENT**
0312	**E_DOOR_CLOSED_WITH_WRONG_COMMAND** The door position signalization (opened, closed) does not match with the door driving direction.
	C1: The door position sensors signalize a closed door when it is expected to be opened and vice versa **A1:**Check for mix-up of the wiring of the door closed and opened position sensors (e.g. KET-S and KET-O)
	C2: No DOOR_CMD = DOOR_CLOSE/ DOOR_LOCK given **A2:**Mismatched KET_S and KET_O
	C3: Door is not reversing **A3:**The direction of the motor is wrong
	C4:KET_S gets active **A4:**The door is reversed just before reaching KET_S but the door reaches this contact because of inertia

Code	Car Door Messages (see also section 7.22, 7.23 and 7.24)
	C5: DoorState=stopped=motor off **A5:**The motor can be connected invert
	C6: Door motor drives in wrong direction **A6:**Check for mix-up of the door motor power wires
0313	**E_DOOR_SHAFT_ERROR** The subsystem door has received a door open command while the car is not detected to be on floor. The command is rejected.
	C1: Different causes **A1:**Check elevator message log for previous reported messages in order to identify the root cause of the problem. If previous error is a Fatal, no action can be taken.
	C2: Door open command outside the door zone **A2:**Check PHS Measure: - Clean PHS - Check PHS attachment - Check the calibration (check there is adequate protection depth across the entire hoistway) - Check car track - Check PHS and PHS2 plugs (on SDIC) and connections - Check function of the PHS and PHS2 with light-emitting diode, renew PHS in case of a fault. - Carry out learning travel Tips & Tricks: In case of preopening and two sides installed please check PHS and PHUET alignment. If this does not help consider to increase preopening delay timer (CF3PA6).

Code	Car Door Messages (see also section 7.22, 7.23 and 7.24)
	C3: Door open command outside the door zone **A3:**Check slip Measure: Stop car with the inspection control in KSE-U stop - Mark the belt and the traction pulley. - Use the inspection control to move five meters downwards and back into the KSE-U stop. - Measure the distance between the mark on the belt and the mark on the traction pulley. In case of deviations of more than 10 mm, contact field technology. Tips & Tricks: See preventative maintenance module 19, K 609755, Traction belts section
0314	**E_DOOR_PRE-OPENING** There was an activation or deactivation failure of the door safety circuit bypass device while the elevator was intended to perform a door pre-opening.
	C1: When moving to the floor, advance door opening has detected a fault **A1:**Check bi-directional light barrier (PHUET) Measure: - Clean sensor and flags - Check the calibration (check there is adequate protection depth across the entire hoistway). - Check car tolerances - Check function of the PHUET with light-emitting diode, renew PHUET in case of a fault. Tips & Tricks: The light-emitting diode is at the bottom in the PHUET

Code	Car Door Messages (see also section 7.22, 7.23 and 7.24)
	C2: Bypass device not working properly **A2:** Check SUET Measure: - Check SUET plug and connections. - Normal mode, perform test drives. Tips & Tricks: SUET printed circuit board next to inspection control under sheet metal cover.
	C3: Bypass device not working properly **A3:** Measure: It is essential to perform measures 1 and 2 BEFORE the preventative HW renewal - Renew printed circuit boards - Set CF8 PA7 back to 1 - Check test drives and advanced door opening Tips & Tricks: Under certain circumstances, the door bridging can be switched off temporarily until the printed circuit boards have been obtained (CF8PA7=0) This MUST be entered in the malfunction log without fail! SUET ID no. 00591811 PHUET ID no. 55505160
0315	**E_DOOR_NOT_RECOVERABLE** The door wasn't able to recover from a door opening or door closing error (typically after 20 repetitive opening and closing attempts). The elevator is blocked.

Code	Car Door Messages (see also section 7.22, 7.23 and 7.24)
	C1: The door could not be recovered after a door opening or door closing fault (generally after 20 repeated attempts to open and close the door). The elevator is blocked. **A1:**Mechanical blocking - Measure - Check the guide rails and clean if necessary - Visual check for mechanical damage Tips & Tricks: It is essential to move to all floors and to check the door opening
	C2: Door has a problem causing repetitive 'door opening' or 'door closing' errors. **A2:**Check why door cannot recover. Check errors 0301 and 0302 for cause and actions.
	C3: The door could not be recovered after a door opening or door closing fault (generally after 20 repeated attempts to open and close the door). The elevator is blocked. **A3:**Check the KET-O/KET-S signal Measure: --Check signal "...OK!" » should flash, self-test OK » Lights without flashing » Control unit defective, go to step 3 --Use the test button to operate the door and observe whether the door limit switch diodes light up --Perform the door teach-in travel --Use the test button to operate the door --Door limit switch diodes light up again, successful measure Tips & Tricks: It is essential to move to all floors and to check the door opening Door limit switch diodes function normally but the fault is still not yet rectified: Notify field technician and put a cross in "other" as measure

Code	Car Door Messages (see also section 7.22, 7.23 and 7.24)
	C4: The door could not be recovered after a door opening or door closing fault (generally after 20 repeated attempts to open and close the door). The elevator is blocked. **A4:**Control unit defective Measure If the OK lamp does not light up or does not flash: » Check the fuse on the Fermator control unit » Check the power supply on the control unit KET-O/KET-S diode still does not light up » Renew device Tips & Tricks: Left door and central door ID 59313502 Right door ID 59313503 Set DIL switch 2 (manual door) to OFF and DIL 3 to right or left opening
	C5: The door could not be recovered after a door opening or door closing fault (generally after 20 repeated attempts to open and close the door). The elevator is blocked. **A5:**Check errors 0301 and 0302 for cause and actions.
0316	**E_DOOR_HEARTBEAT_ERROR** The communication to the door node respectively car node (controller) has broken.
	C1: The control has lost the connection to the SDIC **A1:**Check the SKC plug on SMIC Measure: Check the plug on the SMIC Measure the supply » SKC pin 5 against ground (or pin 7) must display 24V, otherwise renew SI1 (T2.5A) » SKC pin 6 against ground (or pin 7) must display 12V, otherwise renew SI4 (T2.5A)
	C2: Door node respectively car node disconnected **A2:**Reconnect node

Code	Car Door Messages (see also section 7.22, 7.23 and 7.24)
	C3: No or bad door node respectively car node power supply **A3:**Check node power supply
	C4: The control has lost the connection to the SDIC **A4:**Check SDIC on car Measure: --Check SKC plug --Voltage monitoring » 24V lights » 12-NSG lights » if an LED is incorrect then the wiring is faulty --Check SDIC LED » 3.3V/5V lights » WDOG flashes » ERROR OFF » if an LED is incorrect, renew SDIC Tips & Tricks: To move the car, see section 4.9.9, Car door access with installation travel
	C5: Data transmission faulty **A5:**Check for correct data line termination (if present) of all devices connected to the data bus. Check shielding of data line (if present).
	C6: Data transmission faulty **A6:**Check general data line connection. Check for correct data line termination (if present) of all devices connected to the data bus. Check shielding of data line (if present). Check for EMC disturbances.
	C7: Door node respectively car node defective **A7:**Replace corresponding node.

Code	Car Door Messages (see also section 7.22, 7.23 and 7.24)
	C8: Door node respectively car node defective **A8:**Renew SDIC Measure: --Renew SDIC --Trigger teach-in travel (menu 116) --Check SW statuses » Notify CF12 PA1 and PA2 to support Tips & Tricks: It is essential to comply with EMC measures Important: If an SDIC 51 is replaced by an SDIC 52 then the following bridge connectors must be set: » Set plug JHC2, ID 55502643 > Bridge pin 1/2 > Bridge pin 3/4 » Set plug 2KTC ID 55504068 > Bridge pin 1/2 SDIC 52.Q (ID 591885), 2 doors and advanced door opening, backward compatible
0317	**E_DOOR_UNEXPECTED_STATE** The subsystem door reported an inconsistent sequence of a door state (e.g. opening -> closed)
	C1: Power supply of car node (e.g. SDIC) faulty **A1:**Check the power supply of car node (e.g. loose contact)
	C2: Mismatch of car node software and elevator main controller software **A2:**Update the software accordingly
0318	**E_DOOR SDIC HEARTBEAT** The car node has recognized a lost of communication to the elevator main control. Note, this error mainly occurs together with other errors. Please check message log first for other reported errors.
	C1: Car node disconnected (e.g. CAN bus) **A1:**Reconnect node

Code	Car Door Messages (see also section 7.22, 7.23 and 7.24)
	C2: Data transmission faulty (e.g. CAN bus) **A2:**Check general data line connection. Check for correct data line termination (jumpers and switches, if present) of all devices (PCBs) connected to the the data bus. Check shielding of data line (if present). Check for EMC disturbances.
	C3: No or bad power supply of elevator main control **A3:**Check power supply
	C4: Mismatch of car node software and elevator main controller software **A4:**Update the software accordingly
	C5: Elevator main control defective **A5:**Replace corresponding hardware
0319	**E_DOOR_MAX_REOPENINGS_EXCEEDED** Due to repetitive activation of a door reopening device the number of door re-openings exceeded the predefined maximum.
	C1: Obstructions/barriers in the door zone or in the slit/gap **A1:**Remove obstacle
	C2: Door closing force limiter feedback signal KSKB is faulty **A2:**Check contact KSKB for dirt or bad mechanical adjustment. Check general wiring KSKB.
0320	**E_DOOR_SHUTTING**
0321	**E_DOOR_TOO_FAST**
0322	**E_DOOR_KETO_NOT_OFF**
0323	**E_DOOR_KETO_ON_UNECPCTED**
0324	**E_DOOR_KETO_ON_WHEN_LOCKED**
0325	**E_DOOR_KETO_ON_WHEN_CLOSED**
0326	**E_DOOR_KETO_OFF_WHEN_OPENED**

Code	Car Door Messages (see also section 7.22, 7.23 and 7.24)
0327	**E_DOOR_KETS_NOT_OFF**
0328	**E_DOOR_KETS_**
0329	**E_DOOR_KETS_OFF_UNEXPECTED**
	C1: The motor does a tooth of a toothed wheel in the wrong direction before going in the right direction - e.g. the door closes with CMD_CLOSE. The door moves in closing direction. As soon as KET_S is reached we are in closed position. If now the door bounces back and KET_S gets inactive for a short time and right before that the SDIC receives the CMD_LOCK we have this constructed situation. **A1:**Move the KET_S switch a little away from the real closed position (door-end position)
	C2:- **A2:**Try to get some (very little) mechanical resistance in the area of the closed door to reduce the bouncing back
	C3:- **A3:**Look if the door closes smoothly - i.e. the door-blade at the bottom shall close not prior to the top (-> so that there is no mechanical friction/tension in opening direction)
0330	**E_DOOR_KETS_OFF_WHEN_CLOSED**
0331	**E_DOOR_KETS_OFF_WHEN_LOCKED**
0332	**E_DOOR_KETS_ON_WHEN_OPENED**
0333	**E_DOOR_D1_DOD_DIP_WRONG** This error does not apply to Schindler 3100/3300/5300
	A1: Check DIP switch settings (type of door 1) according schematics.
0334	**E_DOOR_D2_DOD_DIP_WRONG** This error does not apply to Schindler 3100/3300/5300
	A1: Check DIP switch settings (type of door 2) according schematics.

Code	Car Door Messages (see also section 7.22, 7.23 and 7.24)
0335	**E_DOOR_KSPT**
	C1: Door opening attempt out of door zone **A1:**Check signal KSPT
0336	**E_DOOR_OVERTEMPERATURE** The door drive (motor) has exceeded its operating temperature.
	C1: Too intensive operation (e.g. too many door cycles per time unit) **A1:**Wait for cool down
	C2: Too hot ambient air temperature (e.g. direct sunlight at glass shaft) **A2:**Wait for cool down
	C3: The temperature feedback signal is faulty **A3:**Check general wiring to temperature sensor. Check operation of temperature sensor.
0337	**E_DOOR_SUET_BOARD_DISCONNECTED** The door safety circuit bypass device is detected as disconnected.
	C1: The door safety circuit bypass device was disconnected by a user. **A1:**Reconnect the safety circuit bypass device (e.g. SUET)
	C2:A door safety circuit bypass feedback signal (e.g. IUET, RFUET) is faulty **A2:**Check general wiring at door safety circuit bypass device (e.g. SUET)
	C3: Failure at door safety circuit bypass device **A3:**Replace corresponding device (e.g. SUET)

8

Code	Car Door Messages (see also section 7.22, 7.23 and 7.24)
0338	**E_DOOR SUET ERROR1 OVERBRIDGING ACTIVATION UNSUCCESSFUL** There was an activation failure of the door safety circuit bypass. Note, this bypass is activated during re-leveling or advanced door opening (pre-opening).
	C1: Door zone detection faulty **A1:**Check door zone signals and sensors (e.g. PHSx, PHUETx).
	C2:A door safety circuit bypass feedback signal (e.g. IUET, RFUET) is faulty **A2:**Check general wiring at door safety circuit bypass device (e.g. SUET)
	C3: Failure at door safety circuit bypass device **A3:**Replace corresponding device (e.g. SUET)
0339	**E_DOOR_SUET_ERROR2_OVERBRIDGING_LOST** The elevator has recognized an unexpected deactivation of the door safety circuit bypass. Note, this bypass is activated during re-leveling or advanced door opening (pre-opening).
	C1: Door zone detection faulty **A1:**Check door zone signals and sensors (e.g. PHSx, PHUETx).
	C2:A door safety circuit bypass feedback signal (e.g. IUET, RFUET) is faulty **A2:**Check general wiring at door safety circuit bypass device (e.g. SUET)
	C3: The car has unintentionally left the door zone while the door safety circuit was bypassed **A3:**Check why the car has left the door zone
	C4: Failure at door safety circuit bypass device **A4:**Replace corresponding device (e.g. SUET)

Code	Car Door Messages (see also section 7.22, 7.23 and 7.24)
0340	**E_DOOR_SUET_ERROR3_OVERBRIDGING DEACTIVATION_UNSUCCESSFUL** There was an deactivation failure of the door safety circuit bypass. Note, this bypass is activated during re-leveling or advanced door opening (pre-opening).
	C1:A door safety circuit bypass feedback signal (e.g. IUET, RFUET) is faulty **A1:**Check general wiring at door safety circuit bypass device (e.g. SUET).
	C2: Failure at door safety circuit bypass device **A2:**Replace corresponding device (e.g. SUET)
0341	**E_DOOR_RPHT_SIGNAL_BLOCKED_ACTIVE** This error message is logged when the RPHT error counter reached its limit value (e.g. PPHT signal got active while the door was closed).
	C1: Sticker on the photo cell **A1:**Remove sticker
	C2: Photo cell defect **A2:**Check photo cell, replace it if necessary
0342	**E_DOOR_RPHT_SIGNAL_RECOVERED_NORMAL_ OPERATION** The RPHT signal has recovered, the RPHT signal is inactive in the state door closed
0343	**E_DOOR_SUET_ERROR4_OVERBRIDGED_WITHOUT_A CTIVATION** There was a failure of the door safety circuit bypass. Note, this bypass is activated during re-leveling or advanced door opening (pre-opening).
	C1: Failure at door safety circuit bypass device **A1:**Replace corresponding device (e.g. SUET)

8

Code	Car Door Messages (see also section 7.22, 7.23 and 7.24)
	C2: Check VRUET signal and verify that SUET is not active during standstill (e.g. recall active) **A2:**If VRUET is always active replace the SDIC board
0344	**E_DOOR_RPHT_SIGNAL_BLOCKED_INACTIVE** There was a failure of the door safety circuit bypass. Note, this bypass is activated during re-leveling or advanced door opening (pre-opening).
0351	**E_DOOR_POSITION_RECOVERY_SUCCESSFUL** The door was able to recover from a position problem
0355	**E_DOOR_UNLOCK_MISUSE_DETECTED** A misuse of the landing door is detected and not permitted: elevator stops any activity. For Russia market. A manual intervention is required for bring back in operation the elevator.
	C1: Door lock monitoring signalize an error **A1:**Check if nobody is in the shaft or on the car roof. Rearm the monitoring. If the error appears again inspect correct closure of every landing door.
0364	**E_DOOR_DRIVE_NO_POWER_SUPPLY** The door is unavailable due to loss of power supply of the door drive.
0368	**E_DOOR_DRIVE_INTERNAL_FAILURE** The door is unavailable due to an severe internal failure of the door drive.
	C1:0x2A 42 eNGT 24VDC Over 10% Limit 0x2B 43 eNGT 24VDC Under 10% Limit Power supply voltage out of range **A1:**Replace Power Supply

Code	Car Door Messages (see also section 7.22, 7.23 and 7.24)
	C2:0x31 49 eOver Current Too high controller gains (velocity control parameter set, Position control parameter set) **A2:**Run AutoSetup
	C3:0x31 49 eOver Current Profile acceleration and/or deceleration too high **A3:**1. Run autosetup 2. Reduce Parameter 39_Speed_Close and 40_Speed Open
	C4:0x32 50 eOver Voltage Too much energy is fed back from the motor during a deceleration phase **A4:**1. Run AutoSetup 2. Reduce parameter 39_Speed_Close and 40_Speed Open
	C5:0x32 50 eOver Voltage The power supply is damaged **A5:**Replace power supply
	C6:0x32 50 eOver Voltage Oscillation of the mechanical system (mass/belt) that leds to energy pulsation of energy. **A6:**1. Check belt tension / 2. Run AutoSetup
	C7:0x38 56 eSoftware Internal Parameter Extreme motion profile parameters. **A7:**1. Check if a parameter was changed recently. If yes, change it back to its default value. 2. Run AutoSetup
	C8:0x38 56 eSoftware Internal Parameter At startup the parameters are corrupted. **A8:**1. Set the parameter No. 169 SetParamToDef to value 1. 2. Run AutoSetup 3. If no success replace DDE-VD35

8

Code	Car Door Messages (see also section 7.22, 7.23 and 7.24)
	C9:0x37 55 eInternal Software An unknown event set the door drive to a state where the determinate behavior of the system cannot be guaranteed anymore. **A9:**Restart door drive by plugging off POWT and CANT (DOOR) on OKR.
	C10:0x39 57 eSensor Position The DDE-VD-35 encoder has encountered an error. **A10:**1. Reset the DDE-VD35 by shortly plugging off POWT and CANT (DOOR) on OKR. 2. Run AutoSetup
	C11:0x3A 58 eCAN Overrun **A11:**1. Check CAN Cable / 2. Check CAN Communication.
	C12:0x3D 61 eCAN Tx COB Id_Collision **A12:**-
	C13:0x3E 62 eCAN PDO Length Too Short **A13:**
	C14:0x3F 63 eCAN Bus Off **A14:**-
	C15:0x40 64 eCAN Rx Queue Overflow **A15:**-
	C16:0x41 65 eCAN Tx Queue Overflow **A16:**
	C17:0x43 67 ePosition Following an Obstacle that slows down or stops the door (only possible if force limit detection KSKB/KOKB) is disabled. **A17:**Check for obstacles or blocking situations, challenge the application setting of KOKB, KSKB.

Code	Car Door Messages (see also section 7.22, 7.23 and 7.24)
	C18:0x43 67 ePosition Following too high acceleration/de-acceleration **A18:**Reduce Parameter 39_Speed_Close and 40_Speed Open
0369	**E_DOOR_MAX_OPENING_FORCE_EXCEEDED** The maximum door opening force was exceeded.
	C1: The door opening is obstructed by an obstacle. **A1:**Verify the doorway and remove any obstacles or dirt.
0371	**E_DOOR_LOCKING_UNLOCKING_OBSTRUCTED** The clutch locking respectively unlocking motion is obstructed.
	C1: Mechanical blockage of the clutch. **A1:**Check the clutch's mechanical condition. Try to lock and unlock the door by means of the HMI and examine the clutch mechanically.
	C2: An obstacle or foreign object inside the clutch obstructs the clutch mechanism. **A2:**1. Remove any foreign object obstacle from the path the clutch has to move on. 2. Check the landing door lock if it is working properly.
	C3: In case of Locking/Unlocking Jam only: Increased friction due to ageing or damage of the mechanical components of the clutch subsystem. **A3:**1. Check the mechanical fitness of the clutch subsystem. 2. Replace clutch if necessary.
	C4: In case of Unlocking Jam only: Blocked landing door lock **A4:**Check landing door lock if it is working properly.

Code	Car Door Messages (see also section 7.22, 7.23 and 7.24)
0374	**E_DOOR_CONTACT_END_CLOSING_RECOVERY_ SUCCESS** The Contact End Door Closing-Closed (KET-S2) recovered from a failure.
0375	**E_DOOR_FINAL_TIMER** The final timer function has been executed. Only for Japan country code.

Code	Drive Subsystem Messages
0401	**E_DRIVE_SAFETY_CHAIN_INTERRUPTED** Safety circuit opened unexpected (e.g. during trip)
	C1: Any safety device has tripped (safety circuit opened) **A1:**Check reason for tripping and resolve problem
	C2: Safety circuit wiring faulty or safety circuit contact(s) defective or bad adjusted **A2:**Check general wiring safety circuit and safety circuit contacts; Check first wiring close to inverter and connectors
	C3: Brief opening of door safety contacts **A3:**Check door safety contacts function and adjustment
0402	**E_DRIVE_TRIP_TOO_LONG** The car hasn't reached the intended destination floor within the legal time limit (typically. 45s). This mechanism protects the driving mechanics in case of a blocked car. E.g. at traction elevator, no damage of the ropes. E.g. at hydraulic elevator, no damage of hydraulic jack/valves.
	C1: Car blocked or moving too slowly **A1:**Use inspection to travel down the hoistway and look and listen out for irregularities Check for object in shaft blocking the car. Check car for obstruction (too much friction at guide rails). Listen also for unusual noises or vibration

Code	Drive Subsystem Messages
	C2: Car blocked or moving too slowly **A2:**Check for object in shaft blocking the car. Check car for obstruction (too much friction at guide rails).
	C3: Drive blocked or turning too slowly **A3:**Check if brake opens right. Check parameter nominal speed at drive.
	C4: Drive blocked or turning too slowly **A4:**Measure: Read out the set nominal speed (CF16 parameter 35, section 5.3.3) Read out the speed (menu 701, section 4.7), speed too slow: » Request support (suspicion of problems with the brakes or defective VF)
	C5: Hoistway information signal(s) faulty **A5:**Check hoistway info wiring and it signals (PHS, KS/KS1, KSE/KSE_U/KSE_D, ..) Measure: --Measure PHS signal on the SDIC plug PHS pin 2, 3 --Use inspection to travel down the hoistway --Observe the signal at each flag --Signal missing: » Renew PHS signaller and perform teach-in travel Tips & Tricks: Test the elevator after the teach-in travel
	C6: Hoistway information signal(s) faulty **A6:**Check hoistway info wiring and it signals (PHS, KS/KS1, KSE/KSE_U/KSE_D, ..)
0403	**E_DRIVE_CONTACTORS_FEEDBACK**
	C1: One of the feedback inputs(1) did switch during traveling, or(2) did not switch after starting, or(3) did not switch after stopping. **A1:**Check: wiring. Check: for contactor fault or input fault in the circuit associated with the input.
	C2: Emergency stop

Code	Drive Subsystem Messages
0404	**E_DRIVE_DIRECTION_ERROR**
	C1: Wrong travel direction **A1:**Check: wiring. Check: why the drive doesn't generate sufficient torque
	C2:FA/ Open loop: Contactors feedback wrong
	C3: Closed loop: tacho defective or motor not energized and car moves slowly by unbalancing situation
	C4: Drive doesn’t generate sufficient torque
0405	**E_DRIVE_SHAFT_INFORMATION**
	A1: Check: CAN cable, Check: CAN wiring
0406	**E_DRIVE_OVERTEMPERATURE** Drive overtemperature detected
	C1: Drive motor, hydraulic pump, hydraulic oil, or hoistway overtemperature due too many trips per time unit or due to too hot ambient air temperature (e.g. direct sunlight at glass shaft) **A1:**Wait for cool down
	C2: Drive overtemperature feedback signal KTHMH/KTHM is faulty **A2:**Check thermal contacts KTHMH/KTHM for right operation. Check general wiring KTHMH/KTHM.
	C3: The ventilation (integrated fan at frequency converter or forced fan) is not working **A3:**Check power supply and mechanics of fan
0407	**E_DRIVE_NOT_READY_DURING_STANDSTILL** The drive (frequency converter, open loop) got technically unavailable while the car was standing still.

Code	Drive Subsystem Messages
	C1: Bad electrical connection from the drive frequency converter to the elevator main contactors connection board (MCCE board) (Not applicable for Schindler 3000) **A1:**Check corresponding cables and connectors (e.g. the drive ready signal)
	C2: General problem at the drive frequency converter **A2:**Check the drive frequency converter local error log for detailed diagnostics. Replace the frequency converter if necessary.
0408	**E_DRIVE_NOT_READY_DURING_TRIP** The drive (frequency converter, open loop) got technically unavailable while the car was moving.
	C1: See 0407 **A1:**See 0407
	C2: See 0407 **A2:**See 0407
0409	**E_DRIVE_SPEED_LIMIT_EXCEEDED_IN_STANDBY** Every time before starting a new trip a consistency check of the signal indicating low speed (typically at v < 0.3m/s) coming from the drive frequency converter (open loop) is performed. The error occurs if the signal is not active.
	C1: Bad electrical connection from the drive frequency converter to the elevator main contactors connection board (MCCE board) (Not applicable for Schindler 3000) **A1:**Check corresponding cables and connectors (e.g. the drive speed limit signal)
	C2: Parameter 'speed limit' is set wrong **A2:**Check parameter 'speed limit' at drive frequency converter
	C3: See 0407 **A3:**See 0407.

8

Code	Drive Subsystem Messages
0410	**E_DRIVE_SPEED_LIMIT_EXCEEDED_DURING_SAFETY CHAIN_BRIDGED** While the car is landing on a floor the signal indicating low speed (typically at v < 0.3m/s) coming from the drive frequency converter (open loop) is expected to change once from inactive to active state. The error occurs if this signal has changed once to active state and changes back again to inactive state (meaning: The car speed has increased during landing).Note, the safety circuit at the door is overbridged at this point of time.Note, the elevator can get blocked if this error appears too many times within a specific period of time (typ. > 3times in 1hour).
	C1: See 0407 **A1:**See 0407
	C2: Parameter 'speed limit' is set wrong **A2:**Check parameter 'speed limit' at drive frequency converter
	C3: Rope slip **A3:**Check driving mechanics and traction
	C4: See 0407, C2 **A4:**See 0407, A2
0411	**E_DRIVE_MOTOR_RESISTOR_CONNECTION** At least one of the drive motor resistor connecting devices, used for a smooth trip start at two speed drives (resistor start-up), does not operate as expected.
	C1: Any connecting device feedback signal (typically. from contactor, e.g. SWAHA, SWAFA) is faulty **A1:**Check general wiring to drive motor resistor/coil connecting devices
	C2: Any connecting device (e.g. SWAHA, SWAFA) is faulty (e.g. stuck contacts, burned coil of contactor) **A2:**Replace corresponding connection device

Code	Drive Subsystem Messages
0430	**E_DRIVE_BRAKE_OPERATION_PARTIAL_FAILURE** *) Partial failure detected at the mechanical drive brake
	C1: Any brake position feedback signal (e.g. KB/KB1) is faulty **A1:**Check brake position sensors (e.g. KB/KB1) for dirt. Check general wiring of brake position sensors.
	C2: The brake control circuit is faulty **A2:**Check contactors controlling the brake (e.g. SB, RB, SF).Check general wiring to brake actuator.Check brake module if present (PCB).
	C3: Brake operates only partially **A3:**Check brake supply voltage
	C4: The brake is mechanically bad adjusted **A4:**Check for available field information. Replace brake if necessary.
0431	**E_DRIVE_BRAKE_OPERATION_FATAL_FAILURE** Fatal failure detected at the mechanical drive brake
	C1: Any brake position feedback signal (e.g. KB/KB1) is faulty **A1:**Check brake position sensors (e.g. KB/KB1) for dirt. Check general wiring of brake position sensors.
	C2: The brake control circuit is faulty **A2:**Check contactors controlling the brake (e.g. SB, RB, SF).Check general wiring to brake actuator.Check brake module if present (PCB).
	C3: See 0430 **A3:**See 0430
	C4: See 0430 **A4:**See 0430
0451	**E_DRIVE_BRAKE_SUPPLY_VOLTAGE_FAILURE** The elevator detected a supply voltage failure for the brake.

8

Code	Drive Subsystem Messages
0452	**DRIVE_TORQUE_CALIBRATION_TIMEOUT**
0453	**E_DRIVE_CDD_ACTIVATION_FAIL** This error is reported in case the activation of the car damping device (CDD) failed.
	C1:CDD feedback signal has a broken wire. **A1:**Check CDD feedback signal contact and wiring to elevator control according schematics. Fix contact or wiring in case the feedback signal does not report activation and deactivation of CDD.
	C2:CDD feedback signal switch has wrong switch logic (normally closed instead of normally open). **A2:**Check switch logic of CDD feedback signal and the connector pin positions according schematics.
0454	**E_DRIVE_CDD_DEACTIVATION_FAIL** This error is reported in case the deactivation of the car damping device (CDD) failed.
	C1: The CDD is mechanically jammed. **A1:**Check the CDD mechanics for obstruction from debris or internal mechanical failure.
	C2: The hardware circuit of the CDD activation output is defect and causes the permanent activation of the CDD. **A2:**Disconnect CDD from activation output signal. In case CDD remains in active position, the CDD device is defect. Otherwise the electronics output circuit is defect or the relay which activates the CDD is defect.
0455	**E_DRIVE_CDD_UNAVAILABLE** This error is reported in case the car damping device (CDD) is deactivated due to failure.
0456	**E_DRIVE_CDD_LOST** This error is reported in case the car damping device (CDD) does not communicate anymore to the controller.

Code	Drive Subsystem Messages
0457	**E_DRIVE_SOFT_STOP_FEEDBACK_WRONG** This error is reported in case the soft stop feedback is not consistent.
	C1:MGB-T board disconnected **A1:**Check soft stop cabling.
	C2: Soft stop function enabled without hardware. **A2:**Check soft stop enable parameter.
	C3: Soft stop hardware failure. **A3:**Replace soft stop hardware
0458	**E_DRIVE_SOFT_STOP_TRANSITION_FAIL** This error is reported in case the soft stop is enabled or disabled and the feedbacks does not change as expected.
0459	**E_DRIVE_SOFT_STOP_RECOVERY_FAILURE** This error is reported in case the soft stop recovery action fails.
0460	**E_DRIVE_SOFT_STOP_RECOVERY_SUCCESS** This error is reported in case the soft stop recovery action is a success.
0461	**E_DRIVE_SOFT_STOP_LOST** This error is reported in case the soft stop circuitry is not anymore detected.
0462	**E_DRIVE_SOFT_STOP_DISCONNECTED** This error is reported in case the soft stop circuitry is disconnected and the function is disabled.
0463	**E_DRIVE_SOFT_STOP_TRANSITION_WHILE_DISABLED** This error is reported in case the soft stop input changes state while the function is disabled.
0464	**E_DRIVE_SOFT_STOP_DISABLED** This error is reported in case the soft stop is disabled and parameter is set to disabled.

Code	Drive Subsystem Messages
0466	**E_DRIVE_SOFT_STOP_SUPPLY_LOST** This error is reported in case the soft stop supply is lost.
0499	**E_DRIVE_UNKNOWN_ERROR** An unknown error was detected by the controller (e.g. unknown error of the ACVF was detected).

Code	Car Load Cell Messages
1101	**E_CLC_NO_FREQUENCY** In case of 5 no frequency within 5h a fatal error is raised and this event is log.
	C1:CLC parameters wrong **A1:**Measure: Read out CF96 and compare with the indicated parameters in the control cabinet » If more than +/-10 points (corresponding to 100 Hz), set the original parameters in the CF97 again Re-calibrate the 0 Kg car load » Perform CF98 calibration according to section 5.4.1 After setting again, perform the function test » The CF95 must display 0 Kg when the car is empty » Stand in the car and use CF95 to read out your own body weight (example: 74 Kg displays 7), tolerance: -1/+2 points Tips & Tricks: See section 5.4.1
	C2: No connection to the car load measuring device **A2:**Mechanical check Measure: --Check screws --Perform 0-load calibration, see section 5.4.1 --Connect plug correctly --If the cable is damaged, jump to action 3 --Perform the load measuring function test » The CF95 must display 0 when the car is empty » Stand in the car and use CF95 to read out your own body weight (example: 74 Kg displays 7), tolerance +/- 1 point

Code	Car Load Cell Messages
	C3: Malfunction of the car load measuring device **A3:**Replace the car load measuring device (e.g. CLC) Renew Digisense, please comply with section 5.4.1 **Renewal requires corresponding expertise, it may be necessary to contact support.** Tips & Tricks: With the 3100, refer to document K 609754.
	C4: Malfunction of the car load signal receiving stage **A4:**Replace corresponding PCB (e.g. SDIC): --Renew SDIC printed circuit board --Trigger teach-in travel (menu 116) --Check SW statuses » Notify CF12 PA1 and PA2 to support Tips & Tricks: It is essential to comply with EMC measures Important: If an SDIC 51 is replaced by an SDIC 52 then the following bridge connectors must be set: » Set plug JHC2, ID 55502643 > Bridge pin 1/2 > Bridge pin 3/4 » Set plug 2KTC ID 55504068 > Bridge pin 1/2 SDIC 52.Q (ID 591885), 2 doors and advanced door opening, backward compatible.
1102	**E_CLC_WRONG_VALUE** The signal from the car load measuring device is present but invalid (e.g. signal out of range)
	C1: The car load measuring device is mechanically bad adjusted **A1:**Check mechanical fixation of the car load measuring device
	C2: The car load measuring signal is faulty **A2:**Check general wiring to the car load measuring device.Check for EMC disturbances.

Code	Car Load Cell Messages
	C3: Malfunction of the car load measuring device **A3:**Replace the car load measuring device (e.g. CLC)
1103	**E_CLC_CALIBRATION_ERROR**
	C1: Car load cell: during the calibration procedure an error occurred **A1:**Check: CLC wiring. Check: proper fixation of CLC. Check: parametrization of CLC. Redo: CLC calibration. Replace: CLC
1104	**E_CLC_OVERBRIDGED**
	C1:DIP switch 1 on SCIC (or ISCPU) board in ON position (car load cell deactivated) **A1:**Check: Switch off DIP switch 1 on SCIC (or ISCPU) board. Check: HMI (LDU) menu 107
1105	**E_CLC_NO_CALIBRATION** The calibration of the CLC is not yet done.
1106	**E_CLC_OPERATION_NOT_SUPPORTED_IN_CURRENT_STATE** It's not possible to display the current car load on the user interface (LDU) while entering the configuration menus CF95, CF98 or CF99.
	C1: The car load measuring device is not calibrated **A1:**Calibrate the car load measuring device
1107	**E_CLC_OPERATION_NOT_SUPPORTED_IN_ERROR_STATE** It's not possible to display the current car load on the user interface (LDU) while entering the configuration menus CF95, CF98 or CF99.
	C1: The car load measuring device is in error state (not working) **A1:**Check error log for previous reported car load measuring device errors.

Code	Car Load Cell Messages
1108	**E_CLC_OUT_OF_RANGE** The signal from the car load measuring device is present but invalid (e.g. signal out of range).
	C1: The parameter car load measuring device type is set wrong and does not correspond with the actual mounted car load measuring device **A1:**Check parameter car load measuring device type for correct setting (e.g. CF08 PA08) Measure: --CF96 read out and compare with the parameters indicated in the control cabinet --If necessary, set the indicated parameters again in the CF97 --Re-calibrate the 0 Kg car load » Perform CF98 calibration according to section 5.4.1 --After setting again, perform the load measuring function test » The CF95 must display 0 Kg when the car is empty » Stand in the car and use CF95 to read out your own body weight (example: 74 Kg displays 7), tolerance 1 max. 2 points
	C2: The car load measuring device is mechanically bad adjusted. **A2:**Check mechanical fixation of the car load measuring device Measure: --Check screws --Perform 0-load calibration, see section 5.4.1 --If the cable is damaged, jump to action 3 --Perform the load measuring function test » The CF95 must display 0 when the car is empty » Stand in the car and use CF95 to read out your own body weight (example: 74 Kg displays 7), tolerance +/- 1 point

Code	Car Load Cell Messages
	C3: Malfunction of the car load measuring device **A3:**Measure: Renew Digisense, please comply with section 5.4.1 **Renewal requires corresponding expertise, it may be necessary to contact support.** Tips & Tricks: With the 3100, refer to document K 609754
	C4: The car load measuring signal is faulty **A4:**Measure Renew SDIC printed circuit board Renewal requires corresponding expertise, it may be necessary to contact support Tips & Tricks: It is essential to comply with EMC measures.
1109	**E_CLC_CALIBRATION_ZERO_LOAD_FREQUENCY_OUT_OF_RANGE** The zero car load calibration has failed due to invalid signal (frequency) from the car load measuring system
	C1: The car load measuring device is mechanically bad adjusted **A1:**Check mechanical fixation of the car load measuring device

Code	Car Load Cell Messages
	C2: The parameter car load measuring device type is set wrong and does not correspond with the actual mounted car load measuring device **A2:**Check parameter car load measuring device type for correct setting (e.g. CF08 PA08) Measure: --Re-calibrate the 0 Kg car load » Perform CF-98 calibration according to section 5.4.1 --Perform the load measuring function test » The CF95 must display 0 Kg when the car is empty » Stand in the car and use CF95 to read out your own body weight (example: 74 Kg displays 7), tolerance 1 max 2 points Tips & Tricks: See section 5.4.1 Entry in service booklet, if fault is repeated then perform measures as in 1101. Car is moved by switching off the load measurement TEMPORARILY with menu 107.
	C3: The car load measuring signal is faulty **A3:**Check general wiring to the car load measuring device.Check for EMC disturbances.
	C4: Incorrect working procedure while calibrating the car load measuring device **A4:**Check for correct working procedure (e.g. correct weight in car)
	C5: Malfunction of the car load measuring device **A5:**Replace the car load measuring device
1110	**E_CLC_CALIBRATION_REFERENCE_LOAD FREQUENCY_OUT_OF_RANGE** The reference car load calibration has failed due to invalid signal (frequency) from the car load measuring system

8

Code	Car Load Cell Messages
	C1: The car load measuring device is mechanically bad adjusted **A1:**Check mechanical fixation of the car load measuring device
	C2: The car load measuring signal is faulty **A2:**Check general wiring to the car load measuring device.Check for EMC disturbances.
	C3: Incorrect working procedure while calibrating the car load measuring device **A3:**Check for correct working procedure (e.g. correct weight in car)
	C4: Malfunction of the car load measuring device **A4:**Replace the car load measuring device
1111	**E_CLC_CALIBRATION_REFERENCE_LOAD_WEIGHT_OUT_OF_RANGE**
	C1: Wrong reference load weight entered. **A1:**Check: Reference load weight value. Redo: CLC calibration/ configuration
1112	**E_CLC_CALIBRATION_SLOPE_OUT_OF_RANGE**
	C1: The car load measuring device is adjusted mechanically incorrectly. **A1:**Check mechanical fixation of the car load measuring device. Redo: CLC calibration/ configuration
	C2: Malfunction of the car load measuring device **A2:**Check and replace the car load measuring device. Redo: CLC calibration/ configuration
1113	**E_CLC_CALIBRATION_RATED_LOAD_WEIGHT_OUT_OF_RANGE**
	A1: Check: Rated load weight configuration file
1132	**E_LMS_ASYMETRIC_OVERLOAD**

Code	Car Load Cell Messages
	C1: One of the two car load sensor is measuring car load too low. **A1:**Verify the measured values of the sensors to identify the faulty sensor. Then locate the fault and fix accordingly.

Code	Frequency Converter Error Messages
1501	**E_FC_OVERCURRENT** The maximum current limit on one or more drive inverter output phases to the drive motor has exceeded
	C1: Short circuit in the motor cables or motor windings **A1:**Check power wires and connectors between drive inverter and drive motor.Check motor for short circuit at windings.
	C2: Driving mechanics inhibited or blocked **A2:**Check driving mechanics for blocking (e.g. gear at geared drive, oiling) Check brake for proper opening.
	C3: Sudden heavy load increase **A3:**Check for impermissible loading
	C4: Drive parameter setting **A4:**Check drive parameter and compare with motor type and its data
	C5: Unsuitable drive motor **A5:**Check drive motor for correct dimensioning in the elevator system
1502	**E_FC_OVERVOLTAGE** Overvoltage at the drive inverter internal DC link detected
	C1: Deceleration is too high **A1:**Adjust the deceleration (proposal 0.5 m/s2)
	C2: High over-voltage at mains **A2:**Check mains voltage for disturbances and tolerances

Code	Frequency Converter Error Messages
	C3:FC failure **A3:**Previous actions are failed: replace FC
1503	**E_FC_EARTH_FAULT** The sum of the phase currents of the drive inverter output to the drive motor is not equal zero
	C1: Earth fault **A1:**Check power wires (insulation) to the motor. Check motor for short-circuit at motor windings
	C2: Creeping current **A2:**Check power wires (insulation) to the motor. Check motor for short-circuit at motor windings
1504	**E_FC_INVERTER_FAULT**
	C1: Vacon frequency converter has detected faulty operation in the gate drivers or IGBT bridge - interference fault (EMC)- component failure **A1:**- Reset the fault and restart. If the fault occurs again replace frequency converter.
1505	**E_FC_CHARGING_CONTACTOR** The drive inverter has detected a operation failure at the internal DC link
	C1:EMC disturbances **A1:**Release elevator from blocking state (reset)
	C2: Drive inverter internal component (e.g. charging contactor) defective **A2:**Replace drive inverter
1506	**E_FC_MC_CURNT_NOT_ZERO** The motor current at standstill is not zero.
1509	**E_FC_UNDERVOLTAGE** Undervoltage at the drive inverter internal DC link detected

Code	Frequency Converter Error Messages
	C1: Failure of the mains supply **A1:**Check mains voltage for disturbances (short breaks) and tolerances
	C2: Automatic evacuation was running **A2:**None, normal behavior
	C3: Drive inverter electronics failure **A3:**Replace drive inverter
1510	**E_FC_INPUT_LINE_SUPERVISION** At least one phase of the drive power supply is missing or insufficient
	C1: No power supply **A1:**Check phases on the JH Measure: --On the JH input, measure voltage at all 3 phases (terminals 1, 3 and 5), if not OK: » Building fuse open --On the JH output, measure voltage (terminals 2,4,6), if not OK: » Screws on the JH loose » JH defective Tips & Tricks: Always measure phase-to-phase, never to ground.

8

Code	Frequency Converter Error Messages
	C2: No power supply **A2:**Check SH Measure: --If the contactor SH has picked up, if not » Request support --On the SH input, check the voltage (terminals 2, 4 and 6), if not OK: » Screws on the SH input loose » Defective input cable --On the SH output, measure voltage (terminals 1,3,5), if not OK: » Screws on the SH output loose » SH defective
	C3: No power supply **A3:**Measure JH1 if present Measure: --On the JH1 input, measure the voltage (terminals 1, 3 and 5), if not OK: » Screws on the JH1 input loose » Defective cable SH-JH1 --On the JH1 output, measure voltage (terminals 2,4,6), if not OK: » Loose screws on JH1 » JH1 defective
	C4: No power supply **A4:**Measure VF supply Measure: --On the VF input, measure the voltage (plug X1 terminals 1,2,3), if not OK: » Screws on the VF terminals 1,2,3 loose » Cable defective --Elevator still does not move, 1510 still active» Phase monitoring defective, renew VF

Code	Frequency Converter Error Messages
	C5: Disturbed power supply **A5:**Check for other power consumers (e.g. powered by the same line) which are decreasing the quality of the drive's power supply. Tips & Tricks: Try to disconnect other devices from the power line and see if the problem persists.
1511	**E_FC_OUTPUT_LINE_SUPERVISION** No current detected at one or more drive inverter output phases to the motor
	C1: Power connection between drive inverter and drive motor bad or missing **A1:**Check power wires and connectors between drive inverter and drive motor
1512	**E_FC_BRAKE_CHOPPER_SUPERVISION** The braking chopper at the drive inverter is not working correctly
	C1: The brake resistor is not installed correctly **A1:**Check presence and wiring of brake resistor
	C2: The brake resistor is broken **A2:**Replace brake resistor
	C3: Optocopuler fails **A3:**Replace opto-coupler
	C4: The brake chopper is broken **A4:**Replace drive inverter (FC)
1513	**E_FC_CONVERTER_UNDER_TEMPERATURE**
	C1: Temperature of heat sink below 10C **A1:**none
1514	**E_FC_CONVERTER_OVERTEMPERATURE**

Code	Frequency Converter Error Messages
	C1: Temperature of heatsink over +75C **A1:**Check: cooling air flow. Check: that sink is not dirty. Check: ambient temperature. Check: that switching frequency is not too high compared with ambient temperature and motor load
1515	**E_FC_MOTOR_STALLED** The load on the drive motor is too high (detected by current measurement on drive inverter output)
	C1: Driving mechanics inhibited or blocked **A1:**Check driving mechanics for blocking (e.g. gear at geared drive, oiling). Check brake for proper opening.
	C2: Sudden heavy load increase **A2:**Check for impermissible loading
1516	**E_FC_MOTOR_OVERTEMPERATURE**
	C1: The Vacon frequency converter motor temperature model has detected motor overheat- motor is overloaded **A1:**Check the THMH sensor on the ACVF. Check the cable connection. Verify the sensor is KTY type. Check: Decrease motor load. Check: the temperature model parameters if the motor was not overheated
1517	**E_FC_MOTOR_UNDERLOAD** The load on the drive motor is too low (detected by current measurement on drive inverter output)
	C1: Driving mechanics broken **A1:**Check driving mechanics for breakage (e.g. gear at geared drive, ropes, etc.)
1518	**E_FC_ANALOGUE_INPUT_FAULT**

Code	Frequency Converter Error Messages
	C1: – Wrong analogue input polarity – Component failure on control board – Irrelevant for Schindler Closed Loop application **A1:**Check: polarity of the analogue input; Check: replace frequency converter
1519	**E_FC_OPTION_BOARD_IDENTIFICATION**
	C1: Reading the frequency converter option board has failed **A1:**Check: installation, if installation is correct replace frequency converter.
1520	**E_FC_10V_SUPPLY_REFERENCE**
	C1:+ 10 V reference shorted on control board or option board **A1:**Check: the cabling from +10V reference voltage
1521	**E_FC_24V_SUPPLY**
	C1:+ 24 V reference shorted on control board or option board **A1:**Check: the cabling from +24V reference voltage
1522	**E_FC_EEPROM**
	C1: Parameter restoring error- interference fault- component failure **A1:**Check: when fault is reset the Vacon frequency converter will automatically load parameter default settings.
	A2: Check: all costumer- specific parameter settings after confirmation and if necessary reload them.
	A3: Check: if the fault occurs again replace converter
1523	**E_FC_CHECKSUM**
	C1: See 1522 **A1:**See 1522
	A2: Check: all parameter settings after reset.

Code	Frequency Converter Error Messages
	A3: Check: if the fault occurs again replace frequency converter.
1525	**E_FC_MICROPROCESSOR_WATCHDOG**
	C1:FC Microprocessor hangup- interference fault- component failure **A1:**Check: reset the fault and restart. Check: if fault occurs again replace frequency converter
1526	**E_FC_PANEL_COMMUNICATION**
	C1: The connection between panel and the Vacon frequency converter is not working **A1:**Check: the panel- FC interface cable
1527	**E_FC_COMMUNICATION_ERROR** An error of internal communication of the drive inverter occurred.
1528	**E_FC_MC_CURNT_DIFF** The control deviation of the current controller exceeds the supervisor parameter is_diff_lim.
1529	**E_FC_THERMISTOR_PROTECTION**
	C1:- Thermistor input of the I/O-expander board has detected increase of the motor temperature- Irrelevant for Schindler Closed Loop application **A1:**Check: motor cooling and loading. Check: thermistor connection (if thermistor input of the expander board is not in use, it has to be bridged)
1531	**E_FC_ENCODER_PULSE_MISSING** Invalid signal from the drive motor encoder
	C1: Encoder signal is faulty **A1:**Check general wiring to the encoder
	C2: Encoder signal is noisy **A2:**Check encoder signals for right termination (terminating resistor)

Code	Frequency Converter Error Messages
	C3: Encoder signal receiving stage defective **A3:**Replace electronics (e.g. option board or even entire drive inverter) of corresponding device
	C4: Encoder defective **A4:**Replace encoder
	C5: Drive Parameter set wrong **A5:**Check drive parameter (e.g. nominal frequency, impulse ratio etc.)
	C6: The rotation direction signaled by the encoder (differential inputs) is contrary to the drive motor mains phase sequence **A6:**Change encoder direction parameter (CF 16, PA 14)
1532	**E_FC_ENCODER_DIRECTION** Wrong rotation direction signal from the drive motor encoder
	C1: The rotation direction signaled by the drive motor encoder (differential inputs) is contrary to the drive motor mains phase sequence **A1:**Change encoder direction parameter (CF 16, PA 14)
1533	**E_FC_SPEED_SUPERVISION** Too high difference between reference and actual car drive motor speed. The actual speed is derived from the incremental encoder information.
	C1: Driving mechanics inhibited or even blocked **A1:**Check brake for proper opening. Check driving mechanics for blocking (e.g. gear at geared drive, oiling). Check for bad weight balancing of car and counterweight.
	C2: Motor encoder parameters set wrong **A2:**Check encoder pulse number and encoder direction.
	C3: In case of Modernization: speed regulation **A3:**Reapply commissioning instruction.

8

Code	Frequency Converter Error Messages
	C4: Malfunction of motor encoder **A4:**Replace the encoder.
	C5: Overtraction condition occurred: the car is lifted with the counterweight stuck or vice versa. **A5:**Verify that car and counterweigh move freely and that movement is not affected by misaligned rails or obstacles.
1536	**E_FC_ANALOG_INPUT_UNDER_CURRENT**
	C1: The current in the analog input line is below 4 mA.Signal source has failed. Control cable is broken. **A1:**Check: mechanical break. Check: make sure that the motor is not running on the current limit.Check: Increase Parameter 11.8 SpeedErrorLim without surpassing 1/3 of rated output frequency.
1537	**E_FC_LN_CHOKE_OVERTEMPERATURE** Overtemperature of line choke
1538	**E_FC_FAN_ERROR** The fan of the inverter is not running
1541	**E_FC_EXTERNAL_FAULT**
	C1: Fault is detected from external fault digital input **A1:**Check: the external fault circuit or device.
1542	**E_FC_TOO_HIGH_OR_TOO_LOW_SPEED_AT_TARGET PHSx_RISING** Too high respectively too low car speed detected while the car is landing on floor
	C1: Malfunction of car position detection **A1:**Check hoistway information system (e.g. vanes, PHS)
	C2: Bad balancing of car and counterweight **A2:**Check balancing according instruction manual (TK)
	C3: Too high traction slip **A3:**Check traction means for enough friction

Code	Frequency Converter Error Messages
	C4: Wear of traction means **A4:**Check traction means for wear
1544	**E_FC_WRONG_PHSx_SEQUENCE**
	C1:FC has received a wrong PHSx logical signal sequence from the car processor - first PHSx is rising, last PHSx- rising missing- too rising or too falling PHSx signals one after each other has been received **A1:**Check: Car HW/ SW errors. Check: PHS light barriers Check: PHS 1/2 supply. Check: EMI. Check: floor (level) flags
	C2: Car processor (SDIC) logical error **A2:**Check: PHS sensors and distances to floor level flags. Check: 24 V supply. Check: SDIC wiring. Check: SDIC board
1548	**E_FC_MOTOR_CURRENT_SUPERVISION**
	C1: Motor current (in one or more phases) below expected value **A1:**Check: wiring between FC and output contactors, main contacts. Check: main contactors, main contacts. Check: wiring between output contactors and motor. Check: motor
1551	**E_FC_SPEED_REFERENCE**
	C1: Trying to change speed on the fly is not accepted by the FC. FC initiates an emergency stop. **A1:**Check: SCIC (S00x) software version (Logical error)
	C2: Elevator controller (EC) logical error.(new speed level= ZERO, EVACUATION or UNKNOWN)
1554	**E_FC_BRAKING_RESISTOR_OVERTEMPERATURE**
	C1:FC (Close Loop) Braking resistor overtemperature **A1:**Check: Brake resistor temperature, wiring and/or the bi-metal itself

Code	Frequency Converter Error Messages
1555	**E_FC_HEARTBEAT** The drive node has recognized a lost of communication to the elevator main control.Note, this error mainly occurs together with other errors. Please check message log first for other reported errors.
	C1: Drive node disconnected (e.g. CAN bus) **A1:**Reconnect node
	C2: Data transmission faulty (e.g. CAN bus) **A2:**Check general data line connection.Check for correct data line termination (jumpers and switches, if present) of all devices (PCBs) connected to the data bus.Check shielding of data line (if present).Check for EMC disturbances.
	C3: No or bad power supply of elevator main control **A3:**Check power supply
	C4: Elevator main control defective **A4:**Replace corresponding hardware
1556	**E_FC_OUTPUT_CONTACTORS_SUPERVISOR_PERSISTENT** One or more drive main contactor does not operate as expected.
	C1: The contactor feedback signal is faulty **A1:**Check on the service interface menu (e.g. 723 or 30623) for the state of the contactor feedback signal.Check general wiring of contactor feedback signal (e.g. to DIN1 and DIN2 of drive) Perform a manual reset respectively perform the persistent fatal error reset procedure to set the elevator back in operation.
	C2:A contactor is defective (e.g. contacts stuck together) **A2:**Replace defective contactor (e.g. SFx, SHx, SB). Perform a manual reset respectively perform the persistent fatal error reset procedure to set the elevator back in operation.

Code	Frequency Converter Error Messages
	C3: The contactor control electronics is defective **A3:**Replace defective electronics (PCB, e.g. MCCE). Perform a manual reset respectively perform the persistent fatal error reset procedure to set the elevator back in operation.
	C4: Some drive types do not have a dedicated logical input (HW) reflecting the state of the safety circuit which is used to determine a failure at one of the drive main contactors. These drives are informed by a data telegram (e.g. CAN) from the elevator main control about the state of the safety circuit. The safety circuit has opened but the mentioned telegram was not transferred. **A4:**Check general data line connection.Check for correct data line termination (if present) of all devices connected to the data bus.Check shielding of data line (if present).Perform a manual reset respectively perform the persistent fatal error reset procedure to set the elevator back in operation.
1557	**E_FC_OUTPUT_CONTACTORS_SUPERVISOR** One or more drive main contactor does not operate as expected.Note, this error message might be logged at certain drive types after opening of the safety circuit. In this case no real contactor error exists unless it is confirmed by subsequent logging of error 1556.
	C1: Safety circuit has opened **A1:**When the safety circuit has opened unexpectedly check elevator message log for previous reported messages in order to identify the root cause of the problem
	C2: The contactor feedback signal is faulty **A2:**Check on the service interface menu (e.g. 723 or 30623) for the state of the contactor feedback signal. Check general wiring of contactor feedback signal (e.g. to DIN1 and DIN2 of drive)
	C3:A contactor is defective (e.g. contacts stuck together) **A3:**Replace defective contactor (e.g. SFx, SHx)

Code	Frequency Converter Error Messages
1559	**E_FC_POSITION_CORRECTION**
	C1: Too high position correction error requested by the EC.FC initiates an stop. **A1:**Check: Par. 11.2 PosCorrectLim and s-curve parameters (max. jerks, max. acceleration, rated speed).
	C2: Too high jerk and/ or acceleration values are used. (Par. 1.10, 1.11, 4.2, 4.3, 4.10...4.13). **A2:**Check: for the closed loop application the max. values for Par. 1.10, 1.11, 4.2, 4.3 4.10...4.13), that limit is imposed by the mechanics
	C3: Car ropes slips over the traction pulley **A3:**Check: elevator mechanics (brake, ropes, pulley, etc.)
	C4:EC shaft image
1564	**E_FC_MECHANICAL_BRAKE_KBKB1** Failure of brake, detected at brake contact KB/KB1.
	C1: The brake is not functioning **A1:**Function test Measure: Reset elevator in normal mode » Press reset » Run menu 101 (section 4.9) » Move to all floors Always request support, even if the elevator (temporarily) functions again Tips & Tricks: Here, attempt to make the elevator available to the customer again, at least temporarily

Code	Frequency Converter Error Messages
	C2: Supply of brake problem, both contactors report wrong state at the same time. Brake is not energized. Brake opens only partially. **A2:**Measure: --Check KB signals in menu 724 (see section 4.7) --Check wiring --Check brake diodes Tips & Tricks: Broken off, chafed through, inadequate insulation, etc.
	C3:VF failure **A3:**Renew VF
	C4: The brake position feedback signal(s) KB/KB1 is/are faulty **A4:**Faulty brake contacts It is necessary to have a special tool and corresponding training Measure: --Move elevator to hoistway head using control travel mode (HMI menu 102) --Set KB contacts, see J 635097 for the setting regulation --If setting is not successful then: » KB/KB1 defective Tips & Tricks: When setting the contacts, it may become apparent that the brakes are no longer OK. In that case, the entire brake must be renewed. Possibly also check the brake voltage.
1565	**E_FC_MECHANICAL_BRAKE_KB2** Failure of brake, detected at brake contact KB2.
1567	**E_FC_STANDSTILL** Encoder signals a motor movement while the elevator is stationary with the brakes applied (in closed loop operation).
	C1: The brake is not fully closed. **A1:**Check the brake (for example, the temperature may be too high).

Code	Frequency Converter Error Messages
	C2: The encoder signal is faulty. **A2:**Check the encoder wiring.
	C3: The OPTAx I/O board on the ACVF is defective. **A3:**Check the I/O board and, if necessary, replace the ACVF.
	C4: The encoder is defective. **A4:**Check the encoder and replace it if necessary.
	C5: Check if the error 1567 has occurred more or less simultaneously with the error 2418 from elevator control. If yes, this may indicate a problem with a version < V2.39 of the ACVF control SW. **A5:**Update the ACVF control SW with the latest available version.
1568	**E_FC_MECHANICAL_BRAKE_PERSISTENT** This error occurs if the drive has reported a KB/KB1 failure and the safety circuit does not open in consequence. This is an inconsistent and dangerous situation which would allow driving the car with a faulty break. The elevator is blocked.
	C1: Too many brake faults (KB and KB1) have occurred within a particular time limit (as a rule, more than three faults per 100 trips). **A1:**Function test Measure: --Reset elevator in normal mode » Press reset » Run menu 101 (section 4.9) » Move to all floors --Always request support, even if the elevator (temporarily) functions again Tips & Tricks: Here, attempt to make the elevator available to the customer again, at least temporarily.

Code	Frequency Converter Error Messages
	C2: Too many brake faults (KB and KB1) have occurred within a particular time limit (as a rule, more than three faults per 100 trips). **A2:**Faulty brake contacts Measure: --Move elevator to hoistway head using control travel mode (HMI menu 102, see section 4.9.3) --Set KB contacts, see J 635097 for the setting regulation --If setting is not successful then: » KB/KB1 defective Tips & Tricks: When setting the contacts, it may become apparent that the brakes are no longer OK. In that case, the entire brake must be renewed. Possibly also check the brake voltage.
	C3: Too many brake faults (KB and KB1) have occurred within a particular time limit (as a rule, more than three faults per 100 trips). **A3:**Defective cable connections Measure: --Check KB signals in menu 725 (see section 4.7) --Check wiring Tips & Tricks: Broken off, chafed through, inadequate insulation, etc.
	C4: Invalid drive frequency converter (FC) software installed. **A4:**Update software of drive frequency converter (FC) and the corresponding software on the elevator main controller (SCPU).
	C5: Too many brake faults (KB and KB1) have occurred within a particular time limit (as a rule, more than three faults per 100 trips). **A5:**Check VF Measure: --Renew VF
1569	**E_FC_UF_CURVE**

8

Code	Frequency Converter Error Messages
1570	**E_FC_MECHANICAL_BRAKE_KB** Failure of brake, detected at brake contact KB.
	C1: Too many brake faults (KB and KB1) have occurred within a particular time limit (as a rule, more than three faults per 100 trips). **A1:**Function test Measure: --Reset elevator in normal mode » Press reset » Run menu 101 (section 4.9) » Move to all floors --Always request support, even if the elevator (temporarily) functions again. Tips & Tricks: Here, attempt to make the elevator available to the customer again, at least temporarily.
	C2: Too many brake faults (KB and KB1) have occurred within a particular time limit (as a rule, more than three faults per 100 trips). **A2:**Faulty brake contacts It is necessary to have a special tool and corresponding training Measure: --Move elevator to hoistway head using control travel mode (HMI menu 102, see section 4.9.3) --Set KB contacts, see J 635097 for the setting regulation --If setting is not successful then: » KB/KB1 defective Tips & Tricks: When setting the contacts, it may become apparent that the brakes are no longer OK. In that case, the entire brake must be renewed. Possibly also check the brake voltage

Code	Frequency Converter Error Messages
	C3: Too many brake faults (KB and KB1) have occurred within a particular time limit (as a rule, more than three faults per 100 trips). **A3:**Defective cable connections Measure: Check KB signals in menu 724 (see section 4.7) Check wiring
	C4: The brake position feedback signal KB is faulty **A4:**Check brake contact KB for dirt. Check general wiring KB.
	C5: Brake opens only partially **A5:**Check brake supply voltage
	C6: The brake is mechanically bad adjusted **A6:**Replace brake
	C7: Defective VF **A7:**Check VF Measure: --Renew VF
1571	**E_FC_MECHANICAL_BRAKE_KB1** Failure of brake, detected at brake contact KB1.
	See C&A of 1570 E_FC_MECHANICAL_BRAKE_KB.
1572	**E_FC_MECHANICAL_BRAKE_KBKB1_PERSISTENT_FATAL** Too many brake failures (KB and KB1) have occurred within a certain limit of trips (typically > 3 failures/100trips). The elevator is blocked.
	See C&A of 1570 E_FC_MECHANICAL_BRAKE_KB.
1573	**E_FC_MECHANICAL_BRAKE_KB_PERSISTENT_FATAL** Too many brake failures (KB) have occurred within a certain limit of trips (typically > 3 failures/100trips). The elevator is blocked.
	See C&A of 1570 E_FC_MECHANICAL_BRAKE_KB.

8

Code	Frequency Converter Error Messages
1574	**E_FC_MECHANICAL_BRAKE_KB1_PERSISTENT_FATAL** Too many brake failures (KB1) have occurred within a certain limit of trips (typically > 3 failures/100trips). The elevator is blocked.
	See C&A of 1570 E_FC_MECHANICAL_BRAKE_KB.
1575	**E_FC_WRONG_PHNRx_SEQUENCE** The drive frequency converter received a inconsistent relevel zone signal sequence from the shaft information system (e.g. two times rising edge from PHNR_U or PHNR_D).
	C1: Bad alignment of the PHNR_U/PHNR_D sensors **A1:**Check positions of the PHNR_U/PHNR_D sensors
	C2: Dirt on vanes **A2:**Clean the vanes in the hoistway
	C3: Bad signals through external light source **A3:**Check proper shielding from sunlight (glass shaft)
	C4: Communication problem, EMC **A4:**Check communication on CAN bus (SDIC)
1576	**E_FC_RELEVELING_DISTANCE_EXCEEDED** The drive frequency converter (FC) wasn't able to level the car on the floor because the maximum relevel distance was exceeded.
	C1: Parameter 'max releveling distance' is set wrong **A1:**Check parameter 'max releveling distance' at drive frequency converter. The value should be 1.3 to 1.5 times bigger than the maximum allowed rope elongation at the specific installation.

Code	Frequency Converter Error Messages
	C2: Bad alignment/position of the PHNR_U/PHNR_D sensors **A2:**Check positions of the PHNR_U/PHNR_D sensors. Check that the displacement of the releveling sensors (PHNR_U/PHNR_D) is according to the schematics. They should be inactive when the car is about 15 mm below or above floor level.
1577	**E_FC_RPM_IDENT_STOPPED** The learning procedure of the drive motor revolution parameter (RPM) was aborted.
	C1: The fitter has cancelled the learning procedure by interaction on the user interface **A1:**None
	C2: Mal manipulation by the fitter. Two consecutive trips in the same direction have been initiated. **A2:**Follow instruction J42101241 (V3 or later) 'Biodyn XX C/P BR Commissioning' or TK.
	C3: The safety circuit has opened unexpectedly **A3:**Check where safety circuit was opened. Get rid of the Problem. Restart the drive motor revolution parameter learning procedure.
1578	**E_FC_RPM_IDENT_FAILED** The learning procedure of the drive motor revolution parameter (RPM) wasn't successful. It was aborted after 10 consecutive trip cycles (travel up and down). Note, no parameter will be overwritten.
	C1:manipulation by the fitter. Different destinations per travel direction entered during the learning procedure. **A1:**Follow instruction J42101241 (V3 or later) 'Biodyn XX C/P BR Commissioning' or TK.

Code	Frequency Converter Error Messages
	C2: Wrong initial value of drive motor revolution parameter (CF16 PA38) set for the learning procedure. **A2:**Follow instruction J42101241 (V3 or later) 'Biodyn XX C/P BR Commissioning' or TK.
	C3: The learning procedure wasn't able to bring the delta trip time (difference of the time measured at a trip in up and down direction) lower than a specific limit. **A3:**Follow instruction J42101241 (V3 or later) 'Biodyn XX C/P BR Commissioning' or TK.
	C4: The learning procedure wasn't able to bring the delta nominal speed (difference of the measured actual car speed compared with the reference car speed) lower than a specific limit. **A4:**Follow instruction J42101241 (V3 or later) 'Biodyn XX C/P BR Commissioning' or TK.
1579	**E_FC_IDENT_LOW_TORQUE** The learning procedure of the drive motor revolution parameter (RPM) has failed. It was aborted after one test cycle (travel up and down) because the needed torque measured by the system is not present. Note, no parameter will be overwritten.
	C1: Bad car balancing or bad absence of general mechanical friction (system with low efficiency) **A1:**Load car with nominal load. Restart the drive motor revolution parameter learning procedure.
1580	**E_FC_CURNT_DIFF** The control deviation of the mains current controller exceeds the supervisor parameter im_diff_lim.
1581	**E_FC_IGBT_OVERTEMPERATURE** Overtemperature of the IGBT power module

Code	Frequency Converter Error Messages
1589	**E_FC_MISSING_SAFETY_CHAIN_SUPPLY** During or while starting a trip (while the motor and brake contactors are activated) the drive has detected a interrupted safety circuit
	C1: Safety circuit signal is faulty **A1:**Check safety circuit wiring between elevator control and drive
	C2: The trip was started before the door has finished bouncing after reaching the closed position **A2:**Increase parameter start delay (menu CF 03)
	C3:JEM on PEBO (SNGL, SEM) is on wrong position **A3:**Switch JEM to position OFF
1590	**E_FC_DRIVE_START_ANGLE_IDENT_FAILED** While starting a trip the drive has detected a failed identification of start angle. Fatal after 3 times/h
1594	**E_FC_HW_ERROR** HW failure of the converter detected.
1595	**E_FC_HW_MISMATCH** The configuration does not match with recognized hardware.
1596	**E_FC_SW_WARNING** The software detected an exceptional behavior of the SW or the HW.
1597	**E_FC_MEMORY_ERROR** Incorrect memory access.
1599	**E_FC_INVERTER_INTERNAL_FAILURE** The drive inverter has an internal problem. See extra info for more information.
	Cause according extra info. Action according extra info.

Code	SEM Messages
1601	**E_SEM_GENERAL_ERROR**

Code	Chip Card (SIM card) Messages
1901	**E_CHIP_CARD_WRONG_DEVICE** The elevator main controller has a software internal problem accessing the SIM card
	C1: Internal elevator main controller software error **A1:**Upgrade elevator main controller software (SDIC)
1902	**E_CHIP_CARD_FILE_NOT_FOUND** The elevator main controller expects specific files on the SIM card. At least one expected file is missing.
	C1:SIM card with wrong software version inserted **A1:**Check SIM card software version. Get right SIM card, insert it and perform a elevator main controller reset.
1904	**E_CHIP_CARD_NOT_FORMATED** The SIM card inserted on the elevator main controller board is not formatted.
	C1: The SIM card is not formatted **A1:**Get right SIM card, insert it and perform a elevator main controller reset.
1905	**E_CHIP_CARD_NO_OR_NO_SCHINDLER_CARD** No or invalid SIM card inserted on the elevator main controller board.
	C1: No SIM card inserted **A1:**Get right SIM card, insert it and perform a elevator main controller reset.
	C2: The SIM card is inserted wrongly **A2:**Remove SIM card, insert it correctly and perform a elevator main controller reset.

Code	Chip Card (SIM card) Messages
	C3: No Schindler SIM card inserted **A3:**Get right SIM card, insert it and perform a elevator main controller reset.
1906	**E_CHIP_CARD_READING_ERROR** There is a problem reading the SIM card. Note, this error typically relates to any other SIM card error.
	C1:SIM card reading error **A1:**Check error log for other SIM card errors. Please refer to corresponding causes and actions. If no other error is reported, replace SIM card.
1907	**E_CHIP_CARD_WRITING_ERROR** There is a problem writing to the SIM card. Note, this error typically relates to any other SIM card error.
	C1:SIM card writing error **A1:**Check error log for other SIM card errors. Please refer to corresponding causes and actions. If no other error is reported, replace SIM card.
1908	**E_CHIP_CARD_WRONG_FILE_SYSTEM_VERSION** Invalid Schindler SIM card inserted on the elevator main controller board.
	C1:SIM card has a wrong file system version respectively a wrong data format. This means the SIM card does not contain the data (e.g. FC parameters) as expected by the elevator main controller (SDIC). **A1:**Get right SIM card, insert it and perform an elevator main controller reset.

Code	Trip Manager Messages
2002	**E_TRIP_LEARNING LEVEL MISSING** The number of floor levels counted during the learning trip in upward direction does not correspond to the one counted with the check in downward direction.

Code	Trip Manager Messages
	C1: Bad alignment of floor sensors with magnets (KS) or PHS flags in shaft **A1:**Check alignment of magnets (KS) or PHS flags.Check position of floor sensor. Check magnet to sensor distance.
	C2: Bad floor sensor signal transmission **A2:**Check general electrical wiring of the floor sensor signal (e.g. connections at KS/PHS, at SDIC, ..)
	C3: Floor sensor(s) defect **A3:**Replace floor sensor(s)
	C4: Faulty input on the interface board (PCB) for the floor sensor **A4:**Replace interface board (SDIC)
2003	**E_TRIP_LEARNING NUMBER OF LEVELS VARY**
	C1: The number of floor levels counted during the learning trip in up direction does not correspond to the one counted during the down direction checking phase. **A1:**Check magnets / PHS flags. Check KS / PHS sensors / cable
2004	**E_TRIP_LEARNING LEVEL OUTSIDE ARRAY LIMITS** Indicates that the number of learned floors is invalid
	C1: The number of learned floors exceeded the max floors supported by the elevator system **A1:**Check the number of door zone indicators in the hoistway (e.g. magnets, vanes)
	C2: The number of learned floors does not correspond with the commissioning data **A2:**Crosscheck the number of door zone indicators in the hoistway (e.g. magnets, vanes) with the allowed number of floors according the commissioning data
2005	**E_TRIP_POSITION_TARGET_NOT_REACHED** The elevator has finished a trip but the car is signalized by the hoistway information system not to be in door zone.

Code	Trip Manager Messages
	C1: Different causes **A1:**Check elevator message log for previous reported messages in order to identify the root cause of the problem.
	C2: Signal of door zone detection faulty **A2:**Check door zone sensors (e.g. PHSx) and wiring.
2006	**E_TRIP_WARNING E_TRIP_POSITION_TARGET_NOT REACHED**
2007	**E_TRIP_POSITION_MOVE_NOT_IN_DOOR_ZONE** The elevator was intended to start a trip while the car was not detected to be on floor. Note, this error only occurs while not in manual trip operation (e.g. inspection)
	C1: Different causes **A1:**Check elevator message log for previous reported messages
	C2: Failure at door zone detection of hoistway information system **A2:**Check door zone detection of hoistway information system (e.g. PHSx, mechanical adjustment, defective sensor etc.)
2008	**E_TRIP_SYNCHRO_ROUGH_POSITION_STATE_ERROR**
	C1: The synchronization or (under certain conditions) the service trip did receive a inconsistent KSE update **A1:**Check KSE magnets.Check KSE magnet switch.Redo learning trip.
2009	**E_TRIP_WARNING_TRIP_SYNCHRO_ROUGH_POSITION_STATE ERROR**
2010	**E_TRIP_LEARNING_DIRECTION_UNKNOWN_RECEIVED**
	C1: During the learning trip, the direction of travel becomes unknown. This can only happen if some serious error in the elevator system occurred. The trip is stopped. **A1:**Start new learning trip

Code	Trip Manager Messages
2011	**E_TRIP_LEARNING_WRONG_MAGNET_ORDER**
	C1: Releveling failure
2012	**E_TRIP_RELEVELING_FATAL_ERROR**
	C1: Releveling failure caused by safety chain opening **A1:**Check: Safety Chain Circuit; SUET
2060	**E_TRIP_LEARNING_MINIMAL_TRAVEL_DISTANCE**
	C1: The distance between two flags (floors) is less than the allowed minimal traveling distance (300mm) **A1:**Check flag distance, PHS position
2061	**E_TRIP_LEARNING_INTOLERABLE_FLAG_LENGTH** Detected a too long or too short door zone.Note, this error is typically reported at the learning travel.
	C1: Tacho factor or drive pulley diameter invalid **A1:**Check corresponding parameter for right values
	C2: Signal of door zone detection faulty **A2:**Check door zone sensors (e.g. PHS)
	C3: Flag length out of range **A3:**Install correct flags
2062	**E_TRIP_LEARNING_UPPER_FLAG_EDGE_ALREADY_SET**
	C1: The EC application tried to set the value for an upper flag that was already set. This can only happen, if we do not allow overwriting of the shaft image (e.g. while traveling upwards, where no value should have been set before). **A1:**Redo learning trip
	C2:EMC disturbance
2063	**E_TRIP_LEARNING_UPPER_FLAG_EDGE_NOT_SET**
	C1: The EC application missed to set the upper flag for a level. **A1:**Redo learning trip

Code	Trip Manager Messages
2064	**E_TRIP_LEARNING_LOWER_FLAG_EDGE_ALREADY_SET**
	C1: The EC application tried to set the value for a lower flag that was already set. This can only happen, if we do not allow overwriting of the shaft image (e.g. while traveling upwards, where no value should have been set before). **A1:**Redo learning trip
	C2: Possible problems with shaft info circuit (PHS, light barrier, magnet switch)
	C3:EMC disturbances
2065	**E_TRIP_LEARNING_LOWER_FLAG_EDGE_NOT_SET**
	C1: The EC application missed to set the lower flag for a level. **A1:**Redo learning trip
	C2: Problems with shaft info circuit? (PHS, light barrier, magnet switch)
	C3:EMC disturbance?
2066	**E_TRIP_LEARNING_INVALID_DOOR_ENTRANCE_SIDE**
	C1: The EC application tried to set a door side that is not allowed (its not the same as 'already set', an invalid value is the problem.). **A1:**Redo learning trip
	C2: Problems with shaft info circuit? (PHS, light barrier, magnet switch)
	C3:EMC disturbance?
2067	**E_TRIP_LEARNING_DOOR_ENTRANCE_SIDE_NOT_SET**
	C1: The EC application missed to set a door side for a level. **A1:**Check flags; Check light barrier cable; Check encoder; Redo learning trip

Code	Trip Manager Messages
2068	**E_TRIP_LEARNING_DOOR_ENTRANCE_SIDE_ALREADY_SET**
	C1: The EC application tried to set a door entrance side that has already been set. **A1:**Check flags; Check light barrier cable; Check encoder; Redo learning trip
2069	**E_TRIP_LEARNING_INVALID_LOWER_FLAG_EDGE**
	C1: The EC application tried to set the height of a lower flag edge bigger than the height of the upper flag. **A1:**Check flags; Check light barrier cable; Check encoder; Redo learning trip
2070	**E_TRIP_LEARNING_INVALID_UPPER_FLAG_EDGE**
	C1: The EC application tried to set the height of a lower flag edge smaller than the height of the upper lower flag. **A1:**Check flags; Check light barrier cable; Check encode; Redo learning trip
2071	**E_TRIP_LEARNING_UPPER_FLAG_EDGE_DIFFER**
	C1: The EC application sets an new upper flag edge height. The difference between the last value and this one is bigger than accepted. **A1:**Check rope slip; Check encoder; Redo learning trip (Warning: Do not change load while performing a learning trip!)
2072	**E_TRIP_LEARNING_LOWER_FLAG_EDGE_DIFFER**
	C1: The EC application sets an new lower flag edge height. The difference between the last value and this one is bigger than accepted. **A1:**Check rope slip; Check encoder; Redo learning trip (Warning: Do not change load while performing a learning trip!)
2073	**E_TRIP_LEARNING_DOOR_ENTRANCE_SIDE_DIFFER**

Code	Trip Manager Messages
	C1: The EC application tried to set a door entrance side. The side was already set, we allowed overwriting but the last value does not correspond with the new one. **A1:**Check rope slip; Check encoder; Redo learning trip (Warning: Do not change load while performing a learning trip!)
2074	**E_TRIP_POSITION_NESTED_MOVE**
	C1:A client requests a move in position mode, but the drive did not have the time to acknowledge the last. **A1:**Reset EC
2075	**E_TRIP_POSITION_CORRECTION_TOO_BIG** The first limit for the correction of the car position (typically 30 mm) during a trip has exceeded. Note, this warning is only applicable for closed loop drives. The current running trip does not get interrupted.
	C1: Insufficient traction **A1:**Check for excessive rope slip
	C2: Mechanical problem with motor tacho **A2:**Check tacho on hoisting machine
	C3: Excessive STM elongation during trip **A3:**Check STM (type and number)
	C4: Problem with floor sensor (flag/photocell) **A4:**Check correct installation/operation of floor sensors in hoistway
	C5: Delayed transmission from floor sensor interface board (SDIC) to the drive frequency converter (FC) **A5:**Check communication on CAN bus (termination)
	C6: Faulty input on the interface board (PCB) for the floor sensor **A6:**Replace interface board (SDIC)

Code	Trip Manager Messages
	C7: Forget to install a blind floor **A7:**Install a blind floor
2076	**E_TRIP_LEARNING_AVERAGE_FLAG_LENGTH_EXCEEDED**
	C1: The calculation of the average flag length after the adjustment of the FC parameter traction "PULLEY_DIAMETER" results in a value bigger than the tolerated limit. **A1:**Check shaft information. Check flags length. Check FC parameter 11.26 "Traction Pulley Diameter-DD"; Redo learning trip
2077	**E_TRIP_LEARNING_FLAG_EDGE_SEQUENCE**
	C1: During the learning trip the same edge is received twice in line.E.g. two times a rising edge with no falling edge in between. **A1:**Check light barrier. Check SDIC board; Check CAN bus (cable, termination, plugs, EMC)
2078	**E_TRIP_FINAL_LEVEL_NOT_FOUND** The terminal floor indication (top or bottom floor) is faulty
	C1: The distance between the signalization of the hoistway end and the corresponding terminal floor (top respectively bottom floor) is out of range **A1:**Check position of terminal floor sensor (top or bottom floor, e.g. PHS); Check position of hoistway end sensor (e.g. KSEx)
	C2: The signal indicating the hoistway end is faulty **A2:**Check hoistway end sensors (e.g. KSEx).Check general wiring to this sensor.
	C3: The signal indicating the terminal floor is faulty **A3:**Check (terminal) floor sensor (e.g. PHS).Check general wiring to this sensor.

Code	Trip Manager Messages
2079	**E_TRIP_PHSx_SIGNAL_PERSISTENTLY_BRIDGED** During a trip the shaft information signal PHS and/or PHS1 do not change their state as expected. They are permanently active (e.g. bridged for test).
	C1:PHS/PHS1 bridge for test still mounted **A1:**Remove bridge. Perform a manual reset respectively perform the persistent fatal error reset procedure to set the elevator back in operation.
	C2: Connection to PHS or PHS1 bad **A2:**Check wiring to PHS/PHS1.Perform a manual reset respectively perform the persistent fatal error reset procedure to set the elevator back in operation.
	C3:PHS or PHS1 sensor defect **A3:**Replace corresponding sensor. Perform a manual reset respectively perform the persistent fatal error reset procedure to set the elevator back in operation.
2080	**E_TRIP_PHNR_SIGNAL_INCONSISTENT** After each landing of the car on a floor a consistency check of the shaft information signals for releveling PHNR_U and PHNR_D gets performed. This error occurs if both of the signals are inactive at this point of time.
	C1: Sensor position of PHNR_U/PHNR_D wrong **A1:**Check these sensors for misalignment
	C2: The shaft information signals PHNR_U/PHNR_D are faulty **A2:**Check sensors PHNR_U/PHNR_D for dirt or defect. Check general wiring PHNR_U/PHNR_D.
2082	**E_TRIP_HOURS_IN_SERVICE_MISMATCH**
2083	**E_TRIP_HOURS_IN_SERVICE_ENTERED_MANUALLY**
2084	**E_TRIP_COUNTER_MISMATCH**
2085	**E_TRIP_COUNTER_ENTERED_MANUALLY**

8

Code	Trip Manager Messages
2086	**E_TRIP_LEARNING_BLIND_FLOOR_INVALID**
	A blind floor flag (PHSB) has been set on a floor with doors (PHS).
2087	**E_TRIP_LEARNING_BLIND_FLOOR_UNEXPECTED**
	A blind floor flag (PHSB) has been detected on a system where no blind floors are expected.

Code	Hoistway Messages
2101	**E_SHAFT_UNDEFINED**
	C1: Shaft is undefined yet. **A1:**Do: Manually reset the system on microprocessor board
2102	**E_SHAFT_LEVEL_MISSING**
	C1: The shaft information contains missing levels. **A1:**check shaft info
2103	**E_SHAFT_NUMBER_OF_LEVELS_VARY**
	C1: The total number of levels stored in the EEPROM differs from the actually measured. **A1:**Redo: learning trip
2104	**E_SHAFT_LEVEL_OUTSIDE_ARRAY_LIMITS**
	C1: Too many floors. While setting up the RAM shaft image at the application startup a level is addressed outside the array limits. [Level <0 or level > 15] **A1:**Check: number of magnets/ flags. Redo: learning trip
2105	**E_SHAFT_OPEN_EEPROM_FILE**
	Code reserved only for internal use
2106	**E_SHAFT_READ_EEPROM_FILE**
	Code reserved, only for internal use.
2107	**E_SHAFT_WRITE_EEPROM_FILE**
	Code reserved, only for internal use.

Code	Hoistway Messages
2108	**E_SHAFT_ERASE_EEPROM_FILE**
	Code reserved, only for internal use.
2109	**E_SHAFT_CREATE_EEPROM_FILE**
	Code reserved, only for internal use.
2110	**E_SHAFT_IMAGE_IN_ERROR_STATE**
	C1: The shaft image object is in error state due to an error at the application startup.
	A1: The system error handling automatically does a synchronization trip.
2111	**E_SHAFT_INVALID_ROUGH_POSITION**
	C1: This error can result from an invalid respectively incomplete signaling of KSE indicator changes.
	A1: See 2110.
2112	**E_SHAFT_INVALID_SHAFT_STAGE**
	C1: This error can result from an invalid respectively incomplete signaling of PHS respectively KS indicator changes.
	A1: See 2110.
2113	**E_SHAFT_INVALID_CURRENT_LEVEL**
	C1: This error can result from an invalid respectively incomplete signaling of PHS respectively KS indicator changes.
	A1: See 2110..
2114	**E_SHAFT_POSITION_LOST_WHILE_STATIONARY**
	The hoistway information system has indicated the car out of the door zone (car not on floor) while the car is expected standing on floor

8

Code	Hoistway Messages
	C1: Failure at door zone detection of hoistway information system **A1:**Check door zone detection of hoistway information system (e.g. PHSx, mechanical adjustment, defective sensor etc.)
	C2: Car has left door zone **A2:**Check suspension means (e.g. ropes for too high elongation). Check drive brake
2115	**E_SHAFT_POSITION_LOST_WHILE_BOUNCING**
	C1:A KSE or level indicator change is signaled while the car is stabilizing. This stabilizing period is actually defined with 2 seconds. **A1:**See 2110.
2116	**E_SHAFT_INCONSTANT_LEVEL_INDICATORS**
	C1: This error can result from an invalid respectively incomplete signaling of PHS respectively KS indicator changes. **A1:**See 2110.
2117	**E_SHAFT_CAR_POSITION_CONTROLLER_IN_SYNCHRONOUS_STATE** Code reserved, only for internal use.
2118	**E_SHAFT_INVALID_TRAVEL_MODE**
	A1: The system error handling automatically does a synchronization trip.
2119	**E_SHAFT_INCONSTANT_KSE_INDICATORS**
	A1: The system error handling automatically does a synchronization trip.
2120	**E_SHAFT_ROUGH_POSITION_HANDLING_DISABLED** Code reserved, only for internal use.

Code	Hoistway Messages
2121	**E_SHAFT_OPERATION_NOT_SUPPORTED_IN_ CURRENT_STATE** Code reserved, only for internal use.
2122	**E_SHAFT_INVALID_SHAFT_IMAGE_FOR_THIS_ SHAFT_TYPE** Code reserved, only for internal use.
2123	**E_SHAFT_IMAGE_FILE_DATA_COULD_NOT_BE_ CONVERTED** Code reserved, only for internal use.
2124	**E_SHAFT_INVALID_SHAFT_IMAGE_VERSION_ NUMBER** Code reserved, only for internal use.
2125	**E_SHAFT_ACCESS_PERSISTENT_MEDIUM**
	C1: Could not write to EEPROM. **A1:**Do: Replace SCIC board (bigger EEPROM needed). Or reduce number of levels.
2126	**E_SHAFT_WRONG_MAGNET_TRANSITION**
	C1:sequence of N->N or S->S **A1:**Check: KS position Do: learning trip
2127	**E_SHAFT_NO_KS1_DETECTION_POSSIBLE**
	A1: Check: KS/ KS1 presence
2128	**E_SHAFT_WRONG_MAGNET_TRANSITION_INFORMATI ON**
	C1:same as 2126 but classified as recoverable instead of fatal **A1:**same as 2126
2129	**E_SHAFT_ROUGH_POSITION_MISMATCH**
	A1: Check: Shaft information, flags length. Do: Learning trip
2130	**E_SHAFT_WRONG_INITIALIZATION**

8

Code	Hoistway Messages
	C1: Shaft image: wrong initialization **A1:**Do: Learning trip
2131	**E_SHAFT_LUET_KS_KS1_INCONSISTENT**
	A1: Check: KS/ KS1 presence and position. Do: Learning trip
2132	**E_SHAFT_INCONSISTENT_KSE_D_U_STATE** The signal indicating the hoistway end is invalid (e.g. inconsistent). Failure at hoistway information.
	C1:A hoistway end position indicator (e.g. KSE, KSE_U, KSE_D) is missing or not at the correct position. **A1:**Check correct position of the corresponding indicator (e.g. magnet and its polarity, vane)
	C2:A hoistway end position indicator (e.g. KSE, KSE_U, KSE_D) is missing or not at the correct position. **A2:**Check KSE Measure: --Check if the magnets are still correctly installed » Distance between magnet switch and magnet must be 8 to 12 mm --Check magnet polarity, uppermost KSE U (uppermost stop) north polarity, bottom south polarity --Check connections on magnet switch --Check KSE connections on the SDIC printed circuit board --Measure on KSE plug » Measure pin 2 KSE D against ground » Measure pin 4 KSE U against ground Tips & Tricks: Car at the lowermost floor: » KSE D=0 and KSE U=1 Car at the uppermost stop: » KSE D=1 and KSE U=0 Car at every other stop: » KSE D=1 and KSE U=1

Code	Hoistway Messages
	C3:A hoistway end position sensor is faulty. **A3:**Renew KSE magnetic switch Measure: --Renew defective KSE --Force teach-in travel » Menu 10 116=1 Tips & Tricks: Magnetic switch ID 00418481 Ring magnet ID 00291284
	C4:A hoistway end position sensor is faulty. **A4:**Check functionality of corresponding sensor. Replace it if necessary.
	C5: Wiring to hoistway end position sensor is faulty. **A5:**Check wiring at the corresponding sensor.
2133	**E_SHAFT_MIX_KSE_AND_KSE_D_U** The elevator control received signals from the hoistway information system which aren't signals of the expected hoistway information system type.

8

Code	Hoistway Messages
	C1: Wiring to hoistway end position sensor is faulty. **A1:**Check KSE Measure: --Check if the magnets are still correctly installed » Distance between magnet switch and magnet must be 8 to 12 mm --Check magnet polarity, uppermost KSE-U (uppermost stop) north polarity, bottom south polarity --Check connections on magnet switch --Check KSE connections on the SDIC printed circuit board --Measure on KSE plug » Measure pin 2 KSE D against ground » Measure pin 4 KSE U against ground Tips & Tricks: Car at the lowermost floor: » KSE D=0 and KSE U =1 Car at the uppermost stop: » KSE D=1 and KSE U =0 Car at every other stop: » KSE D=1 and KSE U =1 Details section 4.9
	C2: Wiring to hoistway end position sensor is faulty. **A2:**Check wiring and correct connection of the corresponding sensor (KSE, KSE_U, KSE_D).
	C3: Wiring to hoistway end position sensor is faulty. **A3:**Renew KSE magnetic switch Measure: --Renew defective KSE --Force teach-in travel » Menu 10 116=1 Tips & Tricks: Magnetic switch ID 00418481 Ring magnet ID 00291284
2134	**E_SHAFT_KSE_AND_TSD_OPTION**

Code	Hoistway Messages
2135	**E_SHAFT_TSD_SIGNAL_CHANGE** The elevator has detected an unexpected change of the signal TSD (state of the presence of the temporary safety device).
	C1: The wiring at the signal TSD is faulty **A1:**Check general wiring
	C2: The elevator main controller board was exchanged by a board from another elevator **A2:**Perform the learning travel. At no success replace elevator main controller board (e.g. SCPU) by a new one from the factory.
2136	**E_SHAFT_ROUGH_POS_MISSMATCH_TOP** Failure at the upper hoistway end detection
	C1: Wiring to the upper hoistway end position sensor is faulty. **A1:**Check wiring at the corresponding sensor.
	C2: The upper hoistway end position sensor is faulty. **A2:**Check functionality of corresponding sensor. Replace it if necessary.
2137	**E_SHAFT_SPEED_SUPERVISION_AT_ROUGH_POS_CHANGE** The car was not decelerating (still traveling in normal speed) while it was reaching the shaft end (at KSE) because the wrong number of floors was counted. Failure in shaft information system.
	C1: Bad alignment of floor sensors (KS/KS1) with magnets in shaft **A1:**Check alignment of magnets. Check position of floor sensors. Check for lost magnets. Check magnet to sensors distance.

Code	Hoistway Messages
	C2: Bad floor sensor signal transmission **A2:**Check general electrical wiring of the floor sensor signal (e.g. connections at KS/KS1, at SDIC, ..)
	C3: Floor sensor (KS/KS1) defect **A3:**Replace corresponding floor sensor
	C4: Faulty input on the interface board (PCB) for the floor sensor **A4:**Replace interface board (SDIC)
	C5: One or more floor indication magnet of the shaft information system is/are mounted with wrong magnetic orientation **A5:**Check entire shaft for the correct magnetic orientation (N,S) of these magnets
2138	**E_SHAFT_CAR_RELEVELING_ZONE_SIGNAL_INVALID** The signal indicating the car releveling zone is invalid (e.g. inconsistent). Failure at hoistway information.
	C1:A car releveling zone indicator is missing or not at the correct position **A1:**Check correct position of the corresponding indicators (e.g. alignment of magnets, vanes)
	C2:A car releveling zone sensor is defective **A2:**Check functionality of corresponding sensor (e.g. PHS, PHNR_U, PHNR_D). Replace it if necessary.
	C3:A car releveling zone sensor signal is faulty **A3:**Check general wiring of corresponding sensors (e.g. PHNR_U, PHNR_D). Check at optical sensors (e.g. PHS, PHNR_U. PHNR_D) vanes and sensors for dirt and proper shielding from sunlight (e.g. at glass shaft).
	C4: Bad data transmission **A4:**Some hoistway information system transfer their data via data bus. Check data bus (e.g CAN) for correct data transmission (e.g. bad shielding of data wire).

Code	Hoistway Messages
2139	**E_SHAFT_ROUGH_POS_MISSMATCH_BOTTOM**
	Failure at the lower hoistway end detection.
2160	**E_SHAFT_MINIMAL_TRAVEL_DISTANCE**
	C1: While validating the RAM shaft image at the application startup, a too small travel distance (< 300 mm) between the levels is detected. **A1:**Redo: Learning trip
2161	**E_SHAFT_INTOLERABLE_FLAG_LENGTH**
	C1: At application startup (reset): While setting up the levels in the RAM shaft image an intolerable flag length is detected. **A1:**Check: for rope slippage. Check: encoder. Redo: learning trip. Warning: do not change the load during learning
2162	**E_SHAFT_UPPER_FLAG_EDGE_ALREADY_SET**
	C1: While setting up the RAM shaft image at the application startup the upper flag edge position of a level is about to be set multiple times. **A1:**Check: rope slip. Check: encoder. Check: flag position. Redo: learning trip
	C2: Problems with shaft info circuit (PHS, light barrier, magnet switch)
	C3:EMC disturbances
2163	**E_SHAFT-UPPER_FLAG_EDGE_NOT_SET**
	C1: While setting the RAM shaft image at the application startup, a missing upper flag edge position is detected. **A1:**Manually erase the shaft image file on the persistent data medium and run the synchronization trip.
2164	**E_SHAFT_LOWER_FLAG_EDGE_ALREADY_SET**

8

Code	Hoistway Messages
	C1: While setting up the RAM shaft image at the application startup the lower flag edge position of a level is about to be set multiple times.E.g. caused by arriving at the flag edge toggling many times. **A1:**Redo: learning trip
	C2:EMC disturbances
	C3: Problems with shaft info circuit.(PHS, light barrier, magnet switch)
2165	**E_SHAFT_LOWER_FLAG_EDGE_NOT_SET**
	C1: While setting the RAM shaft image at the application startup a missing lower flag edge position is detected. **A1:**Redo: learning trip
	C2: Problems with shaft info circuit (PHS, light barrier, magnet switch)
	C3:EMC disturbances
2166	**E_SHAFT_INVALID_DOOR_ENTRANCE_SIDE**
	C1: While setting up the RAM shaft image at the application startup an invalid door entrance side is detected. **A1:**Redo: learning trip
	C2: See 2165
	C3: See 2165
2167	**E_ESHAFT_DOOR_ENTRANCE_SIDE_NOT_SET**
	C1: While setting up the RAM shaft image at the application startup a missing door entrance side is detected. **A1:**Redo: learning trip
	C2: Problems with shaft info circuit (PHS, PHUET, light barrier, magnet switch)
	C3:EMC disturbances
2168	**E_SHAFT_DOOR_ENTRANCE_SIDE_ALREADY_SET**

Code	Hoistway Messages
	C1: While setting up the RAM shaft image at the application startup the door entrance side of a level is about to be set multiple times. **A1:**Redo: learning trip
	C2: Problems with shaft info circuit (PHS, PHUET, light barrier, magnet switch)
	C3:EMC disturbances
2169	**E_SHAFT_INVALID_LOWER_FLAG_EDGE**
	C1: While setting up the RAM shaft image at the application startup an invalid lower flag edge position is detected. **A1:**Redo: learning trip
	C2: See 2165
	C3: See 2165
2170	**E_SHAFT_INVALID_UPPER_FLAG_EDGE**
	C1: While setting up the RAM shaft image at the application startup an invalid upper flag edge position is detected. **A1:**Redo: learning trip
	C2: See 2165
	C3: See 2165
2171	**E_SHAFT_UPPER_FLAG_EDGE_DIFFER** Code reserved, only for internal use.
2172	**E_SHAFT_LOWER_FLAG_EDGE_DIFFER** Code reserved, only for internal use.
2173	**E_SHAFT_DOOR_ENTRANCE_SIDE_DIFFER** Code reserved, only for internal use.
2174	**E_SHAFT_INTOLERABLE_DISCREPANCY_BETWEEN_FLAG_EDGES** Code reserved, only for internal use.

Code	Hoistway Messages
2175	**E_SHAFT_NO_PRECISE_LEVEL_POSITION_SHAFT_IMAGE_FILE** Code reserved, only for internal use.
2176	**E_SHAFT_NEGATIVE_FLAG_HEIGHT** Code reserved, only for internal use.
2177	**E_FIRST_HOISTWAY_END_UP_SENSOR_REACHED** The car reaches the first hoistway end sensor at the last floor and the controller performed a stop.
	C1: This event appears frequently in the log: KLS-U sensor broken. **A1:**Check KLS-U sensor
	C1: This event appears frequently in the log: KLS-U sensor not in the right place **A1:**Check in the shaft if it is properly installed
	C1: This event appears frequently in the log: landing problems. **A1:**Check the reason of frequent landing problems.
2178	**E_FIRST_HOISTWAY_END_DOWN_SENSOR_REACHED** The car reaches the first hoistway end sensor at the last floor and the controller performed a stop.
	C1: This event appears frequently in the log: KLS-U sensor broken. **A1:**Check KLS-U sensor
	C1: This event appears frequently in the log: KLS-U sensor not in the right place **A1:**Check in the shaft if it is properly installed
	C1: This event appears frequently in the log: landing problems. **A1:**Check the reason of frequent landing problems.

Code	Hoistway Messages
2190	**E_SHAFT_NO_LEVEL_POSITION_SHAFT_IMAGE_FILE** Code reserved, only for internal use.

Code	FA Messages
2200 to 2205	**FA drive errors (2 speed systems) Not used with Schindler 3100/3300/5300** If one of these error occurs please contact hotline Locarno or a specialists.

Code	Frequency Converter Alarm Messages
2315	**E_FC_ALARM_MOTOR_STALLED**
	C1:FC (Closed Loop) Motor stalled **A1:**Check motor. Check electromechanical brake MGB
	C2: The motor stall protection has tripped e.g. electromechanical brake (MGB) has not opened. Note: You can program if this condition generates A15 or F15 or nothing.
2316	**E_FC_ALARM_MOTOR_OVERTEMPERATURE** The car drive hoisting machine has exceeded its operating temperature
	C1: The heat dissipation is not working **A1:**Check operation of heat dissipation device (e.g. fan or forced ventilation) if present
	C2: Too hot ambient air temperature (e.g. direct sunlight at glass shaft) **A2:**Wait for cool down
	C3: The temperature feedback signal is faulty **A3:**Check general wiring to temperature sensor.Check operation of temperature sensor.
	C4: Too intensive operation (e.g. too many trips per time unit) **A4:**Wait for cool down
2317	**E_FC_ALARM_MOTOR_UNDERLOAD**

8

Code	Frequency Converter Alarm Messages
	C1:FC (Closed Loop) Motor underload **A1:**Check gear box
2324	**E_FC_ALARM_HISTORY_MAYBE_LOST**
	C1: The values in the Fault history, MWh-counters or operating day/hour counters might have been changed in a previous mains interruption. **A1:**Does not need any actions. Take a critical attitude to these values
2328	**E_FC_ALARM_APPLICATION_CHANGE_FAILED**
	C1: Application change failed **A1:**Choose the application again and press the enter button
2330	**E_FC_ALARM_UNBALANCED_CURRENTS**
	C1: Unbalanced current fault, the load on the segments in not equal. **A1:**Replace frequency converter
2342	**E_FC_ALARM_TOO_HIGH_OR_TOO_LOW_SPEED_AT_ TARGET_PHS**
	C1:FC (Closed Loop) Wrong speed at target (last PHSx rising edge) **A1:**Check flags position Check PHS1/2 sensors. Redo learning trip
2345	**E_FC_ALARM_CONVERTER_OVERTEMPERATURE**
	C1: Temperature of heatsink over programmable temp. value (e.g. +40?C) **A1:**Check the cooling air flow and the ambient temperature
2349	**E_FC_ALARM_STATE_MACHINES_SUPERVISION**
	C1:FC (Closed Loop) Internal logical error **A1:**Check FC states. If error occurs repeatedly replace FC
2354	**E_FC_ALARM_BRAKING_RESISTOR_ OVERTEMPERATURE**

Code	Frequency Converter Alarm Messages
	C1: Brake resistor bimetal temperature switch. The brake resistor temp monitoring circuit has opened. If elevator is executing a trip it can usually be finished. New trip cannot be started before bimetal temp switch is closed again (5 min.) **A1:**Check brake resistors temperature. Check wiring and/or the bimetal temperature switch itself
2357	**E_FC_ALARM_OUTPUT_CONTACTORS_SUPERVISOR**
	C1:FC (Closed Loop) Output contactors failure in standstill **A1:**Check output contactors feedback inputs SH1_STATE and SH1_STATE (DIA1, DIA2) and/or corresponding n.c. SH/SH1 auxiliary contacts. Check also ("external") test jumper XTHS and XTHS1 Position & Wiring (drive module)
2358	**E_FC_ALARM_PWM_ENABLE_INPUT_WRONG_STATE**
	C1:FC (closed Loop) Wrong PWM input state **A1:**Check PWM_ENABLE input (DIB4) and/or corresponding auxiliary n.o. SH1 contact.
2364	**E_FC_ALARM_MECHANICAL_BRAKE_KBKB1**
	C1:FC (Close Loop) Mechanical brake KBKB1 contactor problem **A1:**Check brake and its contacts. Check menu 724
2370	**E_FC_ALARM_MECHANICAL_BRAKE_KB**
	C1:FC (Close Loop) Mechanical brake KB contactor problem **A1:**Check brake and its contacts. Check menu 724
2371	**E_FC_ALARM_MECHANICAL_BRAKE_KB1**
	C1:FC (Close Loop) Mechanical brake KB1 contactor problem **A1:**Check brake and its contacts. Check menu 724
2375	**E_FC_ALARM_ELEVATOR_PHNRx_SEQUENCE_ ERROR**

Code	Frequency Converter Alarm Messages
2376	**E_FC_ALARM_ELEVATOR_RELEVELING_DISTANCE_ EXCEEDED**
2379	**E_FC_ALARM_RPM_IDENT_LOW_TORQUE**

Code	Frequency Converter Warning Messages
2401	**E_FC_PROXI_UNUSED_WARNING**
2402	**E_FC_HEARTBEAT_TIMEOUT** The communication to the drive node (controller) has broken
	C1: Drive node disconnected (e.g. CAN bus) **A1:**Cable visual check Measure: --Check VF plug --Check CAN plug on VF --Check CAN bus termination switch on SMIC, must be OFF (see section 2.4) Tips & Tricks: How do I access the VF? --Switch off the main switch --Pull out plug SKC pin 8 to travel past a floor --Use the PEBO (emergency evacuation, SEM11/12, KA 7.2.2) to position the car --If the car is at the uppermost floor then load the car with weights and use PEBO to position the car further down --Check shielding of data line (if present).
	C2: Drive node disconnected (e.g. CAN bus) **A2:**Reconnect node
	C3: No or bad drive node power supply **A3:**Check node power supply

Code	Frequency Converter Warning Messages
	C4:VF fault **A4:**Renew VF Measure: Renew VF Tips & Tricks: Check encoder direction: CF=16, PA=14 Check phase sequence: CF=16, PA=15
	C5:SMIC fault **A5:**Renew SMIC 61 Measure: --See section 4.15.1, section SMIC 61 Tips & Tricks: --SMIC 63 (ID 594226) replaces SMIC 61 and is backwards compatible. --Important: If an SMIC 61 is replaced by an SMIC 63 then the following bridge connectors must be set: » Set plug KP (ID 997037) > Bridge pin 1/2
	C6: Data transmission faulty (e.g. CAN bus) **A6:**Check general data line connection. Check for correct data line termination (jumpers and switches, if present) of all devices (PCBs) connected to the the data bus. Check shielding of data line (if present).
	C7: Drive node defective **A7:**Replace corresponding node
2403	**E_FC_TRUE_START_TIMEOUT** The elevator main controller does not receive the expected acceleration confirmation data telegram from the drive within the expected time (e.g. 2.5 s) after a start command.
	C1: Different causes **A1:**Check elevator message log for previous reported messages in order to identify the root cause of the problem.

8

Code	Frequency Converter Warning Messages
	C2: See 2402, C3 **A2:**See 2402, A3
	C3: The elevator main controller and the drive are out of sync. **A3:**Perform a reset of the entire system
2404	**E_FC_MOVE_CMD_TIMEOUT_ERROR** The elevator main controller does not receive the expected data from the drive node within the expected time after a request.
	C1: Different causes **A1:**Check elevator message log for previous reported messages in order to identify the root cause of the problem.
	C2: The elevator main controller and the drive are out of sync. **A2:**Perform a reset of the entire system
2405	**E_FC_DRIVE_PHASE**
	C1: Drive (FC) reports phase inconsistent **A1:**Check CAN cable. Check CAN cable termination
	C2: The FC phase (state) is not consistent with the previous one. (e.g. if after standstill, decelerating is sent).There has either missing a message or the FC software has a bug.Has nothing to with electrical motor phase connections.
2406	**E_FC_PROXY_WARNING_DISTANCE_ZERO_MOVE_ RQST**
	C1: The FC has received a zero distance move request.This is an EC application (internal) error.
2407	**E_FC_PROXY_WARNING_DIRECTION_NONE_MOVE_ RQST**
	C1: The FC has received a no direction move request.This is an EC application (internal) error.

Code	Frequency Converter Warning Messages
2408	**E_FC_PROXY_WARNING_DIRECTION_INVERSION_ MOVE_RQST**
	C1:EC application (internal) error.If within a trip from one move request to another the direction changes.
2409	**E_FC_PROXY_WARNING_ZERO_LEVELS_MOVE_RQST**
	C1: The FC has received a zero level move request.This is an EC application (internal) error.
2410	**E_FC_PROXY_WARNING_NESTED_MOVE_RQST**
	C1:EC application (internal) error. If a move command tries to override another one.
2411	**E_FC_PROXY_WARNING_MOVE_RQST_WHILE_ STOPPING**
	C1:EC application (internal) error. EC does not stick to the given time-outs.
2412	**E_FC_PROXY_WARNING_MOVE_RQST_WHILE_ UNAVAILABLE**
	C1:EC application (internal) error. EC does not stick to the given time-outs.
2413	**E_FC_PROXY_WARNING_MOVE_RQST_WITH_WRONG _FC_MODE**
	C1:EC application (internal) error. After startup the enable delay is too long.
2414	**E_FC_PROXY_WARNING_RQST_TO_RECOVER_ AFTER_FATAL**
	C1:EC application (internal) error. EC tried to recover from a fatal error.
2415	**E_FC_PROXY_WARNING_SAFETY_CHAIN_WILL_BE_ DISABLED**
	C1: Due to FC error the safety chain will be disabled (via RH1 relay).
2416	**E_FC_PROXY_WARNING_GENERIC_LOGICAL_ERROR**

8

Code	Frequency Converter Warning Messages
	C1:EC application (Internal) error. Collects the rest of unmentioned errors.
2417	**E_FC_PROXY_WARNING_HIGH_LOAD_UNBALANCE**
	C1:EC application (Internal) error.
2418	**E_FC_PROXY_WARNING_DRIVE_BECAME_ UNAVAILABLE**
	C1: Drive (FC) became unavailable (e.g. command. lost or other error happened) **A1:**Check FC parameters. Restart system and reset FC errors. Change FC
2419	**E_FC_PROXY_WARNING_DRIVE_PHASE_BECAME_ UNKNOWN**
	C1: Drive (FC) state transition inconsistency or communication lost cause the "Drive-phase" to be unknown. **A1:**Check CAN bus.
2420	**E_FC_PROXY_WARNING_UNKNOWN_FC_ERROR_ RECEIVED**
	C1: Unknown FC error received **A1:**Version compatibility?
2421	**E_FC_PROXY_WARNING_UNKNOWN_FC_ALARM_ RECEIVED**
	C1: Unknown FC alarm received. **A1:**Version compatibility?
2422	**E_FC_PROXY_WARNING_RECOVER_FROM_ERROR_ LOGIC**
	C1:EC application (Internal) error.
2423	**E_FC_CMD_STOP_TIMEOUT**
	C1: Drive (FC) does not confirm the stopping request. **A1:**Check CAN cable. Check CAN cable termination

Code	Frequency Converter Warning Messages
	C2: The EC sends a move command to the FC and the FC does not answer. **A2:**Check if FC is running
2424	**E_FC_PROXY_WARNING_STATIC_MOVE_RQST_ WHILE_NOT_IN_STAND_BY**
	C1:EC request a move while drive was not in standby. **A1:**Reset EC.Report repeated occurrences.
2425	**E_FC_PROXY_WARNING_DYNAMIC_MOVE_RQST_ WHILE_DECELEREATING**
	C1:EC requested a dynamic move while drive was decelerating. **A1:**Reset EC. Report repeated occurrences.
2426	**E_FC_PROXY_WARNING_DYNAMIC_MOVE_RQST_ WHILE_IN_STAND_BY**
	C1:EC requested a dynamic move while drive was in standby (=not dynamic). **A1:**Reset EC. Report repeated occurrences.
2427	**E_FC_PROXY_WARNING_FC_SW_VERSION_ UNKNOWN_YET**
2428	**E_FC_PROXY_WARNING_FC_HARDWARE_VERSION_ UNKNOWN_YET**
2429	**E_FC_PROXY_PARAMETER_DOWNLOAD_FAILED**
	C1: Drive (FC) reports a parameter download failure **A1:**Check FC parameter values
2430	**E_FC_PROXY_PARAMETER_COMPARE_FAILED**
	C1: Drive (FC) reports parameter value inconsistent **A1:**Check FC parameter values
2431	**E_FC_PROXY_PARAMETER_FC_DATA_NOT_PRESENT**
	C1:fc data not present
2432	**E_FC_PROXY_PARAMETER_SET_NOT_COMPLIANT**

Code	Frequency Converter Warning Messages
	C1:fc data not compliant
2433	**E_FC_PROXY_PARAMETER_WRONG_FC_SW_VERSION**
	C1:wrong FC SW-Version
2434	**E_FC_PROXY_LAST**

Code	EEPROM Messages
2601	**E_EEPROM_INSUFFICIENT_SPACE**
	C1:insufficient space **A1:**Do: persistent fatal error clearing procedure
2602	**E_EEPROM_DATA_RECOVERY_FAILURE**
	C1:data recovery failure **A1:**Do: persistent fatal error clearing procedure
2603	**E_EEPROM_RANGE_ERROR**
	C1:range error **A1:**Do: persistent fatal error clearing procedure
2604	**E_EEPROM_ACCESS_TO_UNKNOWN_FILE**
	C1: Faulty elevator control software **A1:**Update elevator control software
2605	**E_EEPROM_SHAFT_FILE_ERROR**
	C1: Faulty EEPROM **A1:**Replace elevator control PCB if failure persists
2606	**E_EEPROM_RV_NR_FILE_ERROR** C1/A1: See 2604
2607	**E_EEPROM_ERROR_LOG_FILE_ERROR** C1/A1: See 2604
2608	**E_EEPROM_STATISTICS_FILE ERROR** C1/A1: See 2604

Code	EEPROM Messages
2609	**E_EEPROM_DRIVE_FILE_ERROR** C1/A1: See 2604
2610	**E_EEPROM_MODERNIZATION_FILE_ERROR** C1/A1: See 2604
2611	**E_EEPROM_BASE_NORMAL_FILE_ERROR** C1/A1: See 2604
2612	**E_EEPROM_PASSWORD_FILE_ERROR** C1/A1: See 2604
2613	**E_EEPROM_TRAFFIC_CTRL_FILE_ERROR** C1/A1: See 2604
2614	**E_EEPROM_LOP_FILE_ERROR** C1/A1: See 2604
2615	**E_EEPROM_COP_FILE_ERROR** C1/A1: See 2604
2616	**E_EEPROM_BASE_SECURE_FILE_ERROR** C1/A1: See 2604
2617	**E_EEPROM_EXT_NORMAL_FILE_ERROR** C1/A1: See 2604
2618	**E_EEPROM_EXT_SECURE_FILE ERROR** C1/A1: See 2604
2619	**E_EEPROM_DOOR_FILE ERROR** C1/A1: See 2604
2620	**E_EEPROM_EMBEDDED_RM_FILE_ERROR** Corrupted data in the persistent memory (EEPROM) of the elevator control (CRC error at remote monitoring file)
2621	**E_EEPROM_DATETIME_FILE_ERROR** Corrupted data in the persistent memory (EEPROM) of the elevator control (CRC error at date and time file)

Code	EEPROM Messages
	C1: Faulty EEPROM **A1:** Replace elevator control PCB if failure persists
2699	**E_EEPROM_DATA_CONVERSION_SUCCESS**

Code	Hydraulic Messages
2701 to 2711	**Hydraulic errors. Not used with Schindler 3100/3300/5300** If one of these errors occurs: Please contact a specialist or hotline Locarno.

Code	Automatic Acceptance Tests Messages
3101 to 3146	**AAT Automatic Acceptance Tests Errors** For further description and solutions refer to documentation Automatic Acceptance Tests Guidelines (J139452 or J41140148)
3101 to 3109	**SGC “Safety gear car”** related errors (see J139452 or J41140148)
3110 to 3114	**AOS “Ascending car overspeed protection”** related errors (see J139452 or J41140148)
3115 to 3119	**HBU “Half brake capability upward”** related errors (see J139452 or J41140148)
3120 to 3124	**HBD “Half brake capability downward”** related errors (see J139452 or J41140148)
3125 to 3128	**RTL “Run time limit”** related errors (see J139452 or J41140148)
3129 to 3133	**FBU “Full brake capability upward”** related errors (see J139452 or J41140148)
3134 to 3138	**FBD “Full brake capability downward”** related errors (see J139452 or J41140148)
3139 to 3144	**CWB “Counterweight balancing”** related errors (see J139452 or J41140148)

Code		Automatic Acceptance Tests Messages
3145 to 3146		**SMDO "Door opening speed"** related errors (see J139452 or J41140148)
3147 to 3153		**SGCE "Safety gear with empty car"** related errors (see J139452 or J41140148)
3154 to 3157		**KNU "KNE top"** related errors (see J139452 or J41140148)
3158 to 3161		**KND "KNE bottom"** related errors (see J139452 or J41140148)
3162 to 3165		**CIB "Car impact on buffer"** related errors (see J139452 or J41140148)
3166 to 3169		**CWIB "Counterweight impact on buffer"** related errors (see J139452 or J41140148)
31	70	**Half brake down insufficient shaft height**
31	71	**Half brake up insufficient shaft height**
31	72	**Safety gear car test unexpected stop**
31	73	**Safety gear with empty car test unexpected stop**
31	74	**AAT STM MONITORING CAR NOT AT LDU FLOOR**
31	75	**AAT STM MONITORING CANCELED**
31	76	**AAT STM MONITORING ABORTED**

Code	CANIO Messages
3201 to 3216	**CANIO PCB errors. Not used with Schindler 3100/3300/5300** If one of these errors occurs: Please contact a specialist or hotline Locarno.

8

Code	TSD Messages
3301	**ELEVATOR_N_KNET_IN_NON_TSD** The elevator has detected a change of the signal KNET (state of triangle key at landing door) which is only used together with TSD (temporary safety device) but the TSD is not detected as present.

Code	TSD Messages
	C1: The wiring at the signal KNET is faulty **A1:**Check general wiring
3302	**ELEVATOR_JREC_IN_TSD** The elevator has detected a change of the signal JREC (state of inspection switch on top-of-car inspection panel) which is only used together with standard top-of-car inspection but the standard top-of-car inspection is not detected as present. TSD (temporary safety device) is detected as present instead.
	C1: The wiring at the signal JREC is faulty **A1:**Check general wiring
	C2: Temporary safety device system: JREC transition was detected.
	C3: The elevator main controller board was exchanged by a board from another elevator. **A3:**Perform the learning travel. At no success replace elevator main controller board (e.g. SCPU) by a new one from the factory.
3303	**ELEVATOR KSR-A IN NON TSD** The elevator has detected a change of the signal KSR_A (state of TSD lever) which is only used together with TSD (temporary safety device) but the TSD is not detected as present.
	C1: The wiring at the signal KSR_A is faulty **A1:**Check general wiring
	C2: The elevator main controller board was exchanged by a board from another elevator **A2:**Perform the learning travel. At no success replace elevator main controller board (e.g. SCPU) by a new one from the factory.

Code	TSD Messages
3304	**E_TSD_FORBIDDEN_CAR_ROOF_ACTION** A forbidden car roof action for the TSD (Temporary Safety Device) was detected. This causes a potentially dangerous situation for the service technician.
	C1:A button on the inspection panel (Stop, Up, Down) was pressed or the TSD lever was activated by the service technician on the car top while the car wasn't in inspection mode. **A1:**Leave the car roof. Perform a elevator main controller reset and activate the TSD correctly!
	C2: Bad connection between the inspection panel and the IO interface board on the car (SDIC) **A2:**Check plug and wiring of the inspection panel on the car top
	C3:KNET input signal not working **A3:**Check operation of KNET input signal on elevator main controller board
	C4: Faulty KNET switch on a landing door **A4:**Check operation of KNET switches
3305	**E_TSD21_UNSAFE_SHAFT_ACCESS** The TDS21 device enters unsafe mode.

Code	HMI Messages
3401	**E_HMI_VALUE_OUT_OF_LOWER_BOUND** While parameterizing, the entered value on the elevator user interface is below the allowed minimum value
	C1:STM manufacturing date is void (=0) and it lower than permitted limit (10100) **A1:**No action
3402	**E_HMI_VALUE_OUT_OF_UPPER_BOUND** While parameterizing, the entered value on the elevator user interface is above the allowed maximum value.

Code	Safety Chain Messages (SIAP)
3501	**E_SAFETY_CHAIN_PIT_CONTACT_OVERBRIDGING_ ACTIVATION** The overbridging of the safety circuit contacts in the pit (e.g. used in fire operation Korea) wasn't successful.
	C1: Overbridging relays defect **A1:**Check function of relays RUESG, RUESG1 and RSG_A
	C2: Wiring from/to overbridging relays bad **A2:**Check wiring of relays RUESG, RUESG1 and RSG_A
	C3: Fieldbus communication problem (please check previous errors in error log) **A3:**Check error 3603 for cause and actions
3502	**E_SAFETY_CHAIN_PIT_CONTACT_OVERBRIDGING_ LOST** The overbridging of the safety circuit contacts in the pit (e.g. used in fire operation Korea) was lost.
	C1: Overbridging relays defect **A1:**Check function of relays RUESG, RUESG1 and RSG_A
	C2: Wiring from/to overbridging relays bad **A2:**Check wiring of relays RUESG, RUESG1 and RSG_A
	C3: Fieldbus communication problem (please check previous errors in error log) **A3:**Check error 3603 for cause and actions
3503	**E_SAFETY_ CHAIN_PIT_CONTACT_OVERBRIDGING DEACTIVATION** The cancellation of the overbridging of the safety circuit contacts in the pit (e.g. used in fire operation Korea) wasn't successful.
	C1: Overbridging relays defect **A1:**Check function of relays RUESG, RUESG1 and RSG_A

Code	Safety Chain Messages (SIAP)
	C2: Wiring from/to overbridging relays bad **A2:**Check wiring of relays RUESG, RUESG1 and RSG_A
	C3: Fieldbus communication problem (please check previous errors in error log) **A3:**Check error 3603 for cause and actions
3504	**E_SAFETY_ CHAIN_PIT_CONTACT_UNEXPECTED_ OVERBRIDGING** An unexpected overbridging of the safety circuit contacts in the pit has occurred.
	C1: Overbridging relays defect **A1:**Check function of relays RUESG, RUESG1 and RSG_A
	C2: Wiring from/to overbridging relays bad **A2:**Check wiring of relays RUESG, RUESG1 and RSG_A
	C3: Fieldbus communication problem (please check previous errors in error log) **A3:**Check error 3603 for cause and actions
3505	**E_SAFETY_CHAIN_KNA_CONTACT_OVERBRIDGING_ ACTIVATION** The overbridging of the safety circuit contact at the car emergency exit (e.g. used in fire operation Korea) wasn't successful.
	C1:SIAP PCB defect (e.g. relays simulating door zone (PHS/PHS1) defect) **A1:**Replace SIAP board
3506	**E_SAFETY_CHAIN_KNA_CONTACT_OVERBRIDGING_ LOST** The overbridging of the safety circuit contact at the car emergency exit (e.g. used in fire operation Korea) was lost.
	C1:SIAP PCB defect (e.g. relays simulating door zone (PHS/PHS1) defect) **A1:**Replace SIAP board

Code	Safety Chain Messages (SIAP)
3507	**E_SAFETY_CHAIN_KNA_CONTACT_OVERBRIDGING_ DEACTIVATION** The cancellation of the overbridging of the safety circuit contact at the car emergency exit (e.g. used in fire operation Korea) wasn't successful.
	C1:SIAP PCB defect (e.g. relays simulating door zone (PHS/PHS1) defect) **A1:**Replace SIAP board
3508	**E_SAFETY_CHAIN_KNA_CONTACT_UNEXPECTED_ OVERBRIDGING** An unexpected overbridging of the safety circuit contact at the car emergency exit has occurred.
	C1:SIAP PCB defect (e.g. relays simulating door zone (PHS/PHS1) defect) **A1:**Replace SIAP board
3509	**E_SAFETY_CHAIN_SIM_DOORZONE_CONTACT_OVER BRIDGING_ACTIVATION_UNSUCCESSFUL** The overbridging of the door safety circuit contacts, allowing to travel the car with opened doors (e.g. used in fire operation Korea) wasn't successful.
	C1: Wiring of door zone simulation bad **A1:**Check connections between SUET board and SIAP board
	C2:SIAP PCB defect (e.g. relays simulating door zone (PHS/PHS1) defect) **A2:**Replace SIAP board
	C3: Door safety circuit overbridging failed **A3:**Check errors 0338 for cause and actions

Code	Safety Chain Messages (SIAP)
3510	**E_SAFETY_CHAIN_SIM_DOORZONE_CONTACT_ OVERBRIDGING_LOST** The overbridging of the door safety circuit contacts, allowing to travel the car with opened doors (e.g. used in fire operation Korea) was lost.
	C1: Wiring of door zone simulation bad **A1:**Check connections between SUET board and SIAP board
	C2:SIAP PCB defect (e.g. relays simulating door zone (PHS/PHS1) defect) **A2:**Replace SIAP board
	C3: Door safety circuit overbridging lost **A3:**Check errors 0339 for cause and actions
3511	**E_SAFETY_CHAIN_SIM_DOORZONE_CONTACT_ OVERBRIDGING_DEACTIVATION_UNSUCCESSFUL** The cancellation of the overbridging of the door safety circuit contacts, allowing to travel the car with opened doors (e.g. used in fire operation Korea) wasn't successful.
	C1: Cancellation of door safety circuit overbridging failed **A1:**Check errors 0340 for cause and actions
3512	**E_SAFETY_CHAIN_SIM_DOORZONE_CONTACT_ UNEXPECTED_OVERBRIDGING** An unexpected overbridging of the safety circuit contact, allowing to travel the car with opened doors (e.g. used in fire operation Korea) has occurred.
	C1:SIAP PCB defect (e.g. relays simulating door zone (PHS/PHS1) defect) **A1:**Replace SIAP board
	C1:SUET PCB defect **A1:**Replace SUET board

Code	Safety Chain Messages (SIAP)
3513	**E_SAFETY_CHAIN_ERROR_NOT_RECOVERABLE** The elevator wasn't able to recover from a opened safety circuit error (typically after 20 repetitive door closing attempts). The elevator is blocked.
	C1: Safety chain is open permanently **A1:**Check SK voltage Measure: --IUSK flashes (see section 4.4) » Use key DUEISK-A to resolve a short circuit and trigger reset > IUSK lights up > Check travel to each floor > If everything is OK then the fault has been resolved <red>Tips & tricks It can happen that water gets into the hoistway when cleaning, thus triggering a short circuit. Feedback is important if this happens to be an intermittent fault.
	C2: Safety chain is open permanently **A2:**Test SMIC Measure: --IUSK flashes in spite of action 1 --Pull out plug KSS on SMIC --Use key DUEISK-A to resolve a short circuit and trigger reset » IUSK does not light up or flash > Renew SMIC Tips & Tricks: Do not forget the following when renewing SMIC » SIM card » Bus termination switch to OFF » Make sure that the SCPU and the CLSD are engaged correctly (push in firmly, printed circuit board clips completely closed) See section 4.4 or installation diagram

Code	Safety Chain Messages (SIAP)
	C3: Safety chain is open permanently **A3:**Check SK segment Measure: --Pull out plugs KSS, SKS, SKC, KBV, ESE on SMIC --Use key DUEISK-A to cancel a short circuit and trigger reset --IUSK lights up » Plug in KSS, SKS, SKC, KBV, ESE in this sequence until IUSK flashes once more » Rectify short circuit in the corresponding segment Tips & Tricks: See section 4.4 or installation diagram
	C4: Safety chain is open permanently **A4:**Check RTS segment Measure: --IUSK lights up, ISPT does not » Fault in passive safety circuit (use diagram to find all contacts) --IUSK and ISPT light up but elevator still does not move --Control reset, if the doors do not close after the reset, then: » Check door drive » Door not OK > Rectify door malfunction --RTS does not light up » Bridge KTS with the bridge set and move car in inspection » Remove bridge » Check hoistway door contact --RTS lights up but ISK does not, continue with action 5 --RTS and ISK light up but elevator still does not move » Reset menu 101.1 and control reset » Elevator remains stationary > Request support Tips & Tricks: RTS and ISK never light up when door opens

Code	Safety Chain Messages (SIAP)
	C5: Safety chain is open permanently **A5:**Check ISK segment Measure: Check all safety contacts (on the car) between RTS and ISK according to the diagram.
3514	**E_SAFETY_CHAIN_PIT_ACTIVATION_CHECK_ UNSUCCESSFUL** The periodic check if the overbridging of the safety circuit contacts in the pit is running (e.g. used in fire operation Korea) has failed.
	C1: Overbridging relays defect **A1:**Check function of relays RUESG, RUESG1 and RSG_A
	C2: Wiring from/to overbridging relays bad **A2:**Check wiring of relays RUESG, RUESG1 and RSG_A
	C3: Fieldbus communication problem (please check previous errors in error log) **A3:**Check error 3603 for cause and actions
3515	**E_SAFETY_CHAIN_PIT_DEACTIVATION_CHECK_ UNSUCCESSFUL** The periodic check if the cancellation of the overbridging of the safety circuit contacts in the pit is running (e.g. used in fire operation Korea) has failed.
	C1: Overbridging relays defect **A1:**Check function of relays RUESG, RUESG1 and RSG_A
	C2: Wiring from/to overbridging relays bad **A2:**Check wiring of relays RUESG, RUESG1 and RSG_A
	C3: Fieldbus communication problem (please check previous errors in error log) **A3:**Check error 3603 for cause and actions

Code	Safety Chain Messages (SIAP)
3516	**E_SAFETY_CHAIN_KNA_ACTIVATION_CHECK_ UNSUCCESSFUL** The periodic check if the overbridging of the safety circuit contact at the car emergency exit is running (e.g. used in fire operation Korea) has failed.
	C1:SIAP PCB defect (e.g. relays simulating door zone (PHS/PHS1) defect) **A1:**Replace SIAP board
3517	**E_SAFETY_CHAIN_KNA_DEACTIVATION_CHECK_ UNSUCCESSFUL** The periodic check if the cancellation of the overbridging of the safety circuit contact at the car emergency exit is running (e.g. used in fire operation Korea) has failed.
	C1:SIAP PCB defect (e.g. relays simulating door zone (PHS/PHS1) defect) **A1:**Replace SIAP board
3550	**E_SAFETY_CIRCUIT_NOT_CLOSED_AT_ISK** Safety circuit not closed at tap 'ISK' when expected to be closed (e.g. before starting a trip)
	C1: At least one car door hasn't closed (e.g. KTC open) **A1:**Check why the corresponding door hasn't closed
	C2: Safety circuit wiring faulty or safety circuit contact(s) defective or bad adjusted **A2:**Check general wiring safety circuit and safety circuit contacts at corresponding safety circuit section
3553	**E_SAFETY_CIRCUIT_NOT_CLOSED_AT_IRTS** Safety circuit not closed at tap 'IRTS' when expected to be closed (e.g. before starting a trip)
	C1: At least one landing door hasn't closed (e.g. KTS open) **A1:**Check why the corresponding door hasn't closed

Code	Safety Chain Messages (SIAP)
	C2: Safety circuit wiring faulty or safety circuit contact(s) defective or bad adjusted **A2:**Check general wiring safety circuit and safety circuit contacts at corresponding safety circuit section
3560	**E_SAFETY_CIRCUIT_NOT_OPENED_AT_ISK** Safety circuit not opened at tap 'ISK' when expected to be opened (e.g. while opening door)
	C1: The safety circuit at the car door is bridged or a car door safety contact is defective (e.g. KTC stuck together) **A1:**Check safety circuit for bridges (e.g. plugs) and corresponding door safety circuit contacts for correct operation
3562	**E_SAFETY_CIRCUIT_NOT_OPENED_AT_IRTS** Safety circuit not opened at tap 'IRTS' when expected to be opened (e.g. while opening door)
	C1: At Bionic Control: The safety circuit at the landing door is bridged or a landing door safety contact is defective (e.g. KTS stuck together) **A1:**Check safety circuit for bridges (e.g. plugs) and corresponding door safety circuit contacts for correct operation
3570	**E_SAFETY_CIRCUIT_OPENED_AT_ISK** Safety circuit opened unexpected (e.g. during trip) at tap 'ISK'
	C1: The car has exceeded the hoistway end limit (e.g. KNE, KNE_U, KNE_D) **A1:**Check why the car has exceeded the hoistway end limit. Check log for possible previous reported messages. Release elevator from blocked state (perform reset procedure).

Code	Safety Chain Messages (SIAP)
	C2: The car has exceeded the maximum speed limit (e.g. KBV) **A2:**Check why the car has exceeded the maximum speed (ascending or descending overspeed). Check log for possible previous reported messages. Release elevator from blocked state (perform reset procedure).
	C3: The car safety gear is engaged (e.g. KF) **A3:**Release the car from engaged safety gear (perform reset procedure)
	C4: The car emergency exit door is not locked (e.g. KNA) **A4:**Close and lock the car emergency exit door
	C5: Safety circuit wiring faulty or safety circuit contact(s) defective or bad adjusted **A5:**Check general wiring safety circuit and safety circuit contacts at corresponding safety circuit section
3571	**E_SAFETY_CIRCUIT_OPENED_AT_ISPT** Safety circuit opened unexpected (e.g. during trip) at tap 'ISPT'
	C1: The car speed governor rope tension is too low (e.g. KSSBV) **A1:**Check why the car speed governor rope tension is too low.
	C2: The pit ladder is not retracted **A2:**Retract pit ladder
	C3: Safety circuit wiring faulty or safety circuit contact(s) defective or bad adjusted **A3:**Check general wiring safety circuit and safety circuit contacts at corresponding safety circuit section
3573	**E_SAFETY_CIRCUIT_OPENED_AT_IRTS** Safety circuit opened unexpected (e.g. during trip) at tap 'IRTS'

Code	Safety Chain Messages (SIAP)
	C1: At least one landing door has opened (e.g. KTS) **A1:**Check why the corresponding door has opened
	C2: Safety circuit wiring faulty or safety circuit contact(s) defective or bad adjusted **A2:**Check general wiring safety circuit and safety circuit contacts at corresponding safety circuit section
3575	**E_SAFETY_CIRCUIT_OPENED_AT_IUSK** Safety circuit opened unexpected (e.g. during trip) at tap 'IUSK'
	C1: The power supply of the safety circuit is faulty **A1:**Check operation of power supply (e.g. fuses, input power etc.)
	C2: The power supply of the safety circuit is faulty **A2:**Check LUEISK Measure: Check LUEISK (see section 4.4) » Use DUEISK to resolve short circuit Check elevator function with test travels If LUEISK is still active: » Severe fault, request support Tips & Tricks: It can happen that water gets into the hoistway when cleaning, thus triggering a short circuit. Feedback is important if this happens to be an intermittent fault. Check safety circuit for short circuit to earth.
	C3: The power supply of the safety circuit is faulty **A3:**Renew SMIC Measure: Renew SMIC Tips & Tricks: Comply with EMC regulations Renewing SMIC entails removing door uprights.
	C4: Earth fault **A4:**Check safety circuit for short circuit to earth.

Code	Safety Chain Messages (SIAP)
3596	**E_SAFETY_CIRCUIT_RECOVERY_SUCCESSFUL**
3597	**E_SAFETY_CIRCUIT_RECOVERY_UNSUCCESSFUL**
3598	**E_SAFETY_CIRCUIT_RECOVERY_DOOR_ SUCCESSFUL**
3599	**E_SAFETY_CIRCUIT_RECOVERY_DOOR_ UNSUCCESSFUL**

Code	Communication Messages
3601	**E_BUS_SCAN_FAILED (CAN BUS)** At startup the controller checks if all bus nodes (e.g. landing operating panels) are communicating with the main controller. This error occurs if this check fails. Note, this error does not occur if an additional node is connected to the bus.
	C1: Bad connection to any bus node (e.g. landing operating panel) **A1:**Check generally bus wiring. Check bus connector on controller main board and at all bus nodes.
	C2: Bad bus power supply **A2:**Check supply voltage on bus for instability
	C3:EMC problems **A3:**Check entire bus for interferences
3602	**E_LOP_COUNT_FAILED** After configuration of the bus nodes (e.g. landing operating panels) a check of all possible node addresses is performed and the corresponding IO function mapping get read. This error occurs if there is a communication problem to any bus node during this phase.
	C1: Bad connection to any bus node (e.g. landing operating panel) **A1:**Check generally bus wiring. Check bus connector on controller main board and at all bus nodes. Redo LOP count (CF00, LE00).

8

Code	Communication Messages
	C2: Bad bus power supply **A2:**Check supply voltage on bus for instability.Redo LOP count (CF00, LE00).
	C3:EMC problems **A3:**Check entire bus for interferences. Redo LOP count (CF00, LE00).
3603	**E_BIOBUS_NODE_DEAD** During operation the communication to a bus node (e.g. landing operating panel) has failed.Note, this error occurs every time the communication got lost to a single bus node.
	C1: Bad connection to any bus node (e.g. landing operating panel) **A1:**Check generally bus wiring. Check bus connector on controller main board and at all bus nodes.
3604	**E_BIOBUS_NODE_ALIVE** Node alive again.
	C1: Defect node on this location **A1:**Check node.
	C2: Bad connection to the node **A2:**Check connection wiring and plugs
4050	**E_AAT_CALL_SIM_START** This event informs that calls are automatically generated and a certain number of trip would be executed.
4051	**E_AAT_CALL_SIM_TERMINATED** This event informs that the automatic calls generator has been disabled since the amount of trip requested has been reached.

Code	Safety Device Messages
4305	**E_SAFETY_DEVICE_CAR_UNINTENDED_MOVEMENT** A elevator safety device has detected that the car has performed a uncontrolled movement (e.g left the door zone with door open).
	C1: Serious failure at traction means or brake **A1:**Check carefully reason for unintended car movement and resolve problem. Release elevator from blocked state by dedicated reset of the tripped safety device.
	C2: Car door safety contact opens at start **A2:**Verify the correct function of KTC, adjust KTC accordingly. Verify the correct installation of the doors and resolve any issue found.
4360	**E_SAFETY_DEVICE_EAS_WARNING**
4361	**E_SAFETY_DEVICE_EAS_ANOMALY**
4362	**E_SAFETY_DEVICE_EAS_ERROR**
4363	**E_SAFETY_DEVICE_EAS_INTERNAL_ERROR**
4364	**E_SAFETY_DEVICE_EAS_RESET_RQST**
4365	**E_SAFETY_DEVICE_EAS_ASSISTED_RESET_RQST**
4366	**E_SAFETY_DEVICE_EAS_RESET_FAIL**
4367	**E_SAFETY_DEVICE_EAS_RESET_SUCCESFUL**
4368	**E_SAFETY_DEVICE_EAS_HB_LOST**
4369	**E_SAFETY_DEVICE_EAS_HB_RESTORED**

Code	Traction Media Messages
4601	**E_TM_RESIDUAL_STRENGTH_WARNING** Traction media residual strength insufficient warning

Code	Traction Media Messages
	C1: The number of bending cycles respectively the trip count approaches the service limits. **A1:**Initiate replacement of traction media before the end of the estimated lifetime.
4602	**E_TM_RESIDUAL_STRENGTH_INSUFFICIENT** Traction media residual strength insufficient
	C1: The maximal permitted bend cycle count our trip count for the traction media had been exceeded. **A1:**Replace traction media.
4603	**E_TM_BENDING_STRESS_COUNT_RESET** Traction media bending stress counter reset
	C1: The maintenance person enters a new STM manufacturing date: the bending counter has been reset (set to 0). **A1:**None
4604	**E_TM_BENDING_STRESS_WARNING_LIMIT_CHANGED** The maximum permitted traction media bending stress count warning limit was changed.
4605	**E_TM_BENDING_STRESS_MAXIMUM_CHANGED** The maximum permitted traction media bending stress count was changed.
4606	**E_TM_AGING_CLOCK_FAILURE** The elevator clock is not running or is delayed.
	C1: The date of the elevator clock is not set or is wrong. This could be caused by a loss of mains and battery power. **A1:**Set the correct date.
4607	**E_TM_MAX_AGE_WARNING** The age of the traction media approached the end of life, initiate its replacement.

Code	Traction Media Messages
	C1: The age of the traction media reached the defined percentage of the service limit. **A1:**Initiate replacement of traction media before the end of the estimated lifetime.
4608	**E_TM_MAX_AGE_EXCEEDED** The maximum permitted age of the traction media is exceeded, it shall be replaced.
	C1: The maximum permitted age of the traction media was exceeded. **A1:**Replace the traction media.
	C2: The elevator clock date is not correct. **A2:**Set the correct date.
4610	**E_TM_RES_STRENGTH_MONITORING_VIRGIN_ STORAGE_DETECT** Either the master or backup storages for traction media insufficient strength monitoring is virgin (empty). The virgin storage is initialized with a copy of the STM Monitoring data.
	C1: The controller founds that one of the redundant media does not contains any data. The controller rebuilds the STM information. **A1:**None
4611	**E_TM_RES_STRENGTH_MONITORING_REF_KEY_ MISMATCH** Reference key mismatch of master and backup storages of traction media insufficient strength monitoring. The reference key consists of the manufacturing date of the oldest traction media and the commission number.
	C1: Commissioning number differs between controller and chipcard. **A1:**Replace chipcard with correct commissioning number.

Code	Traction Media Messages
4612	**E_TM_RES_STRENGTH_MONITORING_DATA_MISMATCH** Data mismatch between master and backup storages of traction media insufficient strength monitoring. The reference keys of both storages are identical. The highest STM trip count value and the lowest STM trip count limit are replicated to both storages.
	C1: The controller found different STM bending counter in the redundant storages. **A1:**None
	C2: The controller found different STM bending limits in the redundant storages. The new value is taken from chipcard. **A2:**None
	C3: The controller found different STM bending warning levels in the redundant storages. The new value is taken from chipcard. **A3:**None
	C4: The controller found different SBPT in the redundant storages. The new value is taken from chipcard. **A4:**None
4613	**E_TM_RES_STRENGTH_MONITORING_BACKUP_MISSING** Missing backup storage of traction media insufficient strength monitoring.
	C1:STM data cannot be saved on chipcard. **A1:**Replace chip card with one with 1k.
4615	**E_TM_DATA_MEMORY_RECOVERY_SUCCESS** Successful recovery of traction media data storage memory.
	C1:A mismatch in STM data has been found and solved. **A1:**Check previous log entries to see the reason of mismatch.

Code	Traction Media Messages
4616	**E_TM_DATA_MEMORY_RECOVERY_UNSUCCESSFUL** Recovery of traction media data storage memory was not successful. The reference key did not match any of the storages or restore failed.
	C1: Reference key entered is wrong. **A1:**Enter the correct reference key for the elevator from the maintenance log.
	C2: Fault of one or both storage devices. **A2:**Replace the faulty storage device. See previous log entry for detail.
	C3: Both storage devices are from a different elevator or virgin. **A3:**Re-enter the STM installation data from the maintenance log.
	C4:STM chipcard option is missing. **A4:**Replace chipcard with one with STM option enabled.
4617	**E_TM_INSTALLATION_DATA_ENTERED** Traction media installation data had been entered successfully.
	C1: The procedure of configuring the STM parameter is terminated. **A1:**Reset the controller.
4618	**E_TM_STRESS_COUNT_ENTERED** The STM stress count (trip or tend cycle counter) was set.
	C1:A new STM bending counter has been entered. The advanced STM configuration has been used. **A1:**None.
4619	**E_TM_AGING_CLOCK_RECOVERY_SUCCESS** The failure of the internal clock had been recovered.
	C1: The maintenance person set the clock. **A1:**None.

Code	Traction Media Messages
4620	**E_TM_AGING_CLOCK_RECOVERY_FAILED** The failure of the internal clock had not been recovered within the permitted time.
	C1: The internal clock had been stopped due to a loss of mains and battery power. **A1:**Set date and time.
	C2: The controller expects soon an STM expiration. **A2:**Set date and time.
	C3: The controller uses all the 210 days granted to set the correct date and time. **A3:**Set date and time.
4621	**E_TM_RETAINER_FAILURE** The failure of the belt retainer has been detected. Only recall, inspection and montage movements are allowed.
	C1: The belt retainer has reported a problem. **A1:**Inspect belt, inspect retainer.
4622	**E_TM_RETAINER_RECOVERY_SUCCESS** The belt retainer has recovered from an error.
4623	**E_TM_ECM_ERROR** The STM conductivity test has failed.
	C1: Ground failure on the belt. **A1:**Inspect the belts.
4624	**E_TM_ECM_WARNING** The conductivity test is out of range.
4625	**E_TM_ECM_UNAVAILABLE** The conductivity device for STM: - Is no more alive (heartbeat lost) - Reports self check failure - Reports calibration error

Code	Traction Media Messages
	C1:ECM calibration failure. **A1:**-
	C2:ECM heartbeat is lost. **A2:**Check communication cable.
	C3:ECM reports a self test failure. **A3:**Replace hardware.
4626	**E_TM_ECM_WARNING** The conductivity device for STM is back in normal monitoring operation.
4627	**E_TM_RES_AGING_MONITORING_DATA_MISMATCH** Data mismatch between master and backup storages of traction media aging monitoring. The reference keys of both storages are identical. The aging limit and the aging warning limit differs on both storages.
	C1: Manufacturing date differs. **A1:**Activate menu 190 and enter the correct belt manufacturing date.
	C2: Aging limit differs. **A2:**Clear the fatal error.
	C3: Aging warning limit differs. **A3:**Clear the fatal error.
4628	**E_TM_TEMPERATURE_OUT_OF_RANGE** The TM are out of temperature.
4629	**E_TM_TEMPERATURE_IN_RANGE** The TM are in the temperature range.
4630	E_TM_BENDING_STRESS_COUNT_ENTERED Traction media bending stress counter entered.
4631	**E_TM_BENDING_STRESS_COUNTER_MISMATCH** This error is generated when there is a mismatch between bending counter in EEPROM and on SIM card.

Code	Traction Media Messages
4632	**E_TM_MANUFACTURING_DATE_MISMATCH** This error is generated when there is a mismatch between manufacturing date in EEPROM and on SIM card.

Code	Overlay Messages
4700	**OVERLAY_NO_ERROR**
4701	**OVERLAY_HEARTBEAT_MISSING** The heartbeat from overlay is missing.
	C1: The main controller did loose for some time the connection to the overlay board. The reason can be that the cabling has a problem or the overlay board has a problem of the Overlay board did perform a reset. In case this error is detected the bionic controller gets out of the group and continues to serve all calls in a simplex fashion. If the connection to the overlay is recovered the information E_OVERLAY_HEARTBEAT_RECOVERED is recorded in the log as well. **A1:**Check the proper cabling of the overlay board.
4702	**OVERLAY_ECU_HEARTBEAT_MISSING** The overlay detects that no heartbeat is received from the ECU.
4703	**OVERLAY_HEARTBEAT_RECOVERED** The elevator control starts to receive the heartbeat from the overlay.
	C1: This information is stored in the log of events if a previously lost connection to the overlay board is now recovered. **A1:**none
4704	**OVERLAY_ECU_HEARTBEAT_RECOVERED** The ECU receives that overlay starts to listen an heartbeat from ECU.

Code	Overlay Messages
4705	**OVERLAY_ELEVATOR_IN_GROUP_MISSING** Overlay component are missing in the group.
	C1: This error is inserted in the log if one or more controllers are no longer part of the group. It is possible that the error is detected only by some elevators of the group. **A1:**To better understand what causes the issue it is advised to check the overlay monitoring menu (HMI menu 30 > 309) as the values there can be a certain help in pinpointing where the group problem might be.
4706	**OVERLAY_ALL_ELEVATORS_ARE_IN_GROUP** Overlay component are all present
	C1:A previously reported error OVERLAY ELEVATOR IN GROUP MISSING is now recovered. **A1:**none

Code	Telemonitoring Messages
5002	**E_RM_NO_PHYSICAL_DATA_CONNECTION_TO_CC** The data connection from the elevator (remote monitoring data communication device, e.g. modem) to the control center hasn't physically established or was physically interrupted
	C1: Connection physically interrupted or bad **A1:**Check data connection (e.g. phone line)
	C2: Data speed negotiation failed (e.g. at modem) **A2:**Check configuration (e.g. country code)
5007	**E_RM_DATA_COMMUNICATION_DEVICE_DEAD** The communication between the elevator control and the remote monitoring data communication device (e.g. modem) has failed

Code	Telemonitoring Messages
	C1: The connection between the elevator control board and the remote monitoring data communication device (e.g. modem) is faulty **A1:**Check data connection
	C2: Remote monitoring data communication device (e.g. modem) defective **A2:**Replace remote monitoring data communication device
5009	**E_RM_CLSD_FAILURE** The communication line sharing device has detected a failure of its internal electronics. Note, the telealarm device is still operable, but back-calls from the control center to the telealarm or telemonitoring device are not possible anymore.
	C1: Device internal failure **A1:**Replace the CLSD PCB
5010	**E_RM_DISABLED_DUE_TO_TERMINAL_ACTIVATION** Communication between modem and controller is not active due to local terminal activated (contact the local field expert).

Code	Car Alarm Messages
8001	**E_CAR_ALARM_BUTTON_PRESSED** Log alarm button states for enhanced diagnostic (misuse)
8002	**E_CAR_ALARM_BUTTON_RELEASED** Log alarm button states for enhanced diagnostic (misuse)
8003	**E_CAR_ALARM_RELAY_ACTIVATED** Log alarm relay states for enhanced diagnostic (misuse)
8004	**E_CAR_ALARM_RELAY_DEACTIVATED** Log alarm relay states for enhanced diagnostic (misuse)
8005	**E_CAR_INVALID_ALARM_DETECTED** Alarm button pressed or relay activated was received during the usual filtering time

Code	Car Alarm Messages
8006	**E_CAR_ALARM_FILTERED** An alarm was received in normal mode but with the door not fully open or with the car not traveling
8007	**E_CAR_VALID_ALARM_DETECTED** An alarm was received during the time when the alarm circuit discriminator is not active.
8008	**E_CAR_ALARM_FORWARDED** An alarm was received while the discriminator was active, the system tried to free the passenger in time but did fail and thus the alarm is automatically forwarded by the system.
8009	**E_CAR_ALARM_DEVICE_OFF_HOOK** The alarm device is on the line for an alarm call.
8010	**E_CAR_ALARM_DEVICE_ON_HOOK** The alarm device is no longer on the line for an alarm call.
8011	**E_CAR_ALARM_TEST_ENABLED** A valid COP alarm button sequence has been executed or the appropriate menu (134) has been activated. The elevator informed telealarm device that next alarm would be a test alarm.
8012	**E_CAR_ALARM_TEST_STARTED** An alarm button pressed has been detected and is a test one.
8013	**E_CAR_ALARM_TEST_END** An alarm button pressed has been detected and was a test one.
8014	**E_CAR_ALARM_TEST_DISABLED** This message means that no test alarm has been performed on the COP keyboard. This event is coming after a predefined time window (usually 30s) where the maintenance person can perform a test alarm.
8017	**E_MAIN_SWITCH_OFF**

Code	Car Alarm Messages
8018	**E_MAINS_POWER_FAILURE** – Phase of power line is lost for at least 1200 ms – Power phase has rotated for at least 1200 ms
8019	**E_CAR_EMERGENCY_LIGHT_DEFECT**
8020	**E_CAR_EMERGENCY_LIGHT_OK** Emergency car light is available
8021	**E_CAR_ROOF_TEMPERATURE_OK** Car roof temperature is in range. See range thresholds CF22PA27 and CF22PA28.
8022	**E_CAR_ROOF_TEMPERATURE_OUT_OF_RANGE** Car roof temperature is not in range. See range thresholds CF22PA27 and CF22PA28
8023	**E_CAR_ALARM_BUTTON_DEFECT**
8024	**E_CAR_ALARM_BUTTON_OK**
8025	**E_ELECTRICAL_INDICATOR_COP_FAIL**
8026	**E_ELECTRICAL_INDICATOR_LOP_FAIL**
8028	**E_SELF_TEST_TRIP_PERFORMED** Self test trip has been triggered. See parameters CF22PA7 and CF22PA8.
8030	**E_ECU_TEMPERATURE_OK** Temperature in the ECU is normal.
8031	**E_ECU_TEMPERATURE_OUT_OF_RANGE** Temperature in the ECU is out of range.
8032	**E_MAIN_SWITCH_ON** Main switches (JH) has been switched on.
8033	**E_MAINS_POWER_OK** Mains power is correctly detected.
8034	**E_CAR_ALARM_TEST_FAILURE** Automated alarm test fails.

Code	Car Alarm Messages
8035	**E_CAR_ALARM_TEST_OK** Automated alarm test passed.
8036	**E_ELECTRICAL_INDICATOR_COP_OK**
8037	**E_ELECTRICAL_INDICATOR_LOP_OK**
8038	**E_DAILY_TESTS_NOT_PERFORMED** This event is log in case of none of the following test has been performed in the previous 24h: – COP indicator – LOP indicator – Alarm – Emergency car light.
8039	**E_CAR_ROOF_TEMPERATURE_BROKEN_MISSING** Temperature sensor provides abnormal values. The temperature is above 120 celsius degree or The temperature is below -40 celsius degree.
	C1: Temperature sensor broken **A1:**Replace sensor
	C2: Temperature sensor missing **A2:**Check cabling
9000	**S_OUT_OF_SERVICE_OPERATION**
9001	**S_PASSENGER_TRAVEL_OPERATION**
9002	**S_INDEPENDENT_OPERATION**
9003	**S_FIRE_OPERATION**
9004	**S_FIREFIGHTER_OPERATION**
9005	**S_EMERGENCY_POWER_OPERATION**
9006	**S_EARTHQUAKE_OPERATION**
9007	**S_EMERGENCY_MEDICAL_TECHNICIAN**
9008	**S_SPRINKLER_OPERATION**
9009	**S_WATER_IN_PIT_OPERATION**

Code	Car Alarm Messages
9010	S_ATTENDED_PASSENGER_TRAVEL_OPERATION
9011	S_PASSENGER_TRAVEL_OPERATION_WITHOUT_ LOAD_MONITOR
9012	S_PASSENGER_RELEASE_TRAVEL_OPERATION
9013	S_POWER_SAVING_MODE
9029	S_MOVE_AROUND_OPERATION
9037	S_NO_OPERATION_DUE_TO_STOP_IN_CAR
9038	S_NO_OPERATION_DUE_TO_STOP_IN_CAR_ FIREFIGHTER
9039	S_NO_OPERATION_DUE_TO_CAR_OVERLOAD
9040	S_NO_OPERATION_DUE_TO_INVALID_ CONFIGURATION_DATA
9041	S_NO_OPERATION_DUE_TO_INVALID_HOISTWAY_ IMAGE
9042	S_NO_OPERATION_DUE_TO_INVALID_LOAD_ CONFIGURATION
9043	S_NO_OPERATION_DUE_TO_INVALID_DRIVE_ CONFIGURATION
9045	S_OUT_OF_SERVICE_STM_MONITORING_FAILURE
9050	S_SERVICE_TECHNICIAN_VISIT
9051	S_INSTALLATION_TRAVEL
9052	S_CONFIGURATION_MODE
9053	S_INSP_MACHINE_ROOM
9054	S_INSP_TOP_OF_CAR
9055	S_INSP_IN_CAR
9056	S_HOISTWAY_ACCESS_CONTROL
9057	S_TEST_TRAVEL
9058	S_TEST_MODE
9059	S_LEARN_TRAVEL

Code	Car Alarm Messages
9060	S_INSPECTION_PREPARATION_TRAVEL
9061	S_OVERSPEED_GOVERNOR_RESET_TRAVEL
9063	S_OPERATION_WITH_DISABLED_CAR_LOAD_ MONITORING
9070	S_ELEVATOR_RECOVERY
9071	S_ELEVATOR_TEMPERATURE_RECOVERY
9072	S_ELEVATOR_CAR_POSITION_RECOVERY
9073	S_ELEVATOR_DOOR_POSITION_RECOVERY
9074	S_ELEVATOR_BACKUP_POWER_RECOVERY
9075	S_NO_OPERATION_DUE_TO_SAFETY_CHAIN_OPEN_ AT_ISPT
9080	S_STOP_SWITCH
9081	S_STOP_SWITCH_TOP_OF_CAR
9082	S_STOP_SWITCH_MACHINE_ROOM
9085	S_STOP_SWITCH_HOISTWAY_HEAD
9086	S_STOP_SWITCH_PIT
9089	S_NO_OPERATION_DUE_TO_DISABLED_MONITOR
9090	S_ELEVATOR_UNKNOWN_STATE
9091	S_ELEVATOR_STARTUP
9092	S_ELEVATOR_SUBSYSTEM_FIRMWARE_DOWNLOAD
9095	S_NO_OPERATION_DUE_TO_POWER_DOWN
9097	S_ELEVATOR_BREAKDOWN_PERSISTENT_LIMITED_O PERATION
9098	S_ELEVATOR_BREAKDOWN
9099	S_ELEVATOR_BREAKDOWN_PERSISTENT

9 Appendix D: Main Menu Structure

Menu	Menu function	Description section
[10_ _ _]	Special commands such as Reset, Open loop travel, Learning travel, Car calls, ...	see 4.9 / 9.1
[20_ _ _]	Automatic (assisted) acceptance tests	see 9.2
[30_ _ _]	System info	see 9.3
[40_ _ _]	Configuration	see 5.3
[50_ _ _]	Diagnostics, Error history	see 4.6 and 8
[60_ _ _]	Statistics	see 9.4
[70_ _ _]	ACVF monitoring	see 4.7

9.1 Commands (Menu 10)

For detailed information refer to section 4.9

9.2 Automatic Acceptance Tests (Menu 20)

Automatic (Assisted) Acceptance Tests

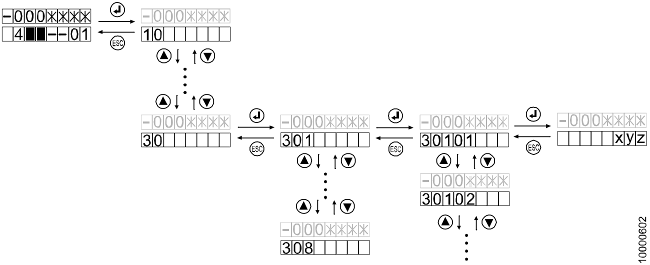
Automatic (assisted) acceptance test are part of the complete acceptance test and are described in the document J 139452 “Inspector’s Guide”.

Some of the tests can be used during the periodic preventive maintenance. These tests are described in the Quick Reference Guide K 609755 “Schindler 3100/300/5300, Preventive Maintenance”.

20	**Automatic (Assisted) Acceptance Test** Description see J 139452 and J41140148		
	- -	Test Trip	
	[__/__/14]	Brake Capability Downward	
	[__/__/15]	Brake Capability Upward	
	[__/__/26]	Car speed for door pre-opening	
	[__/__/32]	Car Impact on Buffer	1)
	[__/__/35]	Safety Gear Car	
	[__/__/37]	Run Time Limit	
	[__/__/42]	Counterweight Balancing	
	[__/__/44]	Ascending Car Overspeed Protection	
	[__/__/48]	Belt monitoring	
	[__/__/62]	Counterweight Impact on Buffer	1)
	[__/__/64]	Half Brake Downward	
	[__/__/65]	Half Brake Upward	
	[__/__/75]	Safety Gear Car (Empty Car)	
	[__/__/88]	KNE Top	
	[__/__/89]	KNE Bottom	

1) Used for oil buffers only

9.3 System Info (Menu 30)



[25787; 08.02.2008]

Overview

301	SW version (CF12)
302	HW version (CF13)
303	Door type
305	I/O inspection (since S00x V9.9)
306	ACVF monitoring, See also section 4.7.
308	ETM(A) Embedded Telemonitoring (Alarm) status
309	Overlay information
320	Telemonitoring menu (Since V9.8)
330	Last 5 STM installation data entries (Since 10.x)
340	Shaft info (Since V11.x)

Detailed descriptions

301	**Software version** – [301-01] SCPU (example: 95 = V9.5) – [301-02] SDIC – [301-03] ACVF – [301-04] SEM (not used, not SEM PCB!) – [301-05] COP #1 (example: 33 = V3.3) – [301-06] COP #2 – [301-08] CLSD (example: 1205 = V1.2.05) – [301-09] CPLD (SMIC61) (example: 108 = V1.08) – [301-10] Release version overlay (xx.xx.xx = “release”.”release”.”subrelease”) – [301-11] COP #3 – [301-12] COP #4 – [301-13] Bootloader (Since V9.83) – [301-14] Door drive side 1 (Varidor only) – [301-15] Door drive side 2 (Varidor only) – [301-16] SA EAS
302	**Hardware version** – [302-01] SCPU – [302-02] SDIC (51..58=SDIC5; 60,63=SDIC51; 61,64=SDIC52; 62,65=SDIC53) – [302-03] ACVF – [302-04] SEM (not used, not SEM PCB!) – [302-05] COP #1 - See table below – [302-06] COP #2 - See table below – [302-07] MMCx (not used) – [302-08] CLSD or ETMA (65..69 = CLSD, 49 = ETMA) – [302-09] SMIC (5 = SMIC5, 6 = SMIC6) – [302-10] Overlay – [302-11] COP #3 - See table below – [302-12] COP #4 - See table below – [301-14] Door drive side 1 (Varidor only) – [301-15] Door drive side 2 (Varidor only) – [301-16] SA EAS

9

0	unknown HW	**10 .. 40**	old COPs, not valid for Schindler 3100/3300/5300
51	COP5_N	**52**	COP5_10
53	COP5B_10 or COP5B_N	**54**	COP4_B (5 floors)
55	COP5 AP (any)	**56**	COP5_N ZLA
57	COP5_10 ZLA	**58**	COP5B_10 ZLA
59	unknown COP5 HW	**80**	SCOPH3
81	SCOPHM3	**82**	SCOPHMH3
83	SCOPMXB3	**90**	COP5B_10 AU
93	COP5 AP with EU fixtures	**94**	COP4_B_EU_8 (8 floors)
95	COP4_B_EU_12 (12 floors)	**96**	FIGS (any)
99	POP1.Q		

Values for 302-05, -06, -11, -12

303	**Door Type** – [303-1] Door Type Side 1 – [303-2] Door Type Side 2
306	**ACVF Monitoring** Same menu structure as menu 70 (701..734). Detailed description see section 4.7. [306-01] Actual elevator speed [306-02] Nominal linear speed [306-03] Encoder speed .. .. [306-36] Encoder offset autotuning result [306-37] Dynamic motor autotuning result

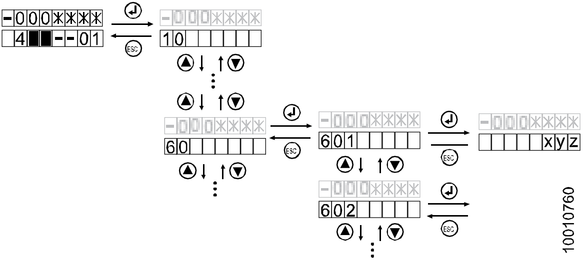
308	**ETM(A) Embedded Telemonitoring (Alarm) Statuses**
	[308-1] ETM(A) configuration status – 0 = ETM(A) not configured – 1 = ETM(A) configured **[308-2]** ETM(A) status – 0 = Undefined – 1 = Normal traffic – 2 = ETM(A) temporarily disabled (service visit) – 3 = ETM(A) temporarily for more than 24 hours – 4 = ETM(A) breakdown first fault. A breakdown was detected for the fist time – 5 = ETM(A) breakdown more faults. More than one breakdown is reported. – 6 = ETM(A) breakdown status active since more than 24 hours **[308-3]** ETM(A) communication status – 0 = Undefined – 1 = Modem initializing (temporary status) – 2 = Idle (modem initialized and ready) – 3 = Modem ringing and connecting (temporary status) – 4 = Connected (negotiation completed, temporary status) – 5 = Modem lost (not responding to "Alive?" polling) – 6 = Modem serial port deactivated (currently not used) – 7 = Modem owned by PPP (data transfer in progress) – 8 = RMP connected: PPP and SSL communication have been successfully established **[308-4]** ETM(A) phone line status – 0 = Undefined – 1 = No operational PSTN connected – 2 = Operational PSTN connected – 3 = Operational GSM connected – 4 = TA device is off-hook **[308-5]** ETM(A) phone line voltage (0..255, 1 = 1 Volt)

9

309	**Overlay information (SW ≥ 9.7x only)** (see also section 4.10, Overlay diagnostics) **[309-1]** Overlay Availability 1 = Persistent storage shows that the overlay was once detected as available. The value can be updated (e.g. cleared) by using the HMI command 10 > 136, in case no overlay is available. **[309-2]** Overlay Heartbeat 1 = The overlay board is properly communicating with the Bionic controller and the board is seen as available. The overlay attached to the controller seems to work properly. (Cabling between the overlay and the controller is ok) **[309-3]** Overlay Components Number = Amount of overlay boards that are "visible" from the Bionic controller. If the number of overlay components is equal to the number of elevators in the group: Ethernet communication among the overlay boards is ok. **[309-4]** Visible Elevators Number = Amount of elevators that are "visible" from the Bionic controller. If the number of visible elevators is equal to the number of elevators in the group: Ethernet communication among the overlay boards is working fine and the connection Overlay <=> Control is ok. The number can only be smaller or equal to the number in [309-3]. **[309-5]** Available Elevators Number = Amount of elevators available to serve group calls. The number can only be smaller or equal to the number in [309-4]. **[309-6]** Max Visible Elevators Number = Maximum number of available elevator since the system startup. This is used to verify the above value and trigger an error in case the number goes below the maximum number.

320	**ETM(A) (Available with SW ≥ 9.8 only) Details** – [320-1] Equipment number – [320-2] Servitel ID – [320-3] Date of last successful telemonitoring call date – [320-4] Date of last successful telealarm call date (not yet used).

9.4 Statistics (Menu 60)



[32339; 12.11.2009]

	Description	
601	Trip counter (1 = 100 trips)	
602	Hours in service (1 = 1 hour)	
603	Door cycle counter, side 1 (1 = 1 cycle)	2)
604	Door cycle counter, side 2 (1 = 1 cycle)	2)
605	Door cumulated moving time, side 1 (1 = 1 hour)	2)
606	Door cumulated moving time, side 2 (1 = 1 hour)	2)
610	Door cycle counter, side 1 (1 = 1 cycle)	3)
611	Door cycle counter, side 2 (1 = 1 cycle)	3)
612	Door cumulated moving time, side 1 (1 = 1 hour)	3)
613	Door cumulated moving time, side 2 (1 = 1 hour)	3)

9

	Description	
614	Door cumulated KSKB, side 1 (1 = 1 cycle)	3)
615	Door cumulated KSKB, side 2 (1 = 1 cycle)	3)
616	Door cumulated PHT, side 1 (1 = 1 cycle)	3)
617	Door cumulated PHT, side 2 (1 = 1 cycle)	3)
619	Clear door statistic data (610..617)	3)
620	Car light: Total on time (1 = 1 hour)	3)
621	Car light switch: Total switch on counter (1 = 1 cycle)	3)
622	Car light: Maximum on time (1 = 1 hour)	3)
623	Car light: Minimum on time (1 = 1 hour)	3)
624	Car light: Average on time (1 = 1 hour)	3)
629	Clear car light statistic (620..624	3)

1) not yet implemented

2) valid for SW < V9.6 only

3) Available if SW ≥ V9.7 and COP5 SW ≥ V3.0 or COP5-AP SW ≥ V1.5)

10 Appendix E: Spare parts

The table below is an extract from the official spare parts lists which can be found in:

– TK Maintenance - Control Bionic 5 / Bionic 6: EJ 604619
– ACVF Biodyn xx C BR TK Maintenance: K 609704

PCB, Object	Remarks	ID
SMIC61	Base PCB. (Can be replaced by SMICE61 or by SMIC(E)63, but KP must be bridged)	594154
SMIC63	Base PCB with KP connector. (Can be replaced by SMICE63)	594226
SMICE61	Base PCB with ETMA support. (Can be replaced by SMICE63, but KP must be bridged)	594303
SMICE63	Base PCB with ETMA support and KP connector.	594305
SMICE74	PCBA main interface controller without KP connector, with e-inspection	560158
SMICE75	PCBA main interface controller with KP connector, with e-inspection	560171
SMICFC	PCBA main interface controller without options (TSD21, NS21, SOA or DM236)	57813435
	PCBA main interface controller with options	57813436
SCPU1	Processor PCB (without SW)	591887
	Processor PCB (with SW)	1)
SEM11	Evacuation PCB (Replaced by SEM21)	594239
SEM21	Evacuation PCB	594157
ETMA-TRI	Triphonie module on the car roof	59700474

PCB, Object	Remarks	ID
ETMA-CAR	Slave ETMA module	59700563
ETMA-CAR-T	Additional car module	59700564
ETMA-MR-PSTN-1	Master module (1 interface to slave module)	59700570
ETMA-MR-PSTN-1-I	Additional master module	59700571
ETMA-MR-PSTN-2-I	Master module (2 interfaces to slave modules)	59700572
ETMA-LND-FF	Slave ETMA module	59700569
CLSD11	Modem / line switching device	594118
BAT (LDU)	Battery for emergency power supply and evacuation, 2 pieces (151x98x96)	432790
DC-AC	DC-AC inverter	55504585
Installation Travel on car	Kit for installation travel on car (with help of recall control and traveling cable)	55505064
Brake Test Connector	Brake test tool for Rel.≥4 (set of two different test connectors, “TEST-MGB”, TEST-MGB1”)	55505065
BESE connector	Substitutes the ESE (when not connected)	258656
GBP Reset Connector	Reset tool for GBP (to be plugged on SMIC.KBV)	55502805
	Car Control Unit CCU and Car Options	
SDIC5	Car interface PCB (For 0-series only. Can NOT be replaced by SDIC51/52/53)	591798
SDIC51	Car interface PCB, limited version (Can be replaced by SDIC52. Needs plug on JHC2 and 2KTC)	591884
SDIC52	Car interface PCB, full version	591885

PCB, Object	Remarks	ID
SDIC53	Car interface PCB, AP version	591886
SUET3	Door overbridging PCB	591811
SIEU1	Re-Leveling PCB. (Can be replaced by SIEU11, if SW ≥V9.7)	594224
SIEU11	Re-Leveling PCB, with additional inputs and outputs.	594306
LC	Fluorescent tube14W/827 HE	55502824
	Fluorescent tube 14W/840 HE	55502825
	Fluorescent tube 21W/827 HE	55503608
	Fluorescent tube 21W/840 HE	55503609
(LC)	Electronic ballast	55502822
LC	Energy saver lamp 14W/827 E27	55504047
	Energy saver lamp 14W/840 E27	55504048
	Energy saver lamp 18W/827 E27	55504196
	Energy saver lamp 18W/840 E27	55504197
GNT	Telealarm TM2-TAM2	59700110
	Car Fixtures	
COP5	COP5 complete (sensitive type, -3..8)	55503651
COP5_10 PI	COP5 complete for PI (sensitive type, 10 keypad)	55503710
COP5_10 KA/KS	COP5 complete for KA/KS (sensitive type, 10 keypad)	55503652
COP5B_10	COP5 complete (push-button type, 10 keypad)	55503653
COP5B_10 AU	COP5 complete (special version Australia, push-button type, 10 keypad)	55503412
COP5B set	Set of default push buttons for COP5B	55503550

PCB, Object	Remarks	ID
COP5B_N	Set of push buttons (-3,-2,-1, 5, blinds) for COP5B_N	55503480
COP5-1N 25 EU	COP5 complete for 25 floors (sensitive type -3..23)	55505240
COP5B-1N 25 EU	COP5 complete for 25 floors (push button type -3..23)	55505241
COP5K	COP key switch unit	55503482
CPI	Car Information Panel	55503481
COP4B	COP4B complete (max. 5 stops, 0..4)	55503970
COP4B_N	Set of push buttons (-2,-1, 5, blinds) for COP4B	55503979
SCOPB4.Q	PCB for COP4B	591897
COP4BE	COP4BE complete (max. 7 stops, 0..6)	55505242
SCOPBE4.Q	PCB for COP4BE	594236
CPI4	CPI4 car position indicator complete	55503990
VCA EU	Voice announcer complete, normal COP	55503509
VCA AP	Voice announcer complete, COP 25 EU	55503799
VCA11	Voice announcer PCB	591838
VCA11 MMC	MMC with language dependent voice file: See J 41322160 “Voice announcer”	
SASA1	Schindler access system PCB	591692
SAS cards	RFID card set (10 pieces) for SAS	55503450
SCOPMXB3	Dual brand COP interface PCB	591858
SCOPH3	Handicapped COP PCB	591854
	Landing Fixtures and Options	

PCB, Object	Remarks	ID
Cable kit	Converter cable: JST 4 poles ↔ WAGO 5 poles (0 series design)	55504168
Key switch cable	Cable to connect key switch to small JST connector pin 4 (only necessary if not ordered initially)	59321674
LOP5_1	LOP sensitive 1 button, JST 4 pin. (0 series, WAGO 5 pin: 59321389 replaced by 55503678 + 55504168)	55503678
LOPM5_1	LOP sensitive 1 button + display, JST 4 pin. (0 series, WAGO 5 pin: 59321390, replaced by 55503679 + 55504168)	55503679
LOP5_2	LOP sensitive 2 buttons, JST 4 pin. (0 series, WAGO 5 pin: 59321391 replaced by 55503680 + 55504168)	55503680
LOPM5_2	LOP sensitive 2 buttons + display, JST 4 pin. (0 series, WAGO 5 pin: 59321392 replaced by 55503681 + 55504168)	55503681
LOP5B_1	LOP mechanical 1 button, JST 4 pin. (0 series, WAGO 5 pin: 59321418 replaced by 55503684 + 55504168)	55503684
LOP5B_2	LOP mechanical 2 buttons, JST 4 pin. (0 series, WAGO 5 pin: 59321419 replaced by 55503685 + 55504168)	55503685
LOP4B	LOP4B with 1 push button	55503950
LOP4B-DM	LOP4B with SLDM4 for DM236, Italy only	55503999
LIN5V	Landing Indicator vertical (Can be replaced by LIN51V or LIN52V, but needs cable kit 55504168)	59321626

PCB, Object	Remarks	ID
Cable kit	Cable kit which is delivered if a LIN5V is replaced by a LIN51V or by a LIN52V.	55504168
LIN51V	Landing Indicator vertical (with input and output) (Can be replaced by LIN52V)	55505330
LIN52V	Landing Indicator vertical (with input, output and magnetic reed contact for configuration)	55506072
SLCUX1	Optional inputs / outputs (PCB only). Can be replaced by SLCUX2	591806
SLCUX2	Optional inputs / outputs (PCB only)	594212
LCUX	Optional inputs / outputs (complete unit with cables)	55502521
SBBD24	Duplex switching PCB	591796
GA	Arrival gong, JST standard 2 poles	59321646
Braille	Braille sticker set	55505112
SLCU2	Dual brand LOP interface (1 button)	591821
LCU2	Dual brand LOP interface (1 button) (complete unit with cables)	55511376
SLCUM2	Dual brand LOP interface (2 buttons and indicator)	591822
LCUM2	Dual brand LOP interface (2 button and indicator) (complete unit with cables)	55511377
BIOGIO1.N	BIO bus general input output PCB	594126
BIOGIO	BIOGIO complete unit in metal box	55505302
	ACVF Frequency Converter	
Cable kit	Upgrade kit: ACVF with option boards → ACVF with integrated I/O boards	59400895

PCB, Object	Remarks	ID
ACVF **Version 2005 with option boards**	Biodyn 12 C BR (complete unit) (Can be replaced by 59400864 + 59400895)	(55501728)
	Biodyn 19 C BR (complete unit) (Can be replaced by 59400865 + 59400895)	(55501729)
ACVF **Version with single board**	Biodyn 9 C BR (complete unit, EN12015:2004 compliant)	59400933
	Biodyn 12 C BR (complete unit) (Can be replaced by 59410012)	(59400864)
	Biodyn 12 C BR (complete unit, EN12015:2004 compliant)	59410012
	Biodyn 19 C BR (complete unit, EN12015:2004 compliant) (Can be replaced by 59400893)	(59400865)
	Biodyn 25 C BR (complete unit, EN12015:2004 compliant)	59400893
	Biodyn 42 C BR (complete unit, EN12015:2004 compliant)	59400868
	Miscellaneous	
ESE	Recall control station	434031
RS232	RS232 cable, Service PC ↔ Control (“CADI cable”)	59700078
RS232	Yellow RS232 connection cable, Service PC ↔ ACVF	55502100

1) Check the SW version on the label

Notes:

Schindler Management Ltd.
Corporate Learning and Development

Zugerstrasse 13
6030 Ebikon, Switzerland
Phone +41 41 445 32 32
Fax +41 41 445 39 33
trainingcenterebikon@ch.schindler.com
www.schindler.com